INTRODUCTION · FOUNDATIONS - ALL DOMAINS

Project Management Professional

A complete, original preparation course for the PMP exam as it has been since 9 July 2026.

EXAM **PMP, 2026 outline** QUESTIONS **180 in 240 min** APPROACHES **40% predictive**

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The PMP rarely asks what you know. It asks **what you would do next**.

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The exam in numbers

ITEM	DETAIL
Questions	180 in total: 170 scored, 10 unscored pretest items you cannot identify
Time	240 minutes, so about 80 seconds per question
Breaks	Two of 10 minutes: after the case-study section, then midway through the rest
After a break	You cannot go back to the questions before it
Delivery	Test centre, or online with a remote proctor
Attempts	Up to three within your one-year eligibility period
Fee	Rose in August 2026. Check the current price on pmi.org before booking

ON THE EXAM

Answer every question. Unscored items look identical, so treat all 180 as real.

The question types you will see

ITEM TYPE	WHAT YOU DO
Case or scenario - NEW	Read one detailed situation, often with a chart, then answer several questions on it
Graphic-based - NEW	Interpret a chart, graph or diagram, then answer
Multiple choice	Pick the single best answer
Multiple response	Pick every correct answer; the question says how many
Matching, enhanced matching	Drag items to pair them, sometimes onto a chart
Point and click	Click the right area of an image
Pull-down list	Choose the answer from a drop-down

ON THE EXAM

The two new types reward reading charts quickly. This course practises exactly that.

What changed on 9 July 2026

UNTIL 8 JULY 2026

The 2021 outline

- People 42, Process 50, Business Env. 8
- 35 tasks
- No case-study sets
- PMBOK 7: 12 principles, 8 domains

FROM 9 JULY 2026

The 2026 outline

- People 33, Process 41, Business Env. 26
- 26 tasks
- Case studies and chart items added
- PMBOK 8: 6 principles, 7 domains

THE HEADLINE

Business Environment more than tripled. Governance, compliance, risk and change now carry a quarter of the marks.

Three domains, twenty-six tasks

DOMAIN	WEIGHT	TASKS	WHAT IT IS ABOUT
I. People	33%	8	Vision, conflict, leading the team, stakeholders, knowledge transfer, communication
II. Process	41%	10	Planning, scope, value, resources, procurement, finance, quality, schedule, status, closure
III. Business Environment	26%	8	Governance, compliance, change, impediments, risk, improvement, organisational and external change

THE APPROACH MIX

About 40% of items are predictive. The other 60% are agile or hybrid, spread through every domain.

A warning about older material

- Material quoting **42 / 50 / 8** was written for the exam that retired on 8 July 2026.
- Many practice banks barely touch **AI, sustainability, cost of quality** or chart reading.
- PMBOK 8 has **40 processes** in five Focus Areas, not 49. The ideas are familiar; the labels changed.
- The judgement those banks teach still holds. This course keeps it and adds what they miss.

> ON THE EXAM
>
> If a resource disagrees with the 2026 outline, the outline wins.

How this course was built

- **1,414 practice questions analysed** to learn which decisions are tested, and how often.
- **Not one of them is reproduced.** Every scenario and practice question here is original.
- Explanations come from PMI's own sources: the 2026 Exam Content Outline, the PMBOK Guide 8th Edition, the Agile Practice Guide.
- The analysis found patterns worth knowing. One: answers that **escalate** were right 6 times in 143.

> ON THE EXAM
>
> Learn the reasoning, not the questions. The exam will not reuse anything you memorise.

One project runs through the course

- **Project Lantern** at Northgate Health replaces patient booking across six clinics.
- Hardware, data migration and regulatory approval run **predictively**. The booking app is built by an **agile** team.
- Teaching examples reuse it, so you watch one hybrid project grow through all 26 tasks.
- Worked scenarios use many industries, so no single setting feels like the answer key.

> ON THE EXAM
>
> One hybrid project, seen from every angle, is exactly what the 60 percent of agile and hybrid questions look like.

Reading the slides

01

Amber

The key fact or rule.

02

Teal

What good practice looks like.

03

Red

The trap, or the tempting wrong answer.

04

The callout

The line to remember for the exam.

ON THE EXAM

When time is short, the callout bars alone are a usable revision set.

When a scenario appears, **pause and commit** to an answer before the walkthrough.

The course map

MODULES	SECTION	WHAT YOU GET
00 - 05	Foundations	The 2026 map, thinking like PMI, approaches, Scrum, Kanban
06 - 16	People	Leadership, teams, conflict, stakeholders, communication
17 - 30	Process	Plan, scope, value, schedule, money, suppliers, quality, closure
31 - 40	Business Env.	Governance, compliance, change, issues, risk, org. change
41	Trends	AI and sustainability in project work
42 - 45	Exam skills	Confusion pairs, numbers and graphs, a full case study
46 - 49	Practice exams	Four timed papers of 20, each opening with a case study

ON THE EXAM

Do the foundations first. Module 02 changes how you read every question.

A study route that moves your score

01

Watch in order

Later modules assume earlier vocabulary.

02

Pause and commit

On every scenario, every time.

03

Sit exam 1

Timed, after module 45. No notes.

04

Rewatch weak areas

Only the domain where misses cluster.

05

Space the rest

One paper every few days, then book.

ON THE EXAM

Revising what you already know feels productive because the scores stay high. It moves nothing.

Before you book

- **Experience:** 36 months leading projects with a bachelor's degree, 60 with secondary education, within the last 10 years.
- **35 hours of project management training** aligned to the outline. Waived for active CAPM holders.
- Books and practice exams alone do not count. Training is subject to audit, so keep evidence.
- After you pass: 60 professional development units every three years to keep it.

> ON THE EXAM
>
> Check how any training you use will be evidenced before you apply, not after an audit asks.

MODULE 01 · FOUNDATIONS - ALL DOMAINS

The 2026 PMP and the PMBOK 8 map

What the exam outline tests, how the new PMBOK Guide is organised, and how the two fit together.

DOMAIN Foundations EXAM WEIGHT All domains MODULE 01 of 49

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The outline is the **syllabus**. The PMBOK Guide is the **textbook**.

How the exam outline is built

01

Domain

A broad area of practice. Three of them, each with a fixed weight.

02

Task

A responsibility of the project manager. Twenty-six in total.

03

Enabler

An example of the work inside a task. Illustrative, not a complete list.

ON THE EXAM

Every exam form covers every task. Nothing on the outline is safe to skip.

Domain I – People: eight tasks

#	TASK	IN ONE LINE
1	Develop a common vision	Everyone understands the same goal, and it stays current
2	Manage conflicts	Find the source, agree a resolution, keep ground rules alive
3	Lead the project team	Set expectations, empower, solve problems, pick a leadership style
4	Engage stakeholders	Identify, analyse, tailor communication, build trust
5	Align stakeholder expectations	Categorise, surface expectations, facilitate agreement, mentor
6	Manage stakeholder expectations	Internal and external customers, satisfaction over time
7	Help ensure knowledge transfer	Identify critical knowledge, gather it, make sharing normal
8	Plan and manage communication	Strategy, transparency, feedback loops, reporting

ON THE EXAM

Stakeholders and communication now live here. In 2021 they were Process tasks.

Domain II - Process: ten tasks

#	TASK	IN ONE LINE
1	Integrated plan and plan delivery	Choose the approach, build and maintain one integrated plan
2	Develop and manage scope	Define it, agree it, break it down
3	Help ensure value-based delivery	Prioritise by value, deliver incrementally, track benefits
4	Plan and manage resources	Plan them from requirements, optimise availability
5	Plan and manage procurement	Contract types, negotiation, supplier performance
6	Plan and manage finance	Budgets, reserves, spend tracking, financial reporting
7	Plan and optimise quality	Quality requirements, compliance, cost of quality
8	Plan and manage schedule	Estimate, baseline, analyse variation
9	Evaluate project status	Metrics, artefacts, progress, status reporting
10	Manage project closure	Acceptance, transition readiness, closing activities

ON THE EXAM

Value-based delivery is a Process task now, not a Business Environment one.

Domain III - Business Environment: eight tasks

#	TASK	IN ONE LINE
1	Define and establish governance	Structure, rules, ethics, success metrics, escalation paths
2	Plan and manage compliance	Security, safety, sustainability, regulation
3	Manage and control changes	Run the change process, communicate, update records
4	Remove impediments, manage issues	Clear blockers, spot when a risk has become an issue
5	Plan and manage risk	Identify, analyse, respond, monitor, communicate
6	Continuous improvement	Use lessons learned, update process assets
7	Support organisational change	Assess culture, plan for the impact of change
8	Evaluate external changes	Regulation, technology, geopolitics, markets

ON THE EXAM

Governance, change, issues and risk moved here. That is most of the jump from 8% to 26%.

Where the 2021 tasks went

MOVED INTO PEOPLE

Relationship work

- Manage communications
- Engage stakeholders
- Knowledge transfer

MOVED INTO BUSINESS ENV.

Control and context

- Governance structure
- Risks, changes, issues
- Impediments and blockers
- Continuous improvement

THE TIEBREAKER

If older material files a topic elsewhere, the judgement is the same. Only the domain label moved.

How the PMBOK Guide 8th Edition is built

01
6 principles
How you think. They guide behaviour.

02
7 domains
Where you focus. Areas of practice.

03
5 focus areas
When. Initiating through closing.

04
40 processes
What you do. Illustrative, not mandatory.

05
Tailoring
Fit all of it to the context.

ON THE EXAM

Principles are how you think. Domains are where you look. Processes are what you might do.

The six principles, and how they show up in answers

PRINCIPLE	IN AN EXAM ANSWER, IT LOOKS LIKE
Adopt a holistic view	Considering the knock-on effects before acting on one symptom
Focus on value	Choosing the option that delivers outcomes, not just outputs
Embed quality into processes and deliverables	Preventing defects rather than inspecting them out
Be an accountable leader	Owning the problem instead of passing it up the line
Integrate sustainability within all project areas	Weighing environmental, social and economic impact
Build an empowered culture	Letting the team decide how to do the work

THE TIEBREAKER

When two answers both work, the one that honours more of these principles usually wins.

Two principles that are new in spirit

- **Integrate sustainability** is new as a core principle. It also appears in the outline's planning, quality, compliance and risk tasks.
- It means more than the environment: social and economic effects count, over the life of the product.
- **Be an accountable leader** sharpens the old stewardship idea: accountability for outcomes, not just for activity.
- Passing a decision up to the sponsor because it is uncomfortable is not accountability.

> ON THE EXAM
>
> Sustainability is no longer a niche topic. Expect it inside ordinary planning and risk questions.

The seven performance domains

DOMAIN	PROCESSES	ABSORBED FROM OLDER EDITIONS
Governance	9	Integration, plus parts of quality and the sourcing strategy
Scope	6	Scope, requirements, and the rest of quality
Schedule	3	Schedule, with four old processes merged into one
Finance	4	Cost, widened to funding and value
Stakeholders	7	Stakeholder engagement and communications, now together
Resources	5	Team and physical resources
Risk	6	Risk, with the two analysis processes merged

ON THE EXAM

Communications and stakeholders are one domain now. Procurement detail moved to an appendix.

Five focus areas: the old process groups, renamed

01
Initiating
Authorise, set direction, identify stakeholders.

02
Planning
Decide how. Half of the forty processes live here.

03
Executing
Do the work, lead the team, engage people.

04
Monitoring and controlling
Compare, correct, manage change.

05
Closing
Accept, transfer, learn, release.

ON THE EXAM

In agile work the focus areas repeat every iteration. They are not a one-way waterfall.

Translating older terms

IN OLDER MATERIAL	IN PMBOK 8
Process groups	Focus areas
Ten knowledge areas	Seven performance domains
Collect requirements	Elicit and analyze requirements
Define activities, sequence, estimate durations, develop schedule	Develop schedule, one process
Qualitative and quantitative risk analysis	Perform risk analysis, one process
Develop team and manage team	Lead the team
Control costs	Monitor and control finances

ON THE EXAM

The exam tests the thinking, not the process names. Recognise both vocabularies and move on.

A project is a temporary initiative in a unique context, undertaken **to create value**.

Two kinds of success

PROJECT MANAGEMENT SUCCESS

Delivered as planned

- On time
- Within budget
- Agreed scope
- Required quality

PROJECT SUCCESS

Worth the effort

- Stakeholders see the value
- Intended outcomes achieved
- Worth the time and money spent
- Judged by the people it serves

THE TIEBREAKER

A project can hit every baseline and still fail. PMI now asks you to own both kinds of success.

M.O.R.E.: how PMI says you get project success

01

Manage perceptions

Value must be seen by stakeholders, not just delivered.

02

Own project success

Accountable for value, beyond managing tasks.

03

Relentlessly reassess

Revisit goals and parameters as things change.

04

Expand perspective

See the organisation and the wider world.

ON THE EXAM

These four habits sit behind many 'what should the project manager do' answers.

From outputs to value

01

Output

The thing produced: an app, a building, a policy.

›

02

Outcome

The change it makes possible: faster booking.

›

03

Benefit

The measurable gain: fewer missed appointments.

›

04

Value

Worth to the organisation and its people.

ON THE EXAM

An output nobody uses produces no outcome. Plan and measure all the way to the right.

Tailoring: nothing is compulsory

- Every principle, process and artefact is **adapted to the context**: size, risk, approach, culture, regulation.
- Tailoring is done deliberately, with the people involved, and revisited as the project learns.
- Using every template 'because it exists' is not good practice. Neither is skipping governance 'because we are agile'.
- Organisational rules still apply. Tailor within them, and propose changes through the right route.

> ON THE EXAM
>
> The best answer usually adapts the process to the project, with agreement, rather than obeying or ignoring it.

How to use the PMBOK Guide for this exam

- Learn the **ideas** behind each domain and principle, well enough to apply them to a story.
- Do not memorise inputs, tools and outputs. The exam does not ask for them.
- Read the Agile Practice Guide alongside it. Sixty percent of items are agile or hybrid.
- Keep the **outline's task list** as your checklist. That is what is weighted and scored.

> ON THE EXAM
>
> Study to apply, not to recite. Every question is a situation, not a definition.

Project Lantern on the map

- **People:** a booking-app team, six clinic managers, patients, a regulator, a hardware supplier.
- **Process:** a hybrid plan, a product backlog, a fixed-price hardware contract, a budget with reserves.
- **Business Environment:** health-data rules, a steering committee, new clinic workflows, a supplier facing tariffs.
- One project touches all three domains every week. So do exam questions.

> ON THE EXAM
>
> Real questions blend domains. Name the main one, but expect the others in the background.

A project to introduce a new expense-claims system finished on time and under budget. Every requirement in the scope statement was accepted.

Three weeks later, most staff are still emailing paper forms to finance. The sponsor thanks the project manager and asks them to close the project so the team can move on.

What should the project manager do?

A Close the project, because every baseline and acceptance criterion was met.

B Raise a change request to add the features users have asked for.

C Record the low adoption in the lessons learned and archive the project.

D Assess with the sponsor why adoption is low, and agree how the intended benefits will be secured.

◆ PAUSE AND DECIDE

What should the project manager do?

A Close the project.

Project management success only. The outputs exist, but the outcome has not happened.

B Raise a change request.

Jumps to a solution before anyone knows why people are not adopting it.

C Record it and archive.

Documents the failure instead of acting while there is still time to fix it.

D Assess with the sponsor, agree how benefits are secured.

Owns project success: finds the cause and protects the value, with the sponsor.

THE SIGNAL

Baselines met, value missing. The exam wants you to own the outcome, not just the output.

> A PMO requires every project to complete its full set of predictive templates, including a detailed schedule baseline and monthly earned value reports.
>
> A project manager is starting a six-week internal project. A small team will build a reporting tool, and the users want to see working versions every week and change priorities as they learn.

What should the project manager do?

A. Complete every PMO template as required, then run weekly demonstrations on top.

B. Propose a tailored approach to the PMO that meets its governance needs with lighter agile artefacts.

C. Skip the templates, because agile projects do not need that documentation.

D. Ask the sponsor to exempt the project from PMO requirements.

◆ PAUSE AND DECIDE

What should the project manager do?

A Complete every template, add demos.

Heavy overhead for six weeks, and a fixed baseline fights the changing priorities.

B Propose a tailored approach to the PMO.

Adapts the process to the context, with agreement, while respecting governance.

C Skip the templates.

Ignores a real organisational rule. Agile still needs governance and records.

D Ask the sponsor for an exemption.

Goes around the PMO instead of working with it, and assumes all or nothing.

THE SIGNAL

The rule does not fit the project. Tailor it, openly and through the right route.

Know the outline's tasks. Think in principles. Tailor everything else.

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MODULE 02 · FOUNDATIONS - ALL DOMAINS

Thinking like PMI

What right answers have in common, the ladder that orders them, and how to read a question before the options.

DOMAIN Foundations EXAM WEIGHT All domains MODULE 02 of 49

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Most wrong answers are good actions **at the wrong moment**.

What the analysis found

PHRASE IN AN OPTION	CORRECT / APPEARED	WIN RATE (BASELINE 26%)
empower	12 / 16	75%
assess	68 / 103	66%
collaborate	29 / 45	64%
sponsor	40 / 367	11%
escalate	6 / 143	4%
replace	1 / 52	2%
overtime	0 / 15	0%

ON THE EXAM

Options that escalate were right 6 times in 143. Options that assess were right two times in three.

What wins, and what loses

WINS FAR ABOVE CHANCE

Understand and involve

- Assess, analyse, evaluate
- Find the root cause
- Meet with, collaborate, facilitate
- Empower, coach, mentor

WINS FAR BELOW CHANCE

Hand off or impose

- Escalate, involve the sponsor early
- Tell, direct, instruct
- Replace, hire, add overtime
- Update, document, inform

THE TIEBREAKER

PMI's project manager understands first, involves people second, and hands problems upward last.

The answer ladder

01
Understand
Get the facts. What actually happened?

02
Assess
Analyse impact and root cause.

03
Collaborate
Work it through with the people involved.

04
Act
Use the plan and your authority to respond.

05
Escalate
Only when it is beyond your authority.

ON THE EXAM

Find the lowest rung the story has not already completed. That is usually the answer.

What each rung sounds like in an option

RUNG	TYPICAL WORDING
1. Understand	Review the details. Clarify with. Find out why.
2. Assess	Assess the impact. Analyse the root cause. Evaluate options.
3. Collaborate	Meet with. Facilitate a discussion. Work with the team to.
4. Act	Implement the planned response. Follow the change process. Coach.
5. Escalate	Escalate to. Refer to the sponsor. Ask management to decide.

ON THE EXAM

Learn to hear the rung in the verb. It lets you rank four options in seconds.

When escalation is the right answer

- The decision exceeds your **authority or an agreed threshold**, and you have already analysed it.
- The governance route says so: a change above your limit goes to the **change control board** or sponsor.
- A **legal, safety or ethical** breach that you cannot resolve yourself.
- Not because the problem is uncomfortable, new, or involves a senior person.

> ON THE EXAM
>
> Escalation arrives with analysis and options attached. It is never the first thing you do.

Read the question word first

THE STEM ASKS	WHAT IT WANTS
What should the PM do FIRST?	The lowest rung. Usually understand or assess
What should the PM do NEXT?	The rung after the one the story says is done
What should the PM have done?	The prevention, usually a planning-stage action
What is the BEST way to…?	The option that works and honours the principles
Which two actions…? (Choose two)	Two independent correct actions, not one repeated

ON THE EXAM

In this bank, 'do next' and 'do first' together appeared over 160 times. Read the tense before the options.

The PMI project manager, as a person

- A **servant leader** who removes obstacles and lets the team decide how to work.
- **Proactive**: plans for risks, sets ground rules early, reads the plan before inventing a fix.
- Talks **face to face and privately** about people problems.
- Never hides bad news, never ignores a problem, never acts unilaterally on someone else's decision.

> ON THE EXAM
>
> When stuck, ask what this person would do. It is rarely the most forceful option.

Sounds decisive, or is decisive?

SOUNDS DECISIVE

Imposes a fix

- Instruct the team to...
- Tell the vendor that...
- Replace the team member
- Direct everyone to work overtime

IS DECISIVE

Solves the real problem

- Find out why it is happening
- Meet the vendor to agree recovery
- Speak privately, then coach
- Re-plan with the team

THE TIEBREAKER

Forceful options feel like leadership under time pressure. On this exam they are usually traps.

The plan often already holds the answer

- A risk has occurred? Check the **risk register** for a planned response before inventing one.
- Someone wants a change? The **change process** or the product backlog decides the route.
- Unclear who gets what information? The **communications plan** says.
- But 'update the register' or 'document it' is usually the second step, not the answer.

> ON THE EXAM
>
> Use the plan to act. Updating the plan after acting is housekeeping.

The agile shift in who decides

- The **product owner** decides priorities. The project manager does not reorder the backlog.
- The **team** decides how to do the work and how much it can take on.
- The project leader **coaches, facilitates and removes impediments**.
- Answers where the PM assigns tasks or reprioritises for an agile team are almost always wrong.

> ON THE EXAM
>
> In agile questions, first ask whose decision it is. Then choose the option that supports that person.

A routine for every question

01

Setting

Predictive, agile or hybrid? Who are you?

02

Question word

First, next, should have, best.

03

Done so far

Which rungs has the story completed?

04

Planted detail

The sentence that rules options out.

05

Eliminate

Unethical, ignoring, unilateral, too early.

ON THE EXAM

Most questions are decided by one planted sentence. Find it before you look at the options.

The ladder is a default, **not a law**. A planted detail can move you up it.

> An agile team's daily stand-up keeps overrunning because two senior developers argue about whether to rewrite a shared component or patch it.
>
> Other team members have stopped raising their own blockers, and the sprint's work has started to slip.

What should the project leader do?

A Decide between the two approaches, so the team can move on.

B Escalate the disagreement to the developers' functional manager.

C Meet with both developers to understand the positions and facilitate a team decision.

D Ask the product owner to choose the approach.

◆ PAUSE AND DECIDE

What should the project leader do?

A Decide the approach.

A technical 'how' decision belongs to the team. Imposing it breaks self-organisation.

B Escalate to the functional manager.

Too high on the ladder. Nobody has tried to resolve it yet.

C Meet both, facilitate a team decision.

Rungs 1 and 3: understand, then collaborate. The team keeps ownership.

D Ask the product owner to choose.

The product owner owns what and why, not how the code is built.

THE SIGNAL

A 'how' disagreement inside an agile team is the team's to solve. The leader facilitates.

> On a predictive factory fit-out, a supplier emails the project manager: a batch of control panels will arrive three weeks late.
>
> The panels feed the installation work package. The sponsor's monthly review is in two days.

What should the project manager do first?

A Assess the delay's impact on the schedule and critical path.

B Inform the sponsor that the project will finish three weeks late.

C Crash the installation work package to recover the time.

D Submit a change request to rebaseline the schedule.

◆ PAUSE AND DECIDE

What should the project manager do first?

A Assess the impact on the critical path.

Rung 2. Until you know whether the work has float, you do not know if there is a delay at all.

B Tell the sponsor it will be three weeks late.

Reports an assumption as fact. A late input is not automatically a late project.

C Crash the installation work.

Spends money on a fix before knowing whether a fix is needed.

D Request a rebaseline.

Far too early, and rebaselining is a last resort, not a first response.

THE SIGNAL

'First' plus a new problem means assess. A late input only delays the project if it is on the critical path.

A new data-retention regulation affects a hybrid records project. The project manager and team have analysed it: two new features are needed, adding 8% to the budget.

The governance plan lets the project manager approve changes up to 5% of budget. Above that, changes go to the change control board.

What should the project manager do next?

A Approve the change and fund it from the contingency reserve.

B Ask the team to absorb the features within the current budget.

C Defer the features to a later release to protect the budget.

D Submit a change request with the impact analysis to the change control board.

◆ PAUSE AND DECIDE

What should the project manager do next?

A Approve it, use contingency.

Exceeds the 5% authority. Acting on it anyway is unilateral.

B Ask the team to absorb it.

Hides a real cost and pressures the team. Unrealistic and not transparent.

C Defer to a later release.

A regulation is not optional. Deferring may put the organisation out of compliance.

D Change request to the board, with analysis.

Rung 5, done properly: analysis complete, decision above your threshold.

THE SIGNAL

Analysis done, and above your threshold. This is the case where going up is correct.

> Two months into a project, a developer in Manila keeps missing the daily call that the rest of the team, in London, holds at 5pm.
>
> The project manager learns that 5pm in London is late at night in Manila, and the developer has been catching up from notes.

What should the project manager have done to prevent this?

A Reported the attendance problem to the developer's line manager.

B Agreed meeting times and working norms with the whole team, allowing for time zones, when forming the team.

C Recorded the missed meetings in the issue log.

D Recruited team members based in the same time zone.

◆ PAUSE AND DECIDE

What should the project manager have done to prevent this?

A Reported it to the line manager.

A reaction, not a prevention, and it blames the person for a scheduling problem.

B Agreed times and norms early, allowing for time zones.

A planning-stage action that removes the cause. This is what the team charter is for.

C Logged it as an issue.

Records the problem after it happens. Prevents nothing.

D Recruited in one time zone.

Usually not the PM's choice, and it avoids rather than manages a distributed team.

THE SIGNAL

'Should have done' points back to planning. Ground rules agreed together prevent most team friction.

> A project's schedule performance index is 0.82 and falling. The sponsor asks the project manager to show the project as green in this month's board report.
>
> The sponsor explains that a red status now would cause a funding review at the worst possible moment.

What should the project manager do?

A Report the project as green, as the sponsor has requested.

B Report the project as amber as a compromise.

C Explain to the sponsor that the report must show the true status, and bring a recovery plan to present with it.

D Report the sponsor's request to the PMO.

◆ PAUSE AND DECIDE

What should the project manager do?

A Report green as asked.

Knowingly false reporting. Breaches honesty, whoever asks for it.

B Report amber as a compromise.

Still inaccurate. Honesty is not negotiable by degrees.

C True status, discussed with the sponsor, with a recovery plan.

Honest, and it gives the sponsor something constructive to take to the board.

D Report the sponsor to the PMO.

Premature. Talk to the sponsor directly first; escalate only if pressure continues.

THE SIGNAL

Ethics answers are not negotiable, but they are still delivered collaboratively. Talk first, then report truthfully.

> A reliable senior analyst has started delivering late, error-filled work over the past month. Colleagues have begun redoing parts of it.
>
> The analyst has not raised any problems in team meetings.

What should the project manager do?

A Meet the analyst privately to understand what has changed.

B Add a mandatory review step for all of the analyst's work.

C Ask the resource manager for a replacement analyst.

D Raise the quality problem at the next team meeting.

◆ PAUSE AND DECIDE

What should the project manager do?

A Meet privately to understand what changed.

Rung 1, in the right setting. A sudden change in a reliable person has a cause.

B Add mandatory reviews.

Treats a symptom and signals distrust, without knowing the cause.

C Request a replacement.

Replacing people won 2% of the time. Nothing has been tried yet.

D Raise it at the team meeting.

Public criticism of an individual damages trust. People problems are private.

THE SIGNAL

A sudden change in a good performer means something changed. Find out privately before acting.

> Midway through a two-week sprint, the product owner asks the developers directly to start three new items a customer has requested.
>
> The items are not urgent. The team is worried it will not meet the sprint goal it committed to.

What should the project leader do?

A Tell the team to add the items, since the product owner owns priorities.

B Escalate to the sponsor that the product owner is disrupting the sprint.

C Tell the product owner that no changes are ever allowed once a sprint starts.

D Meet the product owner to discuss the sprint goal, and add the items to the product backlog for refinement.

◆ PAUSE AND DECIDE

What should the project leader do?

A Tell the team to add them.

The PO owns the product backlog, but the developers own the sprint's work plan.

B Escalate to the sponsor.

A coaching conversation has not happened yet. Far too early.

C No changes, ever.

Too rigid. Scope can be renegotiated if it does not endanger the sprint goal.

D Meet the PO, protect the goal, route items to the backlog.

Rung 3: collaborates and coaches, and uses the backlog as the change route.

THE SIGNAL

Non-urgent new work goes to the product backlog. The sprint goal is protected by conversation, not by rules.

> Four months into a building refurbishment, the client's new head of facilities management attends a site meeting for the first time.
>
> She says the planned plant-room layout will make future maintenance difficult. She was not in the stakeholder register.

What should the project manager do?

A Explain that the design is baselined and cannot be reopened.

B Meet her to understand her concerns, then add her to the stakeholder register and plan her engagement.

C Escalate her objection to the sponsor for a decision.

D Raise a change request to redesign the plant room.

◆ PAUSE AND DECIDE

What should the project manager do?

A. The design is baselined.
Dismisses a stakeholder who will live with the result. Baselines can change.

B. Meet her, then register and engage her.
Rungs 1 and 3: understands the concern, then brings her properly into the project.

C. Escalate to the sponsor.
Nobody knows yet whether her concern is valid or what it would cost.

D. Raise a change request now.
Commits to a redesign before the concern is understood. It may come later.

THE SIGNAL

A newly discovered stakeholder is engaged, not argued with or escalated. Understand before you change anything.

Understand, assess, collaborate, act, and only then **escalate**.

MODULE 03 · FOUNDATIONS - ALL DOMAINS

Development approaches, life cycles and tailoring

Predictive, iterative, incremental and adaptive; how to choose between them; and how to fit any of them to the project in front of you.

DOMAIN Foundations EXAM WEIGHT All domains MODULE 03 of 49

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The approach is chosen, not inherited. It follows the uncertainty.

Four life cycles

LIFE CYCLE	WHAT IT DOES
Predictive	Scope, time and cost are planned early. Work flows through phases to one main delivery
Iterative	Scope is set early, but the solution is refined through repeated cycles of feedback
Incremental	Finished pieces are delivered one after another, each adding usable capability
Adaptive, or agile	Both at once: short, fixed timeboxes that deliver finished increments and refine what comes next

ON THE EXAM

Agile is iterative AND incremental. That combination is what makes it adaptive.

How the four differ

	REQUIREMENTS	ACTIVITIES	DELIVERY	AIMS TO OPTIMISE
Predictive	Fixed early	Once for the project	Single	Cost and control
Iterative	Dynamic	Repeated until right	Single	Correctness of solution
Incremental	Dynamic	Once per increment	Frequent, smaller	Speed
Adaptive	Dynamic	Repeated until right	Frequent, small	Value from feedback

ON THE EXAM

Read a scenario for two clues: how fixed the requirements are, and how often something is delivered.

Iterative or incremental?

ITERATIVE

Sketch, then refine the whole

- A rough version of everything
- Improved in each pass
- Nothing usable until the end
- Feedback shapes the solution

INCREMENTAL

Finish one piece at a time

- One complete section first
- Then the next section
- Each piece usable on arrival
- Speed to first value

THE TIEBREAKER

Iterative improves the same thing repeatedly. Incremental adds new finished things.

Uncertainty points to the approach

REQUIREMENTS ARE...	HOW TO BUILD IT IS...	USUALLY SUITS
Clear and stable	Well understood	Predictive
Clear and stable	Uncertain	Iterative, prototypes, spikes
Uncertain or changing	Well understood	Incremental or adaptive
Uncertain or changing	Uncertain	Adaptive, with short feedback loops

ON THE EXAM

The more uncertainty on either axis, the shorter the feedback loop you want.

What else shapes the choice

FACTOR GROUP	EXAMPLES
The product	Innovation, requirement certainty, ease of change, deliverable in pieces?
Safety and rules	Safety needs, regulation, approvals that need a fixed design
The project	Stakeholder availability, schedule constraints, how funding is released
The organisation	Culture, structure, agile experience, team size and location
Contracts	Fixed price favours fixed scope; outcome-based allows change

ON THE EXAM

A product that cannot be delivered in pieces, like a bridge, limits how adaptive the build can be.

When predictive is the better choice

- Requirements are **well understood and unlikely to change**.
- Change after build is **expensive or impossible**: construction, hardware, physical installation.
- Regulators or contracts need an **approved design and baseline** before work starts.
- Not because 'that is how we always do it', and not because agile feels unfamiliar.

> ON THE EXAM
>
> Predictive is not old-fashioned. It is the right tool when the answer is already known.

When adaptive is the better choice

- Requirements are **unclear, evolving or discovered** by using the product.
- The product can be **delivered in usable slices**, and early value matters.
- Customers or users can **give frequent feedback**, and the team can act on it.
- Not when nobody from the business can take part. Agile without feedback is just unplanned work.

> ON THE EXAM
>
> Adaptive needs an engaged customer. Without feedback, short cycles have nothing to learn from.

Hybrid is not one thing. It is a continuum, chosen piece by piece.

Common ways approaches are combined

PATTERN	EXAMPLE
Adaptive build, predictive roll-out	An app is built in sprints, then deployed to 40 branches on a fixed plan
Side by side	Hardware installed predictively while an agile team builds the software
Mostly predictive, adaptive parts	A building project uses iterations only for its interior design choices
Mostly adaptive, predictive parts	An agile product team with one fixed regulatory submission milestone

ON THE EXAM

Choose the approach per workstream, by its uncertainty. Then plan how the pieces integrate.

Project Lantern on the continuum

- **Predictive:** kiosk hardware, network upgrades in six clinics, data migration, the regulator's approval.
- **Adaptive:** the patient booking app, built in two-week iterations with clinic staff and patient panels.
- **Integration points:** a fixed go-live window per clinic, and a release plan that lines up with it.
- The steering committee sees one integrated plan, with a roadmap for the app and a baseline for the rest.

> ON THE EXAM
>
> One integrated plan, two ways of working inside it. That is what 'hybrid' means on the exam.

Delivery cadence

CADENCE	WHAT IT MEANS	TYPICAL FIT
Single delivery	Everything arrives at the end	Predictive construction, events
Multiple deliveries	Several planned releases or phases	Phased roll-outs, incremental builds
Periodic delivery	Releases on a fixed rhythm	A release at the end of every iteration
Continuous delivery	Increments released as soon as they are ready	Digital products with automated pipelines

ON THE EXAM

Cadence is a separate choice from life cycle. Match it to how soon users can benefit.

The agile mindset: four values

WE VALUE...	...OVER
Individuals and interactions	Processes and tools
Working software	Comprehensive documentation
Customer collaboration	Contract negotiation
Responding to change	Following a plan

> ON THE EXAM
>
> The items on the right still have value. The items on the left have more. Not 'instead of'.

Agile principles the exam leans on

- **Early and continuous delivery** of valuable work satisfies the customer.
- **Welcome changing requirements**, even late, when they give a competitive advantage.
- **Motivated, self-organising teams**, given support and trusted to get the job done.
- **A sustainable pace**, and regular reflection to become more effective.

> ON THE EXAM
>
> An answer that burns the team out or imposes their method contradicts the principles, whatever it promises.

Tailoring, in four moves

01

Select the approach

From uncertainty and the suitability factors.

02

Tailor for the organisation

Respect governance, policy and required controls.

03

Tailor for the project

Size, risk, team, stakeholders, product.

04

Keep improving

Review what works; adjust in retrospectives.

ON THE EXAM

Tailoring never ends at kickoff. The team keeps adjusting how it works as it learns.

What can be tailored

ELEMENT	FOR EXAMPLE
Life cycle and cadence	Phase gates, iteration length, release rhythm
Processes	How much change control, how formal the risk process is
Engagement	Who attends which reviews; how decisions are made
Tools	Boards, scheduling software, collaboration platforms
Methods and artefacts	Which documents, metrics and reports are produced

ON THE EXAM

Tailor with the people affected, and within organisational rules. Unagreed tailoring is just rule-breaking.

Updated agile vocabulary

- **Backlog refinement** replaces 'backlog grooming' in the Agile Practice Guide, Second Edition.
- **Daily coordination meeting** is the guide's framework-neutral name for the daily stand-up.
- Iterations are not always called sprints, and facilitators are not always called scrum masters.
- Questions may use either name. Recognise the practice, not the label.

> ON THE EXAM
>
> The exam is framework-neutral. Expect 'iteration', 'team facilitator' and 'coordination meeting' as often as Scrum terms.

Reading the approach from the question

WORDS IN THE SCENARIO	POINTS TOWARDS
Baseline, phase gate, WBS, change control board	Predictive
Iteration, backlog, product owner, increment, demo	Adaptive
Fixed regulatory milestone alongside iterations	Hybrid
No clue at all	Judge from uncertainty and the principles

ON THE EXAM

Most questions do not name the approach. Decide it from the clues before reading the options.

> A retailer wants a mobile app for a new subscription service it has never offered before. Nobody is sure which features customers will value.
>
> The marketing director wants to see something working within a month, and customers from a pilot group have agreed to try early versions.

Which approach should the project manager recommend?

A Predictive, with a thorough requirements phase to remove the uncertainty first.

B Adaptive, with short iterations and regular reviews with the pilot group.

C Incremental, with every increment's scope fixed at the start.

D Continuous flow with no reviews, so the team can release as fast as possible.

◆ PAUSE AND DECIDE

Which approach should the project manager recommend?

A Predictive, long requirements phase.

Customers do not know yet. Their needs are discovered by using the product, not by asking.

B Adaptive, short iterations, pilot reviews.

High uncertainty, early value wanted, and willing users to give feedback: all three fit.

C Incremental, fixed scope per increment.

Delivers early, but freezes scope just when feedback should be reshaping it.

D Continuous flow, no reviews.

Fast, but throws away the feedback that resolves the uncertainty.

THE SIGNAL

Unclear requirements, early value wanted, feedback available. Every clue points to adaptive.

A city council is replacing street lighting on a bridge with smart lamps, and launching a public dashboard showing energy use and air quality from the new sensors.

The lamp installation needs a road-closure permit for fixed dates. The dashboard's features are undecided, and residents' groups want a say.

How should the project manager structure the delivery approach?

A Fully predictive, so that the permit dates are protected.

B Fully adaptive, so that residents can shape the whole project.

C Predictive for everything, adding dashboard features later through change requests.

D Predictive for the installation and adaptive for the dashboard, within one integrated plan.

◆ PAUSE AND DECIDE

How should the project manager structure the delivery approach?

A Fully predictive.

Protects the permit but forces guesses about dashboard features nobody has decided.

B Fully adaptive.

Road closures on fixed dates cannot be iterated. Physical work needs a firm plan.

C Predictive, changes later.

Plans to be wrong, then pays for it through a stream of change requests.

D Each workstream by its uncertainty, one plan.

Fixed physical work planned firmly; uncertain digital work learned in cycles.

THE SIGNAL

Two workstreams, two levels of uncertainty. Choose per workstream, then integrate.

> A bank adopted Scrum for a project to meet a new reporting regulation. The requirements are set out in the regulation, the deadline is fixed, and no requirement has changed in six months.
>
> The sponsor complains that the ceremonies cost time and have produced no useful feedback. The team agrees.

What should the project manager have done?

A Assessed the project's characteristics with stakeholders when selecting the approach.

B Trained the team more thoroughly in Scrum before starting.

C Used shorter sprints to create more opportunities for feedback.

D Asked the PMO to mandate Scrum so the sponsor could not object.

◆ PAUSE AND DECIDE

What should the project manager have done?

A Assessed the project's characteristics when selecting.

Fixed, known requirements point to predictive. A suitability assessment would have shown it.

B Trained the team better.

The team is not the problem. The approach does not fit the work.

C Shorter sprints.

More feedback loops on requirements that never change adds cost, not value.

D Have the PMO mandate Scrum.

Uses authority to silence a valid concern. Approach follows the work, not a mandate.

THE SIGNAL

Agile is not automatically better. Requirements fixed by law, with no feedback to gain, suit a predictive plan.

> An insurer's standard method requires a formal design review board before any release. The board meets monthly.
>
> A product team now releases small increments every two weeks. Each release waits up to three weeks for the board, and customers are noticing the delays.

What should the project manager do?

A. Release increments without the board's review, since the risk of each one is small.

B. Keep waiting for the monthly board, because it is organisational policy.

C. Work with the board to agree a tailored review that meets its control objectives at the team's cadence.

D. Ask the sponsor to have the design review requirement removed.

◆ PAUSE AND DECIDE

What should the project manager do?

A Skip the review.

Breaks a governance rule unilaterally. Small risk is the board's call, not yours.

B Keep waiting.

Complies, but accepts lost value when the rule could be adapted properly.

C Agree a tailored review with the board.

Tailors for the organisation: keeps the control's purpose, fits the cadence.

D Ask the sponsor to remove it.

Goes around the board, and removing a control is not the same as fitting it.

THE SIGNAL

When a control fights the cadence, redesign the control with its owners. Do not skip it or suffer it.

> A museum is designing a new permanent gallery that will open once, on a fixed date. Curators are unsure how visitors will respond to the layout and displays.
>
> The team builds rough mock-ups, tests them with visitor groups, and reworks the whole design several times before construction begins.

Which life cycle best describes the design work?

A Iterative

B Incremental

C Predictive

D Continuous delivery

◆ PAUSE AND DECIDE

Which life cycle best describes the design work?

A Iterative

The whole design is refined through repeated feedback, with one final delivery.

B Incremental

Nothing finished is delivered in pieces. Visitors see nothing until opening day.

C Predictive

The design is deliberately reworked, not fixed early and followed.

D Continuous delivery

There is one opening, not a stream of releases.

THE SIGNAL

Repeated refinement of the whole, one delivery at the end: iterative, not incremental.

A predictive project is rolling out a customer relationship system. The core platform is on track, but the sales team's reporting needs keep changing as they use the first release.

Twenty change requests about reports are waiting for the change control board, which meets every three weeks.

What should the project manager do next?

A Continue processing each report change through the change control board.

B Freeze the reporting requirements until the platform roll-out is finished.

C Switch the whole project to Scrum immediately.

D Propose to stakeholders an adaptive approach for the reporting workstream, within existing governance.

◆ PAUSE AND DECIDE

What should the project manager do next?

A Keep using the board for each change.

The process works, but a three-week loop cannot keep pace with needs that change weekly.

B Freeze reporting requirements.

Stops the churn by stopping the learning. Users get reports they no longer want.

C Switch everything to Scrum now.

The platform is on track. Changing its approach adds risk for no gain.

D Adaptive for reporting, within governance.

Moves only the uncertain workstream along the continuum, with agreement.

THE SIGNAL

The evidence changed for one workstream only. Move that workstream, not the whole project.

Choose by uncertainty, per workstream, then tailor with the people affected.

MODULE 04 · FOUNDATIONS - ALL DOMAINS

Scrum in practice

Accountabilities, events, artefacts and commitments, and where a project manager fits in a framework that has no project manager.

DOMAIN Foundations EXAM WEIGHT All domains MODULE 04 of 49

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Scrum is deliberately incomplete. It gives **structure for learning**, not instructions for work.

Empiricism: three pillars

01

Transparency

The work and the process are visible to those doing and receiving it.

02

Inspection

Progress and artefacts are checked often, to spot problems early.

03

Adaptation

When something is off, the process or product is adjusted quickly.

ON THE EXAM

No transparency, no useful inspection. No inspection, no informed adaptation. Each pillar needs the one before.

The Scrum Team: three accountabilities

ACCOUNTABILITY	OWNS	DOES NOT OWN
Product owner	Maximising value; the product backlog and its order; the product goal	How the developers build it
Scrum master	The team's effectiveness; Scrum being understood and used well	Assigning tasks or managing people
Developers	The sprint backlog and plan; quality, meaning the definition of done	Product priorities

ON THE EXAM

One team, no hierarchy, typically ten people or fewer. Each accountability has a clear edge.

Product owner or scrum master?

PRODUCT OWNER

What and why

- Decides the order of the backlog
- Represents stakeholders' needs
- Clarifies items and their value
- One person, not a committee

SCRUM MASTER

How well the team works

- Coaches the team and the organisation
- Removes impediments
- Ensures events happen and stay useful
- A true leader who serves

THE TIEBREAKER

A priority question goes to the product owner. A process or impediment question goes to the scrum master.

The developers

- **Self-managing**: they decide internally who does what, when and how.
- **Cross-functional**: between them they have every skill needed to create value each sprint.
- Accountable for the **sprint backlog**, for meeting the definition of done, and for adapting their plan daily.
- 'Developers' means everyone doing the work: testers, designers, analysts. Not only programmers.

> ON THE EXAM
>
> An option where someone outside the developers assigns their tasks contradicts Scrum.

The sprint: a container for everything else

01

Sprint planning

Agree why, what and how for this sprint.

02

Daily scrum

Developers inspect progress to the goal.

03

The work

Build the increment; refine the backlog.

04

Sprint review

Inspect the outcome with stakeholders.

05

Retrospective

Inspect how the team worked; pick improvements.

ON THE EXAM

Sprints are fixed length, one month or less, and a new one starts as soon as the last ends.

Five events and their timeboxes

EVENT	MAX TIMEBOX (1-MONTH SPRINT)	PURPOSE
Sprint	1 month or less	Container for all the work
Sprint planning	8 hours	Sprint goal, items, plan
Daily scrum	15 minutes	Inspect progress, adapt the plan
Sprint review	4 hours	Inspect outcome with stakeholders
Retrospective	3 hours	Improve how the team works

ON THE EXAM

A timebox is a maximum. If the purpose is met early, the event ends early.

Sprint planning: three questions

- **Why is this sprint valuable?** The whole Scrum team agrees a sprint goal.
- **What can be done this sprint?** Developers select items, guided by their past performance and capacity.
- **How will the chosen work get done?** Developers plan it, often breaking items into small tasks.
- Nobody outside the developers decides how much work they take on.

> ON THE EXAM
>
> The sprint goal is the commitment. The selected items are a forecast of how to reach it.

The daily scrum

- A **15-minute event for the developers**, at the same time and place each working day.
- Purpose: inspect progress towards the sprint goal, and adapt the plan for the next day.
- Not a status report to a manager, and not the only time the team talks.
- The Agile Practice Guide calls it, more generally, a **daily coordination meeting**.

> ON THE EXAM
>
> If developers are reporting to someone rather than planning with each other, the daily scrum has been hijacked.

Sprint review or retrospective?

SPRINT REVIEW

Inspects the product

- What was achieved this sprint
- With key stakeholders present
- What to do next; backlog adapted
- A working session, not a gate

SPRINT RETROSPECTIVE

Inspects the process

- How the team worked
- The Scrum team only
- Most helpful changes identified
- Improvements may enter the next sprint

THE TIEBREAKER

Stakeholder feedback on the product: review. Team friction or process problems: retrospective.

Three artefacts, three commitments

ARTEFACT	WHAT IT IS	ITS COMMITMENT
Product backlog	The ordered list of what is needed to improve the product	Product goal: the long-term objective
Sprint backlog	The sprint goal, the items selected, and the plan to deliver them	Sprint goal: the single objective for the sprint
Increment	A usable step towards the product goal, added to all prior increments	Definition of done: the quality standard

ON THE EXAM

Work that does not meet the definition of done is not part of the increment. It returns to the product backlog.

Definition of done, or acceptance criteria?

DEFINITION OF DONE

One standard for everything

- Applies to every item and increment
- Quality: tested, reviewed, documented
- Owned by the Scrum team
- Set by the organisation if it has one

ACCEPTANCE CRITERIA

Specific to one item

- Written for one backlog item
- What this feature must do
- Agreed with the product owner
- Different for every item

THE TIEBREAKER

An item is finished only when it meets its own acceptance criteria AND the definition of done.

When the sprint meets reality

- Scope can be **clarified and renegotiated with the product owner** as the team learns.
- No changes that would **endanger the sprint goal**, and quality does not decrease.
- Only the **product owner can cancel a sprint**, and only if the sprint goal becomes obsolete.
- The sprint is never extended to finish work. Unfinished items go back to the product backlog.

> ON THE EXAM
>
> New ideas mid-sprint go to the product backlog. The current sprint changes only if the goal survives.

Backlog refinement

- An **ongoing activity**, not a formal Scrum event: items are broken down, detailed and sized.
- The **product owner** is accountable for the backlog; the **developers** size the work.
- Items near the top are small and clear enough to be **ready** for a sprint.
- Older material calls it 'grooming'. The Agile Practice Guide now uses 'refinement'.

> ON THE EXAM
>
> Most sprint planning problems are really refinement problems that were left too late.

Five Scrum values

VALUE	LOOKS LIKE
Commitment	The team commits to its goals and to supporting each other
Focus	Attention stays on the sprint's work and the path to the goals
Openness	Work and challenges are made visible, not hidden
Respect	People respect each other as capable, independent professionals
Courage	Doing the right thing, and raising difficult problems

ON THE EXAM

Hiding a problem to look good in the review breaks openness and courage, whatever the short-term gain.

Where the project manager fits

- Scrum has **no project manager** accountability. Exam questions still say 'project manager' or 'project leader'.
- Inside a Scrum team, read that person as a **servant leader**: facilitating, coaching, removing impediments.
- Around the team: governance, budget, contracts, risks, stakeholders beyond the product, and other workstreams.
- Never: assigning developers' tasks, reordering the backlog, or deciding how much the team takes on.

> ON THE EXAM
>
> In an agile question, the PM serves the team and connects it to the organisation. They do not direct it.

Recognising Scrum going wrong

WHAT YOU SEE	WHAT SHOULD HAPPEN
Daily scrum is a status report	Developers plan together; status from boards
Several managers set priorities	One product owner orders one backlog
Sprint extended to finish work	Unfinished work returns to the backlog
Stakeholders task developers directly	Requests go to the product owner
Retros skipped: 'too busy'	Keep them: they fix the busyness

ON THE EXAM

Restore the Scrum rule, and do it by coaching, not by decree.

Project Lantern's app team in Scrum terms

- **Product owner:** Northgate's head of patient access. One person, backed by a clinic advisory group.
- **Scrum master:** an agile coach. **Developers:** seven people, including a tester and a UX designer.
- **Product goal:** patients book, move or cancel any appointment in under two minutes.
- The project manager runs the hybrid plan, the supplier contract and the steering committee, and never assigns app tasks.

> ON THE EXAM
>
> Scrum inside, project management around it. That is how most hybrid exam scenarios are drawn.

> A project manager has started attending a Scrum team's daily scrum. Each developer now reports yesterday's hours to the project manager, who asks follow-up questions.
>
> The meeting regularly runs to forty minutes, and developers say they no longer use it to plan their day.

What should the project manager do?

A. Enforce the fifteen-minute limit with a timer, keeping the updates brief.

B. Replace the daily scrum with written status reports to save time.

C. Return the daily scrum to the developers, and get status from the team's visible boards instead.

D. Ask the scrum master's manager to take over running the meeting.

◆ PAUSE AND DECIDE

What should the project manager do?

A Enforce fifteen minutes.

Fixes the length but keeps the wrong purpose. It is still a status report.

B Replace it with written reports.

Removes the developers' daily planning, which is the whole point of the event.

C Give it back; use visible boards for status.

Restores the purpose and still meets the PM's need for information.

D Ask the scrum master's manager.

Escalates a problem the project manager created and can fix directly.

THE SIGNAL

The project manager caused the problem. The fix is to step back, not to tighten control.

> Two days before the end of a sprint, the developers realise one item will not pass performance testing, which is part of their definition of done.
>
> The product owner had hoped to demonstrate the feature to an important customer at the sprint review.

What should happen to the item?

A It is not included in the increment, and returns to the product backlog for the product owner to reorder.

B The sprint is extended by two days so the item can be finished.

C It is shown as done at the review, and performance is fixed next sprint.

D Performance testing is removed from the definition of done for this sprint.

◆ PAUSE AND DECIDE

What should happen to the item?

A Not in the increment; back to the backlog.

Follows the definition of done and lets the product owner decide what comes next.

B Extend the sprint.

Sprints are fixed length. Extending them destroys the cadence and the forecast.

C Present it as done.

Breaks transparency and openness, and misleads the customer about quality.

D Drop the performance standard.

Lowers quality to hit a date. Quality does not decrease during a sprint.

THE SIGNAL

Undone means undone. Protect the definition of done and let the product owner re-plan.

> Halfway through a sprint to build a loyalty-points feature, the company announces it is closing the loyalty scheme at the end of the quarter.
>
> Everything in the current sprint is loyalty work. The developers ask the project leader what to do.

What should happen?

A The scrum master cancels the sprint and schedules a new one.

B The developers finish the sprint, because they committed to the work.

C The product owner may cancel the sprint, since its goal is now obsolete, and a new sprint is planned.

D The sprint sponsor submits a change request to stop the loyalty work.

◆ PAUSE AND DECIDE

What should happen?

A The scrum master cancels it.

Wrong person. Only the product owner has the authority to cancel a sprint.

B Finish the committed work.

The commitment is to the goal, and the goal no longer has any value.

C Product owner may cancel; plan anew.

An obsolete sprint goal is exactly when cancellation is allowed, by the product owner.

D Sponsor raises a change request.

Imports predictive change control into a framework with its own route.

THE SIGNAL

Sprint goal obsolete: the product owner may cancel. Nobody else can.

> A council's new digital services team is using Scrum. Five department heads jointly act as 'product owner', and each sends the developers their own priorities.
>
> The developers keep switching work, and little is finished in each sprint.

What should the project manager do?

A Let the developers choose which department's priorities to follow each sprint.

B Set the backlog order personally, to settle the disagreement.

C Escalate the conflict to the council's chief executive.

D Work with the sponsor and department heads to appoint one product owner who represents all of them.

◆ PAUSE AND DECIDE

What should the project manager do?

A Developers choose.

Hands a value decision to the people who should not own priorities.

B The PM sets the order.

The PM becomes a sixth voice and takes a decision that belongs to a product owner.

C Escalate to the chief executive.

Too high, too early. Nobody has yet tried to fix the structure with the sponsor.

D Agree one product owner who represents all.

Rung 3: collaborates to fix the root cause, a committee acting as product owner.

THE SIGNAL

Priorities from many voices must flow through one product owner. Fix the structure, not the symptoms.

> At a sprint review, a regional sales director sees the new ordering screen and asks for a bulk-upload option. It sounds valuable.
>
> She asks the developers to add it to the current sprint, which has four days left.

What should the project leader recommend?

A Add the feature to the current sprint, because stakeholder feedback is welcome.

B Capture it as a product backlog item for the product owner to order.

C Decline it as out of scope for the project.

D Submit it as a formal change request to the change control board.

◆ PAUSE AND DECIDE

What should the project leader recommend?

A. Add it to this sprint.
Bypasses the product owner and threatens the current sprint goal.

B. Product backlog item for the PO.
Welcomes the feedback and routes it to the person who owns priorities.

C. Decline as out of scope.
Rejects valuable feedback. Adaptive scope expects new ideas like this.

D. Formal change request.
Predictive machinery. In Scrum the product backlog is the change mechanism.

THE SIGNAL

In Scrum, the product backlog is the change control. New ideas enter it; the product owner orders it.

> A new Scrum team has skipped its last three retrospectives because it felt too busy. The same build problems keep causing delays each sprint.
>
> Team members now describe the retrospective as a waste of time.

What should the project leader do?

A Stop scheduling retrospectives until the team is less busy.

B Identify the improvements personally and tell the team what to change.

C Coach the team on the retrospective's purpose, and facilitate a short one focused on the build problems.

D Report the team's non-compliance with Scrum to the PMO.

◆ PAUSE AND DECIDE

What should the project leader do?

A Stop retrospectives for now.

Removes the one event designed to fix the recurring problem.

B Tell the team what to change.

Takes improvement away from the team, so it will not own the changes.

C Coach, then a short, focused retrospective.

Rebuilds the habit by making it immediately useful on a real, painful problem.

D Report them to the PMO.

Compliance pressure, not coaching. Nobody has helped the team yet.

THE SIGNAL

A team too busy to improve stays busy. Show the retrospective's value on the problem that hurts most.

Priorities to the product owner. The work to the developers. The system to the scrum master.

MODULE 05 · FOUNDATIONS - ALL DOMAINS

Kanban, Lean, XP and scaling agile

Flow instead of timeboxes, waste and pull, the engineering practices behind quality, and what changes when many teams share one product.

DOMAIN Foundations EXAM WEIGHT All domains MODULE 05 of 49

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Scrum manages work in timeboxes.
Kanban manages **the flow** of work.

Lean thinking in five ideas

- **Value** is defined by the customer. Everything else is a candidate for removal.
- The **value stream** is every step from request to delivery, including the waiting.
- **Flow**: work should move steadily, without stopping, batching or looping back.
- **Pull**: start work when there is capacity downstream, not because it arrived.
- **Continuous improvement**, done with respect for the people doing the work.

> ON THE EXAM
>
> Lean asks one question of every step: does the customer value this? If not, shrink it or remove it.

Seven wastes in knowledge work

WASTE	LOOKS LIKE
Partially done work	Code written but not tested; designs never built
Extra features	Functions nobody asked for, or nobody uses
Relearning	Solving the same problem twice because nothing was shared
Handoffs	Knowledge lost each time work passes to another group
Delays	Waiting for approvals, decisions, environments
Task switching	People spread across many items, finishing few
Defects	Rework, and the cost of finding problems late

ON THE EXAM

Delays and task switching are the wastes most scenarios describe. Both are fixed by limiting work in progress.

Value stream mapping

- Draw every step from request to delivery, with the **working time** and the **waiting time** for each.
- Done with the people involved in each step, not from a desk.
- Typical finding: a few days of work spread across weeks of waiting between steps.
- Improve the **biggest wait** first. Working faster rarely helps if the work is mostly waiting.

> ON THE EXAM
>
> If a scenario says work takes days but delivery takes weeks, map the value stream before changing anything.

A simple Kanban board

01	02	03	04	05
Options	**Ready**	**Doing**	**Review**	**Done**
Ideas and requests. No limit.	Refined and agreed. Limit 5.	In progress. Limit 3.	Checked by a peer. Limit 2.	Meets the agreed policy.

ON THE EXAM

The numbers are WIP limits. A full column means nobody starts new work there; they help finish.

Kanban's core practices

PRACTICE	WHAT IT MEANS
Visualise the work	Every item and every step is on the board
Limit work in progress	Explicit limits per column, or for the whole system
Manage flow	Watch how items move; remove blockages quickly
Make policies explicit	Write down the rules, such as what 'done' means for each column
Implement feedback loops	Regular meetings to review flow, service and risks
Improve collaboratively	Evolve the system in small, agreed steps

ON THE EXAM

Kanban starts with what you do now and improves it. No new roles are required on day one.

A motorway at 70% capacity flows. At 100% it stops.

Little's Law: the arithmetic of flow

- **Average cycle time = average work in progress / average throughput.**
- 12 items in progress, finishing 4 a week: each item takes about 3 weeks.
- Halve the WIP to 6, same throughput: about 1.5 weeks. Same people, faster delivery.
- Adding work without adding capacity only makes every item slower.

> ON THE EXAM
>
> To deliver faster without more people, reduce work in progress. The arithmetic shows why.

Flow metrics

METRIC	MEASURES	USEFUL FOR
Lead time	Request to delivery	What the customer experiences
Cycle time	Work starts to work finishes	How efficient the team's process is
Throughput	Items finished per period	Forecasting how much will be delivered
Work in progress	Items started but not finished	Spotting overload early

ON THE EXAM

Lead time includes the wait before work starts. Customers feel lead time; teams control cycle time.

Scrum or Kanban?

SCRUM

Timeboxed batches

- Fixed-length sprints
- Defined accountabilities
- Commits to a sprint goal
- Measures velocity per sprint

KANBAN

Continuous flow

- No iterations; pull as capacity frees
- No required roles
- Commits through WIP limits
- Measures cycle time and throughput

THE TIEBREAKER

Predictable product work that benefits from a planning rhythm: Scrum. Unpredictable arrival and urgent items: Kanban.

Extreme programming: the engineering practices

PRACTICE	WHY IT HELPS
Pair programming	Two people, one task: fewer defects, shared knowledge
Test-driven development	Write the test first, then the code to pass it
Continuous integration	Merge and test many times a day; problems surface early
Refactoring	Improve the design of working code without changing behaviour
Small releases	Get working software to users often
Collective ownership	Anyone can improve any part; no single point of failure

ON THE EXAM

When a scenario describes late integration surprises or recurring defects, XP practices are usually the fix.

Spikes: buying knowledge

- A **spike** is a short, timeboxed piece of research or prototyping to reduce uncertainty.
- Use it when the team cannot estimate or choose an approach, because it lacks knowledge.
- The output is knowledge, a prototype or a decision, not a finished feature.
- Not an excuse for open-ended research. The timebox is the point.

> ON THE EXAM
>
> Estimates that vary wildly usually mean missing knowledge. A spike buys that knowledge cheaply.

Agile-adjacent: design thinking and Lean Startup

- **Design thinking**: empathy with users, and cheap prototypes that invite feedback while change is still cheap.
- **Lean Startup**: treat each idea as a hypothesis. Build, measure, learn, then persevere or pivot.
- A **minimum viable product** here is an experiment: the smallest thing that tests the hypothesis.
- The Agile Practice Guide, Second Edition, treats these as complementary disciplines, not agile methods.

> ON THE EXAM
>
> Unsure whether anyone wants it? Test the idea cheaply before building it properly.

Other names you may meet

NAME	REMEMBER IT FOR
Scrumban	Scrum's rhythm combined with Kanban's flow and WIP limits
DSDM	Fixed time and cost, flexible features; the origin of MoSCoW prioritisation
Crystal	A family of methods scaled to team size and criticality
Feature-driven development	Building and tracking work feature by feature

> ON THE EXAM
>
> You will not be tested on the details. Recognise the name and its one big idea.

Scaling: many teams, one product

- The core problem is **dependencies and integration** between teams.
- **Scrum of Scrums**: representatives meet regularly to coordinate cross-team work and blockers.
- Frameworks: SAFe, LeSS, Scrum@Scale, and PMI's own Disciplined Agile.
- Common ingredients: one product backlog, aligned cadences, shared definition of done, continuous integration.

> ON THE EXAM
>
> The exam tests the problem, not the framework. Cross-team dependencies need a coordination mechanism.

Project Lantern: more than one agile style

- The **app team** uses Scrum, with XP practices: continuous integration and test-driven development.
- The **go-live support team** uses Kanban, with an expedite lane for clinic-stopping faults.
- A weekly **coordination meeting** links the app team, the kiosk supplier and the data migration team.
- Each team works the way that suits its work. The project manager integrates them.

> ON THE EXAM
>
> Different work, different approach, one project. Integration is the project manager's job.

Matching the approach to the work

THE SCENARIO DESCRIBES	REACH FOR
Product features, planned in batches, with regular stakeholder reviews	Scrum
Unpredictable arrivals, urgent interruptions, support or operations	Kanban
Recurring defects, late integration problems, fragile code	XP practices
An idea nobody is sure customers want	Lean Startup, an MVP experiment
Work that takes days of effort but weeks to deliver	Value stream mapping
Several teams breaking each other's work	Scaling: coordination, shared DoD, CI

ON THE EXAM

Name the problem first. The right practice follows from the problem, not from a favourite framework.

> An IT support team of five uses a Kanban board. Customers complain that requests take weeks, even simple ones.
>
> The board shows eighteen items in the In Progress column. Team members say they are busy all day but rarely finish anything.

What should the project manager do?

A Facilitate the team in setting work-in-progress limits for its board columns.

B Request two more team members to handle the volume.

C Move the team to two-week sprints with a sprint goal.

D Ask each team member to work on the oldest items first.

◆ PAUSE AND DECIDE

What should the project manager do?

A Facilitate setting WIP limits.

Eighteen items for five people is the cause. Limits force finishing, so cycle time drops.

B Request more people.

Adds cost without fixing the overload. They would also start too much.

C Move to sprints.

Changes the framework without fixing the actual problem, too much in progress.

D Oldest first.

A sensible policy, but it does not stop people starting new work.

THE SIGNAL

Busy all day, finishing little, many items open: too much work in progress.

> A team tracks its flow. Over the last quarter it has averaged 20 items in progress at any time, and finished an average of 5 items per week.
>
> The product owner asks how long a newly started item will typically take to finish.

What is the best estimate of the average cycle time?

A 1 week

B 4 weeks

C 15 weeks

D 25 weeks

◆ PAUSE AND DECIDE

What is the best estimate of the average cycle time?

A 1 week

That would need throughput of 20 a week, or WIP of 5.

B 4 weeks

Little's Law: 20 items in progress divided by 5 finished per week.

C 15 weeks

Subtracts instead of divides. The law is a ratio.

D 25 weeks

Adds the numbers. It has no meaning as a time.

THE SIGNAL

Cycle time = WIP / throughput. Check the units: items divided by items per week gives weeks.

> During refinement, developers estimate a story to connect the app to a payment provider none of them has used before.
>
> Estimates range from 3 to 40 points, and nobody can explain the difference with confidence.

What should the team do?

A Run a short, timeboxed spike to investigate the provider, then estimate again.

B Use the average of the estimates so planning can continue.

C Use the highest estimate to be safe.

D Ask the product owner to set the estimate.

◆ PAUSE AND DECIDE

What should the team do?

A Timeboxed spike, then re-estimate.

The spread comes from missing knowledge. A spike buys it cheaply.

B Average the estimates.

Averages ignorance. The number looks precise and means nothing.

C Take the highest.

Pads the plan without reducing the uncertainty at all.

D Product owner sets it.

Estimates belong to the people doing the work, not the product owner.

THE SIGNAL

Wildly different estimates plus unfamiliar technology: a knowledge gap. Spike it.

> In a hospital's facilities department, small change requests take about two days of actual work, but six weeks from request to completion.
>
> Each request passes through four departments for sign-off. Staff in every department say they are working flat out.

What should the project manager do first?

A Add another technician to complete the work faster.

B Map the value stream with the four departments to find where requests wait.

C Ask each department head to approve requests more quickly.

D Buy a workflow tool to automate the approvals.

◆ PAUSE AND DECIDE

What should the project manager do first?

A. Add a technician.

The work is two days. Speeding up the work barely touches six weeks of waiting.

B. Map the value stream together.

Rungs 2 and 3: finds where the waiting is, with the people who own each step.

C. Ask for faster approvals.

Pressure without understanding. Everyone is already busy, so nothing changes.

D. Buy a workflow tool.

A solution before the diagnosis. It may automate the wrong steps.

THE SIGNAL

Days of work, weeks of elapsed time: the problem is waiting. Find it before fixing anything.

> Four Scrum teams build one online banking product. Each team's increment passes its own tests.
>
> At every sprint review, features from different teams conflict with each other, and fixing them takes most of the next sprint.

What should the project manager do?

A Let each team release its own features independently.

B Assign a coordinator to each team to report progress daily to the project manager.

C Add a separate integration sprint after every three development sprints.

D Help the teams set up regular cross-team coordination, a shared definition of done and continuous integration.

◆ PAUSE AND DECIDE

What should the project manager do?

A Release independently.

The features interact in one product. Separate releases push conflicts onto customers.

B Daily reports to the PM.

Adds reporting, not integration. The conflicts still appear late.

C Integration sprints.

Batches integration, so conflicts pile up for weeks before anyone sees them.

D Coordination, shared DoD, continuous integration.

Surfaces conflicts daily instead of every sprint, at the source.

THE SIGNAL

Teams fine alone, broken together: an integration problem. Integrate continuously and coordinate across teams.

> An infrastructure operations team adopted two-week sprints. Urgent incidents arrive unpredictably and must be handled within hours.
>
> Almost every sprint is disrupted, planned improvement work rarely finishes, and the team finds sprint planning pointless.

What should the project manager recommend?

A Lengthen the sprints to four weeks so there is more room for incidents.

B Stop planning, and handle work in whatever order it arrives.

C Move to Kanban, with an expedite lane for incidents and WIP limits for improvement work.

D Plan all improvement work in a predictive schedule for the year.

◆ PAUSE AND DECIDE

What should the project manager recommend?

A Four-week sprints.

More time to be disrupted. Incidents still arrive unpredictably.

B No planning at all.

Loses control entirely; improvement work would never be protected.

C Kanban with an expedite lane and WIP limits.

Handles urgent arrivals explicitly while still protecting and finishing planned work.

D A yearly predictive plan.

Even less able to absorb interruptions than two-week sprints.

THE SIGNAL

Unpredictable, urgent arrivals break timeboxes. Flow with explicit policies absorbs them.

Stop starting, **start finishing**. Then choose the practice that fits the problem.

MODULE 06 · DOMAIN 1 - PEOPLE

Leadership styles and servant leadership

Choosing a style that fits the person and the moment, leading without formal authority, and what 'servant leader' really asks of you.

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The right leadership style is the one this person needs **right now**.

Management or leadership?

MANAGEMENT

Positional, about things

- Maintains control
- Focuses on systems and structure
- Relies on position and control
- Asks how and when

LEADERSHIP

Relational, about people

- Inspires trust
- Focuses on relationships
- Relies on influence
- Asks what and why

THE TIEBREAKER

Projects need both. But on this exam, when people are the problem, leadership behaviours win.

Common leadership styles

STYLE	WHAT IT LOOKS LIKE	FITS WHEN
Directive	Leader decides and instructs	Crisis, safety, or someone brand new to a task
Participative	Leader involves the team in decisions	Buy-in matters and time allows
Delegating	Team decides within agreed boundaries	Skilled, motivated, experienced people
Transactional	Rewards and corrections tied to targets	Routine work with clear standards
Transformational	Inspires through vision and growth	Change, innovation, long engagements
Servant	Serves the team so it can perform	The default for agile teams, and most exam answers

ON THE EXAM

Every style has a place. The exam's default is servant leadership, shifted by the situation.

Transactional or transformational?

TRANSACTIONAL

Exchange

- Clear targets and rewards
- Corrects deviations
- Keeps things running
- Motivation from outside

TRANSFORMATIONAL

Inspiration

- A compelling shared vision
- Develops each person
- Encourages new thinking
- Motivation from within

THE TIEBREAKER

Transactional keeps a stable system running. Transformational changes what people believe is possible.

Situational leadership: match style to readiness

THE PERSON IS...	THEY NEED	STYLE
New to the task, but keen	Clear instructions and close guidance	Directing
Learning, and losing confidence	Direction plus encouragement	Coaching
Skilled, but hesitant or unsure	Support and shared decisions	Supporting
Skilled and confident	Freedom and trust	Delegating

ON THE EXAM

Readiness is per task, not per person. An expert in one area can be a beginner in another.

Reading readiness from a scenario

- 'Recently joined', 'first time', 'enthusiastic but unfamiliar': guide closely, with clear steps.
- 'Making mistakes', 'frustrated', 'starting to doubt': explain, encourage, stay close.
- 'Experienced but reluctant', 'capable, lacks confidence': involve them, reassure, share the decision.
- 'Expert', 'has done this many times': delegate, and stay available.
- Delegating to a beginner feels respectful. It is actually abandonment.

> ON THE EXAM
>
> The planted detail is usually one adjective about the person. Read it before choosing a style.

The leader exists to help the team succeed, **not the other way round**.

What a servant leader actually does

- **Removes impediments** the team cannot remove itself: approvals, access, missing people.
- **Shields the team** from interruptions and conflicting demands.
- **Communicates the vision** again and again, so decisions stay aligned.
- **Provides what the team needs** to work: tools, information, a good environment.
- **Grows people** through coaching, listening and opportunities.

> ON THE EXAM
>
> Every item on this list is active. A servant leader is busy, just not busy directing the work.

Servant leadership or laissez-faire?

SERVANT LEADERSHIP

Present and working

- Clears the path actively
- Sets boundaries and a vision
- Checks in and coaches
- Steps in when the team is stuck

LAISSEZ-FAIRE

Absent

- Leaves the team alone
- No boundaries or vision
- Waits to be asked
- Lets problems grow

THE TIEBREAKER

'Let the team handle it' is right only when the team can handle it and you are still supporting them.

Where a project manager's power comes from

POWER	COMES FROM	ON THE EXAM
Formal, or legitimate	Your role and title	Real, but often limited for a PM
Reward	Ability to give something valued	Useful; only if genuinely yours to give
Coercive	Ability to penalise	Almost always the wrong answer
Expert	Knowledge and skill	Strong and durable
Referent	Trust, respect and relationships	Strong and durable
Informational	Access to information others need	Useful; never hoard it

ON THE EXAM

Expert and referent power last longest. Coercive power buys compliance and costs trust.

Leading without authority

- In a **matrix organisation**, team members often report to functional managers, not to you.
- Build relationships early, before you need anything from anyone.
- Negotiate priorities **with the functional manager**, not around them.
- Show people how the project helps their own goals and growth.
- Threats, going over people's heads, and promises you cannot keep all backfire.

> ON THE EXAM
>
> Influence is built before it is needed. Use relationships and expertise, and work with line managers.

Empowerment: agree who decides

LEVEL	WHAT IT MEANS	EXAMPLE
Tell	The leader decides and explains	A regulatory deadline that cannot move
Consult	The leader decides after asking the team	Choosing a supplier with team input
Agree	The leader and team decide together	The team's working agreement
Delegate	The team decides, within boundaries	How to build a feature; which tool to use

ON THE EXAM

Empowerment is not giving away every decision. It is being clear about who decides what, and why.

Setting expectations at team level

- Agree **outcomes and standards**, not minute-by-minute tasks.
- Make **roles and decision rights** clear from the start. Module 07 covers how.
- Create a **working agreement** with the team, so norms are shared, not imposed.
- Revisit expectations when the team, the work or the approach changes.

> ON THE EXAM
>
> Many 'people problems' in scenarios are really unclear expectations. Clarify before you correct.

Be an accountable leader

- A PMBOK 8 principle: own **outcomes**, not just activity.
- You can delegate **authority and tasks**. You cannot delegate your accountability.
- When something goes wrong, take responsibility outwards, then fix the cause with the team privately.
- Blaming a team member in front of stakeholders destroys trust and solves nothing.

> ON THE EXAM
>
> Credit goes to the team in public. Accountability stays with you.

Represent the voice of the team

- Carry the team's concerns upward: unrealistic dates, missing resources, unclear direction.
- Bring **evidence and options**, such as velocity data and scope trade-offs, not complaints.
- Report back honestly, including when the answer is no.
- Silently accepting an impossible demand is not loyalty. It sets the team up to fail.

> ON THE EXAM
>
> Advocacy with data and options is leadership. Passing the pressure straight down is not.

Leadership across approaches and moments

- **Agile teams**: servant leadership by default; the team owns how work is done.
- **Predictive work**: more coordination and planning, still delivered through people, not orders.
- **Hybrid**: both at once, often in the same week, for different workstreams.
- **Crisis or safety risk**: act directively and immediately, then return to your usual style.

> ON THE EXAM
>
> Safety overrides style. When people could be hurt, act first and consult afterwards.

Project Lantern: three people, three styles

- A **new clinic coordinator**, keen but new to go-live checklists: direct, with clear steps and close support.
- A **senior developer** who has built booking systems before: delegate, and stay available.
- The **kiosk supplier's engineer**, who does not report to you at all: influence through relationship and shared goals.
- Same project manager, same week. Style follows the person and the task.

> ON THE EXAM
>
> One leader, several styles in the same week. That flexibility is what the exam is testing.

Scenario signals and the leadership move

THE SCENARIO SAYS	THE MOVE
New team member, first time on this task	Direct: clear steps, close guidance
Team constantly interrupted by outside requests	Shield: route requests through the right channel
Team member ignores the PM, reports to someone else	Influence: relationship, and agree priorities with their manager
Team asks the PM to make a technical choice	Empower: facilitate their decision within constraints
Management imposes an impossible date	Advocate: evidence and options, upward
Someone could be injured	Act directively, now

ON THE EXAM

Match the move to the signal. Most leadership questions reduce to one of these six.

A recent graduate joins a testing team and is excited about the project. She has never written automated test scripts before, and her first scripts are due next week.

Other team members are experienced and working independently.

Which approach should the project manager take with the graduate?

A Delegate the scripts fully, to show trust in her abilities.

B Give clear instructions and close guidance, then reduce it as her skill grows.

C Share decisions with her and offer encouragement, as with a capable colleague.

D Leave her to learn from the other testers as they work.

◆ PAUSE AND DECIDE

Which approach should the project manager take with the graduate?

A Delegate fully.

Right for an expert. For a beginner it is abandonment, however kindly meant.

B Clear instructions, close guidance, then reduce.

High commitment, low competence on this task: directing, then moving on.

C Share decisions, encourage.

The supporting style, for someone skilled but hesitant. She lacks the skill, not the confidence.

D Learn from the others.

Laissez-faire. Nobody is accountable for her learning, and the deadline is next week.

THE SIGNAL

Keen but new to the task: direct now, and plan to step back as competence grows.

> An agile team's work keeps being interrupted by a finance department head, who asks developers directly for urgent report changes.
>
> The developers feel they cannot refuse someone so senior, and the team missed its last two iteration goals.

What should the project leader do?

A Tell the developers to refuse any request that does not come from the product owner.

B Accept the requests, because the department head outranks the product owner.

C Escalate the department head's behaviour to the sponsor.

D Meet the department head to understand the needs, and agree that requests go through the product owner.

◆ PAUSE AND DECIDE

What should the project leader do?

A Developers refuse.

Pushes a senior conflict onto junior people. The leader should shield them.

B Accept, because of seniority.

Seniority does not change who owns priorities, and the team keeps failing.

C Escalate to the sponsor.

Nobody has spoken to the department head yet. Too early.

D Meet, understand, agree the channel.

Shields the team and still meets the head's real needs through the backlog.

THE SIGNAL

The team is being pulled in several directions. The servant leader shields it, by talking to the source.

> In a matrix organisation, three engineers assigned to a project keep prioritising work for their functional manager. Project tasks slip each week.
>
> The project manager has no formal authority over the engineers or their appraisals.

What is the best way for the project manager to gain the engineers' commitment?

A Build relationships with the engineers, and agree project priorities with their functional manager.

B Warn the engineers that their project performance will be reported negatively.

C Promise the engineers bonuses if project tasks finish on time.

D Ask the sponsor to order the engineers to put the project first.

◆ PAUSE AND DECIDE

What is the best way for the project manager to gain the engineers' commitment?

A Relationships, and agree priorities with their manager.

Referent power plus working with the line manager: fixes the real conflict of priorities.

B Warn of negative reports.

Coercive, and not even the PM's power to use. It buys resentment.

C Promise bonuses.

A reward the PM does not control. A promise that cannot be kept destroys trust.

D Sponsor orders them.

Goes over the functional manager's head before talking to them.

THE SIGNAL

No formal authority: use relationships and expertise, and settle priorities with the line manager.

> A self-organising team must choose between two test automation tools. Both meet the organisation's security and budget rules.
>
> The team is split, and asks the project manager to make the choice for them.

What should the project manager do?

A Choose the cheaper tool, to protect the budget.

B Ask the sponsor to decide, since it is a spending decision.

C Facilitate a team decision, confirming the constraints and a way to decide.

D Let the most senior developer choose.

◆ PAUSE AND DECIDE

What should the project manager do?

A Choose the cheaper one.

Both already fit the budget. Takes a decision that belongs to the team.

B Ask the sponsor.

Escalates a decision already inside agreed limits.

C Facilitate the team's decision.

Empowers the team, which will use the tool daily, within clear boundaries.

D Senior developer chooses.

Creates a hierarchy inside a self-organising team and sidelines the others.

THE SIGNAL

Inside the boundaries, the team decides. The leader's job is to help it decide well, not to decide for it.

> A defect in a release the project manager approved has caused incorrect invoices for several customers. The sponsor is angry.
>
> In a meeting with senior stakeholders, the sponsor asks the project manager who is responsible.

How should the project manager respond?

A Name the developer who wrote the faulty code, so the sponsor knows where the problem lies.

B Take responsibility, explain the corrective plan, then work on the root cause with the team privately.

C Explain that the quality assurance team should have caught the defect.

D Point out that the requirements from the business were unclear.

◆ PAUSE AND DECIDE

How should the project manager respond?

A. Name the developer.
Public blame destroys trust, and the PM approved the release.

B. Own it, give the plan, fix the cause privately.
Accountable leadership: responsibility outward, root cause inward.

C. Blame the QA team.
Shifts blame to another group. Still not accountability.

D. Blame unclear requirements.
Blames the business in front of the business. Defensive, and it fixes nothing.

THE SIGNAL

The project manager approved the release. Accountability cannot be delegated downwards.

> Senior management moves a product launch six weeks earlier without consulting the team. The team's measured velocity shows the planned scope cannot be finished by the new date.
>
> The team is demoralised and asks the project manager to push back.

What should the project manager do?

A. Ask the team to work extra hours to meet the new date.

B. Accept the date and reduce testing to fit the time available.

C. Tell management that the new date is impossible.

D. Work with the team on options, such as reduced scope, and present them to management with the velocity data.

◆ PAUSE AND DECIDE

What should the project manager do?

A Extra hours.

Overtime was never a winning answer in the bank. Unsustainable, and burns the team.

B Cut testing.

Trades hidden quality risk for a date. Quality is not a silent variable.

C Say it is impossible.

Right about the facts, but gives management nothing to decide with.

D Options and data, presented upward.

Represents the team with evidence, and gives management a real choice.

THE SIGNAL

Advocate upward with evidence and options. Neither absorb the pressure nor simply refuse.

Read the person, pick the style, **serve** the team, and own the outcome.

MODULE 07 · DOMAIN 1 - PEOPLE

Building the team: charter, roles, ground rules

Turning a group of people into a team: shared agreements, clear roles, and knowing what stage the team is in.

DOMAIN I - People EXAM WEIGHT 33% MODULE 07 of 49

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A team is not people with the same deadline. It is people with **shared agreements**.

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The kickoff: the first chance to build the team

- Share the **purpose and vision**: why this project matters and what success looks like.
- Introduce **people and roles**, including who decides what.
- Start the **working agreements** the team will use, and plan how to finish them.
- For hybrid or distributed teams, bring everyone together, in person if at all possible.

> ON THE EXAM
>
> A good kickoff sets up the charter, the roles and the vision in one go. A missing kickoff shows up later as friction.

What goes in a team charter

SECTION	ANSWERS THE QUESTION
Values	What matters to us as a team?
Working agreements	How do we behave day to day?
Decision-making	How are decisions made, and by whom?
Communication	Which channels, how fast, and when?
Conflict process	What do we do when we disagree?
Meeting norms	When we meet, how do we run it?

ON THE EXAM

The charter is the team's own agreement about how it works. It is the first thing to consult when behaviour goes wrong.

Who writes it, and when

- The **team writes it together**. The project manager facilitates.
- Early: at or soon after the kickoff, before habits set.
- It is a **living document**: revisited when people join, when the work changes, or in retrospectives.
- A charter written by the project manager and handed out is a set of rules, not an agreement.

> ON THE EXAM
>
> Ownership is the whole point. People keep agreements they helped make.

Team charter or project charter?

PROJECT CHARTER

Authorises the project

- Issued by the sponsor
- Formally starts the project
- Names the project manager and their authority
- High-level scope, objectives, risks

TEAM CHARTER

How the team works

- Created by the team
- Values and working agreements
- Decisions, communication, conflict
- Updated as the team changes

THE TIEBREAKER

Authority and objectives: project charter. Behaviour and working norms: team charter.

Ground rules that work

- **Specific and observable**: 'reply to direct messages within four working hours', not 'be responsive'.
- **Agreed by the team**, and few enough to remember.
- **Visible**: on the team board, or pinned in the team's channel.
- Vague rules like 'respect each other' cannot be kept or broken, so they cannot be enforced.

> ON THE EXAM
>
> If you cannot tell whether a rule was broken, it is not a ground rule yet. Make it observable.

When a ground rule is broken

01

Talk privately

Understand why. There may be a reason.

02

Refer to the agreement

The team's rule, not your preference.

03

Team revisits

If the rule does not work, change it together.

04

Escalate

Only if a pattern persists after all of this.

ON THE EXAM

The first move is always a private conversation. Public naming and immediate escalation both lose.

A RACI chart for Project Lantern

ACTIVITY	CLINIC MANAGER	PRODUCT OWNER	PROJECT MANAGER	KIOSK SUPPLIER
Define booking rules	C	A, R	I	–
Install kiosks	C	I	A	R
Clinic go-live decision	A	C	R	I
Train reception staff	R	C	A	I

ON THE EXAM

R does the work. A owns the outcome and signs off: exactly one per activity. C is asked. I is told.

RACI rules the exam uses

- **Exactly one A** per activity. Two accountable people means nobody is.
- At least one **R**, so someone actually does the work.
- Too many **C**s slow every decision; too many **I**s flood inboxes.
- Build it **with** the people named in it, and share it.

> ON THE EXAM
>
> Two people who both think they approve something: the RACI has two A's, or does not exist. Fix that first.

Beyond the RACI

- **Role descriptions**: the role, its authority, its responsibilities, and the competence it needs.
- The **resource management plan** records how roles are staffed, developed and released.
- Organisation charts show reporting lines; the RACI shows who does what on the project.
- On agile teams, the accountabilities are fixed by the framework and the team decides the rest.

> ON THE EXAM
>
> Authority must match responsibility. Someone responsible for a result needs the authority to achieve it.

How teams develop: Tuckman's stages

STAGE	WHAT YOU SEE	WHAT THE LEADER DOES
Forming	Polite, unsure, looking to the leader	Give direction, purpose and structure
Storming	Disagreement, friction, testing limits	Coach; surface and resolve conflict
Norming	Agreements settle, trust grows	Support; step back a little
Performing	Self-directed, effective, interdependent	Delegate; remove obstacles
Adjourning	Work ends, the team disbands	Recognise, reflect, release well

ON THE EXAM

Storming is normal, not failure. The leader's job is to help the team through it, not to stop it happening.

Tuckman on the exam

- **Storming is healthy**. A team that never disagrees has usually not engaged yet.
- Teams can **move backwards**: a new member, a new leader or a big change can return them to forming or storming.
- Teams can **stall** in a stage. A team stuck storming needs facilitated conflict resolution.
- **Adjourning** links to closure: recognise contributions and release people properly.

> ON THE EXAM
>
> A performing team that suddenly struggles after someone joins has moved back a stage. Re-form, don't blame.

Another view: the Drexler-Sibbet model

- Seven stages: **orientation, trust building, goal clarification, commitment**, then **implementation, high performance, renewal**.
- The first four create the team. The last three sustain it.
- Each stage has a question: why am I here, who are you, what are we doing, how will we do it?
- An unresolved early question, such as unclear goals, resurfaces later as poor performance.

> ON THE EXAM
>
> When a team underperforms, go back to the earliest question it has not answered. Often it is 'what are we doing?'.

Agile team composition

- **Cross-functional**: between them, every skill needed to deliver a finished increment.
- **Generalising specialists**, also called T-shaped people: deep in one skill, able to help in others.
- **Small, stable and dedicated**: moving people between teams costs productivity.
- When one skill becomes a bottleneck, others learn to help, rather than waiting in a queue.

> ON THE EXAM
>
> A single-specialist bottleneck is fixed by broadening skills in the team, not by adding a queue.

Onboarding someone new

- Walk them through the **charter, roles and vision**, then invite them to question it.
- Pair them with a **buddy**, and give them a clear, achievable first task.
- Let the team **revisit its agreements** with the newcomer included.
- Expect a short dip in performance. Plan for it rather than being surprised by it.

> ON THE EXAM
>
> A newcomer changes the team. Update the agreements together, rather than asking them simply to fit in.

Project Lantern: an extract from the team charter

- **Decisions:** technical choices by the developers; booking rules by the product owner; clinic dates by clinic managers.
- **Communication:** urgent issues by phone; everything else in the project channel, answered within half a day.
- **Meetings:** start on time, have an agenda, and end with decisions and owners written down.
- **Disagreement:** raise it with the person first, then at the retrospective, then with the project manager.

> ON THE EXAM
>
> Short, specific and owned by the team. That is what a useful charter looks like.

Scenario signals and the team-building move

THE SCENARIO SAYS	THE MOVE
Nobody agrees how decisions are made	Facilitate a team charter
Two people both think they approve something	Clarify the RACI: one accountable person
Someone keeps breaking an agreed rule	Talk privately, refer to the agreement
New team, lots of arguing	Storming: facilitate, clarify roles and goals
Performing team struggles after a change	Moved back a stage: re-form together
Work queues behind one specialist	Cross-skill: others help and learn

ON THE EXAM

When a people problem appears, check first whether the agreement or role was ever clear.

> A team's charter says direct messages are answered within four working hours. One analyst routinely replies after two days, and colleagues have started complaining to the project manager.
>
> The analyst's work, when it arrives, is good.

What should the project manager do first?

A Raise the analyst's slow replies at the next team meeting, so everyone is clear on the rule.

B Report the analyst's behaviour to their functional manager.

C Speak to the analyst privately, refer to the agreement, and find out what is causing the delay.

D Remove the response-time rule, since the analyst's work is good anyway.

◆ PAUSE AND DECIDE

What should the project manager do first?

A Raise it at the team meeting.

Singles out one person in public. Damages trust and causes defensiveness.

B Report to the functional manager.

Escalation before a single conversation has happened.

C Private talk, refer to the agreement, find the cause.

Rung 1, privately: the delay may have a fixable cause, like workload or notifications.

D Drop the rule.

Changes the team's agreement unilaterally, and ignores colleagues' complaints.

THE SIGNAL

A broken ground rule: private conversation first, anchored on the team's own agreement.

> Three weeks into a new project, team members argue frequently about technical approaches and who should do what. Meetings overrun, and little is decided.
>
> Two team members have told the project manager they are frustrated and wonder if the team will work.

What should the project manager do?

A Treat this as a normal stage of development, and facilitate discussions to clarify roles and agree approaches.

B Separate the people who argue most onto different work packages.

C Ask human resources to mediate between the team members.

D Make the technical and role decisions to end the arguments quickly.

◆ PAUSE AND DECIDE

What should the project manager do?

A Normal stage; facilitate roles and approaches.
Storming is expected. Facilitation moves the team towards norming.

B Separate the arguers.
Avoids the conflict, so it never resolves and the team never forms.

C Ask HR to mediate.
Treats a normal team stage as a personnel problem. Too far, too soon.

D Decide everything.
Ends the arguments but takes ownership away, and skips building agreements.

THE SIGNAL

New team, frequent disagreement about roles and approach: storming. Facilitate through it.

> On a building fit-out, the client's facilities manager and its finance director both believe they approve design changes.
>
> They have approved contradictory changes twice this month, and the contractor has had to rework completed work.

What should the project manager do?

A Follow whichever approval arrives most recently.

B Ask the contractor to stop work until the two managers agree.

C Work with the sponsor and both managers to agree one accountable approver for design changes, and record it.

D Let the design team decide which approvals to follow.

◆ PAUSE AND DECIDE

What should the project manager do?

A. Latest approval wins.
Keeps the contradiction and the rework going, just more randomly.

B. Stop the contractor.
Halts all work, costing time and money, without fixing who decides.

C. Agree one accountable approver, and record it.
Fixes the root cause: two A's for one activity.

D. Design team decides.
Puts the team between two senior stakeholders, deciding something that is not theirs.

THE SIGNAL

Two people who both approve the same thing: the RACI has two A's. One accountable person per activity.

> An agile team of seven has one tester. Developers finish items early in each iteration, then the items queue for testing, and several miss the iteration.
>
> The tester works long hours and is close to exhaustion.

What should the project leader do?

A Help the team spread testing skills, with developers pairing with the tester and taking on testing work.

B Request a second tester from the resource manager.

C Ask the tester to work extra hours near the end of each iteration.

D Reduce the amount of testing so items finish on time.

◆ PAUSE AND DECIDE

What should the project leader do?

A Spread testing skills; pair with the tester.

Cross-skilling removes the queue, protects the tester, and fits agile team design.

B Request a second tester.

Possible later, but adds cost and keeps a handoff queue in place.

C More overtime.

The tester is already near exhaustion. Unsustainable, and it breaks an agile principle.

D Test less.

Trades quality for speed, and defects will cost more later.

THE SIGNAL

Work queues behind one specialist while others wait: broaden skills in the team.

A high-performing team gains a new senior engineer from another department. Within two weeks, meetings become tense and the team's throughput drops.

The newcomer questions several of the team's working practices, and some members feel criticised.

What should the project manager do?

A Ask the new engineer to adapt to the team's existing way of working.

B Ask the resource manager to move the new engineer to another team.

C Tell the team to ignore the criticism and keep working as before.

D Facilitate a session where the team, including the newcomer, revisits its working agreements.

◆ PAUSE AND DECIDE

What should the project manager do?

A. Newcomer must adapt.
Ignores that the team has changed, and dismisses possibly useful ideas.

B. Move the newcomer.
Removing people after two weeks was a losing answer. Nothing has been tried.

C. Ignore the criticism.
Suppresses the conflict. It will return, larger.

D. Revisit the agreements together.
The team has moved back a stage. Re-forming with the newcomer included fixes it.

THE SIGNAL

A performing team struggling after someone joins has moved back a stage. Re-form, together.

> Two months into a project, a team spread across two offices keeps reopening decisions made in earlier meetings.
>
> People arrive unprepared, and several say they are unsure who is entitled to make which decisions.

What should the project manager have done to prevent this?

A. Required weekly written status reports from every team member.

B. Facilitated a team charter at the start, covering decision-making and meeting norms.

C. Supervised team meetings more closely to stop decisions being reopened.

D. Escalated the reopened decisions to the sponsor for a final ruling.

◆ PAUSE AND DECIDE

What should the project manager have done to prevent this?

A Weekly status reports.

Adds reporting. Does nothing about who decides or how meetings run.

B A team charter at the start.

Agrees decision rights and meeting norms before friction appears.

C Closer supervision.

Control instead of agreement. The confusion about decision rights remains.

D Escalate the decisions.

Reactive, and pushes team decisions onto the sponsor.

THE SIGNAL

'Should have done' plus confusion about who decides: the missing piece was a team charter.

Agree how you work. Make roles clear.
Expect storming, and facilitate through it.

MODULE 08 · DOMAIN 1 - PEOPLE

Virtual, distributed and diverse teams

Leading people across distance, time zones, cultures and languages, and making different perspectives an advantage.

DOMAIN I - People EXAM WEIGHT 33% MODULE 08 of 49

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Distance does not break teams. Unspoken assumptions do.

Kinds of team arrangement

ARRANGEMENT	WHAT IT MEANS	MAIN CHALLENGE
Co-located	Everyone in one place	Little, except space and noise
Distributed	Several sites, often in different countries	Time zones, culture, handoffs
Fully remote	Everyone works from separate locations	Trust, isolation, informal contact
Hybrid	Some in the office, some remote, varying by day	Fairness between the two groups

ON THE EXAM

Hybrid is often the hardest. Two groups form without anyone intending it.

Virtual teams: what you gain and what you lose

GAINS

Reach and flexibility

- Access to scarce skills anywhere
- Lower travel and office costs
- Flexible working for people
- Follow-the-sun coverage

LOSSES

Signal and connection

- Fewer informal conversations
- Less body language and context
- Time-zone gaps delay answers
- Isolation and weaker trust

THE TIEBREAKER

The losses are real but predictable. Good answers deliberately replace what distance removes.

Synchronous or asynchronous?

MODE	EXAMPLES	USE IT FOR
Synchronous	Video calls, phone, live workshops	Relationships, conflict, complex decisions, feedback
Asynchronous	Chat, shared documents, recorded demos, virtual boards	Updates, status, reference information, across time zones

ON THE EXAM

Updates can wait for the reader. Relationships and disagreements need people present at the same time.

Working across time zones

- Find the **overlap window** and protect it for work that needs everyone live.
- **Rotate** inconvenient meeting times, so the same people do not always suffer.
- Move routine updates to **asynchronous** channels, and record every decision where all can see it.
- Always scheduling for the largest or most senior group quietly excludes everyone else.

> ON THE EXAM
>
> Fairness across time zones is agreed with the team, and written into the working agreement.

Extra working agreements for distributed teams

- **Core hours** and the overlap window each person commits to.
- **Response times** by channel: urgent by phone, routine within a stated time.
- **Where decisions live**, so nobody relies on memory of a call.
- **Calendars** of local holidays and working weeks, which differ by country.

> ON THE EXAM
>
> The team charter from module 07 matters twice as much when people are apart. Add these to it.

Building trust at a distance

- Meet in person at the start if at all possible. Early relationships carry the team through later strain.
- Use video for relationship conversations, and keep regular **one-to-ones**.
- Make time for **informal contact**: virtual coffee, a social channel, team-building.
- **Recognise** contributions visibly, so remote work is not invisible work.

> ON THE EXAM
>
> Trust grows from contact and reliability. At a distance, both have to be designed in.

Match the channel to the message

MESSAGE	BEST CHANNEL
Performance concern, conflict, bad news	Face to face, or a private video call
Complex problem needing discussion	Video call or live workshop
Quick question, coordination	Chat or phone
Decision record, formal notice, detail	Written: email or a shared document

ON THE EXAM

The more emotional or complex the message, the richer the channel it needs.

Cultural differences that show up at work

- **Directness**: some cultures say 'no' plainly; others signal it gently.
- **Hierarchy**: in some, disagreeing with a senior person openly is uncomfortable.
- **Context**: some rely on explicit words, others on relationship and setting.
- Learn the norms, **ask rather than assume**, and avoid judging one style as right.

> ON THE EXAM
>
> Silence in a meeting may not mean agreement. Create other ways for people to raise concerns.

Working across languages

- Use **plain language**. Avoid idioms, jokes and acronyms that do not travel.
- **Follow up in writing**, with examples, after important verbal agreements.
- **Check understanding** by asking people to play back what they heard, not 'is that clear?'.
- 'Yes' on a call may mean 'I heard you', not 'I agree' or 'I understood'.

> ON THE EXAM
>
> Misunderstanding across languages is prevented by playback and written examples, not by longer calls.

Diversity as an advantage

- The outline asks you to **support the team's varied experiences, skills and perspectives**.
- Diverse teams spot more risks and more options, if everyone's view actually gets heard.
- Use **round-robin** input, silent brainstorming or anonymous tools so quieter voices count.
- Use **inclusive language**, which the Agile Practice Guide's second edition emphasises throughout.

> ON THE EXAM
>
> Diversity only helps when the process lets every perspective in. Design meetings for that.

Running fair hybrid meetings

- The trap: the room talks to itself, and remote people become an audience.
- **Remote-first**: if anyone is remote, everyone joins from their own device.
- Use one shared digital board and record decisions in one visible place.
- No decisions in the corridor afterwards. If it was not recorded, it was not decided.

> ON THE EXAM
>
> Hybrid fairness is a design choice. Put everyone on the same footing, and decide only in the shared space.

Project Lantern at a distance

- The **kiosk supplier's engineers** work from another country, five hours ahead.
- The app team is **hybrid**: three in the Northgate office, four remote.
- Agreements: a daily two-hour overlap window, remote-first meetings, decisions logged in the project channel.
- Supplier specifications are confirmed in writing, with worked examples, after every design call.

> ON THE EXAM
>
> Every agreement here makes an assumption explicit. That is the whole technique.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
One region always joins at an unsocial hour	Agree an overlap window; rotate; go asynchronous
Some members never raise concerns	Other channels: one-to-ones, anonymous, round-robin
Sensitive message to a remote colleague	Private video call, not email or chat
Remote members feel left out of decisions	Remote-first meetings; decisions recorded in one place
Work built to a different understanding	Written confirmation with examples; playback
New virtual team, little collaboration	Kickoff, team-building, a charter with norms

ON THE EXAM

In every case, make the unspoken explicit. None of these answers blames the distance.

> A team spans the UK, India and the USA. The daily coordination meeting is always held at a time that suits the US members, which is midnight in India.
>
> The India-based members have started turning their cameras off and contributing less.

What should the project manager do?

A Remind all members that attendance at the daily meeting is mandatory.

B Hold separate meetings for each region, without sharing the outcomes.

C Excuse the India-based members from the meeting and send them the minutes.

D Work with the team to agree an overlap window, rotate times, and move updates to asynchronous channels.

◆ PAUSE AND DECIDE

What should the project manager do?

A Attendance is mandatory.
Enforces the unfairness. Engagement will keep falling.

B Separate regional meetings, unshared.
Splits the team into silos with no coordination at all.

C Excuse them; send minutes.
Turns part of the team into readers of decisions made without them.

D Agree overlap, rotate, go asynchronous.
Fair, agreed with the team, and keeps everyone included.

THE SIGNAL

One group always carrying the inconvenience: agree fairness with the team.

> Several members of a multinational team come from cultures where openly disagreeing with senior colleagues is uncommon. They rarely speak in team meetings.
>
> Problems they were aware of have surfaced late twice, causing rework.

What should the project manager do?

A Tell these team members that they are expected to speak up in meetings.

B Learn about their norms, and create other ways to raise concerns, such as one-to-ones and anonymous input.

C Invite only senior team members to decision meetings to save time.

D Report the late problems to the team members' functional managers.

◆ PAUSE AND DECIDE

What should the project manager do?

A. Tell them to speak up.
Asks people to act against deep norms, without making it any safer to do so.

B. Learn norms; open other channels.
Respects the culture and still gets the information out early.

C. Only seniors in decision meetings.
Removes exactly the people who know about the problems.

D. Report to their managers.
Punishes a cultural difference. Makes speaking up even less likely.

THE SIGNAL

Silence was not agreement. Change the channel, not the person.

> A remote team member's recent work has contained repeated errors, and a client has noticed. The project manager needs to raise it with him.
>
> He works from home, in the same time zone, and the project manager is drafting an email.

How should the project manager raise the issue?

A In a private video call.

B In a detailed email, so there is a written record.

C In a direct message in the team chat channel.

D During the team's weekly video call.

◆ PAUSE AND DECIDE

How should the project manager raise the issue?

A Private video call.

Rich, two-way and private: right for a sensitive performance conversation.

B Detailed email.

Loses tone, invites misreading, and gives no chance to understand the cause.

C Chat message.

Even thinner than email for a sensitive topic.

D The weekly team call.

Public. Performance is discussed privately.

THE SIGNAL

Sensitive and personal: the richest private channel available. A written note can follow.

> A new fully remote team of eight has been assembled from four organisations. Most members have never met.
>
> After a month, work is passed around by email, nobody asks each other for help, and several members say they feel isolated.

What should the project manager do?

A Increase the frequency of individual status reports.

B Assign each person a fixed area so they depend on each other less.

C Bring the team together for a kickoff and team-building, and agree working norms in a charter.

D Escalate the poor collaboration to the four organisations' managers.

◆ PAUSE AND DECIDE

What should the project manager do?

A. More status reports.
More reporting, no more connection. Isolation continues.

B. Reduce interdependence.
Avoids the problem, and turns a team into eight individuals.

C. Kickoff, team-building, a charter.
Builds the relationships and agreements the team never had.

D. Escalate to four sets of managers.
Nothing has been tried yet, and it is the PM's job to build the team.

THE SIGNAL

People who have never met, and feel isolated: build the team deliberately. It will not happen by itself.

> On a hybrid team, meetings include five people in a conference room and four joining remotely.
>
> Remote members say they cannot follow side conversations, and that decisions are often changed in the corridor after meetings end.

What should the project manager do?

A Require all team members to work from the office on meeting days.

B Stop holding hybrid meetings and use email for all decisions.

C Ask remote members to interrupt more when they cannot hear.

D Agree remote-first meeting norms with the team, and record decisions only in a shared channel.

◆ PAUSE AND DECIDE

What should the project manager do?

A Everyone in the office.

May be impossible, and overrides the organisation's working arrangements.

B Decide everything by email.

Thin channel for decisions, and loses discussion entirely.

C Remote people interrupt more.

Puts the burden on the disadvantaged group, and changes nothing structural.

D Remote-first norms; decisions in one shared place.

Puts everyone on the same footing and closes the corridor loophole.

THE SIGNAL

Two classes of participant: redesign the meeting so everyone joins the same way, and decide in one place.

> An offshore supplier delivered a reporting feature built to a different interpretation of the requirements. The requirements were agreed on a video call.
>
> English is a second language for most of the supplier's team, and they had confirmed they understood.

What should the project manager do to prevent this happening again?

A. After each requirements call, confirm in writing with examples, and ask the supplier to play back its understanding.

B. Begin looking for a supplier whose team are native English speakers.

C. Hold longer requirements calls to cover every detail.

D. Apply the contract's penalty clause for the incorrect delivery.

◆ PAUSE AND DECIDE

What should the project manager do to prevent this happening again?

A Written confirmation, examples, playback.

Turns 'yes, understood' into evidence of understanding, before work starts.

B Switch to native speakers.

Discriminatory in effect, and the problem is the process, not the people.

C Longer calls.

More talk does not create shared understanding. Nothing is checked.

D Apply the penalty.

Punishes a failure the process allowed, and damages the relationship.

THE SIGNAL

'They confirmed they understood' is the planted detail. A yes on a call is not proof of understanding.

Make the unspoken **explicit**: times, channels, norms and understanding.

MODULE 09 · DOMAIN 1 - PEOPLE

Performance, motivation, coaching and emotional intelligence

What motivates people, how to handle underperformance, coaching and mentoring, emotional intelligence and psychological safety.

DOMAIN I - People EXAM WEIGHT 33% MODULE 09 of 49

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People do their best work when they feel **safe**, capable, and that the work matters.

Motivation theories you may meet

THEORY	CORE IDEA
Maslow's hierarchy	Basic needs first: security before belonging, belonging before esteem and growth
Herzberg's two factors	Hygiene factors prevent dissatisfaction; motivators create satisfaction
McGregor's X and Y	Assumptions about people shape how they are managed
McClelland's needs	People are driven by achievement, affiliation or power, in different mixes
Vroom's expectancy	Effort depends on believing it leads to results, and that the results are valued
Pink's drive	Autonomy, mastery and purpose drive knowledge work

ON THE EXAM

The exam rarely names a theory. It describes a situation that one of these explains.

Herzberg: hygiene factors or motivators?

HYGIENE FACTORS

Prevent dissatisfaction

- Pay and working conditions
- Equipment and tools
- Policies and job security
- Relationships with supervisors

MOTIVATORS

Create satisfaction

- Achievement
- Recognition
- Interesting work
- Responsibility and growth

THE TIEBREAKER

Motivators cannot fix a hygiene problem. Pizza and awards do not compensate for broken laptops.

Theory X or Theory Y?

THEORY X ASSUMES

People avoid work

- Need close supervision
- Motivated by pay and fear
- Avoid responsibility
- Leads to micromanagement

THEORY Y ASSUMES

People want to contribute

- Self-directed if committed
- Motivated by meaning and growth
- Seek responsibility
- Leads to empowerment

THE TIEBREAKER

Agile and servant leadership assume Theory Y. Answers built on Theory X usually lose.

Autonomy, mastery and purpose

- **Autonomy**: control over how the work is done. Agile teams self-organise.
- **Mastery**: getting better at something that matters. Learning, pairing, feedback.
- **Purpose**: knowing why the work matters. A clear vision, and seeing the users benefit.
- For knowledge work, bigger bonuses tied to targets can reduce creativity rather than increase it.

> ON THE EXAM
>
> A demotivated knowledge worker usually lacks autonomy, growth or purpose. Money is rarely the fix.

Recognition and rewards

- **Timely and specific**: soon after the contribution, naming what was valuable.
- **Fit the person**: some value public praise, some private thanks, some development.
- **Team-based** for team results, especially in agile work.
- Win-lose individual competitions damage collaboration inside a team.

> ON THE EXAM
>
> Reward the behaviour you want more of. For collaborative work, that means rewarding the team.

Emotional intelligence: four areas

AREA	ABOUT	LOOKS LIKE
Self-awareness	Knowing your own emotions	Noticing you are frustrated before you reply
Self-management	Controlling your reactions	Pausing, staying calm under pressure
Social awareness	Reading others' emotions	Empathy; noticing someone has gone quiet
Relationship management	Handling relationships well	Influence, conflict handling, coaching

ON THE EXAM

Self first, then others. You cannot manage a tense meeting until you have managed yourself.

Emotional intelligence under pressure

- When someone is angry, **stay calm, listen, and acknowledge** their concern first.
- Move heated discussions somewhere **private**, and come back with facts.
- Notice changes in behaviour, such as withdrawal or irritability, and **check in privately**.
- Defending yourself, blaming others, or matching the other person's tone all escalate the situation.

> ON THE EXAM
>
> Acknowledge the emotion before addressing the facts. It is what lets the facts be heard.

A team is safe when people can speak up, admit mistakes and disagree **without fear**.

How leaders build psychological safety

- **Admit your own mistakes** and uncertainty first. It gives others permission.
- **Invite dissent** explicitly: 'What might we be missing?'
- **Thank people for bad news**. How you react the first time decides whether there is a second time.
- Run **blameless** reviews: look at the system, not the culprit.
- One public put-down can undo months of effort.

> ON THE EXAM
>
> Safety is built by the leader's reactions, not by announcements. Model it, and protect it.

Coaching or mentoring?

COACHING

Improve a specific performance

- Focused on a skill or goal
- Mostly asks questions
- Shorter term
- Coach need not be an expert in the role

MENTORING

Grow over the long term

- Focused on career and development
- Shares experience and advice
- Longer term relationship
- Mentor is experienced in the field

THE TIEBREAKER

A specific skill gap: coaching or training. Long-term growth and career: mentoring.

Acting on mentoring opportunities

- The outline asks you to **organise and act on mentoring opportunities**, as part of aligning stakeholders.
- Pair less experienced team members with experienced ones, inside the project.
- Offer to mentor people who want to grow, including future project leaders.
- Mentoring can run both ways: a new stakeholder can be mentored on how the project works.

> ON THE EXAM
>
> Mentoring is part of the job, not a favour. Look for the opportunity in the scenario.

Handling underperformance

01

Private conversation

Describe what you observed, and listen.

02

Find the cause

Skill, clarity, motivation, workload or personal?

03

Agree a plan

Expectations, plus support: training, coaching, rebalancing.

04

Follow up

Check progress and recognise improvement.

05

Escalate

To the functional manager, only if it persists.

ON THE EXAM

Understand before you act. Formal warnings and replacements come last, not first.

Match the response to the cause

THE CAUSE	THE RESPONSE
Missing skill	Training, coaching or pairing
Unclear expectations	Clarify the goal, the standard and the role
Low motivation	Understand why; reconnect with purpose, growth or recognition
Too much work	Rebalance, reprioritise, protect a sustainable pace
Personal difficulty	Support, flexibility, and signposting to help

ON THE EXAM

The same symptom, late or poor work, has five different causes. The fix depends on which one.

Giving feedback that helps

- **Situation, behaviour, impact**: when it happened, what exactly was done, and what effect it had.
- **Timely**, while the event is fresh. Not saved up for an appraisal.
- Praise in public if the person welcomes it. **Correct in private**, always.
- Vague feedback, such as 'be more proactive', gives the person nothing to change.

> ON THE EXAM
>
> Specific and timely beats general and delayed. Corrective feedback is always private.

Sustainable pace and burnout

- Warning signs: long hours becoming normal, rising errors, irritability, withdrawal, sick days.
- Respond by **rebalancing work** and **adjusting scope or dates**, with stakeholders.
- Sustainable pace is an agile principle, and people's wellbeing is part of accountable leadership.
- More overtime to recover a late project is a short-term loan at a very high rate.

> ON THE EXAM
>
> An exhausted team is a risk to the project. Treat it as one.

Project Lantern: keeping people at their best

- The clinic coordinators are **paired with a go-live veteran** from the first clinic.
- The app team's retrospectives are **blameless**; the project manager shares their own misjudgements first.
- Patient panel feedback is shown to the developers every sprint, so they **see the purpose**.
- When the data migration team hit weeks of late nights, scope was re-sequenced with the sponsor.

> ON THE EXAM
>
> Mentoring, safety, purpose and pace: four levers, used on one project.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
A good performer's work suddenly slips	Private conversation; find the cause
Team unhappy about equipment or conditions	Fix the hygiene factor, not more rewards
Nobody raises problems; mistakes are hidden	Build psychological safety; blameless reviews
Someone wants to grow into a new role	Mentoring and stretch opportunities
A stakeholder is angry in a meeting	Stay calm, acknowledge, follow up privately with facts
A team result, an individual prize	Recognise the team

ON THE EXAM

Safe, capable, purposeful: find which of the three is missing, and restore it.

> A developer who has always met commitments has missed three in a row since the team adopted a new cloud platform.
>
> He has not raised any problems, and his colleagues seem comfortable with the platform.

What should the project manager do?

A Issue a formal warning about the missed commitments.

B Meet him privately to understand the cause, and arrange training or pairing if it is a skills gap.

C Reassign his platform work to colleagues who are comfortable with it.

D Report the missed commitments to his functional manager.

◆ PAUSE AND DECIDE

What should the project manager do?

A Formal warning.

Punishes before understanding. The timing points to a skills gap, not attitude.

B Private conversation, then training or pairing.

Finds the cause privately and matches the response to it.

C Reassign his work.

Avoids the gap instead of closing it, and sidelines him on the new platform.

D Report to his manager.

Escalates before a single conversation has happened.

THE SIGNAL

A reliable person slips right after a change: suspect a skills gap, and ask privately.

> A team complains repeatedly about slow, outdated laptops that crash during builds. Morale is falling.
>
> The project manager has arranged team lunches and a monthly recognition award, but the complaints continue.

What should the project manager do next?

A Increase the value of the monthly recognition award.

B Organise a team-building day to lift morale.

C Work with management to get the laptops upgraded or replaced.

D Ask the team to focus on delivery and avoid complaining.

◆ PAUSE AND DECIDE

What should the project manager do next?

A Bigger award.

A motivator cannot fix a hygiene factor. It has already failed once.

B Team-building day.

Nice, and irrelevant to crashing laptops.

C Get the laptops fixed.

Removes the source of dissatisfaction. Also a real productivity impediment.

D Stop complaining.

Dismisses a legitimate problem and damages trust.

THE SIGNAL

Dissatisfaction from conditions: fix the hygiene factor. Motivators will not compensate.

> At a recent sprint review, a senior manager sharply criticised a developer who reported a defect she had found in her own code.
>
> Since then, nobody has raised problems in retrospectives, and two defects were found only by customers.

What should the project leader do?

A. Ask each team member directly, in the next retrospective, to name one problem.

B. Cancel retrospectives until the team is ready to contribute.

C. Rebuild safety: blameless retrospectives, anonymous input, and a private word with the senior manager.

D. File a formal complaint about the senior manager with human resources.

◆ PAUSE AND DECIDE

What should the project leader do?

A Put each person on the spot.
Pressure without safety. People will name trivial problems, or none.

B Cancel retrospectives.
Removes the forum just when it is needed most.

C Blameless, anonymous input, private word with the manager.
Rebuilds safety in the team and addresses its cause.

D Formal HR complaint.
Premature. A private conversation has not been tried.

THE SIGNAL

Problems stopped surfacing after someone was punished for raising one: rebuild psychological safety.

> A capable business analyst on the project tells the project manager she would like to become a project manager within two years.
>
> She asks for advice on how to get there.

What should the project manager do?

A Recommend a certification course and leave her to arrange it.

B Offer to mentor her, and give her a small workstream to lead on the project.

C Arrange a coach to improve her analysis skills.

D Suggest she speaks to her functional manager about career plans.

◆ PAUSE AND DECIDE

What should the project manager do?

A Recommend a course only.

Useful, but hands off. It misses the chance to mentor.

B Mentor her; give her a workstream.

Acts on a mentoring opportunity, with real experience attached.

C A coach for analysis skills.

Coaching targets a current skill. Her goal is long-term growth into a new role.

D Send her to her manager.

Deflects a request the PM is well placed to help with.

THE SIGNAL

Long-term growth into a new role: mentoring, plus a real opportunity to practise.

> In a steering meeting, a regional director raises his voice at the project manager about a two-week delay to his region's go-live.
>
> Part of the delay was caused by his own team's late approval of the site plans.

How should the project manager respond?

A. Stay calm, acknowledge his concern, and arrange a private follow-up with the facts and a recovery plan.

B. Point out that his own team's late approvals caused much of the delay.

C. End the meeting until he is ready to discuss the issue calmly.

D. Report his behaviour to the sponsor after the meeting.

◆ PAUSE AND DECIDE

How should the project manager respond?

A Calm, acknowledge, private follow-up with facts.

Self-management first, then empathy; the facts come later, privately.

B Blame his team.

True, but said in public and in anger it escalates and humiliates.

C End the meeting.

Disrupts every other attendee and leaves the concern hanging.

D Report him to the sponsor.

Escalates a relationship the PM has not tried to repair.

THE SIGNAL

An angry stakeholder in public: manage yourself, acknowledge them, and bring the facts privately.

> An agile team delivered a major release early through a collective effort. Several members covered for each other and paired on the hardest problems.
>
> Senior management wants to name one developer 'employee of the quarter' for the release.

What should the project manager recommend?

A Support the award, since recognition motivates people.

B Ask the team to vote for the one person who should receive it.

C Decline any recognition, to avoid favouritism.

D Recommend recognising the whole team for the release.

◆ PAUSE AND DECIDE

What should the project manager recommend?

A Support the individual award.

Singles out one person for a collective result, and undermines collaboration.

B Team votes for one person.

Turns colleagues into competitors for recognition they earned together.

C No recognition.

Wastes a real motivator. The team did something worth recognising.

D Recognise the whole team.

Rewards the collaborative behaviour that produced the result.

THE SIGNAL

A team result earns team recognition. Reward the behaviour you want repeated.

Safe, capable, purposeful. **Find what is missing** before you act.

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MODULE 10 · DOMAIN 1 - PEOPLE

Problem solving and team decisions

Finding root causes instead of treating symptoms, and choosing the right way for a group to decide.

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Most problems in a scenario are symptoms. The right answer treats **the cause**.

A simple problem-solving sequence

01

Define

State the problem precisely, with facts.

02

Find the cause

Analyse why it happens, not just what.

03

Generate options

Several ways to address the cause.

04

Decide

Choose, using an agreed method.

05

Act and check

Implement, then verify it worked.

ON THE EXAM

Skipping straight from 'define' to 'act' is the most common error, in real projects and in exam options.

Symptom or root cause?

TREATING THE SYMPTOM

Fast, and it comes back

- Fix each defect as it appears
- Add overtime when late
- Separate people who argue
- Add another approval step

TREATING THE CAUSE

Slower, and it stays fixed

- Find why defects are created
- Find why estimates are wrong
- Find what the argument is really about
- Find why errors get through

THE TIEBREAKER

A recurring problem means the cause is still there. Look for the option that finds it.

Five whys: an example from Project Lantern

- Why did the booking app crash at go-live? **The appointment service timed out.**
- Why? **It was flooded with requests from the kiosks.**
- Why? **The kiosks retry every second when they get no reply.**
- Why? **The retry setting was never tested under load.**
- Why? **Load testing did not include the kiosks.** That is the cause to fix.

> ON THE EXAM
>
> Keep asking why until you reach something you can change in the process. The number five is a guide, not a rule.

Two more root-cause tools

- **Cause-and-effect diagram**, also called fishbone or Ishikawa: groups possible causes by category.
- Categories are often people, process, technology, materials, environment and measurement.
- **Pareto analysis**: rank causes by frequency. Often a few causes explain most of the problems.
- Use them **with the team**. The people doing the work usually know the causes.

> ON THE EXAM
>
> Root-cause analysis is a team activity. The project manager facilitates; the team supplies the knowledge.

Ways a group can decide

METHOD	WHAT IT MEANS	FITS WHEN
Unanimity	Everyone actively agrees	Rare; very high-stakes, small groups
Consensus	Everyone can support it, even if it was not their first choice	Commitment matters; the usual team default
Majority	More than half agree	Speed matters; the minority can accept it
Plurality	The largest group wins, even under half	Many options, low stakes
Autocratic	One person decides, perhaps after input	Emergencies, or it is clearly their decision

ON THE EXAM

Consensus is not unanimity. It means everyone can live with and support the decision.

Quick techniques teams use

TECHNIQUE	HOW IT WORKS	GOOD FOR
Fist of five	0 to 5 fingers of support	Checking consensus, surfacing concerns
Roman voting	Thumbs up, sideways or down	A quick yes, unsure or no
Dot voting	Dots placed on options	Prioritising a longer list
Delphi	Anonymous expert rounds, summarised	Estimates where experts sway each other
Weighted criteria	Score options on weighted criteria	Complex choices, such as suppliers

ON THE EXAM

Pick the technique by purpose: support, priority, avoiding influence, or criteria.

Fist of five in practice

- **5**: fully support, will champion it. **4**: support. **3**: can live with it.
- **2**: have concerns to discuss. **1**: serious concerns. **0**: cannot support it.
- Anyone showing **2 or fewer** explains their concern, and the team discusses it.
- Vote again. Proceed when **everyone shows 3 or more**.

> ON THE EXAM
>
> The low votes are the point. They surface concerns that a show of hands would bury.

Diverge, then converge

- **Diverge**: generate many options without judging them.
- **Converge**: evaluate, combine and choose.
- Start with **silent, individual** idea generation, then share. It stops the first idea anchoring everyone.
- Criticising ideas while generating them shuts the quiet voices down.

> ON THE EXAM
>
> Separate generating from judging. Mixing them produces fewer and safer ideas.

Groupthink and the loudest voice

- **Groupthink**: a group agrees quickly to keep harmony, and nobody tests the idea.
- The **highest-paid person's opinion** can end a discussion before evidence is heard.
- Counter them: invite dissent, assign a devil's advocate, run a pre-mortem, gather input anonymously.
- A **pre-mortem** asks: 'It is a year from now and this failed. Why?'

> ON THE EXAM
>
> Fast, unchallenged agreement on something important is a warning sign, not a success.

When to decide

- **Reversible** decisions: decide quickly with the information available, and adjust later.
- **Irreversible** decisions: take more care, gather more evidence, involve more people.
- Lean thinking's **last responsible moment**: delay a commitment until just before waiting would cost you.
- Weeks of debate over an easily reversed choice is waste. Timebox it.

> ON THE EXAM
>
> Match the effort to the reversibility. Cheap-to-change decisions deserve speed, not perfection.

Record and communicate decisions

- Keep a **decision log**: what was decided, why, by whom, and on what date.
- Share decisions with everyone affected, including those who were not in the room.
- Reopen a decision only when **new information** appears, and say what it is.
- Decisions that live only in memory get relitigated in every meeting.

> ON THE EXAM
>
> A decision nobody recorded will be made again. Log it, share it, and reopen it only for new facts.

Project Lantern: a team decision

- Question: should clinics **go live one at a time**, or three at once?
- The team **diverged** silently, listing risks and benefits, then shared.
- A **pre-mortem** surfaced a risk nobody had said aloud: support staff stretched across three sites.
- **Fist of five**: two people showed a 2. After discussion, one clinic at a time won support from all.

> ON THE EXAM
>
> The method surfaced the decisive risk. A quick majority vote would have chosen three at once.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
A problem keeps recurring	Root-cause analysis with the team
Need to check team support quickly	Fist of five or Roman voting
Experts sway each other's estimates	Delphi: anonymous rounds
Instant agreement, nobody challenges	Invite dissent; pre-mortem
Teams interpret the vision differently	Break down where understanding diverged
Weeks of debate on a reversible choice	Timebox it and decide

ON THE EXAM

Fix causes, not symptoms, and choose a decision method that fits the purpose.

> For four sprints in a row, the team has found similar integration defects late in the sprint. Each time, developers fix them individually, working late.
>
> The team is tired, and the product owner is losing confidence in the release date.

What should the project leader do?

A Facilitate a root-cause analysis of the defects with the team.

B Add a senior developer's review before any code is merged.

C Request an extra tester for the rest of the release.

D Ask the developers to fix defects faster so the sprint is not affected.

◆ PAUSE AND DECIDE

What should the project leader do?

A Facilitate root-cause analysis.

Rung 2 with the team: similar defects every sprint means a cause nobody has found.

B Add a senior review.

A new gate that catches symptoms later, and creates a bottleneck.

C Extra tester.

Finds the defects, but does not stop them being created.

D Fix faster.

The symptom, again, with more pressure on a tired team.

THE SIGNAL

'Similar defects, four sprints running': a recurring problem. Find the cause before adding people or gates.

> A team has discussed changing its code branching strategy for twenty minutes. Most people seem positive, but two have said little.
>
> The project leader wants to check support quickly and make sure any concerns come out.

Which technique should the project leader use?

A. A majority vote by show of hands.

B. Decide personally, since most people seem positive.

C. Fist of five, discussing any vote of two or fewer.

D. The Delphi technique with external experts.

◆ PAUSE AND DECIDE

Which technique should the project leader use?

A. Majority vote.
Quick, but buries the two quiet people's concerns under the majority.

B. Decide personally.
A team working-practice decision, taken away from the team.

C. Fist of five.
Fast, checks support, and gives low votes a reason to speak.

D. Delphi with outside experts.
Slow, and aimed at expert estimates, not team consensus.

THE SIGNAL

Check support fast AND surface concerns: fist of five does both.

> A project manager needs duration estimates for a novel piece of engineering work. Five internal experts will estimate it.
>
> In past estimating sessions, the most senior expert spoke first and the others tended to adjust towards her figure.

Which approach should the project manager use?

A. Ask the most senior expert to provide the estimate, since the others defer to her anyway.

B. Hold an open meeting and average the figures the experts give.

C. Take a majority vote among the five experts.

D. Use the Delphi technique, with anonymous rounds summarised by a facilitator.

◆ PAUSE AND DECIDE

Which approach should the project manager use?

A The senior expert decides.

Throws away four experts' knowledge.

B Open meeting, average.

The anchoring still happens before anyone speaks their number.

C Majority vote.

Estimates are not a vote, and the influence problem remains.

D Delphi.

Anonymity removes the senior expert's pull; rounds still converge.

THE SIGNAL

Experts influencing each other: anonymity. That is exactly what Delphi is for.

> In a short meeting, a team agreed unanimously to select a supplier recommended by the sponsor's preferred contact. Nobody asked any questions.
>
> The contract is large, and switching suppliers later would be expensive.

What should the project manager do?

A Accept the decision, since it was unanimous.

B Before confirming, invite challenge, for example with a pre-mortem on the choice.

C Overrule the team and choose the supplier personally.

D Escalate the selection to the procurement board to decide.

◆ PAUSE AND DECIDE

What should the project manager do?

A Accept, it was unanimous.

Unanimous without questions, on an irreversible choice, suggests groupthink.

B Invite challenge; pre-mortem.

Tests an expensive, hard-to-reverse decision before committing.

C Overrule the team.

Replaces one untested view with another.

D Escalate to procurement.

Avoids the problem rather than testing the decision properly.

THE SIGNAL

Instant, unchallenged agreement on an irreversible choice: test it before you commit.

> Two agile teams are building parts of the same customer portal. They have delivered features that conflict: one assumes customers log in first, the other assumes they browse anonymously.
>
> Both teams say they followed the product vision.

What should the project manager do first?

A Choose one team's interpretation and ask the other to rework.

B Escalate the conflict to the sponsor to decide.

C Work with both teams to find where their understanding of the vision diverged, then restate it with examples.

D Give the teams separate product backlogs to avoid future conflicts.

◆ PAUSE AND DECIDE

What should the project manager do first?

A. Pick one interpretation.
Treats the symptom; the vision is still ambiguous for the next feature.

B. Escalate to the sponsor.
Too early. Nobody has yet found where the misunderstanding lies.

C. Find where understanding diverged; restate with examples.
The outline's own enabler: break down the situation to find the root cause.

D. Separate backlogs.
Hides the conflict inside one product, making it worse.

THE SIGNAL

Two teams, one vision, conflicting work: the vision is ambiguous. Find where, and fix the vision.

> A team has spent three weeks debating which user-interface component library to use. Both candidates are well supported.
>
> The developers agree that switching libraries later would take about two days.

What should the project leader do?

A Let the debate continue until the team reaches full agreement.

B Escalate the choice to the architecture board.

C Commission a detailed comparison study of the two libraries.

D Help the team timebox the decision, choose with what it knows, and revisit only if problems appear.

◆ PAUSE AND DECIDE

What should the project leader do?

A Debate until full agreement.

Three weeks already spent on a two-day decision. More is waste.

B Escalate to the architecture board.

Adds delay to a reversible, team-level choice.

C Detailed comparison study.

More analysis for a choice that costs two days to undo.

D Timebox, decide, revisit if needed.

Matches effort to reversibility, and keeps the decision with the team.

THE SIGNAL

Cheap to reverse, and already weeks of debate: timebox it and decide.

Find the cause. Choose a method that fits. Match the effort to the reversibility.

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MODULE 11 · DOMAIN 1 - PEOPLE

Managing conflict

Where conflict comes from, reading its context and intensity, the five ways of handling it, and when each one is right.

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Conflict is not the problem. Unmanaged conflict is.

Task conflict or relationship conflict?

TASK CONFLICT

About the work

- Disagreement on approach or priorities
- Can improve decisions
- Keep it respectful and focused
- Encourage it, within ground rules

RELATIONSHIP CONFLICT

About the people

- Personal friction and blame
- Damages trust and performance
- Often grows from unresolved task conflict
- Address it early and privately

THE TIEBREAKER

Healthy teams disagree about the work and respect each other. Protect the first, prevent the second.

Where project conflict comes from

SOURCE	EXAMPLE
Scarce resources	Two workstreams need the same specialist this week
Schedules and priorities	Whose work goes first when time is short
Technical opinions	Rewrite or patch; which tool; which design
Procedures and roles	Who approves; who is responsible; which process applies
Cost	Budget allocation between teams or scope items
Personal work styles	Detail-focused versus big-picture; fast versus careful

ON THE EXAM

Most project conflict is about resources, priorities and procedures, not personalities. Look there first.

Analyse the context before acting

- **Importance**: how much does the outcome matter to the project and to each party?
- **Relationship**: will these people need to work together afterwards?
- **Time**: is there time for a full discussion, or must something happen today?
- **Intensity**: is this still a problem to solve, or has it become personal?
- **Safety**: is anyone's safety, or a legal obligation, at stake?

> ON THE EXAM
>
> The same disagreement needs different handling at different intensities and time pressures.

How intense is it? Five levels of conflict

LEVEL	WHAT YOU HEAR	WHAT HELPS
1. Problem to solve	'We see this differently'	Collaborate on the problem
2. Disagreement	Self-protection; guarded talk	Build safety, then problem-solve
3. Contest	Taking sides; 'always' and 'never'	Facilitate; separate facts from positions
4. Crusade	Defending the faction; principles over facts	Structured mediation; lower the heat
5. World war	Winning means destroying the other side	Separate the parties; formal intervention

ON THE EXAM

The higher the level, the more structure is needed. Most exam scenarios sit at levels 1 to 3.

The five ways of handling conflict

MODE	ALSO CALLED	WHAT HAPPENS
Collaborating	Problem solving, confronting	A solution meeting both sides' real needs
Compromising	Reconciling	Each side gives something up
Smoothing	Accommodating	Stress agreement; often concede
Forcing	Directing, competing	One view imposed by authority
Withdrawing	Avoiding	Step back, postpone, leave it

ON THE EXAM

Collaborating is usually best. But every mode has a situation where it is the right answer.

Two dimensions behind the five modes

ASSERTIVENESS

Concern for my goals

- High: forcing, collaborating
- Low: smoothing, withdrawing
- Middle: compromising

COOPERATIVENESS

Concern for their goals

- High: smoothing, collaborating
- Low: forcing, withdrawing
- Middle: compromising

THE TIEBREAKER

Collaborating is high on both. That is why it produces win-win outcomes, and why it takes the most time.

When each mode is the right answer

MODE	RIGHT WHEN
Collaborating	The issue matters, the relationship matters, and there is time to explore
Compromising	Time is short, both sides have equal power, and collaboration has stalled
Smoothing	Keeping the relationship matters more than this issue, or to calm things first
Forcing	Safety, legal or ethical obligations, or a true emergency
Withdrawing	Emotions are too high right now, or the issue is trivial or will resolve itself

ON THE EXAM

Forcing is right for safety and law. Withdrawing is right as a brief cooling-off. Neither is a long-term fix.

Who resolves it, and in what order

01

The parties

Encouraged to resolve it directly, as the charter says.

02

Facilitated

The project leader brings them together, privately.

03

Team process

Ground rules, retrospective, decision method.

04

Escalate

Functional manager, sponsor or HR, if unresolved.

ON THE EXAM

The people in conflict own it first. The leader facilitates; escalation is the last step.

Facilitating a conflict conversation

- Meet **privately**, in a neutral setting, with both parties if they are ready.
- Separate **positions** from **interests**: what they demand, and why they need it.
- Agree the facts, then explore options that meet both sides' interests.
- Agree what happens next, write it down, and follow up.
- Taking sides, or ruling on who was right, turns you from facilitator into combatant.

> ON THE EXAM
>
> Interests are usually compatible even when positions are not. Find the interests.

Conflict with and between external stakeholders

- The outline asks you to **communicate conflict management principles** to external stakeholders too.
- Anchor disagreements on the **agreed objectives**, the contract, or regulation, not on opinions.
- With suppliers, use the **contract's dispute process**, but talk first.
- Siding automatically with the most senior or most powerful stakeholder is not resolution.

> ON THE EXAM
>
> With external parties, objective criteria settle more disputes than authority does.

Project Lantern: a conflict, resolved

- The data migration lead and the app team both needed the **test database** in the same week.
- **Positions**: each wanted exclusive use for five days.
- **Interests**: migration needed a clean copy for one overnight run; the app team needed daytime testing.
- Resolution: app testing by day, migration's run overnight, with a refresh each morning. Both got what they needed.

> ON THE EXAM
>
> Incompatible positions, compatible interests. That is what collaborating finds.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Two people disagree; there is time	Meet them, find interests, collaborate
Deadline today; collaboration has stalled	Compromise
Someone wants to bypass a safety or legal step	Force: the requirement is not negotiable
Voices raised, personal remarks	Pause, cool off, then meet privately
Vigorous but respectful technical debate	Healthy task conflict: protect it
Two powerful stakeholders disagree	Facilitate on objective criteria, not rank

ON THE EXAM

Read the context, then choose the mode. Collaborate by default; the other four have specific triggers.

> The development lead and the test lead are arguing about access to the only performance-test environment. Each has booked it for the same week.
>
> Both have raised the issue separately with the project manager, and the schedule is at risk.

What should the project manager do first?

A Decide who gets the environment, based on the critical path.

B Meet both leads to understand what each needs, and facilitate a solution together.

C Escalate the conflict to both leads' functional managers.

D Tell the leads to resolve it between themselves by the end of the day.

◆ PAUSE AND DECIDE

What should the project manager do first?

A. Decide, using the critical path.
Forcing, before anyone knows whether both needs could be met.

B. Meet both; facilitate together.
Collaborating: finds the interests behind two booked weeks.

C. Escalate to managers.
Too early. Nobody has facilitated a conversation yet.

D. Sort it out yourselves.
Withdrawing. They have already failed to, and brought it to the PM.

THE SIGNAL

A resource conflict with time to talk: collaborate. Positions clash; interests may not.

> On a construction project, a subcontractor's foreman wants to skip a scheduled scaffold safety inspection to hit a milestone on Friday.
>
> The site safety officer insists the inspection must happen. The two are arguing on site.

How should the project manager handle the conflict?

A Require the inspection to go ahead as scheduled, before work continues.

B Agree a shorter, partial inspection as a compromise.

C Emphasise the shared goal of the milestone to calm both parties.

D Leave the foreman and safety officer to agree between themselves.

◆ PAUSE AND DECIDE

How should the project manager handle the conflict?

A Require the inspection.

Forcing is correct when safety is at stake. The requirement is not negotiable.

B Partial inspection.

Compromises on safety. A half inspection is a failed inspection.

C Emphasise the milestone.

Smoothing, and it signals that the milestone outranks safety.

D Let them agree.

Withdrawing from a safety decision the PM must own.

THE SIGNAL

Safety is the one place where forcing is the right answer. Recovery planning comes afterwards.

> Two department heads both need the same data specialist full-time for the next two weeks. Both needs are legitimate and both deadlines are firm.
>
> The project manager has facilitated a discussion, but no creative solution has emerged, and a decision is needed by tomorrow.

Which approach is most appropriate now?

A Assign the specialist to the more senior department head.

B Postpone the decision until one of the deadlines moves.

C Stress how much the two departments agree on, to reduce tension.

D Agree a split of the specialist's time that both department heads can accept.

◆ PAUSE AND DECIDE

Which approach is most appropriate now?

A Seniority decides.
Forcing by rank. Ignores that both needs are legitimate.

B Postpone.
Withdrawing, when a decision is needed tomorrow.

C Stress agreement.
Smoothing. Pleasant, but it does not allocate the specialist.

D Agree a split.
Compromise fits: collaboration tried, equal claims, time nearly out.

THE SIGNAL

Collaboration tried, equal claims, deadline tomorrow: compromise is the right mode.

> In a design review, a disagreement between the lead architect and a senior developer becomes heated. Voices are raised and the remarks turn personal.
>
> Other attendees look uncomfortable, and the meeting has thirty minutes left.

What should the project manager do?

A Press for a decision now, since everyone is present.

B Call a short break, then meet the two privately once things have calmed.

C Support the architect, since the design is the architect's responsibility.

D Note the behaviour and report it to human resources.

◆ PAUSE AND DECIDE

What should the project manager do?

A. Press for a decision now.
Decisions made in anger are poor, and it escalates in front of others.

B. Break, then meet privately.
Temporary withdrawing to cool off, followed by private collaboration.

C. Back the architect.
Takes sides mid-conflict and turns it into a contest.

D. Report to HR.
Premature and formal. One heated meeting needs a conversation first.

THE SIGNAL

When it turns personal, pause first. Withdrawing briefly is right; withdrawing permanently is not.

An agile team regularly debates design choices vigorously in refinement sessions, but always respectfully, and its delivery is strong.

A newly appointed sponsor observes one session, thinks the team is dysfunctional, and asks the project manager to stop the debates.

What should the project manager do?

A Explain to the sponsor that respectful debate is healthy and improves decisions, and share the team's results.

B Ask the team to stop debating in refinement sessions.

C Replace the most outspoken team members.

D Require all design decisions to be made by majority vote.

◆ PAUSE AND DECIDE

What should the project manager do?

A Explain healthy debate; show results.

Protects valuable task conflict, and manages the sponsor's perception with evidence.

B Stop the debates.

Suppresses the task conflict that is producing good decisions.

C Replace outspoken members.

Punishes people for doing their job well.

D Majority votes only.

Replaces reasoning with counting, and weakens design quality.

THE SIGNAL

Respectful, work-focused disagreement with strong results is healthy task conflict. Protect it.

> The marketing director wants a product's packaging to claim it is '100% recyclable'. The legal counsel says the claim cannot be supported in two target markets.
>
> Both are senior stakeholders, and each has asked the project manager to support their position.

What should the project manager do?

A Support the marketing director, who is also the project sponsor's peer.

B Let the packaging design team choose between the two positions.

C Facilitate a meeting of both, using the regulations and product objectives to find wording that meets both interests.

D Escalate the disagreement to the chief executive.

◆ PAUSE AND DECIDE

What should the project manager do?

A. Side with marketing.
Sides by status, and ignores a real compliance risk.

B. Designers choose.
Pushes a senior, legal-risk decision onto a team that cannot own it.

C. Facilitate on regulation and objectives.
Collaborates on objective criteria: likely a claim that is both appealing and lawful.

D. Escalate to the chief executive.
Too early. The two have not yet been brought together.

THE SIGNAL

Two senior stakeholders, opposite positions: bring them together and anchor on objective criteria.

Protect task conflict. Collaborate by default. **Force only for safety and law.**

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MODULE 12 · DOMAIN 1 - PEOPLE

Shared vision and stakeholder analysis

Building a vision everyone can repeat, finding every stakeholder, and working out who needs what kind of attention.

DOMAIN I - People EXAM WEIGHT 33% MODULE 12 of 49

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A vision nobody can repeat in their own words is **not shared**.

What a common vision does

- Gives every decision a **filter**: does this move us towards the vision?
- **Aligns** teams, stakeholders and suppliers who never meet each other.
- Lets empowered teams decide well without asking permission each time.
- The outline asks you to **ensure it is shared, promote it, and keep it current**.

> ON THE EXAM
>
> Empowerment without a shared vision produces confident teams heading in different directions.

Tools for creating a shared vision

TOOL	HOW IT WORKS
Vision statement	One or two sentences on who it is for, what it does, and why it matters
Elevator statement	For [who], who [need], our [product] is [what], that [benefit], unlike [alternative]
Product box	The team designs the box it would sell in: name, three benefits, one picture
Vision board or canvas	One page: target group, needs, product, business goals
Project canvas	A one-page view of purpose, objectives, stakeholders and constraints

ON THE EXAM

Create the vision together, with key stakeholders. A vision handed down is understood, not shared.

Promoting the vision and keeping it current

- **Repeat it**: at kickoff, in planning, at every review, when onboarding people.
- Make it **visible**: on the team board, at the top of the backlog, in status reports.
- **Revisit it** when the market, regulation or organisation changes. M.O.R.E. calls this relentlessly reassessing.
- A vision that no longer matches reality is worse than none: it confidently points the wrong way.

> ON THE EXAM
>
> When the world changes, update the vision with stakeholders, then tell everyone what changed and why.

Spotting a misunderstood vision

- Teams deliver work that **conflicts**, each sure they followed the vision.
- Stakeholders are **surprised** at reviews by what was built.
- Priorities keep **changing direction** as different people interpret the goal.
- Test it: ask people to restate the vision, then break down exactly where their versions differ.

> ON THE EXAM
>
> The outline's enabler: break down the situation to identify the root cause of the misunderstanding.

Identifying stakeholders

- Start **early**, as soon as the project is authorised, and keep going **throughout**.
- Sources: the charter, the business case, organisation charts, contracts, past projects' lessons learned.
- Techniques: interviews, brainstorming, asking each stakeholder 'who else should we talk to?'
- A stakeholder you miss does not disappear. They appear later, usually objecting.

> ON THE EXAM
>
> Identification is continuous. New stakeholders appear as the project reaches new people and places.

The stakeholder register

PART	TYPICAL CONTENTS
Identification	Name, role, organisation, contact details, location
Assessment	Requirements, expectations, interests, influence, attitude
Classification	Internal or external; category; position on analysis grids
Engagement	Current and desired engagement; preferred communication

ON THE EXAM

The register is a living document. It is updated whenever a stakeholder, or their position, changes.

Handle the register with care

- Assessment notes can be **sensitive**: 'resists the project', 'low interest', 'influences the board'.
- Share the parts people need, such as names, roles and contact details, widely.
- Keep **sensitive assessments restricted** to those who need them.
- A stakeholder who reads a blunt assessment of themselves becomes a harder stakeholder.

> ON THE EXAM
>
> Transparency is a value, but so is discretion. Share the register in layers.

The power and interest grid

QUADRANT	STRATEGY	IN PRACTICE
High power, high interest	Manage closely	Involve in decisions; regular, rich engagement
High power, low interest	Keep satisfied	Brief, relevant updates; consult on what affects them
Low power, high interest	Keep informed	Regular information; listen to their concerns
Low power, low interest	Monitor	Light touch; watch for changes in position

ON THE EXAM

The trap is the high-power, low-interest group: bury them in detail and they disengage or turn hostile.

Other analysis models

MODEL	WHAT IT ADDS
Power/influence, impact/influence	Grid variations with other axes
Salience model	Power, legitimacy, urgency: who first
Directions of influence	Upward, downward, outward, sideways
Stakeholder cube	Three attributes in one view
Personas	Fictional profiles of user groups

ON THE EXAM

The model matters less than the habit: analyse before you engage.

The salience model

- **Power**: can they impose their will on the project?
- **Legitimacy**: is their involvement appropriate and recognised?
- **Urgency**: do they need attention now, and is their claim time-sensitive?
- Stakeholders with **all three** are definitive: they need attention first.

> ON THE EXAM
>
> A regulator with an urgent, legitimate demand outranks a louder stakeholder with only one attribute.

Engagement assessment: Project Lantern

STAKEHOLDER	UNAWARE	RESISTANT	NEUTRAL	SUPPORTIVE	LEADING
Clinic managers		C		D	
Reception staff union rep	C		D		
Head of patient access				C	D
Health data regulator			C, D		

ON THE EXAM

C is where they are now. D is where you need them. The gap drives the engagement plan.

Closing the engagement gap

- **Unaware**: inform them. They cannot support what they do not know about.
- **Resistant**: understand why, privately. Resistance usually protects a real interest.
- **Neutral**: show how the project serves their goals.
- Not everyone needs to lead. Aim for the engagement the project actually needs from each person.

> ON THE EXAM
>
> Resistance is information. Find the interest it protects before trying to change it.

Categorising stakeholders

- The outline asks you to **categorise stakeholders**, as the first step in aligning expectations.
- By relationship: internal or external; customer, supplier, regulator, sponsor, user.
- By stance: supporters, neutrals, resistors.
- By need: what they require from the project, and what they fear from it.
- Categories let you tailor communication and engagement per group, not one message for all.

> ON THE EXAM
>
> One message for every stakeholder satisfies none of them. Categorise, then tailor.

Personas: stakeholders on agile projects

- A **persona** is a realistic, fictional profile of a type of user: goals, frustrations, context.
- Built from research with real users, not invented at a desk.
- Keeps the team designing for real needs when users cannot be in every conversation.
- Lantern: 'Maria, 78, books by phone, struggles with small screens, needs large text'.

> ON THE EXAM
>
> Personas complement real user contact. They never replace it.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
People give different answers about the goal	Create the vision together with key stakeholders
Project just authorised	Identify stakeholders from charter, business case, interviews
Powerful executive, little interest	Keep satisfied: brief, relevant updates
Key stakeholder is resistant	Understand their concerns privately; plan engagement
Sensitive notes on a shared register	Restrict assessments; share the rest
The market has changed the goal	Revisit and update the vision, then communicate it

ON THE EXAM

Vision first, then find everyone, analyse them, and tailor your engagement to each.

> Two weeks into a new project, the project manager asks five team members and two key stakeholders what the project is for.
>
> The seven answers range from 'cutting costs' to 'improving customer experience' to 'replacing an old system'.

What should the project manager do?

A Send everyone the approved business case to read.

B Start delivery and let a shared understanding develop over time.

C Facilitate a workshop with key stakeholders and the team to create a shared vision.

D Ask the sponsor to write and circulate a vision statement.

◆ PAUSE AND DECIDE

What should the project manager do?

A Send the business case.

Information is not alignment. People read the same document differently.

B Let it develop.

Divergent understanding produces conflicting work, not convergence.

C Workshop to create the vision together.

Builds a vision people can repeat, because they helped shape it.

D Sponsor writes it.

A vision handed down is understood at best, not shared.

THE SIGNAL

Seven people, several different purposes: the vision is not shared. Create it together.

> A project manager has just received a signed project charter for a new warehouse management system.
>
> The charter names the sponsor and three departments. No other stakeholder work has been done.

What should the project manager do next?

A Identify stakeholders using the charter, business case, organisation charts and interviews, and record them in a register.

B Write the communications management plan for the three departments named.

C Send a kickoff meeting invitation to the three department heads.

D Wait until requirements are gathered to see who else is involved.

◆ PAUSE AND DECIDE

What should the project manager do next?

A Identify stakeholders broadly; start the register.

The first step after authorisation. Everything else depends on it.

B Communications plan for three departments.

Plans communication before knowing who needs it.

C Invite three heads to kickoff.

Assumes the charter's list is complete. Warehouse staff, IT and suppliers are missing.

D Wait for requirements.

Requirements come from stakeholders. Waiting reverses the order.

THE SIGNAL

Just authorised: identify stakeholders first, widely. The charter's list is a starting point, not the list.

> The chief financial officer can stop the project's funding, but shows little interest in its day-to-day progress.
>
> She has complained that the weekly twenty-page status reports she receives waste her time.

How should the project manager engage her?

A Keep sending the full weekly reports so she has complete information.

B Keep her satisfied: short, relevant updates, and consult her on decisions that affect funding.

C Remove her from all distribution lists and monitor her only.

D Invite her to the team's daily coordination meeting.

◆ PAUSE AND DECIDE

How should the project manager engage her?

A Keep the full reports.

Ignores her feedback and the exact trap of the high-power, low-interest group.

B Keep satisfied: short, relevant, consult on funding.

High power, low interest: the grid's strategy, fitted to what she cares about.

C Monitor only.

Treats a stakeholder who can stop funding as if she had no power.

D Daily meetings.

Even more detail, for someone who wants less.

THE SIGNAL

High power, low interest: keep satisfied. Brief, relevant, and consulted on what matters to them.

> The engagement assessment matrix shows the head of operations as resistant. The project needs her to be supportive before the roll-out begins.
>
> She has attended meetings but raised few specific objections.

What should the project manager do next?

A Ask the sponsor to instruct her to support the project.

B Stop inviting her to meetings until she is more positive.

C Send her more frequent and detailed progress reports.

D Meet her privately to understand her concerns, then plan actions to address them.

◆ PAUSE AND DECIDE

What should the project manager do next?

A Sponsor instructs her.

Compliance, not support. The real concern stays hidden and resurfaces at roll-out.

B Exclude her.

Turns quiet resistance into open opposition.

C More reports.

Information does not address a concern nobody has identified.

D Meet privately; understand; plan actions.

Resistance usually protects an interest. Find it, then address it.

THE SIGNAL

Resistant, with no stated objection: the concern is hidden. Find it privately before trying to change it.

> The stakeholder register is on a shared drive that the whole project can read. It includes the project manager's notes that a vice president 'opposes the project and influences the board'.
>
> A team member asks for the full register to plan a presentation to that vice president's department.

What should the project manager do?

A Share the full register, since project information should be transparent.

B Restrict the sensitive assessment notes, and share the information the team member needs.

C Delete all assessment notes from the register.

D Refuse all access to the register.

◆ PAUSE AND DECIDE

What should the project manager do?

A. Share it all.

Sensitive judgements about a senior person, on an open drive, risk serious harm.

B. Restrict sensitive notes; share what is needed.

Balances transparency with discretion, and fixes the open-drive exposure.

C. Delete the notes.

Loses analysis the project needs to engage the stakeholder well.

D. Refuse all access.

Blocks legitimate work. Most of the register is not sensitive.

THE SIGNAL

Share the register in layers. Assessments of individuals stay restricted.

> A product vision promised 'the only app that lets customers compare every local energy tariff'. A large competitor has just launched exactly that, for free.
>
> The team is carrying on with its existing priorities, and the product owner is unsure what to change.

What should the project manager do?

A Continue with the existing priorities, since the vision was approved.

B Let the development team decide a new direction among themselves.

C Wait for the next quarterly review before changing anything.

D Work with the product owner and key stakeholders to revisit the vision, then communicate the update.

◆ PAUSE AND DECIDE

What should the project manager do?

A Continue; it was approved.

Follows a vision that no longer matches reality. Work loses value daily.

B Team decides the direction.

Product direction belongs with the product owner and stakeholders.

C Wait for the quarterly review.

Keeps building towards an obsolete goal for weeks.

D Revisit the vision together; communicate it.

Keeps the vision current, with the right people, then aligns everyone.

THE SIGNAL

The world changed the vision's premise. Revisit it now, with the right people, and tell everyone.

A vision people can repeat. Every stakeholder found. **Attention matched to each.**

MODULE 13 · DOMAIN 1 - PEOPLE

Engaging stakeholders and managing expectations

Turning analysis into engagement, building trust and influence, surfacing hidden expectations, and keeping customers satisfied.

DOMAIN I - People EXAM WEIGHT 33% MODULE 13 of 49

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Satisfaction is what was delivered,
compared with what was expected.

The engagement cycle

01

Identify

Find stakeholders and analyse them.

02

Plan engagement

Decide how to move each towards the engagement needed.

03

Manage engagement

Carry out the plan: meet, involve, respond.

04

Monitor engagement

Check whether it is working, and adjust.

ON THE EXAM

Engagement is a cycle, not a launch event. Stakeholders' positions change, so the plan must too.

The stakeholder engagement plan

- For each stakeholder or group: **current and desired engagement**, from the assessment matrix.
- **Strategies and actions** to close the gap: meetings, involvement in decisions, demonstrations.
- **Who** engages them, and how often.
- Some parts are sensitive, just like the register. Share it with judgement.

> ON THE EXAM
>
> The engagement plan says how to change engagement. The communications plan says how to move information.

Building trust

- **Consistency**: do what you said you would, especially small things.
- **Transparency**: share the real status, including problems.
- **Bad news early**, with facts and options. Never let a stakeholder be surprised.
- **Listen** more than you present, and show that what you heard changed something.

> ON THE EXAM
>
> Trust is built in small, kept promises and lost in one hidden problem.

Influencing without authority

- Start from **their interests**: what does this stakeholder need, and fear?
- Use **evidence**: data, pilots and demonstrations persuade better than argument.
- **Involve** them in shaping decisions. People support what they helped create.
- Build **allies**: people they already trust can carry the message.
- Pressure through their manager wins compliance, and loses trust.

> ON THE EXAM
>
> Influence starts with understanding the other person. Pressure is the last tool, not the first.

Requirements or expectations?

REQUIREMENTS

Stated and agreed

- Written down and approved
- Testable against acceptance criteria
- Change through the agreed process
- What the project must deliver

EXPECTATIONS

Often unstated

- Assumptions in people's heads
- Include 'obvious' things nobody wrote
- Shift as people learn and talk
- How the result will be judged

THE TIEBREAKER

A project can meet every requirement and still disappoint. The gap is unstated expectations.

Surfacing expectations early

- Ask directly: 'What would make you say this project succeeded? What would disappoint you?'
- Use **prototypes, mock-ups and demonstrations**. People react to things more reliably than to documents.
- Write down **assumptions** and confirm them. An assumption log is an expectations log.
- Revisit expectations as the project learns. They move, even when requirements do not.

> ON THE EXAM
>
> Show something early. A prototype surfaces expectations that no interview will.

When stakeholders expect different things

- Bring them together and make the **conflict and trade-offs visible**.
- Anchor on the **project objectives and business case**, not on who is loudest.
- Use prioritisation techniques to find a sequence that serves both, such as an MVP first.
- The decision belongs to the **sponsor or product owner**. The PM facilitates it.

> ON THE EXAM
>
> Facilitate alignment. Do not quietly pick a side, and do not promise everyone everything.

Internal or external customers?

INTERNAL CUSTOMERS

Inside the organisation

- Departments that receive the output
- Operations teams who will run it
- Colleagues depending on your deliverable
- Judged by fitness for their work

EXTERNAL CUSTOMERS

Outside the organisation

- Clients and end users
- Paying customers of a product
- Partners and the public
- Judged by value, experience and contract

THE TIEBREAKER

Both have expectations the outline asks you to monitor. Operations teams are customers too.

Monitoring satisfaction

METHOD	WHAT IT TELLS YOU
Reviews and demonstrations	Direct reactions to real work, every iteration
Surveys and satisfaction scores	Trends over time, across many people
Usage and support data	What people actually do, not what they say
One-to-one conversations	The concerns people will not raise in a group
Informal signals	Tone of emails, attendance, questions that stop being asked

ON THE EXAM

Monitor, then respond. A satisfaction measure nobody acts on is just a report.

Gold plating is not managing expectations

- **Gold plating**: adding features or quality the customer did not ask for, hoping to delight.
- It adds cost, risk and testing, and quietly resets expectations for next time.
- Deliver what was agreed. If an extra is valuable, propose it through the change process or the backlog.
- Exceed expectations by delivering **reliably and transparently**, not by surprise additions.

> ON THE EXAM
>
> Unrequested extras are a scope problem, not a gift. Route good ideas through the proper channel.

Expectations on agile projects

- **Frequent reviews** keep expectations and reality close, every iteration.
- A visible **backlog and roadmap** show what is coming, and what is not.
- Explain early that **scope is adaptive**: dates and budgets may be fixed while features flex.
- A stakeholder who expects every original idea delivered will be disappointed by a successful agile project.

> ON THE EXAM
>
> Agile manages expectations by showing work often. Set the expectation that scope will adapt, at the start.

Delivering bad news

- **Early**: as soon as you know, even before you have a full recovery plan.
- **Factual**: what happened, what is known, what is not yet known.
- **With next steps**: options being assessed, and when you will report back.
- Waiting until you have a perfect plan lets the stakeholder hear it from someone else.

> ON THE EXAM
>
> No surprises. Share what you know now, and when you will know more.

Keeping the sponsor engaged

- The sponsor **champions** the project, provides **funding**, makes key **decisions**, and resolves issues beyond the PM's authority.
- Agree early how the sponsor wants to be engaged: format, frequency, decision thresholds.
- Bring decisions, not just information: options, a recommendation, and the deadline.
- A disengaged sponsor is a serious risk. Address it directly, with the sponsor, first.

> ON THE EXAM
>
> Sponsor engagement is a key concept in PMBOK 8. A disengaged sponsor is the PM's problem to raise.

Project Lantern: engagement in action

- **Clinic managers**, resistant: each co-designs their clinic's go-live checklist.
- **Reception staff**: see prototypes early, and their feedback visibly changes the screens.
- **Patients**: a panel tests each release; satisfaction and usage are tracked from the first clinic.
- **Sponsor**: a fortnightly thirty-minute decision meeting, with options prepared in advance.

> ON THE EXAM
>
> Different stakeholders, different engagement. Each action is chosen to close a specific gap.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Requirements met, customer still unhappy	Unstated expectations: surface them early with prototypes
Two stakeholders want opposite things	Facilitate trade-offs against objectives; right person decides
Team added unrequested features	Gold plating: deliver what was agreed; route extras properly
Sponsor missing meetings, decisions stalled	Meet the sponsor; agree how they want to engage
A delay is coming; no plan yet	Tell stakeholders now, with facts and when options follow
Product owner happy, users are not	Get direct user feedback and usage data

ON THE EXAM

Close the gap between expected and delivered, early and often. That is expectation management.

A customer formally accepted a new booking website that meets every documented requirement. A week later, the customer complains that it works poorly on mobile phones.

Mobile use was never written into the requirements. The customer says it was 'obviously' expected.

What should the project manager have done to prevent this?

A. Surfaced and confirmed expectations early, including unstated ones, using prototypes and demonstrations.

B. Written a more detailed contract that excluded anything not listed.

C. Added mobile support at no charge as soon as the complaint arrived.

D. Recorded the issue in the lessons learned register at closure.

◆ PAUSE AND DECIDE

What should the project manager have done to prevent this?

A Surface expectations early with prototypes.

A mobile demo in week two would have exposed the 'obvious' expectation.

B Tighter contract.

Protects legally but not the relationship. The customer is still disappointed.

C Add it free.

A reaction, not prevention, and uncontrolled scope.

D Lessons learned at closure.

Records the problem after it has cost the relationship.

THE SIGNAL

Requirements met, customer unhappy, 'it was obvious': an unstated expectation nobody surfaced.

> The head of sales wants a new customer portal launched in March with every planned feature. The head of operations wants a later launch, to allow full stability testing.
>
> Both have asked the project manager to plan around their preference.

What should the project manager do?

A Plan for the March launch, because sales generates revenue.

B Promise both: a March launch with full testing, by adding resources.

C Bring both together to examine trade-offs against the project objectives, so the decision-maker can choose.

D Escalate the disagreement to the chief executive.

◆ PAUSE AND DECIDE

What should the project manager do?

A Side with sales.

Quietly picks a side. Operations' valid concern is simply overruled.

B Promise both via more resources.

Overpromises. More people rarely compress testing without new risk.

C Examine trade-offs together; decision-maker chooses.

Facilitates alignment on objectives, perhaps a March MVP with a later full release.

D Escalate to the chief executive.

Premature. The two have not been brought together.

THE SIGNAL

Conflicting expectations: facilitate trade-offs against the objectives. Do not pick a side or promise both.

> During a routine check, the project manager finds that a developer has built two extra reporting features the customer never requested.
>
> The developer believes the customer will be delighted. The features took three days, and testing has not started on them.

What should the project manager do?

A. Discuss it with the developer, assess the impact, and propose the features through the change process if they add value.

B. Keep the features as a bonus for the customer.

C. Tell the customer about the extras as a goodwill gesture.

D. Remove the features immediately without discussing them.

◆ PAUSE AND DECIDE

What should the project manager do?

A Discuss, assess, route through change.

Handles gold plating properly, and still gives a good idea a fair hearing.

B Keep as a bonus.

Gold plating accepted: untested code, and reset expectations.

C Announce as goodwill.

Commits the project to supporting unrequested, untested work.

D Remove without discussion.

Wastes the effort and the learning, and damages trust with the developer.

THE SIGNAL

Unrequested extras are gold plating. Talk, assess, and route good ideas through the proper channel.

> The project sponsor has missed three steering committee meetings in a row. Two decisions that need her approval are now overdue, and the team is waiting.
>
> Her assistant says she is very busy with a merger.

What should the project manager do?

A Make the two decisions, since the team cannot wait.

B Escalate the sponsor's absence to her manager.

C Meet the sponsor, even briefly, to agree how she can engage, such as delegating decisions within set limits.

D Cancel the steering committee until the merger is complete.

◆ PAUSE AND DECIDE

What should the project manager do?

A Make the decisions.
Acts beyond authority. The decisions are the sponsor's.

B Escalate to her manager.
Goes over the sponsor's head before speaking to her.

C Meet her; agree a workable engagement.
Addresses the risk directly, and adapts to her real constraints.

D Cancel the committee.
Removes governance while decisions are already overdue.

THE SIGNAL

A disengaged sponsor is a risk. Raise it with the sponsor first, and adapt the engagement to reality.

> A key supplier has just told the project manager that a component will arrive late, which will push the go-live back about three weeks.
>
> The next steering committee is in two weeks. The project manager would like to present a full recovery plan at that meeting.

What should the project manager do?

A Wait for the steering committee, and present the delay with a full recovery plan.

B Inform key stakeholders now of what is known, and when recovery options will follow.

C Tell only the project team until the recovery plan is ready.

D Keep the delay quiet while trying to recover the time.

◆ PAUSE AND DECIDE

What should the project manager do?

A Wait two weeks with a full plan.

Tempting, but stakeholders may hear it elsewhere, and lose two weeks to plan.

B Inform now; say when options follow.

No surprises: early, factual, and with a clear next step.

C Team only.

Keeps decision-makers uninformed about a material change.

D Keep it quiet.

Hides a known problem. A breach of honesty and trust.

THE SIGNAL

Material bad news goes out early, even without a full plan. Say what is known and when you will know more.

> An agile team delivers a working increment every two weeks, and the product owner is very satisfied with progress.
>
> Meanwhile, support tickets from end users are rising, and usage data shows most users abandon the new workflow halfway through.

What should the project leader do?

A Rely on the product owner's satisfaction, since the product owner represents the users.

B Ask the team to add more features to make the product more attractive.

C Escalate the support tickets to the sponsor.

D Gather direct feedback from end users and share the usage data with the product owner to adjust the backlog.

◆ PAUSE AND DECIDE

What should the project leader do?

A Trust the product owner's view.

The data says the users disagree. A proxy can drift from the people it represents.

B Add more features.

More of what users already abandon. Treats a usability problem with scope.

C Escalate the tickets.

The product owner, who owns the backlog, has not even seen the data.

D Direct user feedback; data to the PO.

Monitors real customer satisfaction and lets the owner respond.

THE SIGNAL

Product owner happy, users struggling: measure the real customers, and give the owner the evidence.

Surface expectations early, align them openly, and never let anyone be surprised.

MODULE 14 · DOMAIN 1 - PEOPLE

Communication strategy

How communication works, choosing methods, planning who needs what, the channels formula, listening, meetings and feedback loops.

DOMAIN I - People EXAM WEIGHT 33% MODULE 14 of 49

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Communication is not what you send. It is **what is understood**.

The basic communication model

01

Encode

The sender turns thoughts into words, images or data.

02

Transmit

Through a medium: email, call, meeting, board.

03

Noise

Anything that distorts it: jargon, culture, distraction.

04

Decode

The receiver interprets it through their own context.

05

Feedback

The receiver confirms what they understood.

ON THE EXAM

Without feedback, the sender cannot know whether the message arrived as meant.

Acknowledgement is not understanding

- The **sender** is responsible for making the message clear, and for checking it was understood.
- The **receiver** is responsible for receiving it fully and acknowledging it.
- 'Received, thanks' means the message arrived. It does not mean it was understood, or agreed.
- For important messages, check understanding: ask for a summary, or discuss the implications.

> ON THE EXAM
>
> Acknowledgement means 'I got it'. It does not mean 'I understood' and it certainly does not mean 'I agree'.

Three communication methods

METHOD	EXAMPLES	BEST FOR
Interactive	Meetings, calls, workshops, video conferences	Complex, urgent or sensitive topics; decisions; confirming understanding
Push	Emails, memos, reports, notifications	Specific recipients who need the information, where no discussion is needed
Pull	Intranet, shared drives, dashboards, knowledge bases	Large audiences, or reference information people find when they need it

ON THE EXAM

Push confirms delivery, not understanding. When understanding matters, choose interactive.

Analysing communication requirements

- **Who** needs information, and **what** exactly do they need?
- **Why** do they need it: to decide, to act, or to be aware?
- **When** and how often, and in what **format and level of detail**?
- Tailor to the stakeholder: executives want decisions and exceptions; teams want detail.

> ON THE EXAM
>
> Start from what each stakeholder needs to do with the information. Then choose format and frequency.

The communications management plan

IT RECORDS	FOR EXAMPLE
Stakeholder requirements	What each stakeholder or group needs to know, and why
Information and format	Status summary, risk report, decision log; level of detail
Method and technology	Meeting, email, dashboard; which tools
Timing and frequency	Weekly, per sprint, at milestones, on exception
Responsibilities	Who sends, who approves, who receives
Escalation and glossary	How issues escalate; shared terms defined

ON THE EXAM

A communications plan is a living document. When a stakeholder's needs change, the plan changes.

The channels formula

- **Channels = n x (n - 1) / 2**, where n is the number of people.
- 5 people: 5 x 4 / 2 = **10** channels. 10 people: 10 x 9 / 2 = **45**.
- Grow from 10 to 13 people: 78 - 45 = **33 more** channels, from just 3 new people.
- Include the project manager in n. Read whether the question asks for the total or the increase.

> ON THE EXAM
>
> Channels grow much faster than headcount. That is why adding people adds communication overhead.

Dimensions of communication

DIMENSION	EXAMPLES
Formal or informal	A contract notice, or a corridor chat
Written or oral	A report, or a briefing
Internal or external	The team, or a client or regulator
Upward, downward, sideways	To the sponsor, to the team, to peer managers
Official or unofficial	An approved status report, or an off-the-record conversation

ON THE EXAM

Contracts, legal notices and formal decisions are formal and written. Relationships are built informally.

Active listening

- **Attend** fully: no laptop, no phone, no rehearsing your reply.
- **Paraphrase**: 'So what you are saying is...'
- Ask **clarifying questions** before offering solutions.
- **Summarise** agreements and next steps at the end.
- Interrupting to fix the problem tells people you were not really listening.

> ON THE EXAM
>
> Paraphrasing is the feedback loop inside a conversation. Use it for anything that matters.

Meetings that earn their time

- A clear **purpose** and **agenda**, shared in advance.
- Only the **people needed** for the decisions or discussion.
- A **timebox**, a facilitator, and decisions and actions recorded with owners.
- Status that could be read should not be read aloud. Move it to a report or dashboard.

> ON THE EXAM
>
> Meet to discuss and decide. Inform through pull and push. Large status read-outs waste everyone's time.

Establishing a feedback loop

- The outline asks you to **establish a feedback loop** in your communication.
- Ask stakeholders whether they get what they need, in the form they need it.
- Short surveys, a question at the end of reviews, or a communications item in retrospectives.
- Watch the signals: unread reports, repeated questions, surprises at reviews.
- Then **adjust the plan**. Feedback nobody acts on is not a loop.

> ON THE EXAM
>
> If you do not know whether your communication works, you have not closed the loop. Ask, then adapt.

Transparency and collaboration

- **Information radiators**: visible boards and charts anyone can see without asking.
- Open backlogs, shared channels and decision logs, available to everyone who needs them.
- **Osmotic communication**: co-located teams overhear useful information. Distributed teams must recreate it deliberately.
- Transparency is the first pillar of empiricism. It is what makes inspection possible.

> ON THE EXAM
>
> Make information easy to find without asking. That is the most scalable communication there is.

What blocks communication

- **Jargon and acronyms** the receiver does not share.
- **Information overload**: so much sent that the important item is missed.
- **Wrong channel**: sensitive news by email, a decision in a chat thread.
- **Cultural and language differences**, and assumptions about shared context.
- Each blocker has a fix: plain language, tailoring, the right channel, and checking understanding.

> ON THE EXAM
>
> When communication fails, look for the blocker before sending more.

Project Lantern: the communications plan in brief

- **Steering committee**: one-page exception report every fortnight; decisions requested in advance.
- **Clinic managers**: a go-live briefing meeting per clinic, then a weekly readiness checklist.
- **Reception staff**: short demonstration sessions with questions, plus a pull knowledge base.
- **Everyone**: a project dashboard, updated automatically, and a monthly two-question feedback survey.

> ON THE EXAM
>
> Different stakeholders, different methods and frequencies, all from one plan, with a feedback loop built in.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Team size changes; how many channels?	n(n-1)/2; check total or increase
Stakeholder gets too much of the wrong thing	Revisit their requirements; update the plan
Understanding must be confirmed	Interactive, not push or pull
'Received' but acted on old information	Check understanding, not just delivery
Long status meetings, low engagement	Status to dashboards; meetings for decisions
No idea if reports are useful	Establish a feedback loop, then adjust

ON THE EXAM

Communication succeeds when the right person understands the right thing in time. Design for that.

> A project team has eight members, including the project manager. To meet a new deadline, four more people join the team.
>
> The project manager is concerned about the extra communication effort.

How many additional communication channels does the change create?

A 4

B 38

C 66

D 28

◆ PAUSE AND DECIDE

How many additional communication channels does the change create?

A 4

Counts people, not channels.

B 38

12 x 11 / 2 = 66, minus 8 x 7 / 2 = 28, gives 38 additional.

C 66

The new total. The question asked for the increase.

D 28

The original total, before anyone joined.

THE SIGNAL

Compute both totals, then subtract. 'Additional' and 'total' are different answers, and both appear as options.

> A regional director complains that she receives every weekly team report and every meeting invitation for a project, which she has no time to read.
>
> Yet she was surprised last week by a decision that affects her region's budget.

What should the project manager do?

A Reduce the number of reports sent to all stakeholders.

B Add an executive summary to the top of every weekly report.

C Continue as now, since she already receives all the information.

D Review her communication requirements with her, and update the communications plan.

◆ PAUSE AND DECIDE

What should the project manager do?

A Cut reports for everyone.

One stakeholder's problem, fixed by changing everyone's communication.

B Executive summaries everywhere.

Helps a little, but still buries the decision she needed among everything else.

C Carry on.

She received everything and still missed what mattered. That is the failure.

D Review her requirements; update the plan.

Tailors to what she actually needs: fewer reports, and decisions affecting her region.

THE SIGNAL

Too much of the wrong thing and too little of the right thing: the plan does not match her needs.

> A new safety procedure for operating a crane comes into force on a construction site next Monday.
>
> Every crane operator and banksman must understand it, and the site manager needs confirmation that they do.

Which communication method should the project manager use?

A Briefing sessions with demonstrations and questions, confirming each person's understanding.

B An email to every operator with the procedure attached.

C Publish the procedure on the site intranet.

D Post the procedure on the site noticeboard.

◆ PAUSE AND DECIDE

Which communication method should the project manager use?

A. Interactive briefings, confirming understanding.
Safety-critical and understanding must be confirmed: interactive.

B. Email.
Push: confirms delivery at best, never understanding.

C. Intranet.
Pull: relies on each person choosing to read it.

D. Noticeboard.
Pull again, and the weakest check of all.

THE SIGNAL

When understanding must be confirmed, and especially for safety, communicate interactively.

> The project manager emailed a design change to a supplier's lead engineer. He replied 'Received, thanks'.
>
> Two weeks later, the supplier delivers parts built to the old design.

What should the project manager have done?

A Resent the email with a read receipt requested.

B Copied the engineer's manager on the email.

C Confirmed understanding, for example by discussing the change and asking him to summarise its impact.

D Relied on the reply, since the engineer acknowledged the change.

◆ PAUSE AND DECIDE

What should the project manager have done?

A Read receipt.

Proves opening, not understanding. The same failure, better documented.

B Copy his manager.

More recipients, no more understanding.

C Confirm understanding.

The sender's responsibility: check the message was understood, not just received.

D Rely on the reply.

That is exactly what happened. Acknowledgement is not understanding.

THE SIGNAL

'Received, thanks' is acknowledgement. For a critical change, the sender must confirm understanding.

A weekly two-hour status meeting has twenty-five attendees. Each workstream lead reads out an update, and most attendees work on laptops.

Several stakeholders say the meeting wastes their time, but they fear missing something important.

What should the project manager do?

A Make attendance mandatory and ban laptops.

B Extend the meeting so every update can be discussed.

C Cancel all project meetings and rely on email.

D Move updates to a shared dashboard, and keep a shorter meeting for decisions with the people needed.

◆ PAUSE AND DECIDE

What should the project manager do?

A Mandatory, no laptops.
Enforces attention on a format that wastes it.

B Extend the meeting.
More of what is already not working.

C Email only.
Loses discussion and decisions entirely.

D Dashboard for status; short decision meeting.
Status becomes pull; the meeting is kept for what needs people live.

THE SIGNAL

Status read aloud to a crowd is the wrong method. Pull for status, interactive for decisions.

> For six months, a project manager has sent a detailed monthly newsletter to all stakeholders. Nobody has ever replied.
>
> The project manager has no idea whether anyone reads it or finds it useful.

What should the project manager do?

A Keep sending it, since nobody has complained.

B Ask key stakeholders for feedback, for example with a short survey, and adjust based on the answers.

C Send it weekly instead, to increase its visibility.

D Stop sending it, since nobody replies.

◆ PAUSE AND DECIDE

What should the project manager do?

A Keep going.
Silence is not evidence of value. The loop is still open.

B Ask for feedback; adjust.
Establishes the feedback loop the outline asks for.

C Send it weekly.
Four times the output, with no idea whether any of it helps.

D Stop it.
A guess in the other direction. Some stakeholders may rely on it.

THE SIGNAL

No idea whether it works: the feedback loop is missing. Ask, then adapt.

Plan for understanding, not delivery. Then close the loop.

MODULE 15 · DOMAIN 1 - PEOPLE

Reporting, transparency and feedback

Understanding reporting requirements, building reports people can act on, honest status, supporting governance, and reporting agile work upward.

DOMAIN I - People EXAM WEIGHT 33% MODULE 15 of 49

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A good report helps someone decide or act. Everything else is noise.

Understanding reporting requirements

- **Who** receives it, and **what decisions** do they make with it?
- **Governance cadence**: steering meetings, phase gates, portfolio reviews.
- **Mandatory** reporting: contracts, regulators, funding bodies, with their required formats.
- **Organisational standards**: PMO templates and definitions, which are process assets.
- Agree all of this early, and record it in the communications plan.

> ON THE EXAM
>
> Some reporting is optional and tailorable. Contractual and regulatory reporting is not.

Kinds of project report

REPORT	ANSWERS
Status	Where are we now against the plan?
Progress	What has been completed since the last report?
Forecast	Where are we heading, on time, cost and scope?
Variance or exception	What is outside agreed tolerance, and what are we doing about it?
Trend	Is performance improving or getting worse over time?
Gate or milestone	Are we ready to proceed to the next phase?

ON THE EXAM

Status without a forecast only tells people where you were. Decision-makers need where you are heading.

Full status or management by exception?

FULL STATUS

Everything, every time

- Every workstream, every metric
- Long, and read by few
- Important items easily buried
- Fits teams and detail-focused audiences

MANAGEMENT BY EXCEPTION

Only what needs attention

- Items outside agreed tolerance
- Decisions required, with options
- Short, and read by the right people
- Fits sponsors and steering committees

THE TIEBREAKER

Agree the tolerances first. Then senior stakeholders see only what breaches them, plus decisions needed.

RAG status that tells the truth

- Define red, amber and green with **agreed, objective thresholds**, not personal judgement.
- **Watermelon reporting**: green on the outside, red on the inside. Problems hidden until they explode.
- Make amber **safe** to report. An early amber is a gift; a late red is a crisis.
- Always pair a colour with the **reason** and the **action**.

> ON THE EXAM
>
> If people are punished for amber, you will only ever see green, until you see red.

Tailoring reports to the audience

- **Sponsor and steering committee**: one page, trends, exceptions, risks, and decisions needed.
- **Team**: detailed progress, blockers, what is next.
- **Regulators and funders**: exactly the format and timing they require.
- Keep detail **available on request**, so a short report never hides anything.

> ON THE EXAM
>
> The outline asks for reports aligned with sponsor and stakeholder expectations. One report rarely fits all.

Honest reporting

- Report the **real** status, including forecasts that are uncomfortable.
- Omitting a known forecast overrun misleads as much as a false number does.
- If someone asks you to change or leave out information, talk to them first, then report accurately.
- Honesty is one of the four values in PMI's Code of Ethics and Professional Conduct.

> ON THE EXAM
>
> Accurate is not negotiable. How you present bad news, with context and options, is where the skill lies.

Supporting governance

- The outline asks you to **support reporting and governance processes**.
- For a **phase gate**: evidence against the exit criteria, risks, the forecast, and a recommendation.
- For a **decision request**: the question, options with impacts, a recommendation, and the deadline.
- For an **escalation**: what threshold was breached, what has been tried, and what is needed.

> ON THE EXAM
>
> Governance bodies decide. Your job is to make their decision easy, well-informed and on time.

Reporting agile work upward

- **Sprint reviews** are the best report: stakeholders see working results.
- **Burnup charts and release forecasts** show progress towards a goal, and when it is likely.
- Report **outcomes and value** delivered, not just the number of items done.
- Never compare teams by **velocity**. Story points are relative within one team.

> ON THE EXAM
>
> Translate agile progress into forecasts and outcomes for governance. Do not turn velocity into a league table.

Data, dashboards and AI

- **Automated dashboards** from the team's own tools reduce effort and errors.
- One **source of truth**, so different reports never show different numbers.
- Generative AI can draft summaries from updates, which saves time.
- But **verify** anything AI produces against the source before sending. You remain accountable.

> ON THE EXAM
>
> Tools can gather and draft. Only the project manager can be accountable for what the report says.

Is anyone using your reports?

- Ask recipients what they use, what they skip, and what they wish they had.
- Watch for signals: decisions made without the report, questions it should have answered.
- Remove what nobody uses. Shorter reports get read.
- This is the feedback loop from module fourteen, applied to reports.

> ON THE EXAM
>
> A report nobody uses is cost without value. Ask, then cut or change it.

Project Lantern: the steering committee page

- **Top**: overall forecast for go-live and budget, with a trend arrow.
- **Middle**: exceptions only, with the threshold breached and the action under way.
- **Bottom**: decisions needed, with options and a recommendation.
- Links to the live dashboard and the app's latest sprint review recording.

> ON THE EXAM
>
> One page: forecast, exceptions, decisions. Detail one click away.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Everything was green, then suddenly red	Objective thresholds; make amber safe to report
Sponsor cannot use a long report	One-page exception summary; detail on request
A phase gate is coming	Evidence against criteria, risks, forecast, recommendation
PMO wants to compare team velocities	Explain why not; offer forecasts and outcomes
Asked to leave out an uncomfortable forecast	Discuss, then report accurately
AI-drafted report contained an error	Verify against the source; correct it openly

ON THE EXAM

Reports exist to support decisions. Make them honest, tailored and easy to act on.

> For five months, every workstream on a programme reported green. This month, one lead reports that go-live will be missed by two months.
>
> It emerges that the team had known about the problem for weeks, but did not want to be the first to report amber.

What should the project manager do to prevent this happening again?

A Require every workstream to report amber until proven otherwise.

B Formally reprimand the lead who delayed the bad news.

C Agree objective status thresholds, and make it clear that early amber is welcomed, not punished.

D Stop using colour ratings in status reports.

◆ PAUSE AND DECIDE

What should the project manager do to prevent this happening again?

A Default to amber.

Makes the colours meaningless in the other direction.

B Reprimand the lead.

Confirms the fear that caused the silence. Next time, even later.

C Objective thresholds; welcome early amber.

Removes judgement from the colours, and removes the fear behind the delay.

D Drop colour ratings.

The colours were not the problem. The fear of reporting was.

THE SIGNAL

Watermelon reporting has two causes: subjective colours and fear. Fix both.

> The sponsor tells the project manager that the fifteen-page monthly status report is too long to read before steering meetings.
>
> She says she often only discovers important issues during the meeting itself.

What should the project manager do?

A Provide a one-page summary of forecasts, exceptions, risks and decisions needed, with detail available on request.

B Keep the full report, because it contains everything she needs.

C Add an appendix with more detail on the main issues.

D Send the full report weekly instead of monthly, so each one is shorter.

◆ PAUSE AND DECIDE

What should the project manager do?

A One-page exception summary; detail on request.

Aligned with how she actually uses reports: fast, decision-focused.

B Keep it as it is.

Everything is in it, and she still misses the important items.

C Add an appendix.

Makes a report that is too long even longer.

D Send it weekly.

The same volume, delivered four times as often.

THE SIGNAL

The sponsor misses important items in a long report: tailor it, by exception, to her decisions.

> A project reaches the end of its design phase. The steering committee must decide at the phase gate whether to approve moving into construction.
>
> The project manager is preparing for the gate review.

What should the project manager present?

A. A demonstration of the design model, to show the progress made.

B. The full detailed schedule for the construction phase.

C. A brief email summary after the meeting.

D. Evidence against the gate criteria, risks, the forecast, and a recommendation with the decision requested.

◆ PAUSE AND DECIDE

What should the project manager present?

A. A design demonstration.
Useful context, but it does not show whether the gate criteria are met.

B. The full construction schedule.
Detail for the team, not a basis for a go or no-go decision.

C. An email afterwards.
Too late. The committee decides in the meeting.

D. Criteria evidence, risks, forecast, recommendation.
Gives governance exactly what it needs to decide.

THE SIGNAL

A phase gate is a decision. Present evidence against the criteria and a recommendation.

> A PMO asks each of four agile teams to report its velocity monthly, so that team productivity can be compared.
>
> The teams estimate in story points, and each team calibrated its points independently.

What should the project manager do?

A Explain that velocity is team-specific, and propose forecasts and outcome measures for comparison instead.

B Provide the velocities as requested, so the PMO can compare the teams.

C Refuse to report any agile metrics to the PMO.

D Ask all four teams to use the same story point scale so velocities can be compared.

◆ PAUSE AND DECIDE

What should the project manager do?

A Explain; offer forecasts and outcomes.

Meets the PMO's real need for visibility with measures that mean something.

B Provide velocity for comparison.

Comparisons are meaningless, and teams will start inflating points.

C Refuse.

Ignores a legitimate governance need for information.

D Standardise points.

Undermines each team's own estimating, and still invites gaming.

THE SIGNAL

Velocity is for a team's own planning, never a league table. Offer governance forecasts and outcomes.

> The project's cost forecast now shows a twelve percent overrun at completion, although spending to date is within budget.
>
> Before the board report, the finance director asks the project manager to show only spend to date, 'to avoid alarming anyone'.

What should the project manager do?

A. Show only spend to date, since that figure is accurate.

B. Report the forecast to the board without discussing it with the finance director.

C. Discuss with the finance director why the forecast matters, then report it accurately with a recovery plan.

D. Report the finance director's request to the ethics committee.

◆ PAUSE AND DECIDE

What should the project manager do?

A Spend to date only.

Every number true, the overall picture false. Omission misleads.

B Report it without discussion.

Right outcome, but skips the conversation and blindsides a colleague.

C Discuss, then report accurately with a plan.

Honest, collaborative, and gives the board something to act on.

D Report to the ethics committee.

Escalates before a single conversation. Too early.

THE SIGNAL

Leaving out a known forecast is misleading. Talk first, then report the truth, with options.

> A project manager used a generative AI tool to summarise team updates into the weekly sponsor report. The sponsor later points out that it gave the wrong date for a key milestone.
>
> The team's own updates had the correct date.

What should the project manager do?

A Stop using AI tools for any project reporting.

B Correct the error with the sponsor, and verify AI-drafted content against the source before sending.

C Explain to the sponsor that the AI tool made the mistake.

D Continue as before, since the error was minor.

◆ PAUSE AND DECIDE

What should the project manager do?

A Stop using AI.

Overcorrects. The tool saves time; the missing step was verification.

B Correct it; verify before sending.

Owns the error and fixes the process. Accountability stays with the PM.

C Blame the tool.

The PM sent the report. Accountability cannot be delegated to software.

D Carry on.

A wrong milestone date to a sponsor is not minor, and it will recur.

THE SIGNAL

AI can draft; the project manager is accountable. Verify against the source before anything goes out.

Honest, tailored, and built for a decision.

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MODULE 16 · DOMAIN 1 - PEOPLE

Knowledge transfer

Finding the knowledge the project depends on, capturing and sharing it, and building a team where sharing is normal.

DOMAIN I - People EXAM WEIGHT 33% MODULE 16 of 49

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Knowledge that lives in one head leaves when that person does.

Explicit or tacit knowledge?

EXPLICIT KNOWLEDGE

Can be written down

- Facts, procedures, specifications
- Documents, diagrams, data
- Easy to store and search
- Transferred by documenting

TACIT KNOWLEDGE

Hard to write down

- Experience, judgement, intuition
- Know-how, context, relationships
- Lives in people
- Transferred by working together

THE TIEBREAKER

Tacit knowledge is not transferred by writing more documents. It moves through people working side by side.

Identifying critical knowledge

- Ask: **what would stop the project** if one particular person were unavailable tomorrow?
- Look for **single points of knowledge**: the only person who understands a system, a supplier, a regulation.
- Record key-person dependency as a **risk**, with a response.
- Act early, while the person is still available and not under pressure.

> ON THE EXAM
>
> Key-person dependency is a risk. Identify it early and respond before it becomes an issue.

Ways to transfer knowledge

FOR EXPLICIT KNOWLEDGE	FOR TACIT KNOWLEDGE
Documentation, wikis, runbooks	Pairing and shadowing
Decision logs and design records	Mentoring and coaching
Lessons learned register	Communities of practice
Knowledge repositories and search	Workshops, storytelling, demonstrations
Training materials and recordings	Job rotation and joint problem-solving

ON THE EXAM

Match the method to the knowledge. Documents for facts; people for judgement.

The lessons learned register

- Started **early**, and updated **throughout** the project, not just at the end.
- Records what happened, its impact, and a **specific recommendation**.
- Fed by retrospectives, reviews, issue resolution and risk responses.
- 'Communication could have been better' is not a lesson. 'Confirm supplier specs in writing with examples' is.

> ON THE EXAM
>
> Lessons are captured as they happen, and applied during the project, not saved for a final meeting.

Lessons learned register or repository?

REGISTER

This project

- Created and used during the project
- Owned by the project team
- Applied to current work
- Updated continuously

REPOSITORY

The organisation

- Holds lessons from many projects
- An organisational process asset
- Searched at the start of new projects
- Receives the register at phase or project end

THE TIEBREAKER

The register is for this project. At phase or project end, lessons move to the repository for everyone else.

Using lessons, not just collecting them

- At the **start** of a project, search the repository and talk to people who ran similar work.
- In **planning**, turn relevant lessons into risks, estimates and process choices.
- In **retrospectives**, check whether last time's improvements were actually applied.
- A repository nobody searches is an archive, not knowledge transfer.

> ON THE EXAM
>
> The value of a lesson is realised only when someone uses it. Search first, then plan.

Fostering an environment for sharing

- **Psychological safety**: people share mistakes and uncertainties only when that is safe.
- **Time**: make knowledge sharing part of the plan, not something done after hours.
- **Recognition**: celebrate people who teach, document and help others.
- **Easy tools**: one searchable place, simple templates.
- People hoard knowledge when being the only expert feels like job security.

> ON THE EXAM
>
> Hoarding is usually a response to fear. Remove the fear, and reward sharing.

Handovers

- **People leaving**: plan an overlap, pair them with a successor, record key decisions and contacts.
- **Suppliers exiting**: knowledge transfer written into the contract, with acceptance criteria.
- **Transition to operations**: runbooks, training, shadowing, and support during early live running.
- Start every handover **early**, and confirm the receiver can actually do the work.

> ON THE EXAM
>
> A handover is complete when the receiver can do the work, not when the documents are delivered.

Communities of practice

- Groups of people who share a discipline or interest, across teams and projects.
- Testers, data engineers, project managers, Scrum masters, all learning from each other.
- Spread tacit knowledge and good practice beyond any single project.
- Especially useful on scaled agile work, where specialists are spread across many teams.

> ON THE EXAM
>
> Communities of practice carry knowledge between projects, which is exactly where it is usually lost.

AI and project knowledge

- AI search and summaries can make large repositories genuinely usable.
- AI can draft meeting notes, decision records and first versions of documentation.
- Verify accuracy, and respect **confidentiality and data privacy** before feeding in project data.
- AI can move explicit knowledge. Tacit knowledge still moves through people.

> ON THE EXAM
>
> Use AI to make explicit knowledge findable. Check its output, and protect sensitive data.

Project Lantern: keeping knowledge in the project

- Only one analyst understood the **old booking system's data**. A colleague now pairs with her on every migration run.
- The first clinic's go-live lessons became a **checklist** used by the next five.
- The kiosk supplier's contract includes **knowledge transfer** to Northgate's IT team, with a sign-off.
- At each phase end, the register's lessons go to Northgate's **repository**.

> ON THE EXAM
>
> Critical knowledge identified, gathered, shared and passed on. All three outline enablers in one project.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Only one person knows a critical system	Key-person risk: pairing and documentation now
New project resembles a past one	Search the repository; talk to its team
Expert know-how will not go into manuals	Tacit: shadowing, pairing, mentoring
Project or phase closing	Move the register's lessons to the repository
People keep knowledge to themselves	Safety, time and recognition for sharing
Supplier or team member leaving	Planned handover, confirmed by the receiver

ON THE EXAM

Identify, gather, share, and never let critical knowledge depend on one person.

> Only one engineer understands how a project's legacy billing interface works. She starts parental leave in two months.
>
> The interface must be changed during the final phase of the project, four months from now.

What should the project manager do?

A Ask her to postpone her leave until the interface work is complete.

B Plan to hire a specialist contractor once she has gone.

C Arrange for her to be available by phone during her leave.

D Treat it as a key-person risk: pair a colleague with her now, and document the interface together.

◆ PAUSE AND DECIDE

What should the project manager do?

A Ask her to postpone leave.

Inappropriate, and possibly unlawful. The knowledge problem is the project's, not hers.

B Hire a contractor afterwards.

Nobody left to learn from. The contractor starts from zero.

C On call during leave.

Intrudes on protected leave and leaves the dependency in place.

D Key-person risk: pair and document now.

Uses the two months available to move both tacit and explicit knowledge.

THE SIGNAL

A known departure date plus one expert: transfer now, while there is time and the person is available.

> A project manager is starting a project to open a new regional distribution centre. The organisation opened a similar centre in another region eighteen months ago.
>
> The project manager is about to begin planning.

What should the project manager do first?

A. Review the earlier project's lessons in the repository, and talk to people who worked on it.

B. Start planning from scratch, since every region is different.

C. Wait until problems arise, then ask the previous team for advice.

D. Ask the new team to share what they remember hearing about the earlier project.

◆ PAUSE AND DECIDE

What should the project manager do first?

A Review lessons; talk to the previous team.

Uses organisational knowledge before planning, when it is most valuable.

B Plan from scratch.

Ignores eighteen months of directly relevant experience.

C Ask when problems arise.

Reactive. The point is to avoid repeating the problems.

D Rely on second-hand memories.

Hearsay instead of the recorded lessons and the people who were there.

THE SIGNAL

A similar past project and planning about to start: search the repository and talk to the people first.

> A senior technician diagnoses complex machine faults quickly, often from the sound a machine makes. He retires in six months.
>
> Two attempts to capture his knowledge in manuals produced long documents that the junior technicians find unhelpful.

What should the project manager do?

A. Ask him to write a more detailed manual with more diagrams.

B. Record a series of video lectures in which he explains his methods.

C. Have junior technicians shadow and pair with him on real faults over the coming months.

D. Hold a detailed exit interview with him before he retires.

◆ PAUSE AND DECIDE

What should the project manager do?

A. A more detailed manual.
The third attempt at a method that has already failed twice.

B. Video lectures.
Better than text, but still one-way. Judgement is learned by doing.

C. Shadowing and pairing on real faults.
Tacit knowledge moves through people working side by side.

D. An exit interview.
One conversation, at the last moment, cannot transfer years of intuition.

THE SIGNAL

Knowledge that will not go into manuals is tacit. Move it through shadowing and pairing.

> A project has reached closure. Its lessons learned register contains thirty specific lessons gathered throughout the project.
>
> The project manager is completing the closing activities.

What should happen to the lessons learned?

A They should be archived with the rest of the project files.

B They should be transferred into the organisation's lessons learned repository.

C They should be deleted, since the project is over.

D They should be emailed to the project team as a record.

◆ PAUSE AND DECIDE

What should happen to the lessons learned?

A Archive with project files.
Stored, but not findable. Future projects will not look there.

B Transfer to the organisational repository.
Turns project lessons into an organisational process asset others can search.

C Delete them.
Throws away exactly what the organisation needs to improve.

D Email them to the team.
Reaches people who already know them, and nobody else.

THE SIGNAL

At closure, the register's lessons move to the organisation's repository, an organisational process asset.

> In a cost-cutting year, several team members have become reluctant to share how they do their work. One says privately that being 'the only one who knows' is the best protection against redundancy.
>
> Knowledge sharing sessions are poorly attended.

What should the project manager do?

A Address the fear, make time for sharing in the plan, and visibly recognise people who share.

B Make documentation mandatory, with penalties for anyone who does not comply.

C Rotate team members into each other's roles without warning, to force sharing.

D Report the reluctant team members to their functional managers.

◆ PAUSE AND DECIDE

What should the project manager do?

A Address the fear; time; recognition.

Fosters the environment for sharing, which is the outline's third enabler.

B Mandatory documentation with penalties.

Adds a second fear to the first. Compliance, not sharing.

C Surprise rotations.

Confirms that the project sees people as replaceable. Deepens the fear.

D Report them.

Punishes a rational response to a real fear.

THE SIGNAL

Hoarding driven by fear: address the fear and reward sharing. Penalties make it worse.

> A supplier's contract to run a new customer platform ends in one month. The supplier's engineers hold most of the operational know-how.
>
> The organisation's own operations team must take over running the platform when the contract ends.

What should the project manager do?

A Ask the supplier to send its documentation after the contract ends.

B Extend the supplier's contract indefinitely to avoid the risk.

C Let the operations team learn the platform once they take it over.

D Agree a structured handover: joint sessions, runbooks and shadowing, with operations confirming readiness.

◆ PAUSE AND DECIDE

What should the project manager do?

A Documents after the contract ends.

Too late, with no leverage, and documents alone miss the tacit part.

B Extend indefinitely.

Avoids the handover rather than doing it, at open-ended cost.

C Learn after takeover.

Operations learns by failing in front of customers.

D Structured handover, readiness confirmed.

Moves explicit and tacit knowledge, and ends when operations can run it.

THE SIGNAL

A handover is finished when the receiver can do the work. Structure it, and let the receiver confirm.

Find the critical knowledge. Move it the right way. **Never depend on one head.**

MODULE 17 · DOMAIN 2 - PROCESS

Business case, charter and the integrated plan

Why a project exists, how it is authorised, and how one integrated plan holds every part together, in any approach.

DOMAIN II - Process EXAM WEIGHT 41% MODULE 17 of 49

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The business case says **why**. The charter says **go**. The plan says **how**.

From need to plan

01

Needs assessment

What problem or opportunity exists?

02

Business case

Options, costs, benefits, a recommendation.

03

Benefits plan

Which benefits, when, measured how, owned by whom.

04

Project charter

Authorises the project and the PM.

05

Integrated plan

How the work will be delivered and controlled.

ON THE EXAM

The first three usually exist before the project does. The project manager uses them, and keeps checking them.

The business case

- The **problem or opportunity**, and why it matters now.
- **Options** analysed, including doing nothing, with costs, benefits and risks.
- The **recommended option** and its success criteria.
- Owned by the **sponsor**. Revisited at gates and whenever the environment changes.
- A business case that no longer holds is a signal to raise, not a document to ignore.

> ON THE EXAM
>
> The outline asks you to assess plans for continued business value. The business case is the yardstick.

The benefits management plan

- **Target benefits** and how each will be measured.
- **When** each benefit is expected, often after the project has closed.
- A **benefit owner**, usually in the business, not the project team.
- The **assumptions and risks** that could stop benefits arriving.

> ON THE EXAM
>
> Benefits often arrive after closure. Someone in the business must own and measure them.

What the project charter contains

SECTION	EXAMPLES
Purpose and objectives	Why the project exists; measurable objectives and success criteria
High-level scope and requirements	What is in, what is out, at a summary level
Summary milestones and budget	Key dates; pre-approved financial resources
High-level risks, assumptions, constraints	The main known uncertainties and limits
Key stakeholders	Who is affected and involved
Authority	The named project manager, their authority level, and the sponsor
Approval requirements	What defines success and who signs off

ON THE EXAM

The charter is deliberately high-level. Detail belongs in the plan.

What the charter does

- **Formally authorises** the project to exist and to use organisational resources.
- Gives the **project manager authority** to apply those resources.
- Issued by the **sponsor** or another initiator outside the project. The PM may help draft it.
- Without a charter, the project manager has no formal authority, and the project has no agreed purpose.

> ON THE EXAM
>
> No charter, no project. Asked to start planning without one? Work with the sponsor to get it first.

Assumptions and constraints

- **Assumptions**: things believed true for planning, without proof. 'The supplier's API will be ready in month three.'
- **Constraints**: limits the project must work within. 'Go-live before the financial year end.'
- Record both in an **assumption log**, and validate assumptions as the project proceeds.
- An assumption proven false is a risk or an issue. Update the log and assess the impact.

> ON THE EXAM
>
> Every assumption is a risk in disguise. Log it, check it, and act when it turns out to be wrong.

Assessing needs, complexity and magnitude

- **Size**: budget, duration, number of people, teams and sites.
- **Uncertainty**: how clear are requirements, technology and the environment?
- **Complexity**: interdependencies, stakeholders, regulation, organisational change.
- The assessment drives the **approach**, the **governance** level, and how much **planning** is enough.

> ON THE EXAM
>
> A small, clear project needs a light plan. A large, complex one needs more structure. Tailor to the assessment.

Critical information, including sustainability

- The outline asks you to **determine critical information requirements**, naming **sustainability** as an example.
- Environmental: carbon, energy, materials, waste, and what happens at end of life.
- Social and economic: effects on communities, workers, suppliers and long-term costs.
- Capture sustainability requirements early, with stakeholders, and build them into objectives and success criteria.

ON THE EXAM

Sustainability is planned in, not added later. It is a PMBOK 8 principle and an outline enabler.

The integrated project management plan

PART	CONTENTS
Subsidiary plans	Scope, schedule, finance, quality, resources, communications, risk, procurement, stakeholder engagement
Baselines	Scope, schedule and cost; together, the performance measurement baseline
Other components	Change management and configuration management plans, life cycle, development approach
Adaptive equivalents	Product vision, roadmap, release and iteration plans, product backlog

ON THE EXAM

One integrated plan, however many parts. A change to one part is checked for its effect on all the others.

Integration: gaps and dependencies

- The outline asks you to **assess plans for dependencies, gaps and continued business value**.
- Check the parts agree: does the schedule assume people the resource plan never provides?
- Check dependencies between workstreams, teams, suppliers and other projects.
- Subsidiary plans written separately, never reconciled, are a classic source of failure.

> ON THE EXAM
>
> The project manager's integrating role: make the parts consistent, and keep them consistent as they change.

Progressive elaboration and rolling wave planning

- **Progressive elaboration**: the plan gains detail as more becomes known.
- **Rolling wave planning**: near-term work in detail, later work at a higher level.
- Detail the next phase or release as it approaches, and keep later phases as milestones and estimates.
- Detailed plans for work two years away are mostly guesses that will need rework.

> ON THE EXAM
>
> Plan in detail what you can see clearly. Keep the rest at the level of confidence you actually have.

Predictive or adaptive planning?

PREDICTIVE

Plan the whole, then control

- Detailed plan and baselines early
- Scope fixed; time and cost estimated
- Change through formal control
- Progress against the baseline

ADAPTIVE

Plan in layers, often

- Vision, roadmap, release, iteration
- Time and cost often fixed; scope flexes
- Change through the backlog
- Progress in working increments

THE TIEBREAKER

Both plan. Predictive plans more up front; adaptive plans more often. Hybrid does both, per workstream.

Recommending an execution strategy

- **Build or buy**: in-house, outsourced, or a mix.
- **Phasing**: one delivery, phased roll-out, pilot first, or continuous releases.
- **Sequencing**: what must come first for risk, value or dependency reasons.
- A **pilot** reduces risk when the change is large and the organisation untested.

> ON THE EXAM
>
> The execution strategy turns the chosen approach into a practical route. Recommend it with reasons.

Data-driven planning decisions

- The outline asks you to **collect and analyse data to make informed project decisions**.
- Use **historical data and benchmarks** from similar projects for estimates.
- Use the team's own measured performance, such as velocity or throughput, once available.
- Plans built on optimism rather than evidence are the most common cause of late projects.

> ON THE EXAM
>
> Base plans on evidence. When evidence is thin, say so, and plan to gather more.

Approval, baselines and change

- The plan is **approved** by the appropriate authority, and baselines are set.
- After that, baselines change only through **change control**.
- The plan itself is **maintained**: updated as the project learns, with changes recorded.
- Quietly editing a baseline to hide a variance is not maintenance. It is misreporting.

> ON THE EXAM
>
> Baselines are for measuring against. They move only through the agreed change process.

Project Lantern: from case to plan

- **Business case**: missed appointments cost Northgate heavily; options included doing nothing and buying a hosted service.
- **Benefit**: missed appointments down by a quarter within a year of the last go-live, owned by the head of patient access.
- **Charter**: signed by the chief operating officer; names the PM with authority over contingency up to an agreed limit.
- **Plan**: hybrid; baselines for hardware and migration, roadmap and backlog for the app, a pilot clinic first.

> ON THE EXAM
>
> Why, go, how: three documents, each doing its own job, all consistent with each other.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Asked to plan, no charter exists	Work with the sponsor to get a charter first
The business case no longer holds	Raise it with the sponsor, with analysis
Two plans contradict each other	Reconcile them; update the integrated plan
Organisation has sustainability commitments	Capture them as requirements and success criteria
Long project, later phases unclear	Rolling wave: detail near-term, elaborate later
A planning assumption proves false	Update the log, assess impact, treat as risk or issue

ON THE EXAM

Why, go, how. Keep all three current and consistent, and the plan evidence-based.

> A department head tells a project manager in a corridor conversation to 'get started on the new supplier portal' and to have a plan ready in two weeks.
>
> No charter or business case has been shared, and nobody has confirmed the budget or the project manager's authority.

What should the project manager do first?

A Start creating the work breakdown structure to save time.

B Work with the department head, as sponsor, to develop and approve a project charter.

C Hold a kickoff meeting with the likely team members.

D Begin contacting possible suppliers for quotes.

◆ PAUSE AND DECIDE

What should the project manager do first?

A Start the WBS.

Plans work with no agreed purpose, scope or authority.

B Develop and approve a charter with the sponsor.

Authorises the project and the PM, and agrees purpose and objectives first.

C Kickoff meeting.

Commits people's time to a project that does not formally exist yet.

D Contact suppliers.

Commercial activity without authority or budget is a real risk.

THE SIGNAL

No charter, no authority. Get the project formally authorised before planning or committing anything.

> A project is building an app to help customers claim a government energy rebate. Halfway through, the government announces the rebate will end in three months.
>
> The business case assumed two years of claims. The project is on schedule and on budget.

What should the project manager do?

A Continue as planned, since the project is on schedule and on budget.

B Cut scope to finish sooner and reduce cost.

C Analyse the impact on the business case and bring it to the sponsor to decide the project's future.

D Ask the team to work faster so the app launches before the rebate ends.

◆ PAUSE AND DECIDE

What should the project manager do?

A Continue as planned.

On time and on budget towards value that no longer exists.

B Cut scope to finish sooner.

A unilateral response that assumes the project should continue at all.

C Analyse; take it to the sponsor.

Assesses continued business value, and puts the decision with its owner.

D Work faster.

Assumes a three-month window is worth the cost and strain. Nobody has checked.

THE SIGNAL

When the business case collapses, the sponsor decides the project's future, with your analysis.

> While reviewing the plan, the project manager notices that the schedule assumes six testers during the final month, but the resource management plan allocates only three.
>
> Both plans were approved by their respective owners.

What should the project manager do?

A Follow the schedule, and request three more testers when the final month arrives.

B Reconcile the two plans with their owners, and update the integrated plan.

C Follow the resource plan, since resources are the harder constraint.

D Escalate the contradiction to the steering committee.

◆ PAUSE AND DECIDE

What should the project manager do?

A Request testers later.

Leaves the gap until it is too late to fix cheaply.

B Reconcile with the owners; update the plan.

Integration: find the gap now, resolve it, keep the plan consistent.

C Follow the resource plan.

Silently accepts a schedule that cannot be met.

D Escalate.

A planning gap the PM can resolve with the plan owners.

THE SIGNAL

Two plans that contradict each other: that is exactly the gap the outline asks you to find and fix.

A project manager is planning an office refurbishment for a company that has publicly committed to reaching net zero emissions.

The contract with the fit-out firm says nothing about sustainability, and the budget is tight.

What should the project manager do?

A Leave sustainability out, since it is not in the contract.

B Plan the refurbishment now and add sustainability measures if budget remains.

C Identify the sustainability requirements with stakeholders, and build them into objectives and success criteria.

D Leave sustainability decisions to the fit-out firm.

◆ PAUSE AND DECIDE

What should the project manager do?

A Leave it out.

Ignores a stated organisational commitment. The contract can be updated.

B Add it if money is left.

Treats sustainability as optional extra, which the principles reject.

C Identify requirements; build them into objectives.

Critical information captured early, where it can shape design and cost.

D Leave it to the contractor.

Hands an organisational commitment to a party with no obligation to meet it.

THE SIGNAL

An organisational sustainability commitment is critical information. Plan it in, from the start.

> A three-year programme will replace a bank's core systems in three phases. Phase one is well understood. Phases two and three depend on vendor decisions not yet made.
>
> The sponsor asks for a fully detailed schedule for all three phases at the start.

How should the project manager respond?

A. Plan phase one in detail, keep later phases at milestone level, and elaborate each as it approaches.

B. Build a fully detailed schedule for all three phases, using assumptions for the unknowns.

C. Plan phase one only, and leave later phases unplanned.

D. Delay the start until the vendor decisions are made.

◆ PAUSE AND DECIDE

How should the project manager respond?

A Rolling wave planning.

Detail where knowledge exists, milestones where it does not, elaborated over time.

B Full detail on assumptions.

False precision. Most of it will be reworked once vendors are chosen.

C No plan for later phases.

Leaves the sponsor with no view of the overall programme.

D Delay the start.

Wastes time on phase one, which is ready to go.

THE SIGNAL

Detail what you know; outline what you do not. Explain to the sponsor why that is more reliable.

> The assumption log records that a partner's data interface would be available in month three. The partner now says it will not be ready until month six.
>
> Several work packages depend on the interface.

What should the project manager do next?

A Continue as planned, since the partner may still deliver earlier.

B Record the delay as the partner's fault in the project documents.

C Rebaseline the schedule to month six straight away.

D Update the assumption log, assess the impact on dependent work, and respond through the risk and change processes.

◆ PAUSE AND DECIDE

What should the project manager do next?

A Carry on.

Ignores new, specific information that a planning assumption is false.

B Record the partner's fault.

Blame does nothing for the dependent work.

C Rebaseline immediately.

Skips impact analysis and the change process; there may be workarounds.

D Update log, assess impact, use risk and change processes.

A false assumption becomes a risk or issue, handled through the proper processes.

THE SIGNAL

A failed assumption is a risk or issue now. Log it, assess it, and act through the proper processes.

Why, go, how. Keep them consistent, and keep checking the why.

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MODULE 18 · DOMAIN 2 - PROCESS

Scope in predictive projects

Eliciting requirements, defining scope and its exclusions, building the work breakdown structure, and the difference between validating and controlling scope.

DOMAIN II - Process EXAM WEIGHT 41% MODULE 18 of 49

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Scope is what the project will deliver, and just as importantly, **what it will not**.

Product scope or project scope?

PRODUCT SCOPE

What is delivered

- Features and functions of the result
- Measured against requirements
- 'The app lets patients reschedule'
- Owned by the customer's needs

PROJECT SCOPE

The work to deliver it

- All work needed to produce the result
- Measured against the plan
- Includes training, testing, migration
- Owned by the project

THE TIEBREAKER

Project scope includes work the customer never sees, such as testing, training and handover. It all counts.

Eliciting requirements

TECHNIQUE	BEST WHEN
Interviews	Individual experts; sensitive or detailed topics
Workshops and focus groups	Several stakeholders must agree, or reconcile differences
Observation, or job shadowing	Users cannot easily describe what they actually do
Questionnaires and surveys	Many people, spread widely, with simple questions
Prototypes	People react better to something they can see and try
Document analysis and benchmarking	Existing systems, rules, contracts, or other organisations' practice

ON THE EXAM

Match the technique to the people. If users cannot describe the work, watch them do it.

Kinds of requirement

TYPE	EXAMPLE
Business	Reduce missed appointments by a quarter
Stakeholder	Reception staff can rebook a patient in under a minute
Functional	The app sends a reminder forty-eight hours before an appointment
Non-functional	Pages load in under two seconds; accessible to screen readers
Transition	Staff are trained before their clinic goes live
Project	Clinic go-lives fall outside flu season

ON THE EXAM

Non-functional and transition requirements are the ones most often missed. Ask about them explicitly.

The requirements traceability matrix

- Links each requirement to its **source**, the **objective** it serves, the **deliverable** that meets it, and the **test** that proves it.
- Shows that every requirement is delivered and tested, and nothing is missed.
- Shows that every deliverable traces to a requirement, which exposes gold plating.
- Makes change impact visible: change a requirement, and see what else it touches.

> ON THE EXAM
>
> Need to prove each requirement was delivered and tested? The traceability matrix is the tool.

The project scope statement

- **Product scope description**: what will be created.
- **Deliverables** and their **acceptance criteria**.
- **Exclusions**: what is explicitly out of scope.
- **Assumptions and constraints** that shape the scope.
- Silence is not exclusion. If something might be assumed, write down that it is out.

> ON THE EXAM
>
> Exclusions prevent more disputes than any other line in the scope statement.

Building a work breakdown structure

01

Project

The whole of the work.

›

02

Major deliverables

Or phases, the main building blocks.

›

03

Sub-deliverables

Broken down until manageable.

›

04

Work packages

The lowest level: estimated, scheduled, costed, assigned.

ON THE EXAM

A WBS is a hierarchy of deliverables, not a list of activities. Activities come later, in the schedule.

WBS rules the exam uses

- The **100 percent rule**: the WBS includes all the work, and only the work, in scope.
- Each level adds up exactly to the level above. Nothing missing, nothing extra.
- Built **with the team**: the people doing the work know what it involves.
- A **WBS dictionary** describes each work package: scope, acceptance criteria, owner, estimates.
- Work missing from the WBS will be missing from the schedule and the budget too.

> ON THE EXAM
>
> If work is found late that was never planned, the 100 percent rule was broken in the WBS.

The scope baseline

- **Scope statement + WBS + WBS dictionary** = the scope baseline.
- Approved, then changed only through **change control**.
- Together with the schedule and cost baselines, it forms the performance measurement baseline.
- It is what 'in scope' and 'out of scope' are measured against.

> ON THE EXAM
>
> Is it in scope? Check the scope baseline. Changing it? Use change control.

Validate scope or control scope?

VALIDATE SCOPE

Formal acceptance

- Customer or sponsor reviews deliverables
- Checks they meet acceptance criteria
- Produces accepted deliverables
- Happens after quality control

CONTROL SCOPE

Managing the boundary

- Monitors scope status throughout
- Compares work to the baseline
- Manages changes through change control
- Prevents scope creep

THE TIEBREAKER

Validate is about acceptance by the customer. Control is about managing changes to the baseline.

From deliverable to acceptance

01

Build

The team produces the deliverable.

›

02

Control quality

The team checks it is correct.

›

03

Validate scope

The customer checks it is acceptable.

›

04

Accepted

Formally signed off, ready for closure.

ON THE EXAM

Quality control first, then validation. Never ask a customer to accept something the team has not checked.

Scope creep and gold plating

- **Scope creep**: uncontrolled additions, often small, often requested directly of team members.
- **Gold plating**: the team adds extras nobody asked for.
- Both are prevented by a clear baseline, a known change process, and team awareness.
- Requests are welcome. They go through change control, where impact is assessed and someone decides.

> ON THE EXAM
>
> Small, direct, undocumented additions are scope creep. The fix is the process, not refusing every request.

Getting agreement on scope

- The outline asks you to **obtain stakeholder agreement on project scope**.
- Walk key stakeholders through the scope statement and WBS, especially the exclusions.
- Resolve conflicting requirements before the baseline, not after.
- Obtain formal approval, then share the baseline with everyone who needs it.

> ON THE EXAM
>
> Agreement means stakeholders understood the boundaries, not just that someone signed a document.

Project Lantern: part of the WBS

- **1 Hardware**: 1.1 kiosks procured; 1.2 kiosks installed; 1.3 network upgraded.
- **2 Data**: 2.1 data cleansed; 2.2 data migrated; 2.3 migration verified.
- **3 People**: 3.1 staff trained; 3.2 clinic readiness confirmed.
- **4 App**: managed through the product backlog, with release milestones in the WBS.
- Excluded: integration with the regional hospital's system, which is a separate project.

> ON THE EXAM
>
> Deliverables, not activities. Hidden work, such as verification and training, included. Exclusions stated.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
A deliverable passed quality control	Validate scope: customer acceptance
A stakeholder asked a team member for an extra	Route it through change control
Work discovered late that was never planned	The WBS broke the 100 percent rule
Prove every requirement was delivered and tested	Requirements traceability matrix
Customer assumed something was included	Exclusions should have been stated
Users cannot describe their work	Observation, or job shadowing

ON THE EXAM

Define it, break it down, agree it, then control it and get it accepted.

> On a predictive project, the team has completed the first major deliverable, a new payroll interface. It has passed the team's quality control testing.
>
> The project manager is deciding what should happen next with the deliverable.

What should the project manager do next?

A Close the project phase, since the deliverable is complete.

B Repeat quality control to be certain it is correct.

C Update the WBS to show the work package as complete.

D Review the deliverable with the customer to obtain formal acceptance.

◆ PAUSE AND DECIDE

What should the project manager do next?

A Close the phase.

Skips acceptance. A deliverable is not finished until the customer accepts it.

B Repeat quality control.

Already passed. Repeating it adds nothing.

C Update the WBS.

Housekeeping, and the WBS does not track status in that way.

D Obtain the customer's formal acceptance.

Validate scope comes after quality control.

THE SIGNAL

Quality control passed means ready for validation. Acceptance comes before closure.

A client's finance manager asked a developer directly for an extra export button on a report screen. The developer added it, since it took only an hour.

The project manager discovers this during a review. The button is not in the scope baseline.

What should the project manager do?

A Talk to the developer and the finance manager, explain the change process, and assess this request through it.

B Leave the button in place, since it took only an hour and the client wanted it.

C Remove the button immediately, since it is out of scope.

D Formally reprimand the developer for exceeding scope.

◆ PAUSE AND DECIDE

What should the project manager do?

A Explain the process; assess the request through it.

Stops the creep at its source and still treats the request fairly.

B Leave it.

Accepts uncontrolled change. One hour today, a pattern tomorrow.

C Remove it immediately.

Reacts without assessment, and may frustrate the client needlessly.

D Reprimand the developer.

Punishes a helpful instinct. Coaching fixes it; blame does not.

THE SIGNAL

A direct, small, undocumented addition is scope creep. Fix the route, not the person.

> Near the end of a system roll-out, the project manager discovers that nobody planned the training of three hundred end users. It is not in the schedule or the budget.
>
> The training will take four weeks and has no funding.

What should the project manager have done to prevent this?

A Built a more detailed schedule for the roll-out phase.

B Added a larger contingency reserve to the budget.

C Decomposed the full scope with the team and stakeholders, so the WBS included all the work.

D Used fewer levels in the WBS to keep it simple.

◆ PAUSE AND DECIDE

What should the project manager have done to prevent this?

A More detailed schedule.

The schedule comes from the WBS. Missing work stays missing.

B More contingency.

Reserves cover uncertainty, not work that was simply forgotten.

C Full decomposition with team and stakeholders.

The 100 percent rule: all the work in scope, including transition work.

D Fewer WBS levels.

Less decomposition makes missing work more likely, not less.

THE SIGNAL

Work found late that was never planned: the WBS did not contain 100 percent of the scope.

> During a compliance audit, an auditor asks the project manager to show, for each regulatory requirement, which business objective it supports, which deliverable meets it, and which test proved it.
>
> The project has two hundred requirements.

Which project artefact provides this information?

A The issue log

B The requirements traceability matrix

C The WBS dictionary

D The project scope statement

◆ PAUSE AND DECIDE

Which project artefact provides this information?

A Issue log

Tracks problems, not requirements.

B Requirements traceability matrix

Links each requirement to objective, deliverable and test.

C WBS dictionary

Describes work packages, not requirement-by-requirement tracing.

D Scope statement

Describes scope and exclusions, not individual traceability.

THE SIGNAL

Requirement to objective to deliverable to test: that chain is the traceability matrix.

> A consultancy is building a new customer database for a client. At the final review, the client asks when the old customer records will be transferred.
>
> The scope statement lists the new database and its features. It says nothing about migrating old data.

What should the project manager have done to prevent this?

A Stated data migration as an explicit exclusion, and confirmed the exclusions with the client.

B Offered to migrate the data free of charge.

C Refused the request, since it is not in the scope statement.

D Written a longer contract covering every possible item.

◆ PAUSE AND DECIDE

What should the project manager have done to prevent this?

A Explicit exclusion, confirmed with the client.
Silence let the client assume. A stated, agreed exclusion prevents it.

B Migrate free.
A reaction, not prevention, and uncontrolled scope.

C Refuse.
A reaction, and it damages the relationship.

D A longer contract.
Length does not create shared understanding. One clear exclusion does.

THE SIGNAL

Something a stakeholder could reasonably assume was included must be explicitly excluded.

> A project will redesign a warehouse picking process. In interviews, pickers give vague answers about how they work, and say 'we just know where things are'.
>
> The warehouse is noisy, and pickers are paid by the number of orders they complete.

Which elicitation technique should the business analyst use next?

A. A detailed questionnaire for all pickers.

B. Longer, more structured interviews.

C. Observation, shadowing pickers as they work.

D. Analysis of the existing process documents.

◆ PAUSE AND DECIDE

Which elicitation technique should the business analyst use next?

A Questionnaire.

Still asks people to describe what they cannot put into words, and costs them paid time.

B Longer interviews.

More of the technique that is already producing vague answers.

C Observation.

Captures the real process, tacit knowledge included, without stopping work.

D Process documents.

Useful context, but 'we just know' suggests practice differs from documents.

THE SIGNAL

When people cannot describe their work, watch them do it.

Define it, exclude clearly, break it down fully, then **control and accept** it.

MODULE 19 · DOMAIN 2 - PROCESS

Scope in adaptive projects

The product backlog, user stories, acceptance criteria, definitions of ready and done, splitting, story mapping and ordering.

DOMAIN II - Process EXAM WEIGHT 41% MODULE 19 of 49

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In adaptive work, scope is **discovered**, not frozen. The backlog is how it is managed.

How scope behaves: predictive or adaptive

PREDICTIVE

Scope fixed, estimate the rest

- Scope defined and baselined early
- Time and cost estimated from scope
- Change through change control
- WBS and scope statement

ADAPTIVE

Time and cost fixed, scope flexes

- Scope discovered and refined
- Timeboxes and team size often fixed
- Change through the backlog
- Backlog, stories and a product goal

THE TIEBREAKER

Adaptive projects often fix time and cost and let scope flex. The highest-value scope is delivered first.

From big ideas to small work

01
Theme
A broad goal: easier appointment booking.

02
Epic
A large body of work: online rescheduling.

03
Feature
A usable capability: reschedule from a reminder.

04
User story
Fits in an iteration: one small slice of value.

05
Task
The team's own breakdown of how to build it.

ON THE EXAM

Only stories near the top of the backlog need full detail. Epics further down can stay large.

User stories

- **As a** patient, **I want** to reschedule from my reminder message, **so that** I do not have to phone the clinic.
- Who wants it, what they want, and **why**. The 'so that' is the value.
- **Card, conversation, confirmation**: a short note, a discussion, and agreed acceptance criteria.
- A story is a placeholder for a conversation, not a complete specification.

> ON THE EXAM
>
> The conversation matters more than the card. Stories are refined by talking, not by writing more.

INVEST: what makes a good story

LETTER	MEANS	TEST
I	Independent	Can it be built and released without waiting on another story?
N	Negotiable	Is the detail still open to discussion?
V	Valuable	Does it deliver value to a user or customer?
E	Estimable	Can the team size it?
S	Small	Can it be finished within one iteration?
T	Testable	Can we tell when it is done?

ON THE EXAM

A story that fails 'small' or 'estimable' needs splitting or a spike, not a longer iteration.

Acceptance criteria

- The conditions a story must meet for the product owner to accept it.
- Often written as **Given, When, Then**:
- **Given** a booked appointment more than 24 hours away, **when** I choose a new slot from my reminder, **then** the old slot is released and I get a confirmation.
- Agreed **before** work starts, so the team knows what done looks like for this item.

> ON THE EXAM
>
> Acceptance criteria agreed after the work is built are just a rejection waiting to happen.

Definition of ready or definition of done?

DEFINITION OF READY

Before work starts

- Story is small enough
- Acceptance criteria agreed
- Dependencies known
- Team understands it well enough to plan

DEFINITION OF DONE

Before work counts as finished

- Meets acceptance criteria
- Tested, reviewed, integrated
- Documentation updated
- Meets quality and non-functional standards

THE TIEBREAKER

Ready controls what enters an iteration. Done controls what leaves it. Both are agreed by the team.

Backlog refinement

- Continuous: items are clarified, split, estimated and reordered, **ahead** of planning.
- Done by the product owner and the developers together.
- A useful backlog is **DEEP**: detailed appropriately, estimated, emergent, prioritised.
- Refinement squeezed into sprint planning turns planning into a marathon.

> ON THE EXAM
>
> Long, painful planning sessions are usually a refinement problem. Refine ahead, continuously.

Splitting stories: vertical slices

SPLIT BY	EXAMPLE
Workflow step	Search for a slot; then choose it; then confirm it
Business rule	Standard appointments first; urgent ones later
Data variation	One clinic first; then all clinics
Happy path first	Successful booking first; error handling next
Interface	Web first; kiosk later

ON THE EXAM

Split vertically, so each slice works end to end. Splitting by layer, such as database then screen, delivers nothing usable.

Story mapping

- Arrange user activities left to right as the **backbone** of the user's journey.
- Stack stories under each activity, most important at the top.
- Draw horizontal lines to slice **releases**: the top slice across the whole journey is the first release.
- Shows stakeholders the whole product, and what arrives when, far better than a long flat list.

> ON THE EXAM
>
> A long flat backlog hides the journey. A story map shows it, and makes the first release obvious.

Ordering the backlog

- **Value**: what matters most to customers and the business.
- **Risk**: tackle the riskiest, most uncertain items early, while there is time to react.
- **Dependencies** and **learning**: what unlocks other work, or answers big questions.
- **Cost of delay**: what loses value if it waits.
- The **product owner** decides the order, informed by stakeholders and the team.

> ON THE EXAM
>
> Order by value, risk and dependency, not by who shouted loudest. Risky items early, not last.

Where non-functional requirements go

- Standards that apply to everything, such as security and accessibility, go in the **definition of done**.
- Standards for one feature go in that story's **acceptance criteria**.
- Larger pieces of work, such as a performance improvement, can be **backlog items** of their own.
- If no story or standard mentions performance, nobody will build for it.

> ON THE EXAM
>
> Non-functional requirements must live somewhere visible. Otherwise they surface only after release.

Changing scope in adaptive work

- New ideas become **backlog items**; the product owner orders them against everything else.
- Within fixed time and cost, adding something usually means something lower moves down or out.
- Formal change requests still apply to **contracts, budgets and governance baselines**.
- Changing the current iteration's work is limited: only if the iteration goal is safe.

> ON THE EXAM
>
> The backlog is the change mechanism for product scope. Contracts and funding still need formal change.

Scope on hybrid projects

- A **WBS** for the predictive work, and a **backlog** for the adaptive work.
- The WBS can hold the adaptive work as a **release or epic** with its milestones.
- Governance baselines are set at the level of releases and outcomes, not individual stories.
- Integration points, such as a hardware-ready date, are dependencies in the backlog order.

> ON THE EXAM
>
> Baseline the release, not every story. Let the backlog flex inside the release boundary.

Project Lantern: the app backlog

- **Story map backbone**: find a clinic, book, get reminded, reschedule, check in at the kiosk.
- **First release slice**: book and cancel standard appointments at one clinic, on web.
- **Definition of done** includes accessibility checks and page loads under two seconds.
- The riskiest item, the booking-system integration, was built in the second sprint, not the last.

> ON THE EXAM
>
> A journey-shaped backlog, a thin first slice, standards in the definition of done, and risk tackled early.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
A story is too big for one iteration	Split it into vertical slices
Work is rejected at review as not what was wanted	Agree acceptance criteria before work starts
Sprint planning takes hours	Refine continuously, ahead of planning
A quality problem appears after release	Put the standard in the definition of done
Risky work is left until the end	Order by value, risk and dependency
Stakeholders cannot see the whole product	Facilitate a story map; slice releases

ON THE EXAM

Small, valuable, testable stories, refined ahead, ordered by value and risk, and finished to a clear standard.

> A backlog item reads 'Build the management reporting system'. The team estimates it would take at least three sprints.
>
> The product owner wants it in the next sprint.

What should the team do?

A Extend the next sprint so the item can be completed in one go.

B Split it into smaller vertical slices, such as one report at a time, each usable on its own.

C Split it by layer: the database first, then the services, then the screens.

D Start it in the next sprint and carry it over for as long as it takes.

◆ PAUSE AND DECIDE

What should the team do?

A Extend the sprint.

Sprints are fixed length. Oversized items are split, not accommodated.

B Vertical slices, one report at a time.

Each slice is small, valuable and testable, and the most important report can come first.

C Split by layer.

Horizontal slices. Nothing is usable until all three are done.

D Carry it over.

Nothing done for three sprints, and no feedback until the end.

THE SIGNAL

An epic dressed as a story: split it vertically, so value arrives every sprint.

> In the last two sprint reviews, the product owner rejected several stories that the developers considered finished. The developers say they built what the stories described.
>
> The stories have short descriptions and no acceptance criteria.

What should the project leader do?

A Ask the product owner to accept the stories, since they match the descriptions.

B Add more testing before each review.

C Write the acceptance criteria for the team personally.

D Facilitate the product owner and developers agreeing acceptance criteria before work starts on each story.

◆ PAUSE AND DECIDE

What should the project leader do?

A Pressure the PO to accept.

The PO owns acceptance. The gap in understanding remains.

B More testing.

Tests against what? Without criteria, nobody knows what to test for.

C PM writes the criteria.

The PM does not own the product's needs. The PO and team must agree them.

D Agree acceptance criteria before work.

Confirmation in the three Cs: a shared definition of done for each story.

THE SIGNAL

Rejection at review with no acceptance criteria: the confirmation was never agreed. Agree it first.

> Sprint planning for a two-week sprint now takes most of a day. The team spends hours clarifying stories, splitting them and estimating for the first time.
>
> The developers say they dread planning day.

What should the project leader do?

A Help the team hold regular refinement sessions ahead of planning, with a definition of ready.

B Extend the planning timebox, since the work needs doing.

C Skip estimation to save time in planning.

D Refine the stories personally before each planning session.

◆ PAUSE AND DECIDE

What should the project leader do?

A Refine ahead, with a definition of ready.

Moves the clarifying and splitting out of planning, continuously.

B Longer planning.

Accepts the symptom. Planning should confirm, not discover.

C Skip estimation.

Removes the information the team needs to forecast its sprint.

D PM refines alone.

Refinement needs the product owner's knowledge and the developers' sizing.

THE SIGNAL

Painful planning is a refinement problem. Refine continuously, ahead of time.

> After the first public release of a ticket-booking app, customers complain that pages take several seconds to load at busy times.
>
> No story, acceptance criterion or team standard mentioned performance.

What should the team do to prevent this in future?

A. Plan a dedicated performance-testing sprint before each major release.

B. Add performance standards to the definition of done, and to acceptance criteria where specific.

C. Buy more powerful servers for the next release.

D. Ask the product owner to take responsibility for performance.

◆ PAUSE AND DECIDE

What should the team do to prevent this in future?

A. A performance sprint before releases.
Batches quality at the end, the opposite of building it in each sprint.

B. Performance in the DoD and criteria.
Makes the non-functional requirement visible and checked on every item.

C. Bigger servers.
Treats a symptom, and may not help if the design is the cause.

D. Make it the PO's job.
Quality is the developers' accountability, through the definition of done.

THE SIGNAL

A quality attribute nobody wrote down: put it in the definition of done, so every item meets it.

> A product owner orders the backlog mainly by the seniority of whoever requested each item. The integration with a payment provider, the most technically uncertain item, sits near the bottom.
>
> The release date is fixed, four months away.

What should the project leader do?

A. Reorder the backlog personally to bring the integration forward.

B. Leave the order alone, since backlog ordering belongs to the product owner.

C. Coach the product owner on ordering by value, risk and dependencies, including bringing risky items forward.

D. Escalate the product owner's ordering to the sponsor.

◆ PAUSE AND DECIDE

What should the project leader do?

A Reorder it personally.

Takes the product owner's decision away. Wrong person.

B Leave it alone.

Right about ownership, wrong to stay silent about a serious risk.

C Coach the PO: value, risk, dependency.

Respects ownership and fixes the ordering logic that creates the risk.

D Escalate.

Too early. A coaching conversation has not happened.

THE SIGNAL

The PO owns the order; the leader coaches how to order. Risky, uncertain items belong early.

> A product's backlog has grown to three hundred items in one long list. Stakeholders say they cannot tell what the first release will contain or whether it will be usable.
>
> The product owner is struggling to explain the plan.

What should the project leader do?

A Sort the backlog by feature name so it is easier to read.

B Replace the backlog with a work breakdown structure.

C Produce a detailed Gantt chart for all three hundred items.

D Facilitate a user story map with the product owner and stakeholders, and slice the first release.

◆ PAUSE AND DECIDE

What should the project leader do?

A Sort alphabetically.

Still a flat list. The journey and the release remain invisible.

B Switch to a WBS.

Abandons adaptive scope instead of making it visible.

C A Gantt chart.

False precision for adaptive work, and it still hides the user journey.

D Story map; slice the first release.

Shows the whole journey and makes a usable first release visible.

THE SIGNAL

A long flat backlog nobody understands: map the user journey and slice releases across it.

Small, valuable, testable stories. Refined ahead. **Ordered by value and risk.**

MODULE 20 · DOMAIN 2 - PROCESS

Value-based delivery

What value means, delivering the smallest valuable thing first, prioritising by value, measuring benefits, and the financial tests for choosing projects.

DOMAIN II - Process EXAM WEIGHT 41% MODULE 20 of 49

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Deliver the most valuable thing first. Then **check that it was valuable**.

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Kinds of value

KIND	EXAMPLE
Financial	More revenue, lower costs, avoided fines
Customer	Faster service, better experience, higher satisfaction
Strategic	New market position, capability for the future
Social and environmental	Community benefit, lower emissions, accessibility
Risk reduction	Compliance, resilience, fewer outages
Learning	Validated knowledge that shapes the next decision

ON THE EXAM

Value is defined by stakeholders, and it is not only money. Identify the components with them, early.

Minimum viable and its relatives

TERM	WHAT IT IS
Prototype	A mock-up to explore a design; not for real use
Minimum viable product	Smallest thing that tests a key hypothesis with real users
Minimum marketable feature	Smallest feature customers value on its own
Walking skeleton	Thin end-to-end architecture, proving the parts connect
Pilot	A real roll-out to a limited group, to reduce risk

ON THE EXAM

An MVP is for learning. An MMF is for delivering value.

When a minimum viable product is the answer

- Nobody is sure customers **want it**, or will **pay for it**, or will **use it** as expected.
- Build the smallest thing that answers that question, with real users.
- It can be very simple: a landing page, a manual service behind a simple front end.
- Building the full product first, then discovering nobody wants it, is the costliest way to learn.

> ON THE EXAM
>
> Big uncertainty about demand: test it cheaply first. Then persevere, or pivot.

Prioritisation techniques

TECHNIQUE	HOW IT WORKS
MoSCoW	Must, Should, Could, Won't have this time
Kano model	Basic needs, performance needs, delighters
Cost of delay	What value is lost for each week an item waits
Weighted scoring	Score options against weighted criteria
Value versus effort	A two-by-two grid: do quick wins first
100-point method	Stakeholders share 100 points across items

ON THE EXAM

The product owner or sponsor decides. The technique makes the trade-offs visible and the decision defensible.

MoSCoW in practice

- **Must**: without it, the release fails or is unlawful or unsafe.
- **Should**: important, but there is a workaround.
- **Could**: desirable; first to drop if time is short.
- **Won't have this time**: agreed out of this release, not forgotten.
- If everything is a Must, nothing is. Test each Must: would we delay the release for this?

> ON THE EXAM
>
> The discipline is in the definitions. Agree what 'Must' means before anyone labels anything.

The Kano model

- **Basic needs**: expected. Nobody praises them; their absence causes anger.
- **Performance needs**: more is better, in proportion. Faster, cheaper, easier.
- **Delighters**: unexpected extras that create delight. Absence is not noticed.
- Over time, delighters become basic needs. Yesterday's surprise is today's expectation.

> ON THE EXAM
>
> Get the basics right first. A delighter never compensates for a missing basic need.

Cost of delay

- The value lost for each unit of time an item is **not** delivered.
- Feature A loses $10,000 a week while waiting; feature B loses $2,000. Both take two weeks.
- Do A first. Delaying B for two weeks costs $4,000; delaying A would cost $20,000.
- With different sizes, divide cost of delay by duration, and do the highest first.

> ON THE EXAM
>
> When items differ in urgency, the one that loses most value while waiting goes first.

Value versus effort

	LOW EFFORT	HIGH EFFORT
High value	Quick wins: do first	Big bets: plan carefully
Low value	Fill-ins: if capacity allows	Money pits: avoid

ON THE EXAM

Quick wins deliver value early and build trust. Money pits should be challenged, however senior the requester.

Finding chances to deliver value sooner

- The outline asks you to **assess opportunities to deliver value incrementally**.
- Could part of the result be released early, to one region, one group, one product line?
- Could a pilot deliver real benefits while reducing risk for the rest?
- Holding finished, valuable work back for one big launch delays benefits for no gain.

> ON THE EXAM
>
> Even predictive projects can release value in stages. Look for it, and propose it.

Measuring benefits

- Measure a **baseline** before delivery. Without it, you cannot show any change.
- Define **measures and targets**: key performance indicators, or objectives and key results.
- Name a **benefit owner** who will keep measuring after the project closes.
- Put the **measurement system** in place as part of the project, not as an afterthought.

> ON THE EXAM
>
> No baseline, no proof of benefit. Measure before you change anything.

Leading or lagging indicators?

LEADING INDICATORS

Early signals

- Predict future results
- Available during the project
- Adoption rate, feature usage, early feedback
- Let you adjust in time

LAGGING INDICATORS

Final results

- Confirm results after the fact
- Often available only after closure
- Revenue, cost savings, missed appointments
- Prove the benefit was real

THE TIEBREAKER

Use both. Leading indicators let you steer; lagging indicators prove the outcome.

Financial tests for choosing projects

MEASURE	CHOOSE THE PROJECT WITH
Net present value (NPV)	The highest NPV. It already accounts for timing
Internal rate of return (IRR)	The highest IRR
Return on investment (ROI)	The highest ROI
Benefit-cost ratio (BCR)	The highest ratio; above 1 means benefits exceed costs
Payback period	The shortest payback
Opportunity cost	The value of the best option not chosen

ON THE EXAM

With NPV, ignore the project length in the question. Duration is already built into the NPV.

Checking value throughout

- The outline asks you to **examine the business value throughout the project**.
- Re-check at reviews, gates and when the environment changes.
- In adaptive work, stop when the remaining backlog is worth less than the cost of building it.
- Continuing because the plan said so, after the value has gone, wastes the organisation's money.

> ON THE EXAM
>
> Finishing early because the valuable work is done is a success, not a failure.

Project Lantern: delivering value early

- **Baseline**: missed-appointment rates measured at every clinic for three months before any go-live.
- **Leading indicator**: the share of bookings made online, tracked weekly from the pilot clinic.
- **Early value**: text reminders went live at all six clinics before the new app, because they were a quick win.
- **Won't have this time**: video consultations, agreed with the sponsor and recorded.

> ON THE EXAM
>
> Baseline first, a quick win early, and clear agreement on what is out of this release.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Choose between projects using NPV	Highest NPV; ignore the duration
Unsure whether customers will want it	An MVP to test demand cheaply
Stakeholders say everything is a must	MoSCoW with agreed definitions
Part of the result is ready early	Assess releasing it early
Nobody can prove the benefit	A baseline should have been measured first
Remaining work worth less than it costs	Recommend stopping or re-scoping to the sponsor

ON THE EXAM

Value first, value proven, and value re-checked. That is value-based delivery.

> A portfolio board can fund one of four projects. The finance team has calculated each project's net present value.
>
> The projects run for different lengths of time.

Based on the financial information, which project should be selected?

A Project 1: NPV $95,000, over 2 years.

B Project 2: NPV $80,000, with an IRR of 14%.

C Project 3: NPV $120,000, over 5 years.

D Project 4: NPV $60,000, with a benefit-cost ratio of 1.4.

◆ PAUSE AND DECIDE

Based on the financial information, which project should be selected?

A NPV $95,000, 2 years.

Shorter, but the duration is already inside the NPV. Lower value.

B NPV $80,000, IRR 14%.

A distracting second measure. The shared basis is NPV.

C NPV $120,000, 5 years.

Highest NPV. Timing is already accounted for.

D NPV $60,000, BCR 1.4.

A good ratio, but the lowest value of the four.

THE SIGNAL

Compare like with like: every option has an NPV, so choose the highest. Ignore the years.

A start-up plans a subscription service delivering ready-to-cook meals to offices. Nobody knows whether offices will pay for it.

The founder wants to build the full ordering platform, delivery app and kitchen system first, which will take a year.

What should the project manager recommend?

A. A minimum viable product, such as a simple order page with manual fulfilment, to test whether offices will pay.

B. Build the full platform as planned, so the launch makes a strong impression.

C. Run a market research survey and build the full platform if the results are positive.

D. Build a clickable prototype of the ordering screens to show investors.

◆ PAUSE AND DECIDE

What should the project manager recommend?

A An MVP with manual fulfilment.

Tests real willingness to pay in weeks, before a year of investment.

B Build everything first.

A year of spending before learning whether anyone will pay.

C A survey, then build.

Surveys measure what people say, not whether they actually pay.

D A clickable prototype.

Tests the design, not the demand. Nobody pays for a prototype.

THE SIGNAL

The big unknown is demand. Test it with real customers and real money, as cheaply as possible.

> Six stakeholders are reviewing forty features for the first release of a new online banking service. Each stakeholder has marked almost all of their own features as essential.
>
> The release date is fixed by a regulatory deadline.

What should the project manager do?

A Decide which features are essential, based on the project manager's own judgement.

B Deliver features in the order they were requested.

C Treat all forty features as equal priority and deliver as many as possible.

D Facilitate a MoSCoW session, first agreeing that 'Must' means the release cannot go ahead without it.

◆ PAUSE AND DECIDE

What should the project manager do?

A The PM decides.

Priorities belong to stakeholders and the sponsor or product owner.

B Order of request.

Arrival order has nothing to do with value.

C All equal priority.

With a fixed date, the least valuable may get built and the essential ones missed.

D MoSCoW, with 'Must' defined first.

Agreeing the definition first breaks the everything-is-essential deadlock.

THE SIGNAL

Everything marked essential with a fixed date: agree what 'Must' means, then sort together.

> A predictive project will roll out a new customer service system to twelve regional offices at once in nine months. One region's configuration is already complete and tested.
>
> That region handles thirty percent of all customer calls.

What should the project manager do?

A Keep the finished configuration until the planned single go-live.

B Assess with stakeholders whether to release to that region early, to realise benefits sooner.

C Reduce the scope for the other eleven regions to match.

D Ask the team to speed up the other regions' configuration.

◆ PAUSE AND DECIDE

What should the project manager do?

A Wait for the big launch.

Holds back value from thirty percent of calls for months, for no gain.

B Assess an early release with stakeholders.

The outline's enabler: assess opportunities to deliver value incrementally.

C Cut others' scope.

Unrelated to the opportunity, and not the PM's decision to make.

D Speed up the others.

Pressure, not value. The opportunity is already sitting there.

THE SIGNAL

Finished, valuable work waiting for a big launch: assess releasing it early, even on a predictive project.

> A project's main objective was to reduce customer calls to the help desk by twenty percent through a new self-service portal. The portal launched successfully.
>
> The sponsor now asks whether call volumes fell. Nobody recorded call volumes before the launch.

What should the project manager have done?

A Estimated the previous call volumes after launch, from staff recollection.

B Surveyed customers after launch about whether they call less.

C Relied on the portal vendor's typical results for similar clients.

D Measured a baseline of call volumes before launch, with a benefit owner to track them afterwards.

◆ PAUSE AND DECIDE

What should the project manager have done?

A Estimate from memory.

A guess after the fact cannot prove a twenty percent change.

B Survey customers later.

Opinions, not call data, and no starting point to compare with.

C Vendor's typical results.

Someone else's organisation, not evidence about this one.

D Baseline first, with a benefit owner.

The only way to prove a change is to measure the starting point.

THE SIGNAL

No baseline, no proof. Measure the starting point before you change anything.

> After eight months, an agile team has delivered all the high-value features of an internal tool, and usage targets have been met.
>
> The product owner says the remaining backlog items are minor, and would cost more to build than they are worth. The budget runs for four more months.

What should the project manager do?

A Continue until every backlog item is built, since the budget is approved.

B Look for new features to add so the team stays fully used.

C Recommend to the sponsor that the work stops, releasing the team and the remaining budget.

D Keep the team on the project at reduced effort until the budget runs out.

◆ PAUSE AND DECIDE

What should the project manager do?

A Build every item.
Spends four months on work worth less than it costs.

B Invent new features.
Keeps people busy, not productive. Value should drive work.

C Recommend stopping to the sponsor.
Value delivered; remaining work not worth its cost. Stopping is success.

D Coast until the budget ends.
Wastes the budget and the team's time.

THE SIGNAL

When remaining work is worth less than it costs, the valuable thing is to stop. That is a success.

Most valuable first. **Measured** against a baseline. Re-checked all the way.

MODULE 21 · DOMAIN 2 - PROCESS

Estimating effort and duration

Analogous, parametric, bottom-up and three-point estimates; story points and planning poker; ranges, history, and holding your ground under pressure.

DOMAIN II - Process EXAM WEIGHT 41% MODULE 21 of 49

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An estimate is a range with a confidence, not a promise.

Effort or duration?

EFFORT

Work required

- Measured in person-hours or person-days
- 40 hours of design work
- Does not depend on calendar
- Drives cost

DURATION

Time it takes

- Measured in working days or weeks
- Depends on who is available, and how much
- One designer at half time: 2 weeks
- Drives the schedule

THE TIEBREAKER

Forty hours of effort is a week for one full-time person, and two weeks for one person at half time.

Estimating techniques

TECHNIQUE	HOW IT WORKS	BEST WHEN
Analogous	Use a similar past project as the basis	Early, quick, little detail known
Parametric	Multiply a unit rate by a quantity	Reliable rates exist: cost per square metre
Bottom-up	Estimate each work package, then add up	Detail is known; accuracy matters
Three-point	Optimistic, most likely, pessimistic	Uncertainty is significant
Expert judgement	Ask people with relevant experience	Almost always, alongside the others

ON THE EXAM

Analogous is fast and rough. Bottom-up is slow and accurate. Parametric is as good as its unit rate.

Analogous and parametric, side by side

- **Analogous**: last year's regional office fit-out cost $400,000. This one is similar, so about $400,000.
- **Parametric**: fit-outs cost about $1,600 per square metre. This office is 300 square metres: about $480,000.
- Parametric is usually more accurate, if the rate is reliable and the work scales with the quantity.
- Analogous is only as good as the similarity. Adjust for known differences.

> ON THE EXAM
>
> Both use history. Analogous uses one whole project; parametric uses a rate derived from many.

Three-point estimates

- **Triangular** average: (Optimistic + Most likely + Pessimistic) / 3.
- **Beta, or PERT**: (Optimistic + 4 x Most likely + Pessimistic) / 6.
- O = 4 days, M = 6, P = 14. Triangular: 24 / 3 = **8**. PERT: (4 + 24 + 14) / 6 = **7**.
- **Standard deviation**, PERT: (P - O) / 6. Here, 10 / 6 = about **1.7 days**.

> ON THE EXAM
>
> PERT weights the most likely value four times. Read which formula the question wants.

Turning a standard deviation into a range

- Plus or minus **1** standard deviation: about **68%** confidence.
- Plus or minus **2**: about **95%**. Plus or minus **3**: about **99.7%**.
- PERT estimate 7 days, standard deviation 1.7: 95% confident of 3.6 to 10.4 days.
- A wider spread between optimistic and pessimistic means more uncertainty, and a wider range.

> ON THE EXAM
>
> Communicate estimates as ranges with confidence. A single number hides the uncertainty.

Estimates narrow as knowledge grows

ESTIMATE	TYPICAL RANGE	WHEN
Rough order of magnitude	-25% to +75%	Initiation, before detail exists
Budget or preliminary	Narrower, as scope firms up	During planning
Definitive	-5% to +10%	Detailed scope and work packages known

ON THE EXAM

The 'cone of uncertainty': early estimates are wide. Refine them as the project learns.

Who estimates, and how honestly

- The **people who will do the work** estimate it. They know what it involves.
- Estimate each item **without hidden padding**. Uncertainty is handled openly, in reserves.
- Record the **basis of estimates**: assumptions, method and confidence.
- Padding hidden in every task gets used up anyway: work expands to fill the time available.

> ON THE EXAM
>
> Honest estimates plus explicit reserves beat padded estimates. Module 26 covers reserves.

Story points

- A **relative** measure of size: effort, complexity and uncertainty combined.
- Often on a Fibonacci-like scale: 1, 2, 3, 5, 8, 13, 20. The gaps grow because uncertainty grows.
- Compare to a reference story: 'this is about twice our login story'.
- Not hours, and not comparable between teams. Each team calibrates its own.

> ON THE EXAM
>
> Story points measure size relative to other stories, for one team. They are not a unit of time.

Planning poker

01

Explain

The product owner describes the story; the team asks questions.

02

Vote privately

Each developer picks a card, all shown at once.

03

Discuss outliers

The highest and lowest explain their reasoning.

04

Re-vote

Repeat until estimates converge.

ON THE EXAM

Outliers are the point. The person who sees risk others missed usually holds the highest card.

Other agile estimating techniques

TECHNIQUE	HOW IT WORKS
T-shirt sizing	Extra small to extra large; quick and rough, good for epics
Affinity estimating	Place many items in size groups on a wall, fast
Ideal days	How long it would take with no interruptions
Throughput forecasting	Count items finished per period; forecast from history

ON THE EXAM

Coarse techniques for large or early items; finer ones near the top of the backlog.

Forecasting from velocity

- **Velocity**: story points completed per iteration, measured, not planned.
- Iterations remaining = points remaining / average velocity.
- 240 points left; velocity has ranged 30 to 40. Forecast: **6 to 8** iterations.
- Forecast as a range, using the team's own recent history, and update it every iteration.

> ON THE EXAM
>
> Use the team's measured velocity, not a hoped-for one, and forecast a range, not a date.

Benchmarks and historical data

- The outline asks you to **utilise benchmarks and historical data**.
- Your own past projects, the lessons learned repository, and industry benchmarks.
- Compare past estimates with actuals, and **calibrate**: if similar work ran 40% over, find out why.
- Ignoring history in favour of optimism is the most common estimating error of all.

> ON THE EXAM
>
> History is the best predictor you have. Use it, calibrate against it, and explain differences.

When someone wants a smaller number

- Cutting an estimate because a sponsor dislikes it does not change the work. It hides the risk.
- Explain the basis of the estimate, and its range and confidence.
- Offer real options: reduce scope, phase delivery, add resources, or accept the risk explicitly.
- The sponsor chooses between real trade-offs. The estimate stays honest.

> ON THE EXAM
>
> Negotiate scope, time, resources or risk. Never negotiate the estimate itself.

Project Lantern: estimating two ways

- **Kiosk installation**: parametric, at a supplier rate per kiosk, checked against the first clinic's actuals.
- **Data migration**: three-point, because data quality was uncertain. PERT 18 days, standard deviation 3.
- **Booking app**: story points and planning poker; release forecast from measured velocity.
- Every estimate recorded with its basis and range, and revised as actuals arrived.

> ON THE EXAM
>
> The right technique depends on the work and what is known. Hybrid projects use several.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Optimistic, most likely, pessimistic given	PERT: (O + 4M + P) / 6, unless told triangular
Quick early estimate, similar past project	Analogous
Points remaining and recent velocity given	Divide; forecast a range
Sponsor wants the estimate cut	Explain the basis; offer real trade-offs
Planning poker votes far apart	Outliers explain; discuss; re-vote
Estimates keep running over on similar work	Calibrate from history, with the team

ON THE EXAM

Honest ranges, the right technique for what is known, and history used to calibrate.

> A team estimates an activity using three points. Optimistic: 6 days. Most likely: 9 days. Pessimistic: 18 days.
>
> The project uses beta distribution (PERT) estimates.

What is the expected duration of the activity?

A 11 days

B 10 days

C 9 days

D 12 days

◆ PAUSE AND DECIDE

What is the expected duration of the activity?

A 11 days

The triangular average: 33 / 3. The question asked for PERT.

B 10 days

(6 + 4 x 9 + 18) / 6 = 60 / 6 = 10.

C 9 days

The most likely value alone. Ignores the risk of overrun.

D 12 days

The midpoint of optimistic and pessimistic. Not a standard formula here.

ON THE EXAM

PERT: optimistic plus four times most likely plus pessimistic, over six. Check which formula is asked.

> A sponsor needs a rough cost for a new warehouse racking project by tomorrow, for a funding discussion.
>
> The company completed a very similar racking project at another site last year. No detailed design exists yet.

Which estimating technique is most appropriate?

A Analogous estimating, based on last year's similar project.

B Bottom-up estimating of every work package.

C Three-point estimating for every activity.

D A Monte Carlo simulation of the full schedule.

◆ PAUSE AND DECIDE

Which estimating technique is most appropriate?

A Analogous.

Quick, uses a very similar past project, and fits a rough early estimate.

B Bottom-up.

Needs work packages that do not exist yet, and takes far longer than a day.

C Three-point per activity.

Needs a list of activities that does not exist yet.

D Monte Carlo.

Needs a detailed schedule model. Heavy for a rough estimate.

THE SIGNAL

Rough, fast, early, with a similar past project available: analogous.

> A Scrum team has 180 story points left in its release backlog. Its velocity over the last four two-week sprints was 28, 32, 30 and 30.
>
> The product owner asks how long the remaining work is likely to take.

What is the best forecast?

A About 6 weeks

B About 9 weeks

C About 15 weeks

D About 12 weeks

◆ PAUSE AND DECIDE

What is the best forecast?

A About 6 weeks

Six is the number of sprints. The sprints are two weeks each.

B About 9 weeks

Not derived from the data given.

C About 15 weeks

Uses a slower velocity than the team has shown.

D About 12 weeks

Average velocity 30; 180 / 30 = 6 sprints of 2 weeks.

THE SIGNAL

Points divided by average velocity gives sprints. Then convert sprints to weeks.

> A bottom-up estimate for a project comes to $2.4 million, with a documented basis. The sponsor says the budget is $1.7 million and asks the project manager to 'find a way to make the estimate fit'.
>
> The scope has not changed.

What should the project manager do?

A Reduce each work package estimate proportionally so the total is $1.7 million.

B Submit the $1.7 million figure, and hold the difference as hidden contingency.

C Explain the basis of the estimate, and present options such as reducing scope, phasing, or more funding.

D Decline to discuss the estimate further, since it is already documented.

◆ PAUSE AND DECIDE

What should the project manager do?

A. Scale every estimate down.
The work has not shrunk. The overrun is merely postponed and hidden.

B. Hide the difference.
Dishonest, and the numbers still do not add up.

C. Explain the basis; offer real options.
Keeps the estimate honest and gives the sponsor genuine choices.

D. Refuse to discuss.
Unhelpful. The sponsor has a real constraint that needs solving together.

THE SIGNAL

Same scope, smaller number requested: negotiate scope, time, funding or risk, never the estimate.

> During planning poker, most developers show 5 for a story. One shows 2, and one shows 13.
>
> The team is keen to move on to the next story.

What should happen next?

A The developers who showed 2 and 13 explain their reasoning, the team discusses, then votes again.

B Take the average of all the votes and move on.

C Use the highest estimate, 13, to be safe.

D The project leader chooses the estimate to save time.

◆ PAUSE AND DECIDE

What should happen next?

A Outliers explain; discuss; re-vote.

Surfaces the risk or shortcut behind the outliers. That is the point of the technique.

B Average the votes.

Throws away the information in the outliers.

C Take the highest.

Pads without understanding. The 13 might be based on a misunderstanding.

D The leader decides.

Estimates belong to the developers doing the work.

THE SIGNAL

Divergent votes carry information. Let the outliers explain before converging.

> Over the last three projects, a team's estimates for data integration work ran about forty percent below the actual effort each time.
>
> The team is now estimating integration work for a new project.

What should the project manager do?

A. Add forty percent to every estimate the team gives.

B. Review the past actuals with the team, understand the causes, and calibrate the new estimates.

C. Tell the team its estimates are unreliable and must improve.

D. Stop estimating integration work and schedule it as fixed blocks of time.

◆ PAUSE AND DECIDE

What should the project manager do?

A. Add forty percent everywhere.
Blanket padding without understanding. The cause may apply to only some work.

B. Review actuals with the team; calibrate.
Uses historical data, finds the cause, and improves estimates on evidence.

C. Tell the team to improve.
Blame without analysis. Nothing changes.

D. Stop estimating.
Abandons planning information because it was inaccurate, instead of improving it.

THE SIGNAL

A consistent pattern in history is a gift. Find its cause with the team, then calibrate.

Honest ranges. The right technique for what is known. History to calibrate.

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MODULE 22 · DOMAIN 2 - PROCESS

Building and controlling the schedule

Dependencies, the critical path, float, compressing and levelling, agile timeboxes, coordinating with operations, and analysing variation.

DOMAIN II - Process EXAM WEIGHT 41% MODULE 22 of 49

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The project finishes when its longest chain of dependent work finishes.

Building a predictive schedule

01

Define activities

The actions needed to produce each work package.

›

02

Sequence

Which must come before which, and why.

›

03

Estimate

Durations, from module 21.

›

04

Develop

Calculate dates, the critical path and float.

›

05

Baseline

Approve it; measure progress against it.

ON THE EXAM

In PMBOK 8, these steps sit inside one process, Develop Schedule. The logic is unchanged.

Types of dependency

RELATIONSHIP	MEANING	EXAMPLE
Finish-to-start	B cannot start until A finishes	Pour concrete, then build walls. The most common
Start-to-start	B cannot start until A starts	Start testing once coding has started
Finish-to-finish	B cannot finish until A finishes	Documentation finishes when coding finishes
Start-to-finish	B cannot finish until A starts	Old system runs until the new one starts. Rare

ON THE EXAM

Leads let a successor start early; lags add waiting time, such as three days for concrete to cure.

Why a dependency exists

KIND	MEANING	CAN IT BE CHANGED?
Mandatory, or hard logic	Required by the nature of the work or a contract	No: you cannot roof before walls
Discretionary, or preferred	Chosen as best practice, not required	Yes, with some added risk
External	Depends on something outside the project	Not by the project alone
Internal	Between activities inside the project	Depends on its nature

ON THE EXAM

Fast tracking only works on discretionary dependencies. Mandatory logic cannot be overlapped.

A worked network

ACTIVITY	DAYS	AFTER	EARLY START	EARLY FINISH	LATE START	LATE FINISH	FLOAT
A	3	–	0	3	0	3	0
B	4	–	0	4	1	5	1
C	4	A	3	7	3	7	0
D	2	B	4	6	5	7	1
E	3	C, D	7	10	7	10	0

ON THE EXAM

Paths: A-C-E is 10 days, B-D-E is 9. The critical path is A-C-E, with zero float. B and D have 1 day.

Float, and what it tells you

- **Total float**: how long an activity can slip without delaying the project end date.
- **Free float**: how long it can slip without delaying the next activity's early start.
- The **critical path** has zero total float. Any delay there delays the project.
- Watch **near-critical** paths: a path with one day of float can become critical overnight.
- Negative float means the path is already later than a required date.

> ON THE EXAM
>
> A delay smaller than the activity's float does not move the end date. It does use up the float.

Crashing or fast tracking?

CRASHING

Add resources

- More people, overtime, faster equipment
- Increases cost
- Only on critical path activities
- Choose the lowest cost per day saved

FAST TRACKING

Overlap activities

- Do in parallel what was sequential
- Increases risk and possible rework
- Only on discretionary dependencies
- Usually no direct extra cost

THE TIEBREAKER

Money but no appetite for risk: crash. No money but some risk accepted: fast track.

Compression rules the exam uses

- Compress only the **critical path**. Shortening other work saves nothing.
- After each change, **recheck**: another path may have become critical.
- Crashing has limits: adding people to late work can slow it down at first.
- Consider options first: re-sequencing, reducing scope, or agreeing a new date.

> ON THE EXAM
>
> Speeding up non-critical activities is the classic wrong answer. It costs money and moves nothing.

Resource levelling or resource smoothing?

RESOURCE LEVELLING

Fix over-allocation

- Adjusts dates to match resource limits
- Can change the critical path
- May extend the end date
- Used when resources are the hard limit

RESOURCE SMOOTHING

Even out within float

- Adjusts only within available float
- Critical path unchanged
- End date unchanged
- May not remove every peak

THE TIEBREAKER

Levelling can move the end date. Smoothing never does, because it stays within float.

Critical chain, in brief

- Builds the schedule with **resource limits** in mind, not just logic.
- Removes safety from individual activities and pools it into **buffers**.
- A **project buffer** protects the end date; **feeding buffers** protect where other paths join.
- Progress is managed by watching how fast buffers are consumed.

> ON THE EXAM
>
> Critical chain pools uncertainty, rather than hiding it in every task. The same idea as honest estimates.

Schedules on agile and hybrid projects

- **Timeboxes**: fixed-length iterations; the team plans work to fit.
- **Release plans and roadmaps** show when capabilities are likely, forecast from velocity.
- **Milestones** link agile work to predictive work on hybrid projects.
- The outline's first schedule enabler: prepare the schedule **based on the selected development approach**.

> ON THE EXAM
>
> Agile schedules fix time and forecast scope. Hybrid schedules join the two at milestones.

Coordinating with other projects and operations

- The outline asks you to **coordinate with other projects and operations**.
- Shared people and equipment across projects: agree priorities early.
- Operational calendars: change freezes, peak seasons, maintenance windows, financial year end.
- A go-live planned inside an operational change freeze will not happen, however good the plan.

> ON THE EXAM
>
> The schedule lives inside the organisation's calendar. Check it before you commit dates.

Showing the schedule

FORMAT	BEST FOR
Bar chart, or Gantt chart	The team and managers: activities against time
Milestone chart	Sponsors and executives: key dates only
Network diagram	Analysing logic, dependencies and the critical path
Roadmap or release plan	Agile work: capabilities over time, as forecasts

ON THE EXAM

Match the format to the audience. Executives rarely need the full Gantt chart.

Controlling the schedule

- Compare actual progress with the **baseline**, regularly.
- For each slip, ask: **is it on the critical path?** Does it exceed the float?
- Update forecasts; take corrective action only where the end date or a milestone is at risk.
- Rebaselining is a change-control decision, not a way to make variance disappear.

> ON THE EXAM
>
> Analyse variation before reacting. A slip within float is information, not a crisis.

Project Lantern: the schedule

- **Critical path**: network upgrade, then kiosk installation, then migration rehearsal, then pilot go-live.
- **External dependency**: the regulator's approval, with a two-week lag before any clinic go-live.
- **Operations calendar**: no go-lives in December, the clinics' busiest month.
- The app's release plan meets the predictive schedule at one milestone per clinic.

> ON THE EXAM
>
> A predictive critical path, an external dependency, an operational constraint, and agile releases joined at milestones.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Durations and predecessors given	Find the longest path; float = late minus early
Late, money available, no extra risk wanted	Crash the critical path, cheapest first
Late, no money, some risk acceptable	Fast track discretionary dependencies
Over-allocated resource on the critical path	Levelling may extend the end date
Go-live clashes with an operational freeze	Coordinate with operations calendars in planning
An activity slips by less than its float	No end-date impact; monitor the reduced float

ON THE EXAM

Always ask: is it on the critical path, and how much float is there?

A network has five activities. A takes 4 days and B takes 6 days; both can start immediately. C (5 days) follows A. D (2 days) follows B. E (3 days) follows both C and D.

The project manager is assessing flexibility in the schedule.

How much total float does activity D have?

A 0 days

B 2 days

C 1 day

D 3 days

◆ PAUSE AND DECIDE

How much total float does activity D have?

A 0 days

D is not on the critical path. Only A, C and E have zero float.

B 2 days

D's own duration, not its float.

C 1 day

A-C-E = 12 days; B-D-E = 11. D can slip 1 day.

D 3 days

E's duration. Not relevant to D's float.

THE SIGNAL

Add up each path. Float on a non-critical path is the critical path length minus that path's length.

> A project is two weeks behind its contractual completion date. The sponsor has approved extra budget to recover, but insists that no additional risk of rework be introduced.
>
> The project manager has identified the critical path activities.

Which schedule compression approach should the project manager use?

A Fast track by overlapping critical path activities.

B Crash the non-critical activities with the lowest crash cost.

C Apply resource levelling across the schedule.

D Crash the critical path activities with the lowest cost per day saved.

◆ PAUSE AND DECIDE

Which schedule compression approach should the project manager use?

A Fast track.

Overlapping adds exactly the rework risk the sponsor ruled out.

B Crash non-critical work.

Spends money and moves the end date not at all.

C Resource levelling.

Resolves over-allocation; it can extend the schedule, not shorten it.

D Crash critical activities, cheapest first.

Uses the budget, adds no overlap risk, and targets the path that matters.

THE SIGNAL

Budget yes, extra risk no: crash, on the critical path, cheapest per day first.

> A software project is three weeks late. There is no budget for additional people or overtime.
>
> The team says it could begin building some components before the detailed design is fully signed off, accepting some risk of rework.

Which technique does this describe, and is it appropriate?

A. Fast tracking; appropriate, since no budget exists and some risk is acceptable.

B. Crashing; appropriate, since it shortens the critical path.

C. Resource smoothing; appropriate, since it keeps the end date.

D. Gold plating; inappropriate, since it adds scope.

◆ PAUSE AND DECIDE

Which technique does this describe, and is it appropriate?

A Fast tracking; appropriate.

Overlapping design and build; no extra cost; risk explicitly accepted.

B Crashing.

Crashing adds resources. There is no budget, and nothing is added.

C Resource smoothing.

Smoothing redistributes work within float; it does not recover lateness.

D Gold plating.

No scope is being added. The work is being overlapped.

THE SIGNAL

Starting the next step before the previous one finishes is fast tracking. No money, some risk: that fits.

> A resource histogram shows the project's only database specialist allocated at 150 percent in weeks five and six. Both of her activities in those weeks are on the critical path.
>
> No other specialist is available.

Which statement is most accurate?

A. Resource levelling will resolve the over-allocation without changing the end date.

B. Resource levelling is likely to extend the end date, because the affected activities are critical.

C. Resource smoothing will remove the over-allocation without any delay.

D. Crashing the activities will reduce her allocation.

◆ PAUSE AND DECIDE

Which statement is most accurate?

A Levelling, no end-date change.

Critical activities have no float. Delaying one delays the project.

B Levelling will likely extend the end date.

One of the critical activities must move later, and there is no float to absorb it.

C Smoothing, no delay.

Smoothing only uses float, and critical activities have none. It cannot help.

D Crashing reduces her allocation.

Crashing adds resources; with no other specialist, it cannot.

THE SIGNAL

Over-allocation on the critical path: levelling moves the end date, and smoothing has no float to use.

> A retail chain's new point-of-sale system was scheduled to go live in late November. Two weeks before, the operations director reminds the project manager that no system changes are permitted from November to early January.
>
> The freeze has been company policy for years.

What should the project manager have done?

A Crashed the schedule to go live before the freeze began.

B Asked the operations director for an exemption from the freeze.

C Checked operational calendars and constraints with operations while building the schedule.

D Proceeded with the go-live, since the project schedule was approved.

◆ PAUSE AND DECIDE

What should the project manager have done?

A Crash to beat the freeze.

A reaction, not prevention, and the freeze may start before any recovery.

B Ask for an exemption.

A reaction, and freezes exist to protect peak trading.

C Coordinate with operations in planning.

The outline's enabler: coordinate with other projects and operations.

D Go ahead anyway.

Breaches a long-standing policy protecting the business at its busiest time.

THE SIGNAL

A long-standing operational constraint discovered late: it should have been in the schedule from the start.

> An activity to configure a reporting tool is running five days late. The schedule shows it has eight days of total float.
>
> The activity owner is worried and asks the project manager whether the project will be late.

What should the project manager do?

A Crash the activity immediately to recover the five days.

B Inform the sponsor that the project will finish five days late.

C Submit a change request to rebaseline the schedule.

D Confirm there is no impact on the end date, and monitor the activity now that its float has fallen to three days.

◆ PAUSE AND DECIDE

What should the project manager do?

A. Crash it.

Spends money on an activity whose delay does not affect the end date.

B. Tell the sponsor it will be late.

Incorrect: a five-day slip within eight days of float does not move the end.

C. Rebaseline.

No baseline date is at risk. Nothing to rebaseline.

D. No end-date impact; monitor the three days left.

Correct analysis of variation, with attention on the shrinking float.

THE SIGNAL

Slip smaller than float: no end-date impact. But the float is now smaller, so watch it.

Find the longest chain. Protect it. Always ask about float.

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MODULE 23 · DOMAIN 2 - PROCESS

Planning and managing resources

Planning people, equipment and materials from requirements; acquiring them in different organisational structures; and optimising, protecting and releasing them.

DOMAIN II - Process EXAM WEIGHT 41% MODULE 23 of 49

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Plan resources from **requirements**, confirm their **availability**, and manage them as a shared asset.

Kinds of resource

KIND	EXAMPLES	MAIN CONCERNS
Team resources	Developers, engineers, analysts, contractors	Skills, availability, motivation, development
Equipment	Cranes, test rigs, laptops, licences	Booking, lead time, maintenance, sharing
Materials	Steel, cabling, components	Lead time, storage, waste, quality
Facilities	Offices, labs, test environments	Availability, capacity, conflicts

ON THE EXAM

Physical resources fail projects too: late materials and double-booked environments are common scenario causes.

The resource cycle

01
Plan
How resources will be defined, acquired, managed, released.

02
Estimate
What is needed, how much, and when.

03
Acquire
Obtain the people and physical resources.

04
Lead and manage
Develop the team; control physical use.

05
Release
Return resources on time, with recognition.

ON THE EXAM

Release is part of the cycle. Holding people after their work is done costs the rest of the organisation.

The resource management plan

- **Roles and responsibilities**: role, authority, responsibility, competence, and the RACI from module 07.
- **Acquisition**: how and when people and physical resources are obtained.
- **Development and recognition**: training, team building, rewards.
- **Physical resources**: ordering, storage, inventory, maintenance.
- **Release**: when and how resources return.

> ON THE EXAM
>
> A good resource plan answers who, with what skills, from where, when, and until when.

Estimating resources from requirements

- Start from the **work**: what skills, how many people or units, and exactly when.
- A **resource breakdown structure** groups needs by category and type.
- Include specialist and certified skills that regulation or quality demands.
- Asking for 'three developers' without skills and timing invites the wrong three people, at the wrong time.

> ON THE EXAM
>
> The outline's enabler: define and plan resources based on requirements. Specific needs get the right resources.

Acquiring team resources

METHOD	WHAT IT MEANS
Pre-assignment	Named people are agreed in advance, often in a bid or charter
Negotiation	Agree people and availability with functional managers or other projects
Acquisition	Hire or contract from outside when skills are not available internally
Virtual teams	Use people in other locations, as module 08 covered
Selection criteria	Weigh availability, cost, skills, experience, attitude

ON THE EXAM

In most organisations, people are negotiated for. Build the relationship with functional managers early.

Organisational structure and the PM's authority

STRUCTURE	WHO CONTROLS RESOURCES	PM AUTHORITY
Functional	Functional managers	Little or none; often a coordinator
Weak matrix	Mostly functional managers	Low
Balanced matrix	Shared between PM and functional manager	Moderate
Strong matrix	Mostly the project manager	Moderate to high
Projectised	The project manager	High; team reports to the PM

ON THE EXAM

The weaker the PM's authority, the more resources depend on negotiation and relationships.

Resource calendars and availability

- A **resource calendar** shows when each person or item is actually available.
- Include holidays, part-time allocations, other project commitments, training and leave.
- Confirm availability with the functional manager before building dates on it.
- A schedule that assumes someone full-time, when they are half-time, is wrong from day one.

> ON THE EXAM
>
> The outline's other enabler: manage and optimise resource needs and availability. Start by knowing the availability.

Optimising resources

- Use **resource histograms** to spot over-allocation and idle time.
- **Levelling** and **smoothing**, from module 22, reshape the schedule to fit.
- **Cross-training** reduces dependence on one specialist.
- Share equipment and environments across teams with an agreed booking process.
- Pushing people to 100 percent utilisation creates queues and burnout, not speed.

> ON THE EXAM
>
> Optimise for flow, not for keeping everyone busy. Module 05's motorway applies to people too.

Managing physical resources

- Plan for **lead times**: long-lead items are ordered early, sometimes before design is final.
- **Just-in-time** delivery reduces storage, damage and theft on constrained sites.
- Track inventory, maintenance and calibration of equipment.
- Reduce **waste**: over-ordering and scrap cost money and harm sustainability goals.

> ON THE EXAM
>
> Too early causes storage, damage and waste. Too late stops the critical path. Plan deliveries to the schedule.

When a resource is taken away

- **Assess the impact**: is the person's work on the critical path? For how long?
- **Negotiate** with the functional manager: partial availability, a substitute, different timing.
- Consider options: re-sequence work, cross-train, or bring in a contractor.
- Escalate only if the impact exceeds your authority and negotiation has failed.

> ON THE EXAM
>
> Impact first, then negotiate. Escalating a resource loss before talking to the functional manager usually loses.

Releasing resources well

- Release people **on time**, as planned, and tell their managers early.
- **Recognise** their contribution, and give feedback to their functional managers for appraisals.
- Capture their knowledge before they go, as module 16 described.
- Return equipment and close bookings, licences and access.

> ON THE EXAM
>
> How you release people decides whether functional managers give you good people next time.

Resources on agile teams

- Prefer **stable, dedicated, cross-functional** teams.
- **Bring work to the team**, rather than assembling a new team for each piece of work.
- Plan by team **capacity**, using measured velocity or throughput.
- Moving people between teams to maximise utilisation destroys velocity and trust.

> ON THE EXAM
>
> Agile resourcing optimises the team's flow, not each person's utilisation.

Project Lantern: resources

- **Structure**: a balanced matrix. Clinic staff and IT engineers report to their own managers.
- **Negotiated**: two network engineers at 60 percent for four months, confirmed in writing with the IT manager.
- **Physical**: kiosks delivered just in time to each clinic, because clinics have no storage.
- **Stable**: the app team stays together for the whole project, and beyond.

> ON THE EXAM
>
> Negotiated people, just-in-time equipment, and a stable agile team, in one hybrid project.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
A functional manager takes a key person	Assess impact; negotiate options first
Schedule assumed someone full-time	Confirm availability with resource calendars
Which structure gives the PM most authority?	Projectised
Materials damaged or stolen in storage	Just-in-time deliveries matched to the schedule
Offered someone without a required certification	Explain the requirement; negotiate or plan training
People moved between agile teams for utilisation	Stable teams; bring work to the team

ON THE EXAM

Plan from requirements, confirm availability, negotiate, and release well.

> In a balanced matrix organisation, a functional manager tells the project manager that a key engineer will move to another project for three weeks, starting Monday.
>
> The engineer is working on a critical path activity.

What should the project manager do first?

A Assess the impact, then meet the functional manager to negotiate options such as partial availability or a substitute.

B Escalate to the sponsor to stop the move.

C Accept the decision and adjust the schedule.

D Hire a contractor immediately to replace the engineer.

◆ PAUSE AND DECIDE

What should the project manager do first?

A Assess impact; negotiate options.

Understands the cost, then solves it with the person who controls the resource.

B Escalate to the sponsor.

Goes over the functional manager's head before a conversation.

C Accept and adjust.

Gives up without assessing impact or exploring options.

D Hire a contractor now.

Commits cost without authority or analysis. Maybe unnecessary.

THE SIGNAL

A resource taken away: assess impact, then negotiate with the functional manager, before anything else.

> Three weeks into execution, the project manager learns that a security specialist, scheduled full-time on the project, is also committed half-time to another project.
>
> Her work is already falling behind.

What should the project manager have done to prevent this?

A Crashed her activities at the start of the project.

B Confirmed her availability with her functional manager and a resource calendar when planning.

C Added more contingency reserve to the budget.

D Included her in the RACI chart.

◆ PAUSE AND DECIDE

What should the project manager have done to prevent this?

A Crash from the start.

Crashing is a response to lateness, not a way to discover availability.

B Confirm availability with a resource calendar.

Plans on real availability, so the schedule is right from day one.

C More contingency.

Money does not create a person's missing half-time.

D Put her in the RACI.

A RACI records responsibility, not availability.

THE SIGNAL

A schedule built on assumed availability is wrong from the start. Confirm it with a resource calendar.

> A project manager is choosing between two job offers. In one, team members report directly to the project manager, who controls the budget and assignments.
>
> In the other, team members report to department heads, who decide their priorities.

The first role describes which organisational structure?

A Projectised

B Weak matrix

C Functional

D Balanced matrix

◆ PAUSE AND DECIDE

The first role describes which organisational structure?

A Projectised

Team reports to the PM, who controls budget and assignments: high authority.

B Weak matrix

Functional managers keep most control. The PM coordinates.

C Functional

This describes the second role, where department heads decide priorities.

D Balanced matrix

Authority is shared, not held by the PM alone.

THE SIGNAL

Team reports to the PM, PM controls budget: projectised. Department heads decide: functional or weak matrix.

> On a city-centre building site with almost no storage space, all the cladding panels were delivered in the first week. Several have been damaged by weather and site traffic, and two pallets were stolen.
>
> The cladding will be installed over the next four months.

What should the project manager do for the remaining materials?

A Rent a larger secure storage area off site for all future deliveries.

B Arrange just-in-time deliveries, scheduled to match the installation sequence.

C Increase the insurance cover and continue as planned.

D Order extra panels to cover future losses.

◆ PAUSE AND DECIDE

What should the project manager do for the remaining materials?

A Rent off-site storage.

Adds cost and handling, and treats the symptom rather than the delivery pattern.

B Just-in-time deliveries to the sequence.

Removes the storage problem at its source, on a constrained site.

C More insurance.

Pays for losses; does not prevent them or the delays they cause.

D Order extra panels.

Plans for waste, adds cost, and harms sustainability goals.

THE SIGNAL

No storage and materials needed over months: deliver just in time to the installation schedule.

> The resource plan for a pipeline project requires welders certified to a specific safety standard, as regulation demands.
>
> The functional manager offers an experienced welder who does not hold that certification, saying 'he is just as good'.

What should the project manager do?

A Accept the welder, since he is experienced.

B Hire a certified welder externally without telling the functional manager.

C Explain the regulatory requirement, and negotiate for a certified welder or a plan to certify this one in time.

D Remove the certification requirement from the resource plan.

◆ PAUSE AND DECIDE

What should the project manager do?

A Accept him.

Breaches a regulatory requirement, whatever his skill.

B Hire externally, secretly.

Bypasses the functional manager and the agreed acquisition route.

C Explain; negotiate a certified welder or certification.

Keeps compliance, works with the manager, and may develop the person too.

D Drop the requirement.

The requirement comes from regulation. It is not the PM's to remove.

THE SIGNAL

A regulatory skill requirement is not negotiable. The route to meeting it is.

> A department head proposes moving developers between four agile teams each week, to wherever demand is highest, so that nobody is ever idle.
>
> The teams' velocities have been stable for months.

What should the project manager recommend?

A Adopt the proposal, since higher utilisation should increase output.

B Assign every developer half-time to two teams, to balance the load.

C Refuse the proposal, since agile teams must never change.

D Explain the value of stable teams, and propose bringing work to the teams based on their capacity.

◆ PAUSE AND DECIDE

What should the project manager recommend?

A Adopt it.

Every move resets team dynamics and knowledge; output falls, not rises.

B Half-time on two teams.

Doubles the task switching and halves each team's stability.

C Refuse outright.

Too absolute, and ignores the manager's real concern about demand.

D Stable teams; bring work to them.

Meets the concern about demand without destroying team performance.

THE SIGNAL

Maximising each person's utilisation damages the team's flow. Move work, not people.

Plan from requirements. Confirm availability. **Negotiate**, then release well.

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MODULE 24 · DOMAIN 2 - PROCESS

Procurement strategy and contract types

Make or buy, procurement documents, the contract types and who carries the risk in each, incentive calculations, and fair source selection.

DOMAIN II - Process EXAM WEIGHT 41% MODULE 24 of 49

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Every contract type answers one question:
who carries the cost risk?

The procurement cycle

01

Plan

Make or buy; strategy; contract type; documents.

02

Conduct

Invite, evaluate and select sellers; award.

03

Control

Manage the relationship and performance; handle changes.

04

Close

Verify completion; settle; archive; lessons.

ON THE EXAM

In PMBOK 8, procurement detail moved to an appendix. On the exam it is still a full task with ten enablers.

Make or buy

- **Capability**: do we have the skills and capacity, now?
- **Cost**: full cost of each option, including management, support and ongoing costs.
- **Control and intellectual property**: do we need to own and control this?
- **Strategy and risk**: is it core to the business? Which option carries less risk?
- Buying what is not core, and building what differentiates you, is a common pattern.

> ON THE EXAM
>
> Compare full costs and strategic fit, not just the headline price.

Procurement strategy and the delivery solution

- **Delivery method**: design then build, design-and-build together, or a turnkey solution.
- **Contract payment type**: fixed price, cost-reimbursable, or time and materials.
- **Phasing**: one contract, or stages with break points and decisions.
- The outline asks you to **plan and manage the procurement strategy** and **develop a delivery solution**.

> ON THE EXAM
>
> The strategy decides how the work will be bought. The contract type decides who carries the risk.

Procurement documents

DOCUMENT	USED WHEN
Request for information (RFI)	You want to learn what the market can offer before specifying
Request for quotation (RFQ)	The need is clearly defined; price is the main decision factor
Request for proposal (RFP)	The solution is complex; you want sellers to propose an approach
Procurement statement of work	Describes the work in enough detail for sellers to respond
Source selection criteria	How proposals will be evaluated and weighted

ON THE EXAM

Exactly defined, price matters: RFQ. Complex, approach matters: RFP. Exploring the market: RFI.

Fixed price or cost-reimbursable?

FIXED PRICE

Seller carries the cost risk

- Scope must be well defined
- Buyer knows the price in advance
- Seller may add risk premium
- Changes need formal agreement

COST-REIMBURSABLE

Buyer carries the cost risk

- Scope can be uncertain
- Buyer pays actual costs plus a fee
- Buyer must monitor costs closely
- Flexible when requirements evolve

THE TIEBREAKER

Clear scope: fixed price. Uncertain scope: cost-reimbursable. Neither is better in general.

Contract types and where the risk sits

TYPE	HOW PAYMENT WORKS	BUYER'S COST RISK
Firm fixed price (FFP)	One set price, defined scope	Lowest
Fixed price incentive fee (FPIF)	Target price, incentive, ceiling	Low
FP with economic price adjustment	Fixed price, indexed for inflation	Low; long contracts
Time and materials (T&M)	Rates for time; materials at cost	Medium; needs a cap
Cost plus incentive fee (CPIF)	Costs plus a shared-saving fee	High, but shared
Cost plus fixed fee (CPFF)	Costs plus a fixed fee	High
Cost plus award fee (CPAF)	Costs plus a judged award	Highest

ON THE EXAM

Top to bottom, cost risk moves from the seller to the buyer.

Choosing the contract type

- Scope **well defined**, buyer wants **cost certainty**: firm fixed price.
- Scope **uncertain**, such as research: cost-reimbursable, often with an incentive.
- **Urgent**, small, or adding people to a team: time and materials, with a not-to-exceed cap.
- **Multi-year** contract exposed to inflation: fixed price with economic price adjustment.
- Want the seller to **control costs**? Add an incentive fee.

> ON THE EXAM
>
> Read the scope clarity and the buyer's priority. Those two clues choose the contract.

How an incentive fee works

- Cost plus incentive fee: target cost **$100,000**, target fee **$10,000**, share ratio **80/20** (buyer/seller).
- Actual cost **$90,000**: an underrun of $10,000.
- Seller's share of the saving: 20% of $10,000 = **$2,000** extra fee. Fee = **$12,000**.
- Buyer pays actual cost plus fee: $90,000 + $12,000 = **$102,000**.
- If costs overrun, the seller's fee shrinks by its share of the overrun.

> ON THE EXAM
>
> The seller's share of any saving is added to its fee; its share of any overrun is taken off.

Time and materials, carefully

- Pays agreed **rates** for time spent, plus materials at cost.
- Fast to set up; good for small, urgent or open-ended work, and staff augmentation.
- Without a cap, cost has no ceiling, and the seller has no incentive to finish quickly.
- Add a **not-to-exceed** limit, and monitor hours and deliverables closely.

> ON THE EXAM
>
> Time and materials is flexible and quick. Always cap it, and always watch it.

Selecting sellers

TECHNIQUE	PURPOSE
Weighted scoring	Score proposals against agreed, weighted criteria
Screening	Remove sellers who fail mandatory requirements
Independent estimate	Your own estimate, to test whether bids are realistic
Bidder conference	Answer all prospective sellers' questions together, fairly
Presentations and references	Test capability and past performance

ON THE EXAM

Lowest price is rarely the only criterion. Agree weighted criteria before seeing any bids.

Fairness in the selection process

- Every prospective seller receives the **same information** at the same time.
- Questions from one seller are answered formally, and the answer is **shared with all**.
- Private conversations with one bidder create an unfair advantage and a legal risk.
- Declare any **conflict of interest**, and decline gifts from bidders.

> ON THE EXAM
>
> Treat all sellers equally. A private answer to one bidder is an ethics problem, not a courtesy.

Sustainability and responsibility in procurement

- Include **sustainability criteria**: emissions, materials, waste, end-of-life.
- Check **labour and ethical standards** in the supply chain.
- Respect local procurement **regulations**, especially in the public sector.
- Criteria written into the selection and the contract are the ones that get met.

> ON THE EXAM
>
> If sustainability matters to the organisation, it belongs in the selection criteria and the contract.

Project Lantern: procurement

- **Kiosks**: bought, not built. Firm fixed price, from a clear specification, via an RFP.
- **Network engineers** for peak weeks: time and materials, with a not-to-exceed cap.
- **Booking app**: built in-house, because patient experience is core to Northgate.
- Kiosk selection criteria included energy use and a take-back scheme for old hardware.

> ON THE EXAM
>
> Buy the commodity, build the differentiator, and match each contract to its scope clarity.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Fully specified work, buyer wants price certainty	Firm fixed price
Uncertain scope; motivate the seller to save	Cost plus incentive fee
Target cost, fee and share ratio given	Seller's share of the variance adjusts the fee
A bidder asks a private question	Answer formally; share with all bidders
Commodity available now; not core	Buy
Exact specification; best price wanted	Request for quotation

ON THE EXAM

Scope clarity chooses the contract. Fairness governs the selection.

> A company needs a warehouse built to a complete, detailed design already approved by its architects. Planning permission is granted.
>
> The finance director insists on knowing the total cost before work begins.

Which contract type is most appropriate?

A Cost plus fixed fee

B Time and materials

C Firm fixed price

D Cost plus award fee

◆ PAUSE AND DECIDE

Which contract type is most appropriate?

A Cost plus fixed fee

Buyer carries cost risk; the total is unknown in advance.

B Time and materials

Open-ended by nature; no certainty of total cost.

C Firm fixed price

Complete design and a need for cost certainty: the classic fit.

D Cost plus award fee

Highest buyer risk, and a subjective fee on top.

THE SIGNAL

Scope fully defined, and the buyer wants certainty: firm fixed price.

> A cost plus incentive fee contract has a target cost of $200,000, a target fee of $20,000, and a share ratio of 70/30 (buyer/seller).
>
> The seller completes the work for an actual cost of $180,000.

What is the total amount the buyer pays?

A $200,000

B $206,000

C $214,000

D $220,000

◆ PAUSE AND DECIDE

What is the total amount the buyer pays?

A $200,000

The target cost. Ignores the fee entirely.

B $206,000

Underrun $20,000; seller's 30% = $6,000; fee $26,000; $180,000 + $26,000.

C $214,000

Gives the seller the buyer's 70% share of the saving.

D $220,000

Target cost plus target fee. Ignores the actual cost.

THE SIGNAL

Underrun times the seller's share, added to the target fee. Then actual cost plus that fee.

A manufacturer is commissioning research into a new battery material. The scope is uncertain, and the work may change direction as results come in.

The manufacturer accepts that it must carry the cost risk, but wants the research firm motivated to keep costs down.

Which contract type is most appropriate?

A Firm fixed price

B Time and materials with no cap

C Cost plus fixed fee

D Cost plus incentive fee

◆ PAUSE AND DECIDE

Which contract type is most appropriate?

A Firm fixed price

Scope too uncertain; sellers would refuse or add a huge premium.

B T&M with no cap

No ceiling and no incentive to be efficient.

C Cost plus fixed fee

Right family, but the fixed fee gives no reason to save.

D Cost plus incentive fee

Buyer carries the risk, and the seller shares any savings.

THE SIGNAL

Uncertain scope plus 'motivate the seller to control cost': cost plus incentive fee.

> During the bidding period for a software contract, one prospective seller emails the project manager privately with a question about a performance requirement.
>
> Proposals are due in ten days.

What should the project manager do?

A Answer through the formal channel, and share the question and answer with all prospective sellers.

B Reply privately, since the seller asked privately.

C Ignore the question to avoid favouring anyone.

D Disqualify the seller for contacting the project manager directly.

◆ PAUSE AND DECIDE

What should the project manager do?

A Formal answer, shared with all.

Equal information for every bidder, and the requirement is clarified for everyone.

B Reply privately.

Gives one bidder an advantage. Unfair and a legal risk.

C Ignore it.

The requirement may be genuinely unclear. Every bid would suffer.

D Disqualify the seller.

Disproportionate. Asking a question is not misconduct.

THE SIGNAL

One bidder's question is everyone's question. Answer formally and share with all.

> An online retailer needs card payment processing for a new store. Its developers estimate nine months to build a compliant system in-house.
>
> Several certified payment providers offer ready-made services that could be integrated in weeks. Payment processing is not a differentiator for the retailer.

What should the project manager recommend?

A. Build in-house, to keep full control of the payment system.

B. Buy a certified payment service, since it is available now and is not core to the business.

C. Hire contractors on time and materials to build it faster.

D. Postpone the decision until the store's design is complete.

◆ PAUSE AND DECIDE

What should the project manager recommend?

A Build in-house.
Nine months and compliance risk for something that does not differentiate.

B Buy the certified service.
Available, compliant, fast, and not core: a clear buy.

C Contractors on T&M.
Still building a commodity, with open-ended cost.

D Postpone.
The decision does not depend on the store's design.

THE SIGNAL

Not core, readily available, compliance already solved: buy it.

> A school trust needs five hundred laptops of a precisely defined specification, with a standard three-year warranty.
>
> It wants the best price from qualified suppliers.

Which procurement document should it issue?

A. A request for information

B. A request for proposal

C. A letter of intent

D. A request for quotation

◆ PAUSE AND DECIDE

Which procurement document should it issue?

A Request for information

For exploring the market. The need is already fully defined.

B Request for proposal

For complex solutions where sellers propose an approach. Not needed here.

C Letter of intent

Signals an intention to contract; it does not invite competitive prices.

D Request for quotation

Exactly defined need, price the deciding factor.

THE SIGNAL

Precisely defined, price matters: RFQ. Complex, approach matters: RFP.

Scope clarity chooses the contract.
Fairness governs the selection.

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MODULE 25 · DOMAIN 2 - PROCESS

Negotiation, suppliers and agile contracts

Negotiating with a strategy, managing supplier performance, handling changes and claims, contracts for adaptive work, and closing agreements.

DOMAIN II - Process EXAM WEIGHT 41% MODULE 25 of 49

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A contract sets the rules. The relationship delivers the work.

Preparing a negotiation strategy

- Know your **objectives** and priorities: what matters most, and what you can trade.
- Know your **BATNA**: the best alternative to a negotiated agreement.
- Set your **walk-away point** from the BATNA, before you start.
- Understand the other side's interests, constraints and likely alternatives.

> ON THE EXAM
>
> Your BATNA is your power in a negotiation. The better your alternative, the stronger your position.

BATNA, worked through

- Supplier A quotes **$500,000** for a data centre fit-out.
- Supplier B has quoted **$460,000** for equivalent work, with similar quality and dates.
- Your BATNA is **contracting Supplier B at $460,000**.
- So agreeing with Supplier A above about $460,000, for the same value, makes no sense.
- The **zone of possible agreement** is the overlap between what each side will accept.

> ON THE EXAM
>
> A BATNA is a real, available alternative. Not a hope, and not simply walking away with nothing.

Positional or principled negotiation?

POSITIONAL

Haggle over positions

- Each side states a position
- Concessions traded reluctantly
- Relationship strained
- Often ends in a split-the-difference

PRINCIPLED

Solve a shared problem

- Separate people from the problem
- Focus on interests, not positions
- Create options for mutual gain
- Use objective criteria

THE TIEBREAKER

Interests, options and objective criteria. The same approach as conflict management in module 11.

The project manager's role in negotiation

- The outline says **participate in** agreement negotiations. Often the PM is one voice among several.
- **Contract authority** usually sits with procurement or legal. Know who can sign and commit.
- The PM brings knowledge of scope, schedule, risks and what the project truly needs.
- A PM who agrees terms without authority can bind the organisation to a bad deal.

ON THE EXAM

Contribute the project's knowledge. Leave commitments to those with contract authority.

Managing suppliers

- Agree **performance measures** in the contract: service levels, quality, delivery, responsiveness.
- Hold regular **performance reviews**, not only when something goes wrong.
- Use **inspections and audits** where the contract allows them.
- Tie **payments** to accepted deliverables or milestones, as agreed.
- Treat the supplier's people as part of the extended team.

> ON THE EXAM
>
> The outline asks you to manage suppliers and contracts. Measure, review, and build the relationship.

Evaluating vendor performance

DIMENSION	EVIDENCE
Quality	Defects, rework, acceptance results
Schedule	Milestones met, delivery dates
Cost	Invoices against the contract, change frequency
Responsiveness	Speed of answers and fixes; communication
Compliance	Contract terms, safety, security, sustainability commitments

ON THE EXAM

Evaluate against the contract, with evidence, and discuss the results with the vendor.

Contract changes

- Change the contract only through its **change clause**, in writing, with the right authority.
- **Constructive changes**: verbal instructions or actions that change the work without a formal change.
- A site instruction to 'use the better material' can become a legitimate claim for extra cost.
- Make sure the whole team knows who may instruct the supplier, and how.

> ON THE EXAM
>
> Informal instructions to a supplier create formal obligations. Route every change through the contract.

Claims and disputes: an escalating path

01

Negotiate

Direct discussion, using the contract's claims process.

02

Mediation

A neutral third party helps the sides agree.

03

Arbitration

A neutral third party decides, usually bindingly.

04

Litigation

Court. The last resort: slow, costly, public.

ON THE EXAM

Negotiation first, always. Alternative dispute resolution before litigation.

Contracts for agile work

APPROACH	HOW IT WORKS
Capped time and materials	Pay for time, up to an agreed maximum
Fixed price per increment	Price each iteration or release separately, then decide the next
Not-to-exceed with early exit	Buyer can stop when enough value is delivered
Graduated fixed price	Rates adjust depending on whether delivery is early or late
Outcome-based	Payment tied to business results achieved
Exchange of scope	Swap unstarted items for new ones of equal size, at no charge

ON THE EXAM

Agile contracts keep flexibility to reprioritise and to stop early, while capping the buyer's risk.

Principles behind agile contracts

- **Collaboration** over negotiation, as the Agile Manifesto puts it.
- **Incremental** commitment: agree the next piece, not everything at once.
- The right to **reprioritise** and to **stop early**, without penalty.
- The contract still needs clear terms for quality, acceptance, intellectual property and payment.

> ON THE EXAM
>
> Collaboration over negotiation does not mean no contract. It means a contract that supports collaboration.

Verifying and closing the agreement

- **Verify objectives are met**: all deliverables accepted against the contract's criteria.
- Settle any open **claims**, and make the **final payment**.
- Give the seller **formal written notice** that the contract is complete.
- **Archive** records; run a procurement audit; record lessons learned.
- Early termination is possible, for convenience or for default, as the contract allows.

> ON THE EXAM
>
> A contract is closed formally, in writing, after acceptance, claims and payment are complete.

Project Lantern: the kiosk supplier

- **Negotiation**: Northgate's BATNA was a second supplier's quote; procurement signed, the PM advised.
- **Measures**: installation dates per clinic, defect rate, response time for faults.
- **Monthly review** with a shared scorecard; payments tied to each clinic's acceptance.
- **Rule**: only the PM or procurement lead may instruct the supplier; clinic staff raise requests through them.

> ON THE EXAM
>
> A prepared negotiation, measured performance, regular reviews, and one channel for instructions.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
What is our BATNA?	The best real alternative if this deal fails
Supplier claims extra cost for a verbal instruction	Constructive change: changes only in writing
Supplier keeps missing dates	Review against the contract; agree corrective action
Evolving product, need to reprioritise and stop early	Incremental or capped agile contract
Dispute over a claim	Negotiate first, then mediation or arbitration
Supplier finished; deliverables accepted	Settle claims, pay, give written notice, archive

ON THE EXAM

Prepare negotiations, measure suppliers, change contracts formally, and close them formally.

> A buyer is negotiating with Supplier A, which quotes $500,000 for a network upgrade. Supplier B has submitted a firm quote of $460,000 for equivalent work, quality and dates.
>
> The project manager is preparing the negotiation strategy.

What is the buyer's best alternative to a negotiated agreement?

A Contracting Supplier B at $460,000.

B Accepting Supplier A's quote of $500,000.

C Walking away and doing nothing.

D Splitting the difference at $480,000.

◆ PAUSE AND DECIDE

What is the buyer's best alternative to a negotiated agreement?

A Supplier B at $460,000.
The best real alternative available if talks with A fail.

B Accept A at $500,000.
That is an agreement with A, not an alternative to it.

C Walk away.
Worse than a real alternative that is already on the table.

D Split the difference.
A possible outcome of the negotiation, not an alternative to it.

THE SIGNAL

The BATNA is what you would actually do if this negotiation failed. Here, a firm quote from B.

> On a fixed price building contract, the buyer's site engineer verbally told the contractor to use a premium floor finish 'because it will look better'.
>
> The contractor has now submitted a claim for the extra cost, citing the instruction.

What should the project manager have done to prevent this?

A Allowed verbal instructions, since they save time on site.

B Refused in advance to pay for any changes on the project.

C Made sure all changes went through the contract's formal written change process, and told the team who may instruct the contractor.

D Terminated the contract when the instruction was discovered.

◆ PAUSE AND DECIDE

What should the project manager have done to prevent this?

A Allow verbal instructions.
Exactly how the constructive change happened.

B Refuse all change payments.
Not enforceable when the buyer's own staff instructed the change.

C Formal written changes; clear authority.
Prevents constructive changes at their source.

D Terminate the contract.
Wildly disproportionate. The contractor followed the buyer's instruction.

THE SIGNAL

A verbal instruction that changes the work is a constructive change. Formal process and clear authority prevent it.

> A logistics supplier has missed four of its last six contracted delivery dates. The goods themselves meet quality standards.
>
> The late deliveries are starting to affect the project's schedule.

What should the project manager do?

A. Review the supplier's performance against the contract with them, agree corrective action, and document it.

B. Terminate the contract for default immediately.

C. Withhold all payments until deliveries are on time.

D. Take no action, since the quality of the goods is good.

◆ PAUSE AND DECIDE

What should the project manager do?

A Review against the contract; corrective action.

Evaluates vendor performance with evidence and fixes it together, on record.

B Terminate now.

Premature, disruptive, and the goods are good. Try correction first.

C Withhold all payments.

Likely breaches the contract, and damages the relationship.

D Take no action.

Ignores a contracted obligation that is now hurting the schedule.

THE SIGNAL

A supplier underperforming against the contract: review with evidence, agree corrective action, document it.

> A charity wants an external agency to build its new donor app. The features will evolve as donors test early versions, and the charity may stop once the most valuable features are live.
>
> The trustees want a clear limit on the maximum spend.

Which contracting approach fits best?

A. A firm fixed price contract for the full specified scope, agreed up front.

B. A cost plus award fee contract with no cap.

C. A contract priced per increment, with a cap and the right to reprioritise and stop early.

D. An informal agreement, since agile values collaboration over contracts.

◆ PAUSE AND DECIDE

Which contracting approach fits best?

A Fixed price, full scope up front.

Freezes features that are meant to evolve. Every change becomes a dispute.

B Cost plus award fee, no cap.

No limit on spend, which the trustees ruled out.

C Per increment, capped, reprioritise and stop early.

Keeps adaptive flexibility and meets the trustees' spending limit.

D Informal agreement.

Misreads the manifesto. Collaboration needs a contract that supports it.

THE SIGNAL

Evolving scope, early exit, and a spending cap: an incremental, capped agile contract.

> A supplier submits a claim for additional payment, arguing that the buyer's delayed approvals caused extra work. The project manager believes the delays were within the contract's agreed turnaround times.
>
> The contract includes a claims procedure and an arbitration clause.

What should the project manager do first?

A Refer the claim to litigation.

B Start arbitration under the contract.

C Stop paying the supplier's invoices until the claim is withdrawn.

D Negotiate directly with the supplier through the contract's claims procedure.

◆ PAUSE AND DECIDE

What should the project manager do first?

A Litigation.
The last resort. Slow, costly, public, and relationship-ending.

B Arbitration.
Comes after negotiation fails, not before it starts.

C Stop paying.
Likely a breach by the buyer, and turns one dispute into two.

D Negotiate via the claims procedure.
The first step: compare the facts against the contract's turnaround times.

THE SIGNAL

Disputes follow a ladder: negotiate, then mediate or arbitrate, and litigate last.

> A supplier has completed its work, and all deliverables have been formally accepted. The supplier has asked for its final payment.
>
> One small claim about travel costs is still open.

What should happen before the contract is closed?

A. Make the final payment and move on, since the deliverables are accepted.

B. Settle the open claim, make the final payment, give formal written notice of completion, and archive the records.

C. Keep the contract open indefinitely in case problems arise later.

D. Close the contract and destroy the procurement records to save storage.

◆ PAUSE AND DECIDE

What should happen before the contract is closed?

A Pay and move on.

Leaves the claim open, which can resurface after closure.

B Settle, pay, written notice, archive.

Complete, formal closure of the agreement.

C Keep it open indefinitely.

Leaves obligations and budget unresolved. Warranties cover later problems.

D Destroy the records.

Records are needed for audits, warranties and future disputes.

THE SIGNAL

Closure is formal: claims settled, final payment, written notice, records archived.

Prepare with a BATNA. Manage the relationship. Change and close formally.

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MODULE 26 · DOMAIN 2 - PROCESS

Budgets, reserves and funding

Building a budget from estimates, contingency and management reserves, funding limits, spend tracking, and the financial challenges ahead.

DOMAIN II - Process EXAM WEIGHT 41% MODULE 26 of 49

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A budget is the estimate, plus money for what might go wrong.

How a budget is built

01

Activity estimates

Cost of each activity.

›

02

Work packages

Activities rolled up.

›

03

+ Contingency

For identified risks.
Gives the cost baseline.

›

04

+ Management reserve

For unidentified risks.
Gives the budget.

ON THE EXAM

Contingency is inside the cost baseline. Management reserve is outside it, but inside the budget.

A budget, worked through

LAYER	AMOUNT	RUNNING TOTAL
Work package estimates	$600,000	$600,000
Contingency reserve, for identified risks	$60,000	$660,000 = cost baseline
Management reserve, for unknown risks	$30,000	$690,000 = project budget

ON THE EXAM

Asked for the cost baseline? Estimates plus contingency. Asked for the budget? Add management reserve too.

Contingency reserve or management reserve?

CONTINGENCY RESERVE

The umbrella you packed

- For identified risks: known unknowns
- Part of the cost baseline
- Used by the PM, per the risk plan
- Sized from risk analysis

MANAGEMENT RESERVE

The rainy-day fund

- For unidentified risks: unknown unknowns
- Outside the baseline, in the budget
- Released by management, via change control
- Often a percentage set by policy

THE TIEBREAKER

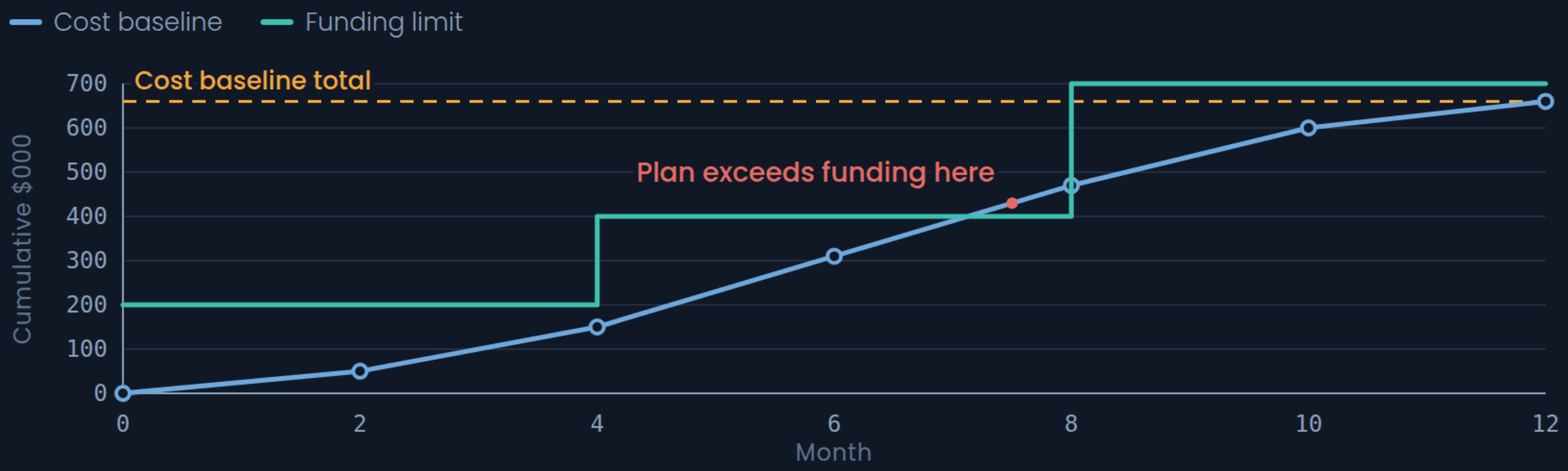
Identified risk occurs: use contingency. Something nobody foresaw: request management reserve.

Quantifying contingency

- The outline asks you to **quantify risk and contingency financial allocations**.
- **Expected monetary value**: probability times impact, summed across risks. Module 36 covers it.
- **Simulation**, such as Monte Carlo, for a confidence level: 'an 80 percent chance of finishing under this cost'.
- A flat percentage with no link to the risk register is a guess dressed as a reserve.

> ON THE EXAM
>
> Contingency comes from risk analysis. The size of the reserve should trace to identified risks.

The cost baseline and funding limits



ON THE EXAM

Spending must stay under the funding limit at every point, not just at the end.

Reconciling the plan with funding

- Compare the cost baseline with **when** funding becomes available.
- Where the plan spends ahead of funding, **re-sequence** work, or delay some spending.
- Or request a change to the **funding profile**, through governance.
- Spending ahead of approved funding, and hoping to be covered later, is not an option.

> ON THE EXAM
>
> Funding limit reconciliation can change the schedule. It is done during planning, not after an overspend.

Analysing financial needs

- **How much**, and **when**: the cash-flow profile, not just the total.
- **Capital or operating** spend: organisations often fund and account for them differently.
- **Currency**, payment terms, and supplier payment schedules.
- **Ongoing costs** after handover: licences, support, maintenance.

> ON THE EXAM
>
> The outline's first finance enabler: analyse project financial needs, including timing, not just size.

Anticipating future finance challenges

- **Inflation** on multi-year work, especially materials and wages.
- **Exchange rates** when costs and budgets are in different currencies.
- **Funding changes**: budget cuts, reprioritisation, a new financial year.
- **Supply shocks**: tariffs, shortages, price rises.
- Treat each as a financial risk: identify it, quantify exposure, and plan a response with finance.

> ON THE EXAM
>
> Anticipate, do not react. Financial exposures are risks, and belong in the risk register.

Planning spend tracking

- Track spend by **control account** or work package, matching the budget structure.
- Track **commitments** as well as **actuals**: an order placed is money already spoken for.
- Agree approval limits: who can authorise what spend.
- Tracking only invoices paid shows spending weeks late.

> ON THE EXAM
>
> Commitments show where spend is heading. Actuals alone show where it has been.

Planning financial reporting

- The outline asks you to **plan financial reporting**.
- What to report: spend, commitments, forecast at completion, reserve use.
- To whom and when: aligned with the **finance calendar** and governance meetings.
- Use the same numbers everywhere, from one source of truth.

> ON THE EXAM
>
> Finance wants forecasts and reserve use, on its own calendar. Plan the reporting to fit.

Managing reserves

- When an **identified risk occurs**, the PM uses contingency, as the risk response plan says.
- When something **unforeseen** happens, request **management reserve** through change control.
- Track reserve drawdown, and report it. A fast-shrinking reserve is an early warning.
- Return contingency when the risk it covered has passed, according to policy.

> ON THE EXAM
>
> Contingency: PM, identified risk, per the plan. Management reserve: management, unforeseen, via change control.

Cost words the exam uses

TERM	MEANING
Direct and indirect costs	Costs of this project's work, and shared overheads allocated to it
Fixed and variable costs	Costs that stay the same, and costs that change with volume
Life-cycle cost	Total cost of ownership, including operating the result
Sunk cost	Money already spent, which cannot be recovered

ON THE EXAM

Sunk cost is ignored in decisions. Decide on future costs and future benefits only.

Finance on agile work

- Fund **teams and timeboxes**: a stable team costs roughly the same each iteration.
- Track **cost per iteration** and **burn rate** against value delivered.
- Fund incrementally, and review at each release whether to keep investing.
- Stopping when enough value is delivered protects the budget, as module 20 described.

> ON THE EXAM
>
> Agile finance fixes the cost of capacity and asks, every release, whether the value justifies continuing.

Project Lantern: the finances

- **Cost baseline**: estimates plus contingency for named risks, such as kiosk price rises and migration rework.
- **Management reserve**: set by Northgate policy, released only by the steering committee.
- **Funding**: quarterly tranches; the second clinic's go-live was moved a month to fit.
- **Currency risk**: kiosks priced in euros, so a price adjustment clause was agreed with finance.

> ON THE EXAM
>
> Contingency traced to risks, reserve held by governance, spending fitted to funding, exposures planned for.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Estimates, contingency and reserve given	Baseline = estimates + contingency; budget adds reserve
An identified risk has occurred	Use contingency, per the risk response plan
Something nobody foresaw has happened	Request management reserve via change control
'We have spent too much to stop'	Ignore sunk cost; decide on future cost and benefit
Planned spend exceeds funding in a period	Reconcile: re-sequence or change the funding profile
Costs exposed to exchange rates or inflation	Treat as a financial risk and plan a response

ON THE EXAM

Know which money is which, who can spend it, and when it arrives.

> A project's work package cost estimates total $800,000. The risk analysis supports a contingency reserve of $80,000.
>
> The organisation's policy adds a management reserve of $40,000.

What is the project's cost baseline?

A $800,000

B $840,000

C $920,000

D $880,000

◆ PAUSE AND DECIDE

What is the project's cost baseline?

A $800,000

The estimates alone. Leaves out contingency, which is part of the baseline.

B $840,000

Adds management reserve instead of contingency.

C $920,000

The project budget: it includes management reserve, which is outside the baseline.

D $880,000

Estimates plus contingency: $800,000 + $80,000.

THE SIGNAL

Cost baseline: estimates plus contingency. Project budget: add management reserve.

> The risk register identified a possible price rise from the steel supplier, with a planned response funded from contingency. The supplier has now raised prices by the expected amount.
>
> The project manager is deciding how to fund the increase.

What should the project manager do?

A Request management reserve from the sponsor.

B Use the contingency reserve, as the risk response plan describes, and update the records.

C Submit a change request to increase the cost baseline.

D Absorb the cost by reducing the steel specification.

◆ PAUSE AND DECIDE

What should the project manager do?

A Request management reserve.

Management reserve is for the unforeseen. This risk was identified.

B Use contingency per the plan.

Exactly what contingency is for: an identified risk with a planned, funded response.

C Change request to raise the baseline.

Not needed. The baseline already contains this contingency.

D Cut the steel specification.

Quietly reduces quality, and may be unsafe or non-compliant.

THE SIGNAL

An identified risk occurring, with a planned response: use contingency. That is what it is for.

A flash flood damages equipment on a construction site in an area never previously affected. Nothing like it appears in the risk register. Repairs will cost $60,000.

The contingency reserve is fully allocated to identified risks that have not yet occurred.

How should the project manager fund the repairs?

A Request management reserve through the change control process.

B Take the money from the contingency reserve.

C Reduce scope elsewhere without telling the sponsor.

D Delay the repairs until the end of the project.

◆ PAUSE AND DECIDE

How should the project manager fund the repairs?

A Management reserve via change control.

An unidentified risk: exactly what management reserve is for.

B Use contingency.

Contingency is committed to identified risks that may still occur.

C Cut scope secretly.

Unauthorised scope change, and hides the problem from governance.

D Delay the repairs.

Damaged equipment may stop work now, and the cost does not go away.

THE SIGNAL

An unforeseen event, not in the register: request management reserve, through change control.

> A project has spent $2 million. Finishing it will cost a further $1.5 million. After a market change, the expected benefits have fallen to $1 million.
>
> The sponsor says, 'We have spent too much to stop now.'

What should the project manager recommend?

A Continue, because $2 million has already been invested.

B Continue, but reduce testing to lower the remaining cost.

C Base the decision on future costs and benefits only, which suggests stopping or re-scoping.

D Speed up the remaining work so the benefits arrive sooner.

◆ PAUSE AND DECIDE

What should the project manager recommend?

A Continue because of what was spent.

The sunk cost fallacy. The $2 million is gone either way.

B Cut testing.

Adds quality risk and still spends more than the benefit.

C Decide on future cost and benefit.

Spending $1.5m for $1m of benefit destroys value. Recommend stopping or re-scoping.

D Speed up the work.

Does not change the fact that the remaining cost exceeds the benefit.

THE SIGNAL

Sunk cost is irrelevant. Compare what you will still spend with what you will still gain.

> The finance team releases project funding in quarterly tranches. The cost baseline shows planned spending in the second quarter $120,000 above that quarter's tranche.
>
> Total funding for the year covers the total baseline.

What should the project manager do?

A Overspend in the second quarter, and ask finance to cover it later.

B Reconcile the plan with funding: re-sequence spending, or request a change to the funding profile.

C Cut $120,000 of scope from the project.

D Take no action, since the annual totals match.

◆ PAUSE AND DECIDE

What should the project manager do?

A Overspend and ask later.

Spends money that has not been released. Not permitted.

B Reconcile: re-sequence or change the profile.

Funding limit reconciliation, done before the overspend happens.

C Cut scope.

Unnecessary: the money exists, just later. Timing is the problem, not total.

D Do nothing.

Matching totals do not help when a quarter's tranche runs out.

THE SIGNAL

The total is fine but the timing is not: reconcile the spending profile with the funding profile.

> A three-year project is budgeted in pounds sterling, but its main equipment supplier invoices in US dollars, in stages over the three years.
>
> The exchange rate has moved by more than ten percent in the past year.

What should the project manager do?

A. Identify the currency exposure as a financial risk, quantify it, and plan a response with the finance team.

B. Ignore the exchange rate, since it may move in the project's favour.

C. Pay the supplier for all three years in advance.

D. Reduce the project's scope now to create a buffer.

◆ PAUSE AND DECIDE

What should the project manager do?

A Identify, quantify, respond with finance.
Anticipates a finance challenge and manages it as a risk.

B Ignore it.
A ten percent swing on multi-year dollar invoices is a real exposure.

C Pay everything upfront.
Removes currency risk but adds supplier and cash-flow risk, and needs authority.

D Cut scope now.
Loses value to cover a risk that has not been analysed.

THE SIGNAL

A foreseeable financial exposure is a risk. Identify, quantify, and plan the response with finance.

Know which money is which, **who can spend it**, and when it arrives.

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MODULE 27 · DOMAIN 2 - PROCESS

Earned value and forecasting

Planned value, earned value and actual cost; variances and indices; forecasting the final cost; and reading the S-curve.

DOMAIN II - Process EXAM WEIGHT 41% MODULE 27 of 49

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For the time spent and money spent, **how much work is actually done?**

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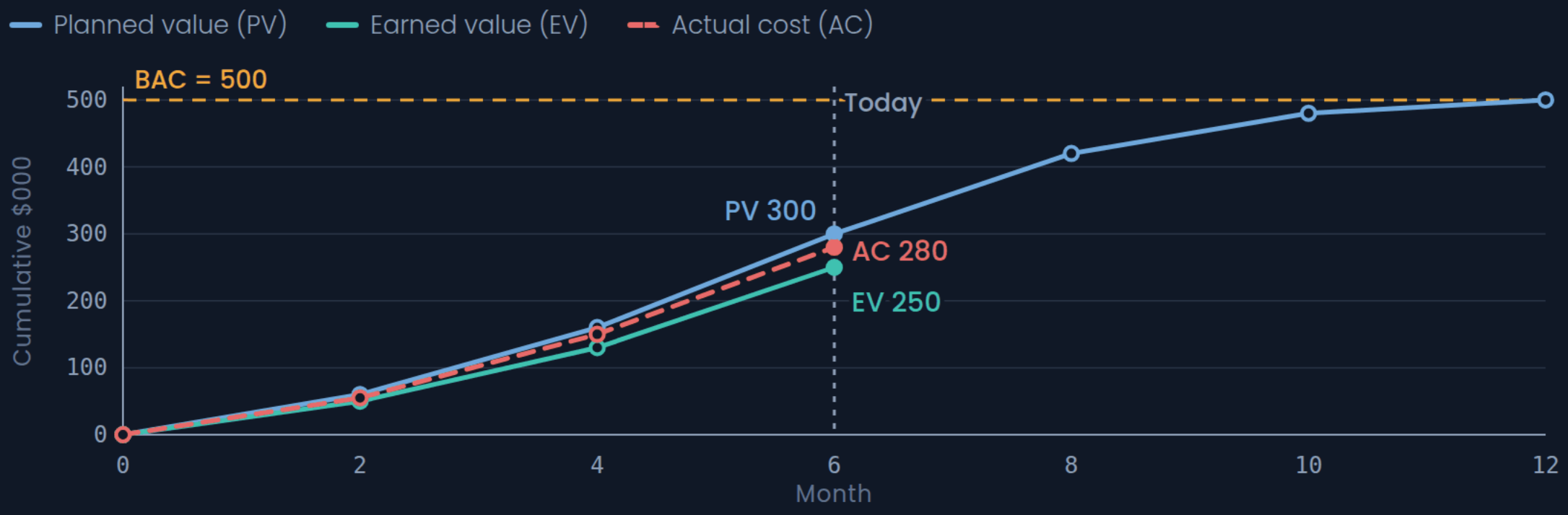
Four numbers to start with

TERM	MEANING	QUESTION IT ANSWERS
BAC, budget at completion	The total planned budget for the work	What should the whole thing cost?
PV, planned value	The budgeted value of work planned by now	What should be done by now?
EV, earned value	The budgeted value of work actually done	What is actually done?
AC, actual cost	What has actually been spent to date	What did it cost?

ON THE EXAM

EV and PV are measured in the budget's money, not in hours or percentages. That is what makes them comparable.

Reading the earned value S-curve



ON THE EXAM

Earned value below planned: behind schedule. Actual cost above earned: over budget.

Variances and indices

MEASURE	FORMULA	BAD NEWS WHEN
Cost variance (CV)	EV - AC	Negative: over budget
Schedule variance (SV)	EV - PV	Negative: behind schedule
Cost performance index (CPI)	EV / AC	Below 1: over budget
Schedule performance index (SPI)	EV / PV	Below 1: behind schedule

ON THE EXAM

Every formula starts with EV. Variance subtracts; index divides. Negative, or below 1, is bad.

The chart's numbers, worked through

- CV = 250 − 280 = **−30**. Over budget by $30,000.
- SV = 250 − 300 = **−50**. Behind schedule by $50,000 of work.
- CPI = 250 / 280 = **0.89**. Getting about 89 cents of work for each dollar.
- SPI = 250 / 300 = **0.83**. Progressing at about 83 percent of the planned rate.

> ON THE EXAM
>
> Schedule variance is measured in money: $50,000 of planned work not yet done.

Interpreting the indices

INDEX	ABOVE 1	EQUAL TO 1	BELOW 1
CPI	Under budget	On budget	Over budget
SPI	Ahead of schedule	On schedule	Behind schedule

ON THE EXAM

Both below 1: over budget and behind. The worst combination, and the most common exam answer.

Forecasting the final cost

FORECAST	FORMULA	USE WHEN
EAC, typical	BAC / CPI	Current cost performance will continue
EAC, atypical	AC + (BAC - EV)	The variance was a one-off; the rest goes to plan
EAC, cost and schedule	AC + (BAC - EV) / (CPI x SPI)	Schedule pressure will also affect cost
EAC, new estimate	AC + bottom-up ETC	The original plan is no longer valid
ETC	EAC - AC	How much more will be spent
VAC	BAC - EAC	Negative: forecast to finish over budget

ON THE EXAM

The question's wording chooses the EAC formula. 'Will continue' or 'typical': BAC / CPI.

Forecasting, worked through

- Typical: EAC = 500 / 0.893 = **560**. Expect to finish $60,000 over budget.
- ETC = 560 - 280 = **280** still to spend. VAC = 500 - 560 = **-60**.
- Atypical, if the overrun was a one-off: EAC = 280 + (500 - 250) = **530**.
- If schedule pressure also drives cost: 280 + 250 / (0.893 x 0.833) = about **616**.

> ON THE EXAM
>
> Same project, three forecasts. The story in the question decides which one is right.

To-complete performance index

- **TCPI = (BAC - EV) / (BAC - AC)**: work remaining divided by money remaining.
- Chart's project: (500 - 250) / (500 - 280) = 250 / 220 = **1.14**.
- The team must deliver $1.14 of work for every remaining dollar to finish on budget.
- A TCPI well above 1, with a CPI below 1, means finishing on budget is unrealistic.
- If a new EAC is approved, use (BAC - EV) / (EAC - AC) instead.

> ON THE EXAM
>
> TCPI above 1 means the future must be more efficient than the plan. Compare it with the current CPI.

What earned value does not tell you

- SPI drifts towards 1 near the end, because EV eventually equals PV, even on a late project.
- It does not show the **critical path**: good SPI can hide a late critical activity.
- It does not measure **quality** or **value**: work can be 'earned' and still be wrong.
- It relies on honest **progress rules**, such as 0/100 or 50/50, agreed in advance.

> ON THE EXAM
>
> Use earned value with critical path and quality checks, never alone.

Earned value on agile work

- A **release burnup** chart plays the same role: planned scope versus completed scope over time.
- Agile earned value can use **story points**: planned points versus completed points, and cost per point.
- Only work meeting the definition of done counts as earned.
- Partially done stories earn nothing. There is no '80 percent done'.

> ON THE EXAM
>
> On agile work, 'earned' means done to the definition of done. Burnups show it; module 29 covers them.

Working with governance on variances

- The outline asks you to **monitor financial variations and work with the governance process**.
- Agree **thresholds** in the plan, for example CPI below 0.95 triggers escalation.
- When a threshold is breached: analyse the cause, forecast, prepare options, then escalate.
- Waiting for the numbers to 'recover by themselves' rarely works. Early indices are strong predictors.

> ON THE EXAM
>
> Variances are measured against agreed thresholds. Breach one, and governance hears about it, with options.

Project Lantern: earned value in practice

- Used for the **predictive workstreams**: kiosks, network and migration.
- Progress rule: **0/100** for each clinic's installation, so nothing is 'nearly done'.
- The app uses a **release burnup** instead, in story points.
- Governance threshold: CPI or SPI below 0.92 goes to the steering committee, with options.

> ON THE EXAM
>
> Earned value where the baseline is predictive; burnups where the work is adaptive; thresholds for both.

Scenario signals and the move

THE SCENARIO GIVES	THE MOVE
PV, EV and AC; asks for status	CPI and SPI; below 1 is bad
'Current performance will continue'	EAC = BAC / CPI
'The overrun was a one-off'	EAC = AC + (BAC - EV)
'What efficiency is needed to finish on budget?'	TCPI = (BAC - EV) / (BAC - AC)
SPI near 1 near the end, milestones late	Check the critical path; SPI misleads late on
An index breaches the agreed threshold	Analyse, forecast, options, escalate per governance

ON THE EXAM

Start every calculation with EV. Let the question's wording choose the forecast.

> At a monthly review, a project's planned value is $400,000, its earned value is $360,000, and its actual cost is $450,000.
>
> The sponsor asks for a one-line summary of the project's status.

Which summary is correct?

A Under budget and ahead of schedule.

B Over budget and ahead of schedule.

C Over budget and behind schedule.

D Under budget and behind schedule.

◆ PAUSE AND DECIDE

Which summary is correct?

A Under budget, ahead.

Both indices are below 1. Both parts are wrong.

B Over budget, ahead.

Right about cost; SPI is 0.9, so behind, not ahead.

C Over budget, behind.

CPI = 360 / 450 = 0.8; SPI = 360 / 400 = 0.9. Both below 1.

D Under budget, behind.

Right about schedule; CPI is 0.8, so over budget.

THE SIGNAL

Compute both indices. Below 1 on cost: over budget. Below 1 on schedule: behind.

> A project has a budget at completion of $1,000,000. Its earned value is $400,000 and its actual cost is $500,000.
>
> The team expects current cost performance to continue for the rest of the project.

What is the estimate at completion?

A $1,100,000

B $1,000,000

C $1,150,000

D $1,250,000

◆ PAUSE AND DECIDE

What is the estimate at completion?

A $1,100,000

The atypical formula, for a one-off overrun. The question says it will continue.

B $1,000,000

The original budget. Ignores a 20 percent cost overrun.

C $1,150,000

Not produced by any standard formula here.

D $1,250,000

CPI = 400 / 500 = 0.8; EAC = 1,000,000 / 0.8.

THE SIGNAL

'Current performance will continue' means the typical formula: EAC = BAC / CPI.

> The same project, with a budget at completion of $1,000,000, earned value $400,000 and actual cost $500,000, is reviewed again.
>
> The team now establishes that the entire overrun came from a one-off equipment failure that will not recur. The remaining work should go to plan.

What is the estimate at completion?

A $1,100,000

B $1,000,000

C $1,150,000

D $1,250,000

◆ PAUSE AND DECIDE

What is the estimate at completion?

A $1,100,000

AC + (BAC - EV) = 500,000 + 600,000. The one-off stays; the rest goes to plan.

B $1,000,000

Ignores the $100,000 already lost to the failure.

C $1,150,000

Not produced by any standard formula here.

D $1,250,000

The typical formula. Assumes the poor performance continues, which it will not.

THE SIGNAL

Same numbers, different story, different formula. A one-off overrun: EAC = AC + (BAC - EV).

> A project has a budget at completion of $200,000. Its earned value is $120,000 and its actual cost is $140,000.
>
> The sponsor insists that the project must still finish within the original budget.

What cost performance must the team achieve on the remaining work?

A 0.75

B 1.33

C 0.86

D 1.17

◆ PAUSE AND DECIDE

What cost performance must the team achieve on the remaining work?

A 0.75

The TCPI formula upside down: money remaining over work remaining.

B 1.33

TCPI = (200 - 120) / (200 - 140) = 80 / 60.

C 0.86

The current CPI, 120 / 140. The question asks about the future.

D 1.17

Not produced by the TCPI formula.

THE SIGNAL

TCPI: work remaining over money remaining. Compare it with the current CPI to judge realism.

> Near the end of a construction project, the schedule performance index is 0.98. However, the building's handover milestone, which is on the critical path, is forecast to be three weeks late.
>
> The sponsor asks why the index looks healthy.

What should the project manager do?

A Report the project as on schedule, based on the SPI.

B Rely on the SPI, since it combines all the schedule information.

C Explain that SPI converges towards 1 late in a project, and report schedule status from critical path analysis.

D Rebaseline the schedule so the SPI and the milestone agree.

◆ PAUSE AND DECIDE

What should the project manager do?

A Report on schedule.

Misleading. The critical milestone is three weeks late.

B Rely on SPI.

SPI ignores the critical path and drifts towards 1 near the end.

C Explain the SPI limit; use critical path status.

Honest, accurate status from the right tool.

D Rebaseline to match.

Hides a real delay by moving the target. Not what baselines are for.

THE SIGNAL

Late in a project, SPI flatters. For schedule truth, look at the critical path.

> The governance plan says that a cost performance index below 0.92 must be escalated to the steering committee. This month the CPI has fallen to 0.88.
>
> The project manager believes a recovery is possible.

What should the project manager do?

A Wait a month to see whether performance recovers before reporting.

B Reduce scope quietly to bring costs back into line.

C Rebaseline the cost so the CPI returns to 1.

D Analyse the cause, update the forecast, prepare recovery options, and escalate as the governance plan requires.

◆ PAUSE AND DECIDE

What should the project manager do?

A. Wait a month.
Breaches the agreed threshold, and early indices rarely recover alone.

B. Cut scope quietly.
An unauthorised change that hides the problem from governance.

C. Rebaseline.
Hides the variance instead of managing it. Needs change control anyway.

D. Analyse, forecast, options, escalate.
Works with the governance process, arriving with analysis and a plan.

THE SIGNAL

A breached threshold goes to governance, promptly, with analysis, a forecast and options. Recovery ideas belong in the options.

Start with EV. **Subtract** for variances, **divide** for indices. Let the story choose the forecast.

MODULE 28 · DOMAIN 2 - PROCESS

Quality and cost of quality

Quality requirements, managing versus controlling quality, the cost of quality, control charts and Pareto, audits, and quality built in.

DOMAIN II - Process EXAM WEIGHT 41% MODULE 28 of 49

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Quality is planned and built in. It cannot be inspected in.

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Quality or grade?

QUALITY

Meets requirements

- Degree to which it fulfils requirements
- Defects mean low quality
- Low quality is always a problem
- 'It does what was agreed, reliably'

GRADE

A category of features

- Level of features or specification
- Basic versus premium
- Low grade can be exactly right
- 'A basic phone, or a flagship'

THE TIEBREAKER

A low-grade product can be high quality. A premium product full of defects is low quality.

Gathering quality requirements

- From **stakeholders**: what does 'good' mean to the people who will use and run it?
- From **standards and regulation**: industry, safety, accessibility, data protection.
- From the **organisation**: quality policy and procedures, which are process assets.
- Non-functional measures: reliability, performance, usability, durability.
- Make each one measurable, so it can be checked.

> ON THE EXAM
>
> Quality requirements that cannot be measured cannot be controlled. Turn 'robust' into a number.

The quality management plan

SECTION	COVERS
Standards and objectives	Which standards apply; measurable quality targets
Roles and responsibilities	Who plans, performs and checks quality work
Activities and tools	Reviews, testing, audits, the tools and techniques to use
Metrics	What is measured, how, and the tolerance for each
Continuous improvement	How lessons feed back into the process

ON THE EXAM

The plan says how quality will be achieved and checked. It is tailored to the project's size and risk.

Manage quality or control quality?

MANAGE QUALITY

Is the process right?

- Also called quality assurance
- Focuses on the process
- Audits, process analysis, prevention
- Builds confidence that quality will be met

CONTROL QUALITY

Is the product right?

- Focuses on deliverables
- Inspection, testing, measurement
- Finds defects before release
- Feeds scope validation

THE TIEBREAKER

Checking whether the team follows its process: manage quality, via audit. Checking a deliverable: control quality.

The cost of quality

CATEGORY	KIND	EXAMPLES
Cost of conformance	Prevention	Training, design reviews, standards, good tools
Cost of conformance	Appraisal	Testing, inspection, audits
Cost of non-conformance	Internal failure	Rework and scrap found before release
Cost of non-conformance	External failure	Warranty claims, recalls, lost customers, liability

ON THE EXAM

Money spent on prevention is almost always cheaper than money spent on failure, especially failure customers see.

Why prevention wins

- A common rule of thumb: a defect costs about **1** to prevent, **10** to fix before release, **100** after release.
- Early investment in reviews, training and standards reduces every later cost.
- Cutting prevention to save money usually moves the cost to failure, where it grows.
- External failure also damages reputation, which no budget line captures.

> ON THE EXAM
>
> Asked where to invest in quality? Prevention first. Asked what to cut? Not prevention.

Quality and sustainability

- The outline pairs them: **manage cost of quality and sustainability**.
- Rework and scrap are waste: materials, energy and effort thrown away.
- Durable, repairable, energy-efficient products are both higher quality and more sustainable.
- Include sustainability measures in quality requirements and acceptance criteria.

> ON THE EXAM
>
> Good quality is often good sustainability. Every defect prevented is waste avoided.

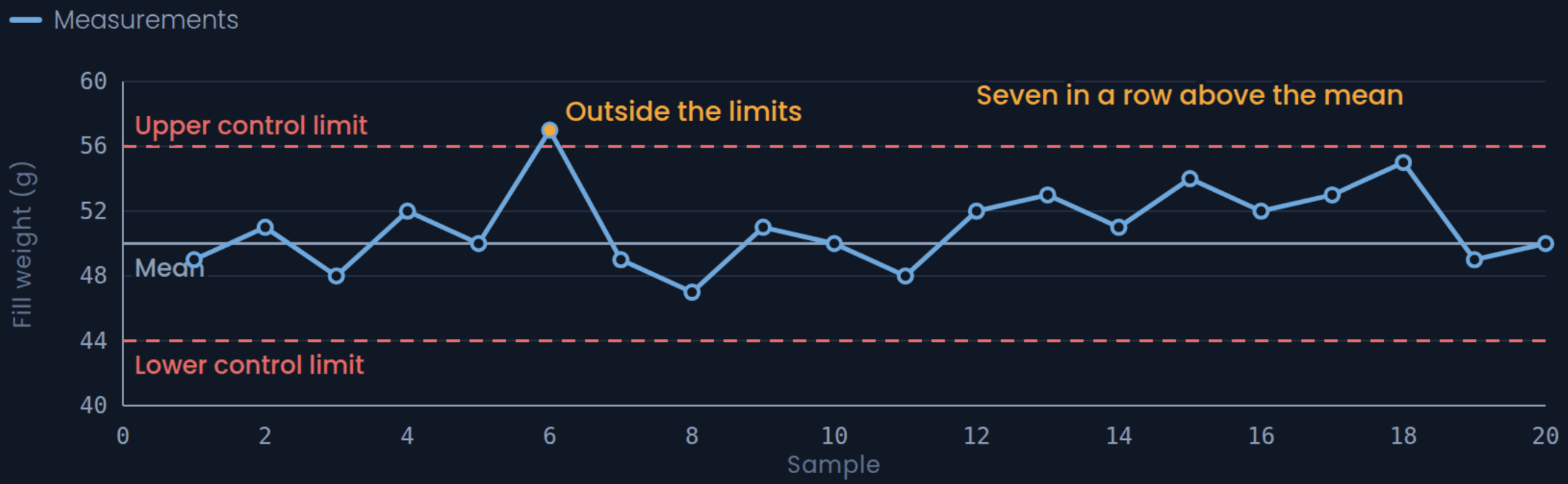
Quality tools

TOOL	USE IT TO
Checklist or check sheet	Make sure steps are done; tally defects by type
Cause-and-effect diagram	Explore possible causes of a problem, by category
Histogram	See how measurements are distributed
Pareto chart	Find the few causes behind most defects
Scatter diagram	See whether two variables are related
Control chart	Tell whether a process is stable over time
Flowchart	Map a process to find where defects arise

ON THE EXAM

Match the tool to the question: which causes, which cause matters most, is the process stable?

A control chart



ON THE EXAM

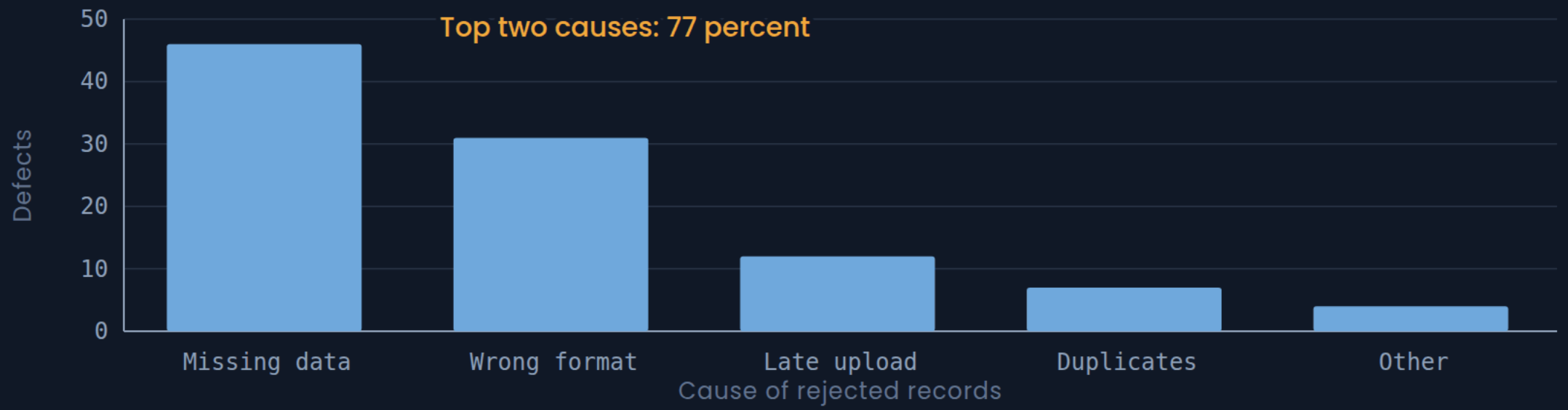
Out of control: a point outside the limits, or seven in a row on one side.

Reading control charts

- **In control**: points vary randomly within the limits. Leave the process alone.
- **Out of control**: a point outside a limit, or seven consecutive points on one side of the mean.
- **Control limits** come from the process itself. **Specification limits** come from the customer.
- Adjusting a stable process after every normal variation, called tampering, makes it worse.

> ON THE EXAM
>
> Out of control means investigate for a special cause. In control means do not tamper.

A Pareto chart of defect causes



ON THE EXAM

Sort causes from most to fewest. Fix the tallest bars first: the vital few behind most defects.

Inspection, sampling and audits

- **Inspection** examines a deliverable against requirements. It is control quality.
- **Statistical sampling** inspects a representative sample when checking everything is impractical.
- A **quality audit** checks whether the project follows its quality processes and policies. It is manage quality.
- Audits also spread good practice and find process improvements.

> ON THE EXAM
>
> Inspection checks the product. An audit checks the process.

Regulatory compliance in quality

- The outline asks you to **help ensure regulatory compliance** as part of quality.
- Identify the regulations that apply to the product: safety, medical, financial, accessibility.
- Build compliance checks into the definition of done and acceptance criteria.
- A deliverable that works but breaches regulation fails quality, however well it performs.

> ON THE EXAM
>
> Regulatory requirements are quality requirements. Module 33 covers compliance in full.

Ongoing reviews and continuous improvement

- **Plan, do, check, act**: plan a change, try it, check the result, then standardise or adjust.
- Conduct **ongoing quality reviews**, not just one at the end.
- On agile teams, quality is built in: definition of done, test-first, continuous integration, pairing.
- Retrospectives are continuous improvement in a two-week rhythm.

> ON THE EXAM
>
> Quality improves in small, repeated cycles. Plan, do, check, act.

Project Lantern: quality

- **Requirements**: bookings confirmed in under two seconds; accessible to screen readers; kiosks repairable on site.
- **Prevention**: design reviews with clinic staff, and test-first development on the app.
- **Control**: a control chart tracks booking response times; a Pareto chart ranks migration errors.
- **Audit**: a mid-project quality audit confirmed each clinic followed the go-live checklist.

> ON THE EXAM
>
> Measurable requirements, prevention first, charts for control, and an audit of the process.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Classify a quality cost	Prevention, appraisal, internal or external failure
Seven points in a row on one side of the mean	Investigate: the process is out of control
Is the team following its quality process?	Quality audit
Premium materials, but defects	Grade is not quality: it has a quality problem
Many defect types, limited time	Pareto: fix the vital few
Sponsor wants to cut reviews and training	Explain cost of quality: prevention is cheapest

ON THE EXAM

Build quality in, measure it, and spend on prevention before failure.

> A manufacturer reviews its quality costs for a new product line. They include staff training, design reviews, final testing, and warranty repairs for products that failed after customers bought them.
>
> The finance director asks how the warranty repairs should be classified.

Which cost of quality category do the warranty repairs belong to?

A Prevention

B External failure

C Appraisal

D Internal failure

◆ PAUSE AND DECIDE

Which cost of quality category do the warranty repairs belong to?

A Prevention

Training and design reviews are prevention. Repairs are not.

B External failure

Defects found after the customer has the product.

C Appraisal

Final testing is appraisal. Repairs come after failure.

D Internal failure

Internal failure is caught before release. These reached customers.

THE SIGNAL

Found before release: internal failure. Found by the customer: external failure, the most expensive.

> A control chart for a packaging line shows every point within the control limits. However, the last seven consecutive measurements all sit above the mean.
>
> The line supervisor says there is nothing to worry about because nothing is outside the limits.

What should the project manager do?

A. Take no action, since all points are within the control limits.

B. Adjust the machine settings after each measurement to bring values back to the mean.

C. Widen the control limits so the process has more room.

D. Investigate for a special cause, since the run of seven indicates the process is out of control.

◆ PAUSE AND DECIDE

What should the project manager do?

A No action.

Misses the second signal. Seven in a row is not random variation.

B Adjust after each measurement.

Tampering. It adds variation rather than removing the cause.

C Widen the limits.

Control limits come from the data, not preference. It hides the signal.

D Investigate for a special cause.

The rule of seven: a run on one side means something has shifted.

THE SIGNAL

Within the limits is not the whole test. Seven in a row on one side also means out of control.

> Final inspections keep finding the same kinds of defects in a construction project's electrical work. The project manager suspects the crews are not following the agreed installation procedures.
>
> The project manager wants to find out whether that is true.

Which activity should the project manager use?

A. A quality audit of the installation process.

B. More final inspections of completed work.

C. A Pareto chart of the defect types.

D. A control chart of defects per week.

◆ PAUSE AND DECIDE

Which activity should the project manager use?

A Quality audit.

Checks whether the process and procedures are being followed. Exactly the question.

B More inspections.

Finds more defects in the product; says nothing about the process.

C Pareto chart.

Ranks defect types, but cannot show whether procedures are followed.

D Control chart.

Shows stability over time, not compliance with procedures.

THE SIGNAL

'Are they following the process?' is an audit question. Inspection checks the product.

> A client ordered a basic, low-cost staff canteen fit-out. The contractor used premium fittings the client had not asked for, and the handover inspection found several leaking taps and faulty lights.
>
> The contractor argues that the fit-out is high quality because the fittings are premium.

Which statement is correct?

A. The fit-out is high quality, because premium fittings were used.

B. The problem is the low grade the client chose.

C. The fit-out has a quality problem, the defects, and an unrequested change of grade.

D. The fit-out is acceptable, because premium fittings outweigh minor defects.

◆ PAUSE AND DECIDE

Which statement is correct?

A High quality because premium.

Confuses grade with quality. Defects mean low quality.

B Low grade was the problem.

Low grade was agreed and is fine. It did not cause the leaks.

C A quality problem, plus an unrequested grade.

Defects are low quality; premium fittings nobody asked for are gold plating.

D Acceptable overall.

Defects fail the requirements, whatever the grade.

THE SIGNAL

Premium is grade, not quality. Defects are always a quality problem.

> A data migration has produced four hundred rejected records across eleven different error types. The team has one week to reduce rejections before the next migration run.
>
> They cannot fix every error type in that time.

Which approach should the project manager use first?

A. A Pareto analysis, to find the few error types causing most rejections.

B. Fix all eleven error types equally, a little at a time.

C. A scatter diagram of rejections against record size.

D. A cause-and-effect diagram for each of the eleven error types.

◆ PAUSE AND DECIDE

Which approach should the project manager use first?

A Pareto analysis.
Finds the vital few error types, so a limited week goes where it matters most.

B Fix all equally.
Spreads effort thinly; the biggest causes get the same attention as the smallest.

C Scatter diagram.
Tests one possible relationship; does not prioritise causes.

D Eleven fishbone diagrams.
Useful later for the top causes. Too slow as a first step.

THE SIGNAL

Many causes, limited time: Pareto first. Then dig into the biggest causes.

> To save money, the sponsor proposes cancelling the planned design reviews and staff training, and relying on thorough final testing instead.
>
> The product is a medical device, where defects found by patients would be very costly.

What should the project manager do?

A Agree, since final testing will catch any defects.

B Agree, and reduce final testing as well to save more.

C Explain the cost of quality, showing that prevention is cheaper than failure, and recommend keeping the reviews and training.

D Agree, and hire extra inspectors to compensate.

◆ PAUSE AND DECIDE

What should the project manager do?

A Rely on final testing.
Testing finds defects late, when fixing them costs far more.

B Cut testing too.
Removes both prevention and appraisal on a medical device. Dangerous.

C Explain cost of quality; keep prevention.
Prevention is cheapest, and external failure here could harm patients.

D More inspectors.
Swaps cheap prevention for expensive appraisal. Quality cannot be inspected in.

THE SIGNAL

Cutting prevention moves cost into failure, where it grows. Advise the sponsor with cost of quality.

Build quality in. **Prevent** before you inspect.
Read the chart before you react.

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MODULE 29 · DOMAIN 2 - PROCESS

Evaluating status: metrics, charts and artefacts

Choosing metrics, reading burndown, burnup and cumulative flow charts, and tailoring, maintaining and making artefacts accessible.

DOMAIN II - Process EXAM WEIGHT 41% MODULE 29 of 49

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Status is not what people say. It is what the evidence shows.

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Kinds of metric

KIND	EXAMPLES
Schedule and cost	SPI, CPI, milestones, burn rate
Flow	Throughput, cycle time, lead time, work in progress
Quality	Defects found, escaped defects, test pass rate
Value	Features used, benefits realised, adoption
Stakeholder	Satisfaction scores, engagement, feedback
Team	Happiness, turnover, sustainable pace

ON THE EXAM

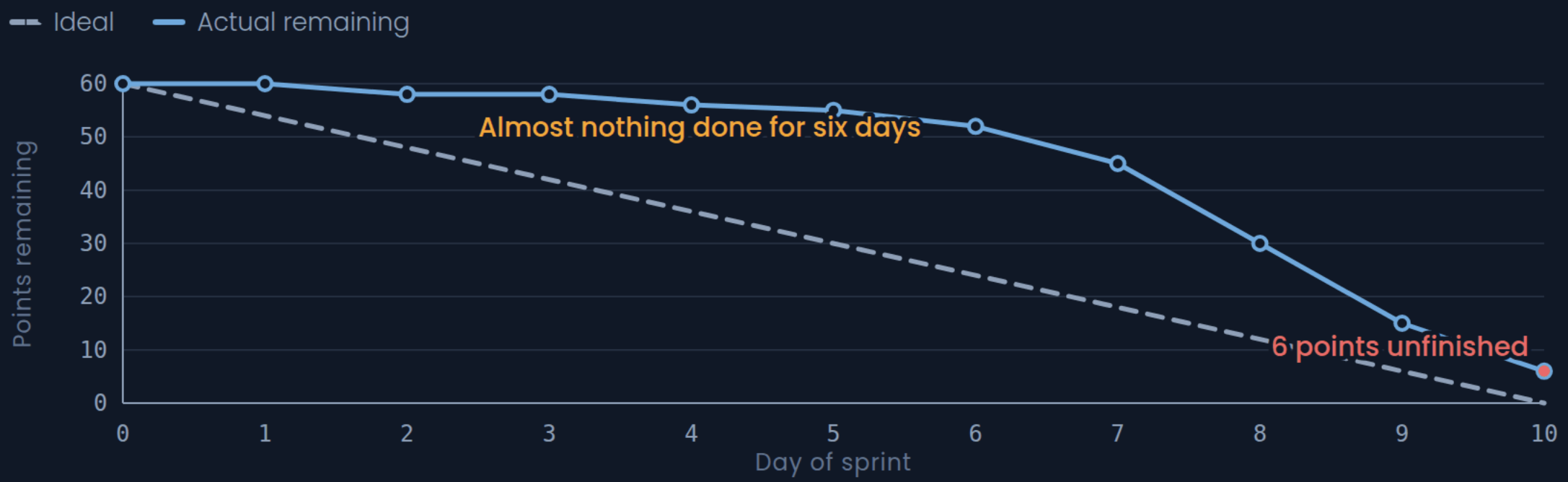
Measure a balanced set. Speed alone, without quality or value, rewards the wrong behaviour.

Metrics that help, and metrics that harm

- Few, relevant, **actionable**: each one should prompt a decision when it moves.
- Agreed **thresholds**, so everyone knows when a number matters.
- **Vanity metrics** look good and change nothing: lines of code, hours logged.
- **Goodhart's law**: when a measure becomes a target, it stops being a good measure.

> ON THE EXAM
>
> Reward outcomes, not the measure. Targets on velocity or points produce inflated points, not faster delivery.

A sprint burndown



ON THE EXAM

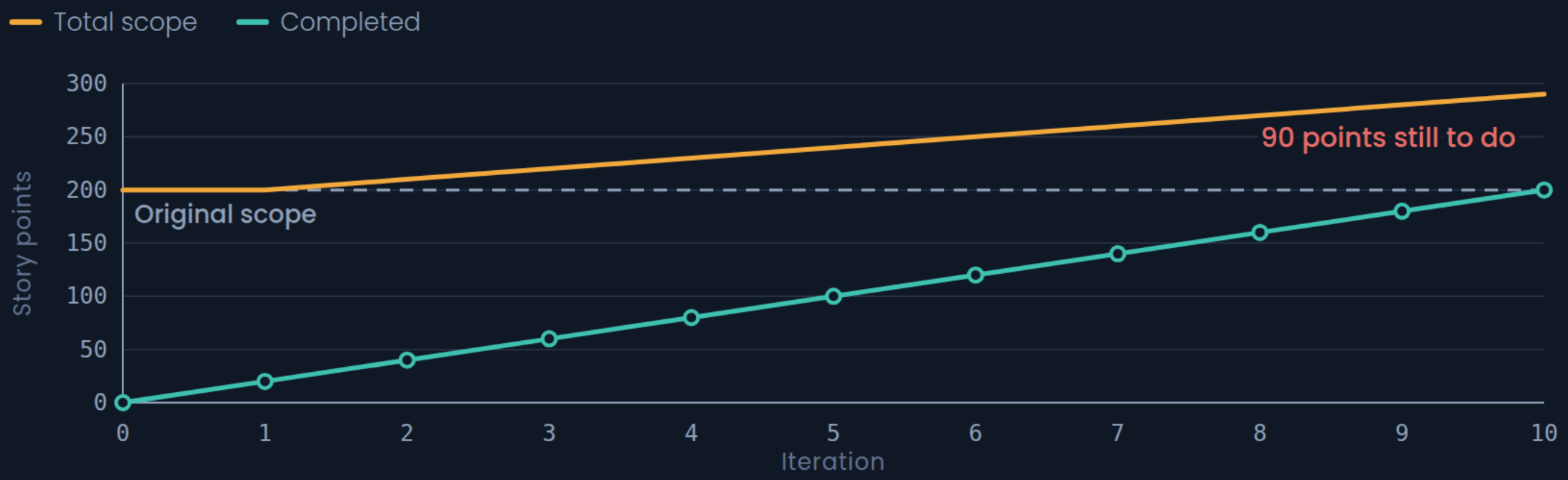
Above the ideal line: behind. Flat for days: nothing reaching done. A late cliff: stories too big.

Reading a burndown

- **Above** the ideal line: behind. **Below** it: ahead.
- **Flat** for several days: nothing is reaching done, perhaps blocked or too large.
- A sudden **rise**: work was added during the sprint.
- A burndown mixes progress and scope change. A rise could be either new work or re-estimation.

> ON THE EXAM
>
> Burndowns show remaining work. When scope changes, they cannot tell you why the line moved.

A release burnup



ON THE EXAM

Two lines: scope and done. If scope had not grown, this release would be finished now.

Burndown or burnup?

BURNDOWN

Work remaining

- One line going down
- Simple and familiar
- Scope changes are hidden in the line
- Common for a single sprint

BURNUP

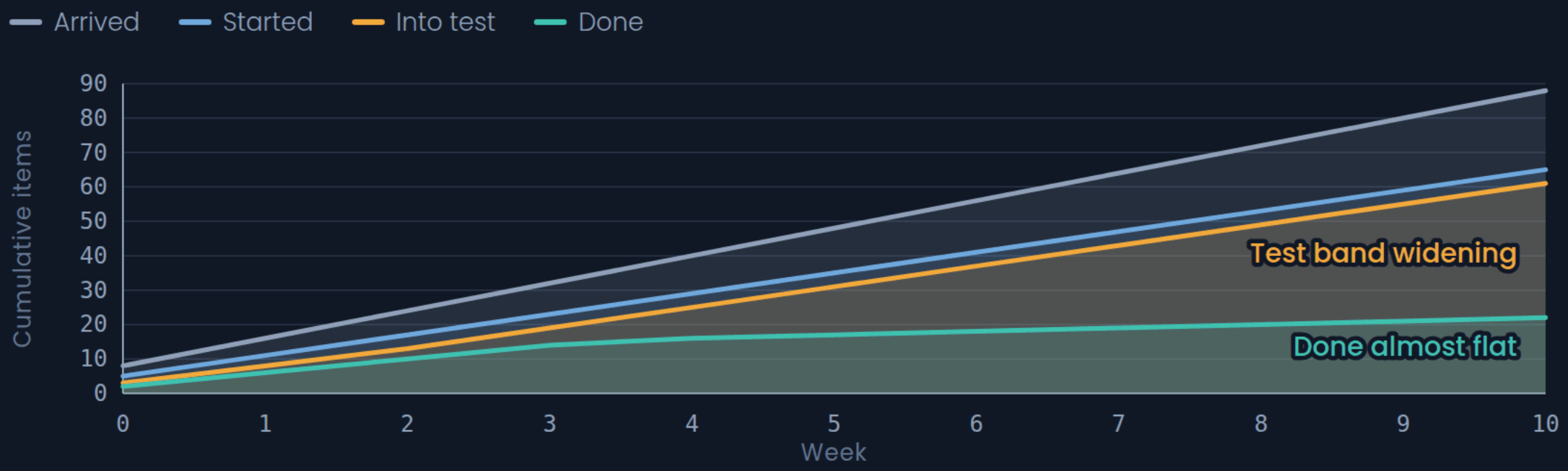
Work done, and total scope

- Two lines going up
- Shows progress and scope separately
- Reveals scope growth clearly
- Common for releases and projects

THE TIEBREAKER

Scope changing and people asking why the date slips? A burnup shows it. A burndown hides it.

A cumulative flow diagram



ON THE EXAM

Band width is work in that state. A widening band is a bottleneck. A flattening 'Done' line means little is finishing.

Reading a cumulative flow diagram

- **Vertical** thickness of a band: how much work is in that stage.
- **Horizontal** distance across the bands: roughly how long items take, the lead time.
- A **widening** band: a bottleneck in that stage, or the stage after it.
- A **flat** top line: no new work arriving. A flat done line: nothing finishing.
- Parallel, steady bands mean smooth flow.

> ON THE EXAM
>
> Look for the band that widens. Then fix flow there, usually by limiting work in progress upstream.

Other status charts, in brief

CHART	SHOWS
Earned value S-curve	Cost and schedule performance together (module 27)
Control chart	Whether a process is stable (module 28)
Velocity chart	Points completed per iteration, and its stability
Cycle time scatter	How long each item took, and the spread
Milestone chart and RAG status	Key dates, and status against thresholds (module 15)

ON THE EXAM

Choose the chart that answers the question being asked. Every chart hides something.

Project artefacts

KIND	EXAMPLES
Strategy	Business case, charter, vision, roadmap
Plans and baselines	Integrated plan, schedule, budget, scope baseline
Logs and registers	Risk, issue, assumption, change, decision, lessons learned, stakeholder
Backlogs and boards	Product backlog, iteration backlog, task board
Visual data	Burnup, burndown, cumulative flow, dashboards
Reports and agreements	Status reports, contracts, team charter

ON THE EXAM

An artefact exists to be used. If nobody uses it, question whether you need it.

Tailoring and maintaining artefacts

- The outline asks you to **identify and tailor** the artefacts the project needs.
- Keep the ones that add value or that governance and regulation require. Lighten the rest.
- Use **version control** and one source of truth, so everyone works from the current version.
- Review and update on a rhythm. An out-of-date register is worse than none: it misleads.

> ON THE EXAM
>
> Tailor the set of artefacts to the project, keep what governance requires, and keep what you keep current.

Accessible artefacts

- New in 2026: the outline asks you to **ensure accessibility of artefacts**.
- **Findable**: people know where they are, and have the right permissions.
- **Readable by everyone**: screen-reader friendly formats, plain language, clear structure.
- **Colour is never the only signal**: label red, amber and green, or use shapes and patterns.
- **Available across time zones**, and in formats that work on the tools people actually have.

> ON THE EXAM
>
> An artefact some team members cannot open, read or interpret is not transparent. Fix access and format.

Analysis and reconciliation

- The outline's first status enabler: **metrics, analysis and reconciliation**.
- When two sources disagree, such as a green burnup and a red earned value, find out why before reporting.
- Check definitions: are both measuring the same scope, with the same 'done'?
- Then **assess the effectiveness** of artefact management: are artefacts used, current and trusted?

> ON THE EXAM
>
> Reconcile conflicting numbers before you report them. Two versions of the truth destroy trust.

Project Lantern: reading the evidence

- The app's **release burnup** showed scope creeping up; the product owner cut two features to hold the date.
- The support team's **cumulative flow** showed faults piling up at supplier review; a daily call cleared it.
- The steering dashboard labels every status with a word as well as a colour.
- The PMO's 40-document template was tailored to 12, agreed with the PMO.

ON THE EXAM

Charts that led to decisions, artefacts everyone can read, and a tailored set agreed with governance.

Scenario signals and the move

THE CHART OR SCENARIO SHOWS	THE MOVE
Burndown flat, then a late cliff	Stories too big or unfinished; split them, finish before starting
Burnup: done steady, scope line rising	Scope growth: manage scope against the date
Cumulative flow: one band widening	Bottleneck: limit WIP upstream, help that stage
Someone cannot read or reach an artefact	Fix accessibility: labels, formats, permissions
A metric became a target and got gamed	Goodhart's law: measure outcomes instead
Documents nobody reads, heavy template	Tailor artefacts, with governance's agreement

ON THE EXAM

Read the shape of the chart, name what it reveals, and act on the cause.

> A team's sprint burndown stays almost flat for the first six days of a ten-day sprint, then drops steeply in the last three days. Several points are still unfinished at the end.
>
> This pattern has repeated for three sprints.

What does the pattern most likely indicate?

A The team is ahead of schedule and finishing early.

B Stories are too large, or work is started but not finished until late.

C The product owner removed scope during each sprint.

D The team's velocity is steadily increasing.

◆ PAUSE AND DECIDE

What does the pattern most likely indicate?

A Ahead of schedule.

Flat above the ideal line means behind, and work is left over.

B Stories too large, or finished late.

Nothing reaches done until the end: a classic burndown shape.

C Scope removed.

Removing scope causes a sudden drop at that moment, not a flat line first.

D Velocity rising.

Unfinished work every sprint says the opposite.

THE SIGNAL

Flat, then a cliff, with leftovers: stories too big, or too much started and too little finished.

A release burnup shows the team completing about twenty points every iteration, steadily. The total scope line has risen in every iteration since the start.

The sponsor asks why the release date keeps slipping, and whether the team is underperforming.

What should the project manager do?

A Tell the sponsor the team is slowing down, and ask it to work faster.

B Add developers to the team to increase velocity.

C Replace the burnup with a burndown to simplify the picture.

D Show that growing scope is the cause, and work with the product owner and sponsor on scope versus date.

◆ PAUSE AND DECIDE

What should the project manager do?

A The team is slowing.

The completed line is steady. Blaming the team misreads the chart.

B Add developers.

Treats a scope problem as a capacity problem.

C Switch to a burndown.

Hides the very scope growth that explains the slip.

D Show scope growth; decide scope versus date.

Reads the chart correctly and puts the trade-off with the decision-makers.

THE SIGNAL

Burnup with steady progress and rising scope: it is a scope decision, not a performance problem.

> A Kanban team's cumulative flow diagram shows the 'In test' band widening steadily for five weeks. The 'Done' line has almost flattened. The 'In progress' band is thin and steady.
>
> The team lead proposes adding two more developers.

What should the project manager do?

A Treat testing as the bottleneck: limit work entering test, and have the team help clear it.

B Add the two developers, as proposed.

C Take no action, since development is flowing smoothly.

D Pull more items into progress to keep developers busy.

◆ PAUSE AND DECIDE

What should the project manager do?

A Testing bottleneck: limit WIP, help clear it.

The widening band locates the constraint; flow improves only there.

B Add developers.

More development feeds a queue that is already stuck at test.

C No action.

Almost nothing is finishing. Smooth development is not the goal; done is.

D Start more work.

Increases work in progress, and lengthens every item's lead time.

THE SIGNAL

A widening band marks the bottleneck. Improve flow there; more work upstream only deepens the queue.

> A team member with red-green colour vision deficiency cannot interpret the project's red, amber and green status dashboard.
>
> Meanwhile, two remote contractors cannot open the risk register at all, because of file permissions.

What should the project manager do?

A. Ask colleagues to explain the dashboard to the team member each week.

B. Print the dashboard and register for everyone.

C. Make the artefacts accessible: add text labels or symbols to status, and fix the contractors' access.

D. Remove the colour coding and stop sharing the risk register.

◆ PAUSE AND DECIDE

What should the project manager do?

A Have colleagues explain it.

A workaround that makes one person dependent, and fixes nothing.

B Print everything.

Out of date quickly, and useless for remote contractors.

C Labels or symbols; fix access.

Ensures accessibility of artefacts, the new 2026 enabler.

D Remove colour; stop sharing.

Throws away useful information and reduces transparency.

THE SIGNAL

Colour must never be the only signal, and artefacts must be reachable by those who need them.

> Management began ranking agile teams by story points completed per sprint, with a bonus for the top team. Within three sprints, every team's velocity has risen sharply.
>
> Customers report no improvement in what is actually delivered.

What should the project manager recommend?

A. Keep the ranking, since velocity has improved.

B. Replace the ranking with measures of outcomes and value, and explain why points became unreliable.

C. Cap the number of points each team may claim per sprint.

D. Standardise story points across teams so the ranking is fair.

◆ PAUSE AND DECIDE

What should the project manager recommend?

A Keep the ranking.

The velocity rise is inflation, not delivery. Customers see no change.

B Measure outcomes; explain the problem.

Goodhart's law in action. Reward what customers actually value.

C Cap the points.

Treats the symptom; teams will game the next rule.

D Standardise points.

Keeps a target on a measure that should never be a target.

THE SIGNAL

When a measure becomes a target, it stops measuring. Velocity rose; delivery did not.

> A small eight-week internal project must use the PMO's standard template, which requires twenty-five documents. The team spends nearly a third of its time on them.
>
> Stakeholders say they read only the status summary and the risk register.

What should the project manager do?

A Produce all twenty-five documents, since the template is mandatory.

B Stop producing the documents, since nobody reads them.

C Outsource the documentation to an administrator.

D Work with the PMO to tailor the artefacts to those that add value, keeping what governance requires.

◆ PAUSE AND DECIDE

What should the project manager do?

A Produce all 25.

A third of an 8-week project spent on documents nobody reads.

B Stop them all.

Breaks a governance requirement unilaterally.

C Outsource them.

Still produces unused documents, now at extra cost.

D Tailor with the PMO.

Identify and tailor artefacts, with governance's agreement.

THE SIGNAL

Heavy artefacts nobody uses: tailor them with governance. Neither obey blindly nor ignore the rule.

Read the shape. Name what it reveals. Act on the cause.

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MODULE 30 · DOMAIN 2 - PROCESS

Closing projects and phases

Closure criteria, formal acceptance, transition readiness, early termination, and closing finances, contracts, people and lessons.

DOMAIN II - Process EXAM WEIGHT 41% MODULE 30 of 49

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A project is finished when it is **accepted, handed over and learned from**. Not when the work stops.

Closing, step by step

01

Confirm criteria

Deliverables meet the agreed closure criteria.

02

Formal acceptance

Sponsor or customer signs off.

03

Transition

Operations or the next phase confirms it is ready.

04

Close out

Contracts, finances, resources, records.

05

Learn and celebrate

Final lessons, final report, recognition.

ON THE EXAM

Acceptance comes before release of people and closing of accounts. Order matters on the exam.

Closure criteria, agreed early

- The outline asks you to **determine criteria to successfully close** the project or phase.
- Set them at the start, in the charter and plan: what 'done' means for the whole project.
- Include deliverables accepted, transition complete, contracts closed, benefits handed to an owner.
- Without agreed criteria, closure becomes a negotiation, and often a dispute.

> ON THE EXAM
>
> Agree how the project will close before it starts. Then closing is checking, not arguing.

Closing a phase or closing the project?

PHASE CLOSURE

A checkpoint

- End of a phase or stage
- Exit criteria reviewed at a gate
- Hand over to the next phase
- Decide: continue, change or stop

PROJECT CLOSURE

The end

- All work finished, or cancelled
- Final acceptance of the whole result
- Hand over to operations
- Everything closed and archived

THE TIEBREAKER

Phases close too, formally, with lessons and acceptance. A phase end is also a decision point.

Obtaining approval of completion

- The outline asks you to **obtain stakeholder approval of project completion**.
- Formal, written sign-off by the sponsor or customer, against the agreed criteria.
- If they will not sign, find out exactly which criterion they believe is unmet.
- New wishes at sign-off are new scope. They go through change control or a new project.

> ON THE EXAM
>
> Acceptance is judged against the agreed criteria. New requests are welcome, but they are new work.

Validating readiness for transition

- The outline asks you to **validate readiness for transition**, to operations or the next phase.
- The receiving team is trained, has runbooks, support contacts and service levels.
- The receiving team **confirms** it is ready. Its word, not the project's, decides.
- A short period of extra support after go-live, sometimes called hypercare, smooths the handover.

> ON THE EXAM
>
> Transition is ready when the receivers say so. Handing over to an unprepared team is not closure.

A transition readiness checklist

AREA	READY MEANS
People	Operations staff trained; support rotas in place
Documentation	Runbooks, known issues, configuration records handed over
Support	Service levels agreed; escalation contacts and suppliers named
Access and tools	Permissions, monitoring and licences transferred
Acceptance	Operations formally accepts responsibility

ON THE EXAM

Ask the receiving team to confirm each line. That confirmation is the readiness validation.

When a project is cancelled

- Cancelled projects still **close formally**.
- Document what was done, what was not, why it stopped, and what can be reused.
- Settle contracts under their termination clauses; close accounts; release people properly.
- Capture lessons: cancelled projects often teach the most.
- Simply stopping and walking away leaves costs, obligations and knowledge scattered.

> ON THE EXAM
>
> Cancellation is not failure to close. Close a cancelled project as carefully as a completed one.

Final lessons learned and retrospective

- Hold a final, facilitated session with the team and key stakeholders, **before** people disperse.
- Blameless: what worked, what did not, what we would do differently.
- Move the lessons into the organisation's repository, as module 16 described.
- A survey sent after everyone has left gets few replies and fewer insights.

> ON THE EXAM
>
> Capture final lessons while the team is still together. Then send them to the repository.

Closing contracts, finances and resources

- **Procurement**: verify deliverables accepted, settle claims, final payments, written notice (module 25).
- **Finances**: final invoices paid, accounts closed, unused reserves returned (module 26).
- **Resources**: people released with recognition and feedback; equipment returned; access removed (module 23).
- **Records**: archived, findable, and retained as policy requires.

> ON THE EXAM
>
> Closing is thorough administration. Every contract, account, person and record ends properly.

The final report

- Performance against the **scope, schedule and cost baselines**, with explanations.
- **Quality** results and any known issues handed to operations.
- **Benefits**: what has been realised so far, what is expected, and who owns tracking them.
- Risks and issues transferred, and a summary of lessons learned.

> ON THE EXAM
>
> The final report ends with benefits and their owner, because most benefits arrive after closure.

Closure on agile work

- Value is released continuously, so each **release** has its own small closure and review.
- The **product** often continues after the project; the team may continue too.
- Hand over the remaining backlog, the definition of done, and the product's operating knowledge.
- Stop when enough value is delivered, as module 20 described. That is a good close.

> ON THE EXAM
>
> Agile work closes releases often, and closes the project when value, not the backlog, says so.

Recognise and celebrate

- Recognise the team's work publicly, and thank stakeholders who helped.
- Give individuals feedback their functional managers can use in appraisals.
- Mark the end. This is the adjourning stage of team development.
- People remember how a project ended. It shapes whether they want to work with you again.

> ON THE EXAM
>
> Adjourning well is part of leading well. Recognition is a closing activity, not an optional extra.

Project Lantern: closing well

- Each clinic's go-live closed as a **phase**, with acceptance by the clinic manager and a short retrospective.
- Operations confirmed readiness line by line; the app team gave **four weeks of hypercare**.
- The kiosk contract closed with written notice after the last warranty claim was settled.
- The final report named the head of patient access as owner of the missed-appointment benefit.

> ON THE EXAM
>
> Phases closed as they finished, operations confirmed readiness, contracts closed formally, benefits owned.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Customer wants a new feature before signing	Accept against criteria; new feature is new scope
Operations team not trained or ready	Validate readiness before handover
Project cancelled	Close it formally anyway
All deliverables complete; what first?	Confirm criteria and obtain formal acceptance
Team keen to leave before lessons	Short final retrospective before release
End of phase one of several	Close the phase formally at the gate

ON THE EXAM

Accept, transition, close out, learn. In that order, for phases and projects, finished or cancelled.

> A project has delivered everything in the scope baseline, and every acceptance criterion has been met and documented.
>
> At the final review, the customer says they will not sign off until an additional reporting feature is added.

What should the project manager do?

A. Add the feature at no charge to obtain the sign-off.

B. Close the project without the customer's sign-off.

C. Show that the agreed criteria are met, seek formal acceptance, and treat the feature as a change request or new work.

D. Refer the dispute to the legal department immediately.

◆ PAUSE AND DECIDE

What should the project manager do?

A Add it free.

Uncontrolled scope, and it teaches the customer that acceptance can be bargained.

B Close without sign-off.

Skips a required closure step and invites later disputes.

C Criteria met; seek acceptance; feature as new work.

Acceptance against what was agreed; the new wish handled properly.

D Go to legal.

Far too early. A conversation about the criteria comes first.

THE SIGNAL

Acceptance is judged against agreed criteria. A new wish at sign-off is new scope.

A new customer service platform is due to go live next week, when the project will hand it to the IT operations team.

The operations team says it has had no training, and no runbooks or support contacts have been provided.

What should the project manager do?

A Work with operations to agree readiness criteria, complete training and runbooks, and hand over only when they confirm readiness.

B Hand over as planned; operations will learn as they go.

C Keep the project team supporting the platform indefinitely.

D Close the project on schedule, since the platform works.

◆ PAUSE AND DECIDE

What should the project manager do?

A Agree criteria; train; hand over when confirmed.
Validates readiness for transition, with the receiving team deciding.

B Hand over anyway.
Operations learns by failing in front of customers.

C Project supports it forever.
The project never closes, and operations never takes ownership.

D Close on schedule.
A working platform nobody can support is not a successful handover.

THE SIGNAL

The receiving team is not ready: validate readiness before handover. Their confirmation decides.

> After a change in company strategy, the sponsor cancels a project that is sixty percent complete.
>
> Several team members have already been offered roles on other projects, and two supplier contracts are active.

What should the project manager do?

A Stop all work immediately and let the team move on.

B Continue until the work is finished, to avoid waste.

C Keep the team together in case a new project is approved.

D Close the project formally: document its status, settle the contracts, capture lessons, release people and archive.

◆ PAUSE AND DECIDE

What should the project manager do?

A Stop and walk away.

Leaves contracts open, costs running, and knowledge scattered.

B Finish anyway.

Overrides the sponsor's decision and spends on work no longer wanted.

C Hold the team together.

Holds people back from real roles for a project that may never exist.

D Formal closure.

Cancelled projects close properly too: status, contracts, lessons, people, records.

THE SIGNAL

A cancelled project is closed as carefully as a completed one.

> All deliverables on a predictive project are complete and have passed quality control. The team is ready to move on to other work.
>
> The project manager is planning the closing activities.

What should the project manager do first?

A Release the team members to their new assignments.

B Confirm the closure criteria are met and obtain formal acceptance from the customer.

C Archive the project documents.

D Hold the final lessons learned session.

◆ PAUSE AND DECIDE

What should the project manager do first?

A Release the team.

Too early. if acceptance reveals a gap, the people who can fix it are gone.

B Confirm criteria; formal acceptance.

The first closing step. Everything else depends on acceptance.

C Archive documents.

One of the last steps, after everything else is closed.

D Final lessons learned.

Important, but it follows acceptance and transition.

THE SIGNAL

'First' at closure: acceptance. Release people and archive only after it.

> At the end of an eighteen-month project, most team members are starting new roles next week. Some suggest skipping the final retrospective because 'everyone is busy'.
>
> The organisation has a lessons learned repository that is rarely updated.

What should the project manager do?

A Hold a short, facilitated final retrospective this week, and add the lessons to the repository.

B Skip the retrospective, since the team is busy.

C Send a survey after everyone has moved to their new roles.

D Write the lessons learned personally, without the team.

◆ PAUSE AND DECIDE

What should the project manager do?

A Short retrospective now; update the repository.

Captures eighteen months of learning while the people are still together.

B Skip it.

Loses the lessons the outline names as a closing activity.

C Survey later.

Few replies once people have moved on, and no discussion.

D PM writes them alone.

One view of eighteen months. The team's knowledge is missing.

THE SIGNAL

Capture final lessons before the team disperses, keep it short, and send it to the repository.

A three-phase infrastructure programme has finished its design phase. The construction phase is due to start next month.

The team is eager to begin construction immediately.

What should the project manager do?

A Start construction immediately to keep momentum.

B Close the whole programme and start a new one for construction.

C Leave closure activities until the end of the whole programme.

D Close the design phase formally: check exit criteria, obtain acceptance, capture lessons, and hand over at the gate.

◆ PAUSE AND DECIDE

What should the project manager do?

A Start construction now.

Skips the gate decision and the design's formal acceptance.

B Close the whole programme.

Confuses phase closure with project closure.

C Close everything at the end.

Loses the design phase's lessons and the gate decision.

D Formal phase closure at the gate.

Phases close too, and the gate decides whether to proceed.

THE SIGNAL

Each phase closes formally, at a gate that decides whether and how to continue.

Accept, transition, close out, learn. In that order.

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MODULE 31 · DOMAIN 3 - BUSINESS ENVIRONMENT

Governance, PMOs and escalation paths

Who decides what, the kinds of PMO, process assets and environmental factors, success metrics, and escalation thresholds.

DOMAIN III - Business Environment EXAM WEIGHT 26% MODULE 31 of 49

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Governance decides **who decides**, and when a decision must go higher.

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What project governance covers

- **Structure**: the bodies and roles with authority, such as the sponsor and steering committee.
- **Decision rights**: who can approve what, up to which limits.
- **Rules and procedures**: policies, standards, templates, ethics.
- **Oversight**: reporting, reviews, gates and audits.
- PMBOK 8 puts Governance first among its seven domains, with nine processes.

> ON THE EXAM
>
> Governance is the frame the project works inside. The PM designs it with the sponsor, early.

Who's who in project governance

ROLE	TYPICAL AUTHORITY
Portfolio board	Chooses and prioritises projects; allocates funding
Sponsor	Owns the business case; champions; key decisions and funding
Steering committee	Oversees direction; decides escalated issues and gates
Change control board	Approves or rejects changes above the PM's limits
PMO	Standards, support, reporting, sometimes control
Project manager	Runs the project within delegated authority
Product owner	Decides product priorities on agile work

ON THE EXAM

Know which body owns which decision. Many exam answers depend on sending it to the right one.

Three kinds of PMO

TYPE	CONTROL	WHAT IT DOES
Supportive	Low	Templates, training, lessons learned, advice; a consultant
Controlling	Moderate	Requires compliance with methods, templates, reporting, audits
Directive	High	Directly manages projects; project managers report to it

ON THE EXAM

Supportive supplies, controlling requires, directive runs. Read how much control the scenario describes.

Portfolio, programme or project?

PORTFOLIO AND PROGRAMME

Above the project

- Portfolio: the right work, for strategy
- Programme: related projects, managed together
- Benefits beyond any one project
- Prioritise and coordinate

PROJECT

Delivering the work

- A temporary initiative to create value
- Its own objectives and scope
- Contributes to programme or portfolio
- Deliver within governance

THE TIEBREAKER

Portfolios choose the right projects. Programmes coordinate related ones. Projects deliver.

Process assets or environmental factors?

ORGANISATIONAL PROCESS ASSETS

Inside, and usable

- Policies, procedures, templates
- Lessons learned repositories
- Historical data and past project files
- The project can use and update them

ENTERPRISE ENVIRONMENTAL FACTORS

Conditions you work within

- Culture, structure, market conditions
- Laws, regulation, standards
- Infrastructure and available skills
- Largely outside the project's control

THE TIEBREAKER

Can the project use it and update it? Process asset. Must the project work within it? Environmental factor.

Tailoring governance

- Match the weight of governance to the project's **size, risk and complexity**.
- Agile governance: frequent reviews of working product, guardrails instead of approvals, funding by outcome.
- Hybrid: steering sees one integrated view; the product owner decides product detail within agreed limits.
- Heavy governance on small, low-risk work slows delivery and adds no protection.

> ON THE EXAM
>
> Governance should be as light as possible and as strong as necessary.

Escalation paths and thresholds

- The outline asks you to **outline governance escalation paths and thresholds**.
- Define **tolerances** for cost, schedule, scope, quality and risk at each level of authority.
- Name **who** decides beyond each tolerance, and **how fast** they must respond.
- Without defined thresholds, every issue either escalates too early or too late.

> ON THE EXAM
>
> Within tolerance, the PM acts. Beyond it, the path is already agreed. That is what makes escalation answers clear.

An example tolerance table

LEVEL	COST	SCHEDULE	SCOPE
Project manager	Up to 5 percent	Up to 2 weeks	Minor, no change to objectives
Sponsor	Up to 10 percent	Up to 6 weeks	Changes within the business case
Steering committee	Beyond 10 percent	Beyond 6 weeks	Changes to objectives or benefits

ON THE EXAM

Read the scenario's numbers against a table like this. The level that owns the decision is the answer.

Defining success metrics

- The outline asks you to **define success metrics** as part of governance.
- Agree them with key stakeholders **at the start**: what will count as success, measured how.
- Cover both kinds of success from module 01: delivered as planned, and worth the effort.
- Use measurable indicators: KPIs, objectives and key results, benefit targets.

> ON THE EXAM
>
> Success agreed at the start prevents arguments at the end about whether the project worked.

Gates and governance reviews

- **Phase gates**: go, no-go or change direction, at the end of a phase.
- **Steering reviews** on a regular cadence, fed by exception reports (module 15).
- **Audits** of process, finance or compliance, sometimes independent.
- Each gate has **exit criteria** agreed in advance, so the decision is about evidence.

> ON THE EXAM
>
> Gates are decisions, not ceremonies. Agree the criteria first, then bring the evidence.

Project Lantern: governance

- **Steering committee**: chief operating officer, finance, clinical lead, IT; monthly, plus each clinic gate.
- **Tolerances**: PM 5 percent and two weeks; sponsor 10 percent; beyond that, steering.
- **Controlling PMO**: requires a monthly exception report and a risk review.
- **Success metrics** agreed at the start: go-lives on time, and missed appointments down by a quarter.

> ON THE EXAM
>
> Clear bodies, clear tolerances, a known PMO role, and success defined before work began.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Change within the PM's tolerance	Decide it; record and communicate
Change beyond the PM's tolerance	Take it, with analysis, to the level that owns it
PMO gives templates and advice only	Supportive PMO
A new external law or market condition	Enterprise environmental factor
Nobody knows who decides an issue	Escalation paths should have been defined
Stakeholders argue about whether it succeeded	Success metrics should have been agreed at the start

ON THE EXAM

Know the structure, the tolerances and the success measures. Then every decision has an obvious home.

> A change request would increase a project's cost by 3 percent and add one week to a non-critical activity. The governance plan gives the project manager authority up to 5 percent of cost and two weeks of schedule.
>
> The team has analysed the change and recommends it.

What should the project manager do?

A Escalate the change to the steering committee for approval.

B Approve the change within delegated authority, then update the plan and inform stakeholders.

C Reject the change to protect the baseline.

D Ask the sponsor to sign the change request.

◆ PAUSE AND DECIDE

What should the project manager do?

A Escalate to steering.

Wastes senior time on a decision already delegated to the PM.

B Approve within authority; update and inform.

Uses the delegated tolerance exactly as governance intends.

C Reject it.

Baselines change through control, not never. The change is analysed and recommended.

D Ask the sponsor to sign.

Unnecessary. The decision sits within the PM's own limits.

THE SIGNAL

Inside your tolerance, decide. Escalating what governance has delegated to you is also a mistake.

> A growing company wants to set up a PMO. Its leaders want it to provide templates, training and lessons learned to project managers.
>
> They do not want it to require compliance or manage projects directly.

Which type of PMO fits best?

A Directive

B Controlling

C Supportive

D A portfolio board

◆ PAUSE AND DECIDE

Which type of PMO fits best?

A Directive

Manages projects directly. Ruled out by the leaders.

B Controlling

Requires compliance. Also ruled out.

C Supportive

Supplies templates, training and lessons, with low control.

D Portfolio board

A governance body that selects projects, not a PMO type.

THE SIGNAL

Supportive supplies, controlling requires, directive runs.

> Halfway through a project to build a customer data platform, the government introduces a new data privacy law that changes how customer consent must be recorded.
>
> The project manager is updating the project documents.

How should the new law be classified?

A An enterprise environmental factor

B An organisational process asset

C A lesson learned

D A project assumption

◆ PAUSE AND DECIDE

How should the new law be classified?

A Enterprise environmental factor

An external condition the project must work within, outside its control.

B Organisational process asset

Assets are the organisation's own, usable and updatable. A law is neither.

C Lesson learned

A lesson is knowledge from experience, not an external rule.

D Project assumption

It is a known fact now, not something assumed to be true.

THE SIGNAL

External, and must be worked within: environmental factor. Internal, usable and updatable: process asset.

> A supplier dispute threatens a 12 percent cost increase. The project manager knows this exceeds their authority, but nobody can say whether it should go to the sponsor, the PMO or the steering committee.
>
> Two weeks have passed without a decision.

What should the project manager have done to prevent this?

A Escalated the dispute directly to the chief executive.

B Decided the dispute personally to avoid delay.

C Waited for the supplier to withdraw its claim.

D Defined escalation paths and thresholds with the sponsor when governance was set up.

◆ PAUSE AND DECIDE

What should the project manager have done to prevent this?

A Go to the chief executive.

Skips every level of governance. Not prevention.

B Decide it personally.

Acts beyond authority, which the PM already knows.

C Wait.

Not prevention, and the delay is already costing time.

D Define escalation paths and thresholds early.

The outline's enabler. With a path agreed, the decision goes straight to its owner.

THE SIGNAL

Two weeks lost because nobody knew who decides: the escalation path was never defined.

> At closure, the sponsor says the project succeeded because it finished on time and on budget. The operations director says it failed, because staff productivity has not improved.
>
> No success measures were written down at the start.

What should the project manager have done?

A Run a satisfaction survey at closure to settle the question.

B Agreed success metrics with key stakeholders at the start, covering delivery and outcomes.

C Measured success only by schedule and budget, since those are objective.

D Let the sponsor define success, since the sponsor owns the business case.

◆ PAUSE AND DECIDE

What should the project manager have done?

A Survey at closure.
Too late, and opinion cannot replace agreed measures.

B Agree success metrics at the start.
Both views could have been reconciled into shared, measurable criteria.

C Schedule and budget only.
Measures project management success, not whether it was worth doing.

D Sponsor alone defines success.
Ignores key stakeholders, including the operations director who must live with it.

THE SIGNAL

Two definitions of success collide at the end: success metrics were never agreed at the start.

> A steering committee asks an agile product team for a detailed monthly Gantt chart of every task. The team plans with a roadmap and backlog, and says a Gantt chart would be obsolete within days.
>
> The committee's real concern is whether the release date is at risk.

What should the project manager do?

A. Produce the detailed Gantt chart each month, as requested.

B. Decline, since agile teams do not produce that kind of report.

C. Agree tailored reporting that answers the committee's concern, such as a release burnup and forecast.

D. Escalate the disagreement to the portfolio board.

◆ PAUSE AND DECIDE

What should the project manager do?

A Produce the Gantt chart.

Effort spent on an artefact that is wrong within days.

B Decline.

Ignores a legitimate governance need for assurance.

C Tailored reporting that answers the concern.

Tailors governance: meets the need, in a form that fits the approach.

D Escalate.

A conversation with the committee has not happened yet.

THE SIGNAL

Governance needs assurance, not a particular chart. Agree the form that gives real assurance.

Clear structure, clear tolerances, **success defined up front**.

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MODULE 32 · DOMAIN 3 - BUSINESS ENVIRONMENT

Ethics and professional conduct

PMI's four values, the 2025 update to the Code, a framework for dilemmas, and the situations the exam uses to test them.

DOMAIN III - Business Environment EXAM WEIGHT 26% MODULE 32 of 49

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On ethics, the right answer is **not negotiable**. But it is still delivered with care.

PMI's four values

VALUE	MEANS	ON THE EXAM, LOOKS LIKE
Responsibility	Owning decisions and consequences	Admit mistakes; report misconduct
Respect	High regard for people and resources	Listen; negotiate in good faith
Fairness	Impartial, objective decisions	Disclose conflicts; treat bidders equally
Honesty	Understanding and acting on the truth	Report accurately; keep commitments

ON THE EXAM

Every ethics question maps to one of these four. Name the value, and the answer usually follows.

Responsibility in practice

- Take on work only within your **competence**, or be open about gaps and how you will close them.
- Keep commitments, and own mistakes openly, then fix them.
- Protect **confidential** information and respect **intellectual property**, including software licences.
- Report unethical or illegal conduct through the appropriate channel.

> ON THE EXAM
>
> Seeing a breach and staying silent is itself a failure of responsibility.

Respect in practice

- Listen to understand other points of view, including in conflict.
- **Negotiate in good faith**; never mislead the other side.
- Respect **cultural norms**, while staying within the Code and the law.
- Never use position or expertise to intimidate, or for personal gain.

> ON THE EXAM
>
> Respect covers the powerless as well as the powerful: suppliers, junior staff, communities.

Fairness in practice

- **Disclose** any real or potential **conflict of interest**, and step back from the decision.
- Give every bidder the same information and the same chance, as module 24 described.
- Allocate opportunities, rewards and work on merit, not favouritism.
- Deciding fairly while hiding a conflict still breaks the Code. The disclosure is the point.

> ON THE EXAM
>
> A conflict of interest is disclosed first, then managed. You do not get to judge your own objectivity.

Honesty in practice

- Report status, forecasts and risks **accurately**, including bad news (module 15).
- Make commitments in good faith, and do not create false impressions.
- Be accurate about your own qualifications and experience.
- Help create an environment where others feel safe to tell the truth.

> ON THE EXAM
>
> Leaving something out to mislead is still dishonest. Accuracy includes completeness.

What the 2025 update brought

- The updated Code took effect on **17 November 2025**. The four values are unchanged.
- It applies to all members, volunteers, certification holders and **certification applicants**.
- Commentary on the update highlights **responsible use of technology and AI**, and **sustainability**.
- Read the current Code on pmi.org. It is short, and the exam is built on the version in force.

> ON THE EXAM
>
> Same four values, applied to today's realities: AI, data and long-term impact.

PMI's Ethical Decision-Making Framework

01

Assessment

Get the facts. Is it legal? Does it fit the Code and your values?

02

Alternatives

List the options, not just the first one.

03

Analysis

Who is affected, and how? Would it survive being made public?

04

Application

Apply ethical principles to choose.

05

Action

Act, take responsibility, and accept the consequences.

ON THE EXAM

Facts before judgement, options before choice. The same discipline as the answer ladder.

Classic dilemmas and the right move

SITUATION	THE MOVE
A bidder offers a valuable gift	Decline; follow the gifts policy; disclose if needed
A relative's company is bidding	Disclose; remove yourself from the evaluation
Asked to hide a problem or alter a report	Talk to the person; report accurately anyway
Unlicensed software or copied material found	Stop the use; inform the right people; put it right
A colleague breaks the Code or the law	Report through the appropriate channel
Confidential data put into a public AI tool	Stop; assess the breach; follow data policy; set guidelines

ON THE EXAM

In every row, the answer is firm on the principle and follows the proper route.

When local custom and the Code differ

- Where local law, organisational policy and the Code differ, follow the **strictest**.
- **Bribery** is not made acceptable by being customary somewhere. Many countries' laws apply abroad too.
- Small, customary courtesy gifts may be acceptable if policy allows and they are disclosed.
- When unsure, ask the organisation's legal or ethics function before acting.

> ON THE EXAM
>
> Respect for culture never extends to breaking the law or the Code.

Project Lantern: an ethics moment

- During kiosk selection, one supplier invited the project manager to a hospitality box at a football match.
- The project manager **declined**, recorded it under Northgate's gifts policy, and told procurement.
- Later, a clinic manager's partner turned out to work for a bidding supplier.
- The clinic manager **disclosed** it, and was removed from that evaluation panel. No drama; just the process.

> ON THE EXAM
>
> Ethics is mostly routine: decline, disclose, record, step back. The drama comes from skipping those steps.

> During the evaluation of bids for a software contract, a bidding supplier's account manager offers the project manager two expensive concert tickets.
>
> The account manager says the tickets are 'just a thank you for your time, nothing to do with the bid'.

What should the project manager do?

A Decline the tickets, and follow the organisation's gifts policy, disclosing the offer if required.

B Accept, since the concert is outside working hours.

C Accept, but make sure the evaluation is not influenced.

D Accept, and give the tickets to team members instead.

◆ PAUSE AND DECIDE

What should the project manager do?

A Decline; follow policy; disclose.

Protects fairness to every bidder, and the appearance of fairness too.

B Accept: outside hours.

Timing does not change who gave it or why.

C Accept, stay objective.

You cannot judge your own objectivity. Other bidders will not believe it either.

D Give them to the team.

The organisation still accepted a valuable gift from a bidder.

THE SIGNAL

A valuable gift from a bidder during evaluation: decline and disclose. Intent does not matter.

> A project manager is on the panel evaluating bids for a construction contract. One of the bidders is a company owned by the project manager's brother-in-law.
>
> The project manager is confident they can evaluate it objectively.

What should the project manager do?

A. Evaluate all bids fairly, without mentioning the relationship.

B. Disclose the relationship to the sponsor and procurement, and withdraw from evaluating.

C. Reject the brother-in-law's bid, to avoid any appearance of bias.

D. Resign from the project.

◆ PAUSE AND DECIDE

What should the project manager do?

A. Evaluate without disclosing.

Hides a conflict of interest. Confidence in your own objectivity is not the test.

B. Disclose and withdraw.

Fairness: disclose the conflict first, then step back from the decision.

C. Reject that bid.

Unfair to the bidder, who may offer the best value. Bias in reverse.

D. Resign.

Disproportionate. Disclosure and withdrawal from this decision are enough.

THE SIGNAL

A conflict of interest: disclose, then step back. Neither hide it nor overcorrect.

> A project manager discovers that several team members have been using unlicensed copies of a design software package to save the project money.
>
> The team lead says 'everyone does it, and the vendor will never find out'.

What should the project manager do?

A Ignore it, since the savings help the budget and the risk is low.

B Quietly buy licences later if the budget allows.

C Stop the unlicensed use, inform the appropriate managers, and obtain proper licences.

D Tell the team not to mention it to anyone.

◆ PAUSE AND DECIDE

What should the project manager do?

A Ignore it.

Knowingly allows an intellectual property breach and a legal risk.

B Buy licences later, quietly.

The breach continues meanwhile, and 'if the budget allows' may be never.

C Stop, inform, license properly.

Responsibility: respects intellectual property and reports the problem.

D Tell the team to keep quiet.

Concealment turns a mistake into deliberate misconduct.

THE SIGNAL

'Everyone does it' is never an ethical defence. Stop it, report it, fix it.

> A team member pasted a client's confidential contract into a free public AI chatbot to get a quick summary. The organisation has a data policy but no specific AI guidance.
>
> The team member did not realise this might be a problem.

What should the project manager do?

A Take no action, since the team member meant well.

B Ban all use of AI tools on the project permanently.

C Delete the chat history and move on.

D Stop further use, assess the breach under the data policy and client obligations, and set clear AI-use guidelines.

◆ PAUSE AND DECIDE

What should the project manager do?

A No action.

Good intent does not undo a possible breach of client confidentiality.

B Ban AI permanently.

Overreacts, and does not deal with the breach that has already happened.

C Delete and move on.

Deleting a chat does not recall data already shared, and hides the incident.

D Stop, assess under policy, set guidelines.

Handles the breach responsibly and prevents the next one.

THE SIGNAL

A confidentiality slip with a new tool: contain it, handle it through policy, then guide everyone.

Firm on the principle. **Careful in the approach.** Decline, disclose, report, record.

MODULE 33 · DOMAIN 3 - BUSINESS ENVIRONMENT

Compliance

Finding every obligation, classifying it, spotting threats, choosing methods, weighing consequences, and measuring whether the project complies.

DOMAIN III - Business Environment EXAM WEIGHT 26% MODULE 33 of 49

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Compliance is designed in, like quality. **Discovering it at go-live** is the most expensive way to comply.

The compliance sequence, from the outline

01
Confirm
What must the project comply with?

02
Classify
Group obligations into categories.

03
Threats
What could cause non-compliance?

04
Methods and approach
How will compliance be built in and checked?

05
Consequences and measure
What if we fail? Are we compliant now?

ON THE EXAM

The outline's enablers read as a sequence. Confirm first; measure throughout.

Compliance categories

CATEGORY	EXAMPLES
Legal and regulatory	Industry regulators, licensing, consumer law
Health and safety	Site safety, equipment safety, working hours
Security	Information security, cyber security, physical security
Data protection and privacy	Personal data handling, consent, retention
Environmental and sustainability	Permits, emissions, waste, sustainability reporting
Contractual and financial	Contract terms, audit, anti-bribery, procurement rules
Standards and internal policy	Industry standards, accessibility, company policies

ON THE EXAM

The outline names security, health and safety, sustainability and regulation explicitly. Expect all four.

Confirming compliance requirements

- Ask the experts early: legal, compliance, security, safety, data protection officers.
- Check sources: regulators, contracts, standards bodies, organisational policies.
- Record obligations in a **compliance register**, with owners and evidence needed.
- Trace each obligation into requirements and acceptance criteria, as module 18 described.
- Assuming another country's rules match your own is a classic, costly mistake.

> ON THE EXAM
>
> Compliance requirements are confirmed with specialists, recorded, owned and traced. Not assumed.

Threats to compliance

THREAT	EXAMPLE
Schedule pressure	Skipping a safety check or security review to hit a date
Suppliers and subcontractors	A vendor that does not meet the contract's standards
Weak security practice	Shared passwords, unpatched systems, test data from production
Lack of awareness	Staff untrained on a rule they must follow
Changing rules	A new regulation mid-project
New tools	Staff using unapproved software or AI with sensitive data

ON THE EXAM

Most compliance failures start as a threat nobody named. Identify threats, then treat them as risks.

Methods to support compliance

METHOD	HOW IT HELPS
Build into requirements and the definition of done	Compliance checked on every item, not at the end
Checklists and approvals	Required steps cannot be skipped quietly
Training and awareness	People know the rules that apply to their work
Security and privacy by design	Protection built into the solution from the start
Supplier clauses and audits	Obligations flow down to the supply chain, with evidence
Records and evidence	Proof of compliance, ready for auditors

ON THE EXAM

The outline asks you to use methods to support compliance. Build them into how work is done.

Security compliance

- **Secure by design**: security requirements from the start, not a test at the end.
- **Least privilege** access; classify data; remove access when people leave.
- Test security, for example with penetration testing, before release.
- Have an **incident response** plan: who does what if a breach happens.
- The outline's risk task names poor IT security as an example risk. Module 36 returns to it.

> ON THE EXAM
>
> Security is a compliance requirement and a risk. Design it in and test it.

Health and safety compliance

- Assess risks, plan safe methods, and use permits for hazardous work.
- Train people and supervise; anyone can stop unsafe work.
- Investigate near misses: they are free lessons.
- Safety is never traded against schedule or cost. Module 11: this is where forcing is right.

> ON THE EXAM
>
> When safety and schedule conflict, safety wins, immediately. Then re-plan the schedule.

Sustainability compliance

- **Environmental permits** and conditions: noise, emissions, waste, water.
- **Reporting obligations**: many organisations must report emissions and sustainability data.
- Organisational **commitments**, such as net zero targets, become project requirements.
- Sustainability is both an outline enabler here and a PMBOK 8 principle.

> ON THE EXAM
>
> Sustainability is now a compliance matter, not just a good intention. Treat it as a requirement.

Consequences of non-compliance

- **Legal**: fines, prosecution, loss of licence, contract termination.
- **Operational**: a system or site shut down, a launch blocked.
- **Financial**: rework, compensation, lost sales.
- **Human and reputational**: people harmed, trust lost.
- Analyse the consequences to **prioritise** effort: highest impact and likelihood first.

> ON THE EXAM
>
> The outline asks you to analyse consequences. That analysis tells you where compliance effort goes first.

Measuring compliance

- The outline asks you to **measure the extent to which the project is in compliance**.
- Audits and inspections, internal and independent.
- Compliance register status: obligations met, evidenced, outstanding.
- Indicators: incidents, near misses, audit findings open and closed.
- Traceability from each regulatory requirement to its evidence.

> ON THE EXAM
>
> Compliance is measured continuously, with evidence. 'We think we comply' is not a measure.

Compliance on agile projects

- Put compliance checks in the **definition of done**, so every increment is compliant.
- Add **compliance items** to the backlog, ordered like any other work.
- Involve regulators, auditors and quality specialists **early and often**.
- 'Agile' is never a reason to skip evidence that regulation requires.

> ON THE EXAM
>
> Agile and regulation work together when compliance is part of 'done'. Documentation that is required is still required.

When a compliance gap is found

- **Contain** it: stop the non-compliant activity or data use.
- **Report** it through the proper channel: compliance, data protection, safety. Regulators if required.
- **Fix** it, including the root cause, and verify the fix.
- Fixing it quietly, or 'after go-live', is concealment. It turns a gap into misconduct.

> ON THE EXAM
>
> Contain, report, fix. Honesty and responsibility from module 32 apply fully here.

Project Lantern: compliance

- **Data protection**: patient data rules; a privacy impact assessment before design was approved.
- **Security**: penetration testing before each clinic go-live; access removed at contract end.
- **Accessibility**: booking pages meet the public-sector accessibility standard.
- **Safety**: kiosk electrical safety certified; installers work under site permits.
- A compliance register, owned by the PM, reviewed monthly with Northgate's data protection officer.

> ON THE EXAM
>
> Each obligation named, owned, built in and evidenced, and reviewed with a specialist.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Unsure which rules apply in a new market	Confirm requirements with specialists; record them
A near miss or unsafe practice	Stop the work, investigate, fix before restarting
A data or security gap found before go-live	Contain, report, fix. Never quietly, never later
A supplier may not meet contract standards	Raise it under the contract; require evidence
Many obligations, little time	Prioritise by consequence and likelihood
Regulated work on an agile team	Compliance in the definition of done and backlog

ON THE EXAM

Confirm, build in, check, and act immediately on gaps. Compliance is never traded for schedule.

> A company is launching a connected home product in the UK, the EU and the USA. The team has experience only of UK rules.
>
> The design phase starts next week, and nobody has confirmed what regulations apply in the other markets.

What should the project manager do?

A Assume the same rules apply as in the UK, and adjust later if needed.

B Launch first, and respond to any regulator feedback afterwards.

C Engage legal and compliance specialists now to identify and record requirements for each market.

D Leave compliance decisions to the electronics supplier.

◆ PAUSE AND DECIDE

What should the project manager do?

A Assume UK rules apply.

Classic, costly assumption. Adjusting after design is expensive.

B Launch, then respond.

Risks products being blocked, fined or recalled.

C Specialists identify and record requirements now.

Confirms compliance requirements before design locks them out.

D Leave it to the supplier.

The organisation remains accountable; the supplier may not know all markets.

THE SIGNAL

Unknown obligations and design about to start: confirm requirements with specialists, now.

> On a construction site, a scaffold board slipped and nearly hit a worker below. Nobody was hurt. The project is two weeks behind schedule.
>
> The site supervisor suggests continuing work and investigating at the end of the month.

What should the project manager do?

A Stop the affected work, investigate the near miss, and correct the cause before restarting.

B Continue work, and investigate at the end of the month as suggested.

C Identify who placed the board and discipline them.

D Record the incident in the monthly safety report.

◆ PAUSE AND DECIDE

What should the project manager do?

A Stop, investigate, correct, then restart.

Safety overrides schedule. The near miss is a warning to act on now.

B Continue; investigate later.

Leaves the hazard in place for a month, under schedule pressure.

C Discipline the individual.

Blame before investigation; the cause may be the system, not the person.

D Just record it.

Recording without acting prevents nothing.

THE SIGNAL

A near miss is a free warning. Stop, investigate, fix. Schedule pressure changes nothing.

> A week before go-live, testers discover that customers' personal data is being stored unencrypted, contrary to the organisation's data protection policy.
>
> The sponsor asks whether the team could fix it quietly after go-live, to protect the launch date.

What should the project manager do?

A Fix it quietly after go-live, as the sponsor suggests.

B Go live, and add encryption to the next release.

C Record it as a known issue in the release notes.

D Contain the risk, report it to the data protection officer, fix it, and reassess the go-live.

◆ PAUSE AND DECIDE

What should the project manager do?

A Fix quietly later.

Concealment, and live customer data stays exposed meanwhile.

B Go live; encrypt next release.

Knowingly launches a policy breach with real personal data.

C Record as a known issue.

Documents the breach without containing or fixing it.

D Contain, report, fix, reassess go-live.

Handles the gap honestly and puts compliance before the date.

THE SIGNAL

A data compliance gap before launch: contain, report, fix. The date is re-decided, not the rule.

> The project's contract requires the supplier to use only security-vetted staff on site. The project manager learns the supplier has brought in a subcontractor whose staff have not been vetted.
>
> The subcontractor's work is going well.

What should the project manager do?

A Take no action, since the work is going well.

B Raise it with the supplier under the contract, require evidence of vetting, and assess the security risk.

C Terminate the supplier's contract immediately.

D Carry out background checks on the subcontractor's staff personally.

◆ PAUSE AND DECIDE

What should the project manager do?

A No action.

Good work does not make unvetted access compliant or safe.

B Raise under the contract; require evidence; assess risk.

Enforces the flowed-down obligation and manages the threat.

C Terminate immediately.

Disproportionate before the supplier has had a chance to comply.

D Vet them personally.

Not the PM's role, and may breach privacy rules itself.

THE SIGNAL

A supplier breaching a compliance clause: raise it under the contract, require evidence, manage the risk.

> A regulator has given a bank three months to meet thirty new obligations. The team cannot fully address all thirty in that time.
>
> The obligations vary widely in their penalties and in how likely a breach is.

What should the project manager do?

A Treat all thirty obligations equally and work through them in order.

B Complete the easiest obligations first, to show quick progress.

C Ignore the obligations with smaller penalties.

D Analyse the consequences and likelihood of each breach, prioritise with the compliance team, and agree the approach with the regulator if needed.

◆ PAUSE AND DECIDE

What should the project manager do?

A All equal, in order.

The highest-consequence obligations may be left until too late.

B Easiest first.

Shows activity, not reduced exposure.

C Ignore small-penalty ones.

Decides non-compliance without authority, analysis or agreement.

D Analyse consequences and likelihood; prioritise; agree.

The outline's enablers: analyse consequences, determine the approach.

THE SIGNAL

More obligations than time: prioritise by consequence and likelihood, with specialists and, if needed, the regulator.

> A team building regulated medical software uses Scrum. The quality and regulatory auditors are worried that agile means the required design documentation and traceability will be missed.
>
> The team wants to keep working in sprints.

What should the project manager do?

A. Build the required documentation and traceability into the definition of done and the backlog, and involve the auditors each sprint.

B. Switch the project to a predictive approach to satisfy the auditors.

C. Reduce documentation, since agile values working software over documentation.

D. Ask the auditors to relax the requirements for this project.

◆ PAUSE AND DECIDE

What should the project manager do?

A Compliance in the DoD and backlog; auditors involved.

Keeps agile, and makes every increment compliant and evidenced.

B Switch to predictive.

Unnecessary. Regulated agile works when compliance is built in.

C Reduce documentation.

Misreads the manifesto. Required documentation is required.

D Ask auditors to relax.

Regulatory requirements are not the auditors' to relax.

THE SIGNAL

Agile and regulation coexist when compliance is part of 'done'. Neither drop agile nor drop the evidence.

Confirm, build in, check, and act at once on gaps.

MODULE 34 · DOMAIN 3 - BUSINESS ENVIRONMENT

Change control across approaches

The change control process, who decides, change in agile and hybrid work, communicating decisions, implementing approved changes, and keeping records straight.

DOMAIN III - Business Environment EXAM WEIGHT 26% MODULE 34 of 49

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Change is welcome. Uncontrolled change is not.

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The change control process

01

Request

The change is written down and logged.

02

Analyse

Impact on scope, schedule, cost, quality, risk, resources.

03

Decide

The right authority approves, rejects or defers.

04

Communicate

Requester and those affected are told why.

05

Implement and update

Do it, verify it, update baselines and documents.

ON THE EXAM

Analysis comes before the decision, and before anyone does the work. That order is the exam's favourite test.

The first thing to do with a change request

- **Analyse the impact**, with the team, before anything else.
- Not: implement it because the requester is important.
- Not: reject it because it was not in the plan.
- Not: send it straight to the sponsor without analysis.
- Then submit it, with the analysis and options, to whoever has the authority to decide.

> ON THE EXAM
>
> 'A stakeholder asks for a change. What should the PM do first?' Almost always: assess the impact.

Kinds of change request

TYPE	PURPOSE
Corrective action	Bring performance back in line with the plan
Preventive action	Stop a future deviation from the plan
Defect repair	Fix a product that does not meet requirements
Update or enhancement	Change scope, requirements, plans or baselines

ON THE EXAM

All four go through change control when they affect baselines. Small corrective actions may sit within the PM's tolerance.

Who decides

- Governance sets the **authority levels**: the tolerance table from module 31.
- A **change control board** decides changes beyond the project manager's authority.
- The project manager facilitates: analyses, recommends, and records the decision.
- Approving a change above your tolerance, however sensible, is acting beyond authority.

> ON THE EXAM
>
> The PM runs the process and recommends. The decision belongs to whoever governance names.

Predictive or adaptive change?

PREDICTIVE

Formal change control

- Scope is baselined early
- Changes go through a change request
- Board or sponsor decides
- Baselines updated after approval

ADAPTIVE

The backlog absorbs change

- Scope is expected to evolve
- New items go into the product backlog
- The product owner orders them
- Time and cost stay fixed; scope flexes

THE TIEBREAKER

In agile work, the backlog is the change mechanism for product scope. No change request is needed for that.

Change on hybrid projects

- Inside a release boundary, the **product owner** reorders the backlog freely.
- Changes that affect a **milestone, budget, contract or governance baseline** go through formal change control.
- Agree the boundary at the start: what the product owner may decide, and what needs the board.
- Formal change requests for every story, or no control at all over budgets, are both wrong.

> ON THE EXAM
>
> On hybrid work, ask: does this change stay inside the agreed boundary? Inside: backlog. Outside: change control.

Keeping the records straight

RECORD	WHAT IT HOLDS
Change log	Every request: description, requester, date, analysis, decision, status
Configuration management	Which version of each deliverable and document is current
Version control	History of changes to code, designs and documents
Baselines	The approved scope, schedule and cost, as updated by approved changes

ON THE EXAM

Change control decides; configuration management keeps track of what exists. Both are needed.

Communicating the status of changes

- The outline asks you to **communicate the status of proposed changes**.
- Tell the requester the decision **and the reason**, promptly, whatever it is.
- Tell everyone affected: the team, suppliers, stakeholders whose work changes.
- Keep the change log visible, so status is easy to find.
- A rejected change nobody explained comes back, often as a request to a team member directly.

> ON THE EXAM
>
> Every decision, approve, reject or defer, is communicated with its reason.

Implementing approved changes

- The outline asks you to **implement approved changes to the project**.
- Plan the work, assign it, and schedule it like any other work.
- Verify and test the change; check nothing else broke.
- Only approved changes are implemented. Work started early is unauthorised, even if later approved.
- Genuine emergencies follow an agreed **emergency change** route, recorded and reviewed afterwards.

> ON THE EXAM
>
> Approval first, then implementation. Emergency changes still get recorded and reviewed.

Updating documentation

- The outline's fourth change enabler: **update project documentation** to reflect changes.
- Baselines: scope, schedule, cost.
- Plans and registers: risk register, requirements traceability, stakeholder register.
- Contracts: amended through their formal change clauses.
- An approved change that never reaches the documents leaves the team working from the old version.

> ON THE EXAM
>
> A change is not finished until every affected document and baseline reflects it.

When there are too many changes

- A flood of change requests is a **symptom**, not just a workload.
- Look for the root cause: unclear requirements, missing stakeholders, or the wrong approach.
- Fix the cause: better elicitation, engaging the missing stakeholder, or moving that work to an adaptive approach.
- Freezing all change stops learning. Processing changes faster treats the symptom.

> ON THE EXAM
>
> Many changes in one area: find out why. Module 03's reporting workstream is the classic case.

Project Lantern: change control

- **App**: the product owner reorders the backlog freely within each release.
- **Boundary**: anything moving a clinic go-live, the budget or the kiosk contract goes to the change board.
- **Change log** visible to all; every decision emailed to the requester with the reason.
- When the regulator added a new consent screen, it was analysed, approved by the board, then built and documented.

> ON THE EXAM
>
> A backlog for product change, a board for baseline change, and a visible log for both.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
A stakeholder requests a change; what first?	Analyse the impact
The board has approved a change	Implement, verify, update baselines and documents, communicate
A change was made without approval	Assess it, take it through change control, coach the team
New story on an agile team, within budget and time	Product owner adds and orders it; no change request
A rejected requester keeps asking	Communicate the decision and the reason
A flood of change requests	Find the root cause

ON THE EXAM

Analyse, decide at the right level, communicate, implement, update. And ask why when changes pile up.

> On a predictive project to build a warehouse management system, the head of logistics asks the project manager to add a barcode-scanning feature that was not in the scope baseline.
>
> She says it is important and should be easy.

What should the project manager do first?

A Ask the team to add the feature, since the head of logistics is a key stakeholder.

B Analyse the impact of the change on scope, schedule, cost and risk with the team.

C Reject the request, since it is not in the scope baseline.

D Send the request straight to the sponsor for a decision.

◆ PAUSE AND DECIDE

What should the project manager do first?

A Just add it.

Uncontrolled change. 'Should be easy' is an assumption, not an analysis.

B Analyse the impact first.

The first step of change control, before any decision.

C Reject it.

Refuses a possibly valuable change without assessing it.

D Straight to the sponsor.

A decision-maker needs the analysis first; it may even be within the PM's tolerance.

THE SIGNAL

A change request, and the question says 'first': analyse the impact.

> The change control board has approved a change to add a second data centre for resilience, with an increase to budget and a four-week extension.
>
> The project manager receives the approval.

What should the project manager do next?

A Close the change request in the change log.

B Re-analyse the change to confirm the board's decision.

C Implement the change, update the baselines and project documents, and communicate the decision.

D Ask the sponsor to sign the change as well.

◆ PAUSE AND DECIDE

What should the project manager do next?

A Close the request.

Premature. Nothing has been implemented or updated yet.

B Re-analyse.

The analysis was done before the decision. Repeating it adds delay.

C Implement, update, communicate.

The outline's enablers after approval: implement, update documents, communicate status.

D Get the sponsor to sign too.

The board had the authority. An extra signature is unnecessary.

THE SIGNAL

After approval: implement, update baselines and documents, and tell everyone affected.

> During a review, the project manager finds that a developer changed a report layout at a customer's direct request. The change was not submitted or approved.
>
> The customer is pleased, but the change affects two other reports that depend on the layout.

What should the project manager do?

A Remove the change immediately and restore the original layout.

B Leave the change in place, since the customer is pleased.

C Formally reprimand the developer.

D Assess the impact, take the change through change control, and coach the team on the process.

◆ PAUSE AND DECIDE

What should the project manager do?

A Remove it at once.

Reacts without analysis, and frustrates a pleased customer.

B Leave it.

Ignores the impact on two dependent reports, and rewards bypassing the process.

C Reprimand the developer.

Blame. Coaching fixes the behaviour; the change still needs deciding.

D Assess, take through control, coach.

Brings the change under control and prevents the next one.

THE SIGNAL

An unapproved change found: assess it, put it through the process now, and coach so it does not recur.

> On an agile product team, the product owner wants to add a new user story to the backlog after feedback at a sprint review. It fits within the release's agreed budget and date.
>
> It does not affect any contract or milestone.

How should the change be handled?

A The product owner adds the story to the backlog and orders it against other items.

B Submit a formal change request to the change control board.

C Refuse the story until the next release is planned.

D Ask the sponsor to approve the new story.

◆ PAUSE AND DECIDE

How should the change be handled?

A The PO adds and orders it.

Inside the agreed boundary, the backlog is the change mechanism.

B Formal change request.

Predictive machinery for a change that stays inside the release boundary.

C Refuse until next release.

Rejects exactly the feedback-driven change agile is designed for.

D Sponsor approval.

Product priorities within the release belong to the product owner.

THE SIGNAL

Agile, inside budget and date, no contract impact: the product owner uses the backlog. No change request.

> Three weeks ago, the change control board rejected a request from the marketing manager. Nobody told her the decision.
>
> She has since asked two developers directly when her change will be ready.

What should the project manager do?

A Ignore it; the developers can tell her it was rejected.

B Tell the marketing manager the decision and the reasons, and make sure the change log shows its status.

C Ask the developers to implement it quietly to keep her happy.

D Resubmit the change to the board on her behalf.

◆ PAUSE AND DECIDE

What should the project manager do?

A Leave it to the developers.

Pushes a governance message onto the team, without the reasons.

B Communicate the decision and reasons; update the log.

The outline's enabler: communicate the status of proposed changes.

C Implement it quietly.

Overrides the board. Unauthorised change.

D Resubmit it for her.

Nothing new has been offered. She has not even heard the decision.

THE SIGNAL

An unexplained decision comes back through side channels. Communicate every decision, with its reason.

> In one month, a predictive project has received forty change requests. Most are clarifications of reporting requirements from the finance department, which was consulted only briefly during planning.
>
> The change control board is overwhelmed.

What should the project manager do?

A Process the requests faster by holding extra board meetings.

B Freeze all changes until the project is complete.

C Address the root cause: engage finance properly to clarify its reporting requirements, and consider an adaptive approach for that area.

D Hire more business analysts to write the change requests.

◆ PAUSE AND DECIDE

What should the project manager do?

A More board meetings.
Treats the symptom; the flood continues.

B Freeze all change.
Stops learning; finance gets reports it cannot use.

C Fix the cause: engage finance; consider adaptive.
Finds why the changes happen, and stops the flood at its source.

D More analysts.
Writes the change requests faster. The cause remains.

THE SIGNAL

A flood of changes from one group: a missed stakeholder or the wrong approach. Fix the cause.

Analyse, decide at the right level,
communicate, implement, update.

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MODULE 35 · DOMAIN 3 - BUSINESS ENVIRONMENT

Impediments and issues

Risk or issue, surfacing blockers, evaluating and prioritising them, choosing an intervention, workarounds, and continual reassessment.

DOMAIN III - Business Environment EXAM WEIGHT 26% MODULE 35 of 49

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An issue is happening **now**. Clear the path quickly, at the right level, then find out why.

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Risk or issue?

RISK

Might happen

- An uncertain future event
- Recorded in the risk register
- Managed with a planned response
- Can be a threat or an opportunity

ISSUE

Is happening

- A current problem or condition
- Recorded in the issue log
- Resolved, or worked around
- Needs action now

THE TIEBREAKER

When an identified risk occurs, it becomes an issue. Use its planned response first.

Impediments, obstacles and blockers

KIND	EXAMPLES
Waiting on others	An approval, a decision, another team's deliverable
Tools and environments	A test environment down, missing access or licences
Information	Unclear requirements, a missing answer from a stakeholder
People	A key person unavailable, a skills gap
Organisational	A policy, process or rule slowing the work

ON THE EXAM

Most impediments are outside the team's control. That is why removing them is a leader's job.

Handling an impediment

01
Surface
Make it visible, early.

02
Evaluate
What does it block, and how much does that matter?

03
Prioritise
Highlight the ones that matter most.

04
Intervene
Choose and apply a strategy.

05
Reassess
Check it is really cleared; review the rest.

ON THE EXAM

This is the outline's enablers in order: evaluate, prioritise and highlight, intervene, reassess.

Surfacing impediments early

- The **daily coordination meeting** asks directly: is anything in your way?
- Mark blocked items visibly on the **board**, with how long they have been blocked.
- Keep an **issue log** for problems beyond a single day or a single team.
- People raise blockers only when it is **safe** to admit being stuck.

> ON THE EXAM
>
> A blocker hidden for three days costs three days. Make raising it normal and safe.

Evaluating impact and prioritising

- Is it on the **critical path**, or does it threaten the **sprint or release goal**?
- How much work, and how many people, does it block?
- What does each day of delay cost?
- Highlight the top items to those who can act: the team, the sponsor, other managers.

> ON THE EXAM
>
> The outline asks you to prioritise and highlight. Fix what blocks the goal first, not what was raised first.

The issue log

FIELD	PURPOSE
Description and date raised	What the issue is, and since when
Impact and priority	What it affects, and how urgent it is
Owner	One named person responsible for resolving it
Actions and due date	What will be done, and by when
Status and resolution	Open, in progress, resolved, and how

ON THE EXAM

Every issue has one owner and a due date. An issue owned by 'the team' is owned by nobody.

Intervention strategies

STRATEGY	WHEN IT FITS
Team resolves it	The team has the skills and authority; the leader supports
Leader removes it	The blocker sits outside the team, within the leader's reach
Negotiate	Another team or manager controls what is needed
Workaround	An unplanned problem needs a fast, temporary fix
Escalate	Resolution needs authority beyond the project manager's
Accept and re-plan	It cannot be removed; adjust the plan and inform stakeholders

ON THE EXAM

The outline asks you to determine and apply an intervention strategy. Match it to where the blocker sits.

Planned response or workaround?

PLANNED RISK RESPONSE

Prepared in advance

- For an identified risk
- Written in the risk register
- Often funded from contingency
- Use it first when the risk occurs

WORKAROUND

Improvised now

- For an unplanned problem
- No response existed
- Fast and temporary
- Log it, then fix the root cause

THE TIEBREAKER

Risk occurred and a response exists: use it. Nothing was planned: work around it, then learn.

When a risk becomes an issue

- Check the **risk register**: was this risk identified, with a response planned?
- If yes, **implement the planned response**, and its contingency if funded.
- Move it to the issue log, and update the risk register: this risk has occurred.
- Check for **secondary risks** the response itself creates.

> ON THE EXAM
>
> 'The risk has occurred. What next?' Consult the register and apply the planned response.

Collaborating to resolve issues

- The outline asks you to **collaborate with relevant stakeholders** on the approach to resolve issues.
- Bring in the people who can actually resolve it, and those it affects.
- Agree the approach together, rather than imposing one.
- Keep affected stakeholders informed until it is closed.

> ON THE EXAM
>
> Issues are resolved with the people who hold the key, not around them.

Reassessing continually, and learning

- The outline asks you to **reassess continually** that impediments are being addressed.
- Review open issues on a rhythm: daily for blockers, weekly for the issue log.
- Watch **ageing**: an issue open for weeks with no progress needs a new strategy or escalation.
- After resolution, find the **root cause** and capture the lesson, so it does not recur.

> ON THE EXAM
>
> An issue log full of old, ownerless items is not issue management. Review, act, close, learn.

Project Lantern: issues in practice

- The risk register had foreseen a kiosk shipping delay; the planned response, a loan kiosk, was used at once.
- An unforeseen network outage at clinic four: a mobile router workaround, logged, then a root-cause fix.
- The app team was blocked by a firewall change; the project manager went to the security lead that day.
- The issue log is reviewed weekly; anything open over two weeks goes to the sponsor.

> ON THE EXAM
>
> Planned response for a foreseen risk; workaround for a surprise; leader removes the external blocker; ageing rule.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
An identified risk has occurred	Consult the register; apply the planned response
An unforeseen problem stops work	Workaround; log it; inform; fix the root cause
Team blocked by another department	Leader engages that department; escalate if needed
Several issues, limited capacity	Prioritise by impact on the goal or critical path
An old, ownerless issue log	Review, assign owners and dates, close or escalate
Someone hid a blocker for days	Help now; build safety to raise blockers early

ON THE EXAM

Planned response if it exists; workaround if not; the leader removes what the team cannot.

> The risk register identified that the company's own test laboratory might be unavailable during product testing. The planned response was to rent an external laboratory, funded from contingency.
>
> The internal laboratory has just been closed for three weeks for repairs.

What should the project manager do next?

A Implement the planned response and rent the external laboratory, then update the risk register and issue log.

B Create a new plan for testing, since the situation has changed.

C Escalate the closure to the sponsor for a decision.

D Record the event in the lessons learned register.

◆ PAUSE AND DECIDE

What should the project manager do next?

A Apply the planned response; update records.
The risk was foreseen and a funded response exists. Use it.

B Make a new plan.
Throws away the response prepared for exactly this.

C Escalate to the sponsor.
The response is already planned and funded, within the PM's authority.

D Lessons learned.
Useful later, but it does nothing to keep testing going.

THE SIGNAL

An identified risk has occurred and its response is planned: apply it. That is what planning was for.

> A shared build server fails, stopping three developers from deploying their work. Nothing in the risk register covers this.
>
> The infrastructure team says a permanent fix will take two days.

What should the project manager do?

A Wait for the infrastructure team's permanent fix.

B Look for the planned response in the risk register.

C Agree a workaround with the team, log the issue, inform affected stakeholders, and follow up on the root cause.

D Escalate the failure to the sponsor immediately.

◆ PAUSE AND DECIDE

What should the project manager do?

A Wait two days.

Three developers idle for two days when a workaround may exist.

B Check the risk register.

The question says nothing there covers it. There is no planned response.

C Workaround, log, inform, root cause.

An unplanned issue: keep work flowing now, then fix it properly.

D Escalate immediately.

A two-day infrastructure fault is within normal project management.

THE SIGNAL

No planned response exists: work around it, log it, inform people, then fix the cause.

> An agile team has been blocked for three days, waiting for the corporate security team to open a firewall port. The request is sitting in the security team's general queue.
>
> The sprint goal is now at risk.

What should the project leader do?

A Let the developers disable the firewall on their test machines to keep working.

B Wait for the security team to reach the request in its queue.

C Raise the delay at the next sprint retrospective.

D Contact the security team lead directly to resolve it, escalating if needed, and keep the blocker visible.

◆ PAUSE AND DECIDE

What should the project leader do?

A Disable the firewall.

Breaches security policy. A workaround that creates a compliance problem.

B Wait in the queue.

Three days lost already, with the sprint goal at risk.

C Raise it at the retrospective.

Useful for learning, but far too late to unblock the sprint.

D Go to the security lead; escalate if needed.

The leader removes a blocker outside the team's control, now.

THE SIGNAL

A blocker outside the team, threatening the goal: the leader removes it, today.

> A project manager has five open issues and time to work on only one or two today. One is blocking a critical path activity. The others are a formatting problem, a meeting-room clash, a query about expenses, and a minor tool licence renewal.
>
> The oldest is the expenses query.

Which issue should the project manager address first?

A The expenses query, because it is the oldest.

B The issue blocking the critical path activity.

C The formatting problem, because it is quickest to fix.

D Whichever issue was raised by the most senior person.

◆ PAUSE AND DECIDE

Which issue should the project manager address first?

A The oldest.
Age is not impact. The expenses query delays nothing important.

B The critical path blocker.
Every day it stays open delays the whole project.

C The quickest.
Feels productive; changes nothing that matters.

D The most senior requester.
Priority comes from impact, not rank.

THE SIGNAL

Prioritise issues by impact on the goal or critical path. Not by age, ease or rank.

> A new project manager inherits an issue log with thirty open issues. Many are months old, most have no owner, and nobody is sure which are still real.
>
> Stakeholders have stopped reading it.

What should the project manager do?

A Archive the whole log and start a new one.

B Add any new issues to the existing log and carry on.

C Review each issue with the team: close the resolved ones, assign owners and dates, escalate stuck critical ones, and set a regular review.

D Report the thirty open issues to the sponsor as they stand.

◆ PAUSE AND DECIDE

What should the project manager do?

A Archive it and start again.
Some of those issues may still be real, and now forgotten.

B Carry on adding to it.
Adds to a log nobody trusts or reads.

C Review, close, assign, escalate, set a rhythm.
Restores a useful log and the reassessment the outline asks for.

D Report it as it stands.
Passes on noise; the sponsor cannot act on stale, ownerless items.

THE SIGNAL

A stale, ownerless issue log: clean it up with the team, then keep it alive with a regular review.

> A developer has been stuck on an integration problem for three days without telling anyone. When asked, she says she did not want to look incompetent in front of the team.
>
> The team's daily meetings have become brief status reports to the project leader.

What should the project leader do?

A. Help her with the blocker now, and work with the team to make raising blockers early normal and safe.

B. Remind her firmly that blockers must be reported within a day.

C. Reassign the integration work to a more experienced developer.

D. Add daily progress reports so hidden problems are caught.

◆ PAUSE AND DECIDE

What should the project leader do?

A Help now; make raising blockers safe.
Clears this blocker and fixes why blockers stay hidden.

B A firm reminder.
Adds to the fear that caused the silence.

C Reassign the work.
Confirms her fear of looking incompetent.

D More progress reports.
More status reporting, when status meetings are part of the problem.

THE SIGNAL

Hidden blockers come from fear. Remove today's blocker, then remove the fear.

Planned response if it exists. **Workaround** if not. Remove what the team cannot.

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MODULE 36 · DOMAIN 3 - BUSINESS ENVIRONMENT

Risk I: plan, identify, analyse

Threats and opportunities, appetite and thresholds, finding and recording risks, qualitative analysis, EMV, decision trees and Monte Carlo.

DOMAIN III - Business Environment EXAM WEIGHT 26% MODULE 36 of 49

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A risk is uncertainty that matters. It can be good news too.

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Threat or opportunity?

THREAT

Uncertainty that could harm

- A supplier may deliver late
- Data may be poorer than expected
- A key person may leave
- Aim: reduce probability or impact

OPPORTUNITY

Uncertainty that could help

- A component may arrive early
- A new tool may speed testing
- A grant may become available
- Aim: increase probability or impact

THE TIEBREAKER

Identify opportunities as deliberately as threats. Both go in the risk register.

The risk management plan

- **Approach, roles and timing**: how risk is managed, by whom, and how often.
- **Categories**: often a risk breakdown structure, such as technical, external, organisational.
- **Definitions** of probability and impact, so everyone scores the same way.
- **Appetite and thresholds**, and how risks are escalated.
- **Funding** for risk responses and reserves, and how risk is reported.

> ON THE EXAM
>
> The outline's first risk enabler is to develop this plan. It makes risk management consistent.

Risk appetite or risk threshold?

RISK APPETITE

How much risk, in general

- The broad level of uncertainty the organisation will accept
- Set by leadership
- Qualitative: cautious, balanced, bold
- Shapes the whole approach

RISK THRESHOLD

The line in a specific measure

- A measurable limit on a specific objective
- 'Any risk over $100,000 impact'
- Triggers escalation or a response
- Turns appetite into action

THE TIEBREAKER

Appetite is the attitude; threshold is the number. A risk beyond a threshold is escalated, whatever its probability.

Identifying risks

TECHNIQUE	HOW IT HELPS
Brainstorming and workshops	Many perspectives, many risks, quickly
Interviews and expert judgement	Depth from people who have seen it before
Checklists and prompt lists	Categories so nothing obvious is missed
Assumption and constraint analysis	Every assumption is a potential risk
SWOT analysis	Strengths and weaknesses inside; opportunities and threats outside
Lessons learned and pre-mortem	What went wrong before; what could make this fail

ON THE EXAM

Identify risks with the team and stakeholders, early and throughout. Never alone at a desk.

Writing a clear risk statement

- **Cause**, **risk**, **effect**: 'Because X, Y may happen, which would lead to Z.'
- 'Because the old booking data has never been cleaned, migration errors may exceed five percent, delaying clinic go-live by up to three weeks.'
- 'Data problems' is not a risk statement. Nobody can score it or respond to it.
- A clear statement makes the probability, impact and response obvious.

> ON THE EXAM
>
> Cause, risk, effect. Vague risks get vague responses.

A risk register entry

FIELD	EXAMPLE: POOR IT SECURITY
Risk statement	Shared supplier admin passwords may let an attacker reach patient data
Category and owner	Security; IT security lead
Probability and impact	Medium; very high
Trigger	A shared account found in an access review
Response and status	Individual accounts, multi-factor login; in progress

ON THE EXAM

The outline names poor IT security as a register example. Each risk: an owner, a trigger, a response.

Qualitative analysis

- Score each risk's **probability** and **impact** using the plan's definitions.
- Combine them in a **probability and impact matrix** to rank risks.
- Consider **urgency** and **proximity**: how soon it could happen, and how soon you must act.
- Check data quality: a score based on a guess should be treated as uncertain itself.
- Quick, cheap, and done for every risk. It decides which few deserve more.

> ON THE EXAM
>
> Qualitative analysis first, for all risks. Quantitative analysis later, only for the few that justify it.

A probability and impact matrix

PROBABILITY	LOW IMPACT	MEDIUM IMPACT	HIGH IMPACT
High	Medium priority	High priority	High priority
Medium	Low priority	Medium priority	High priority
Low	Low priority	Low priority	Medium priority

ON THE EXAM

High probability and high impact first. But a low-probability, very high-impact risk still needs a response.

Expected monetary value

- **EMV = probability x impact**. Threats negative, opportunities positive.
- Threat: 30% x −$50,000 = **−$15,000**. Opportunity: 20% x +$40,000 = **+$8,000**.
- Together: −$15,000 + $8,000 = **−$7,000**.
- Summed across risks, EMV helps size the contingency reserve (module 26).

> ON THE EXAM
>
> Keep the signs: threats subtract, opportunities add. Most EMV errors are sign errors.

A decision tree: build or buy?

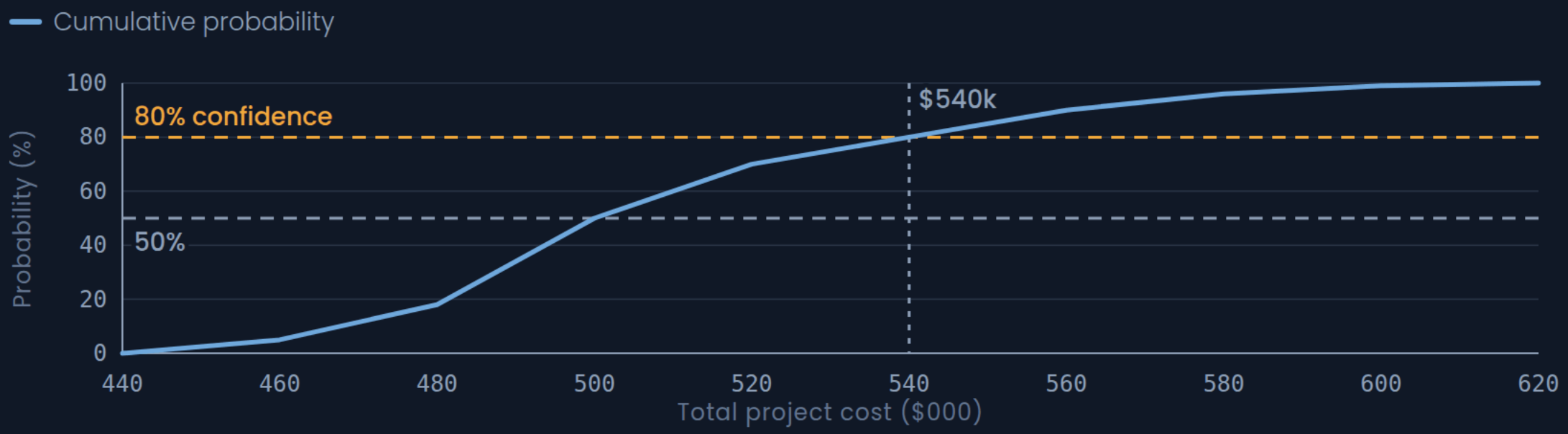
OPTION	COST	OUTCOMES	EMV
Build	-$200k	60% chance of $500k benefit; 40% chance of $300k	-200 + 300 + 120 = $220k
Buy	-$300k	90% chance of $450k benefit; 10% chance of $250k	-300 + 405 + 25 = $130k

ON THE EXAM

Compute EMV for each branch, including its cost. Choose the highest. Here: build, at $220k.

A Monte Carlo result

Probability of finishing at or below each cost



ON THE EXAM

Read across from the confidence level, then down to the cost. P80 here is $540,000.

Sensitivity analysis and quantitative choices

- **Sensitivity analysis** shows which risks move the outcome most, often as a **tornado diagram**.
- The widest bar at the top is the risk to manage first.
- Use quantitative analysis when the stakes, size or uncertainty justify the effort.
- Running simulations on every small project is effort without value.

> ON THE EXAM
>
> Qualitative for everything, quantitative for the few that matter, sensitivity to find the biggest drivers.

Risk on agile projects

- Short iterations and frequent reviews **reduce** risk by exposing problems early.
- A **risk-adjusted backlog** puts risk-reducing work, such as spikes, near the top.
- **Spikes**: timeboxed experiments to answer a risky question before committing.
- Risks are discussed in planning, daily meetings and retrospectives, not only in a register.

> ON THE EXAM
>
> Agile manages risk by learning fast. Risky, uncertain work goes early, not last.

Project Lantern: identifying and analysing risk

- A risk workshop with clinic managers, IT, the supplier and a patient representative.
- Forty-two risks identified, including three **opportunities**, such as a regional grant for digital health.
- Qualitative scoring for all; **Monte Carlo** on the migration and kiosk costs only.
- The contingency reserve was set at the P80 figure, and the steering committee agreed it.

> ON THE EXAM
>
> Wide identification, opportunities included, quantitative analysis only where it mattered.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Probabilities and impacts; total EMV?	Sum probability x impact; threats negative
A risk exceeds the plan's threshold	Escalate as the plan says, even if unlikely
Choose between options with uncertain outcomes	Decision tree: highest EMV, costs included
Many risks, little time	Qualitative first; quantitative for the top few
PM wrote the risk register alone	Identify with the team and stakeholders
Budget wanted at a confidence level	Read the Monte Carlo curve at that level

ON THE EXAM

Find risks widely, write them clearly, score them all, and quantify the few that matter.

> A project has three identified threats: a 20% chance of a $50,000 loss, a 10% chance of a $100,000 loss, and a 50% chance of a $10,000 loss.
>
> It also has one opportunity: a 25% chance of a $40,000 saving.

What is the project's total expected monetary value from these risks?

A −$25,000

B −$15,000

C −$35,000

D +$15,000

◆ PAUSE AND DECIDE

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What is the project's total expected monetary value from these risks?

A -$25,000

The threats only. The opportunity was left out.

B -$15,000

-10,000 - 10,000 - 5,000 + 10,000 = -15,000.

C -$35,000

Treats the opportunity as another loss. A sign error.

D +$15,000

The right size with the wrong sign.

ON THE EXAM

Threats negative, opportunities positive, then add. Every distractor is a sign or omission error.

> The risk management plan says any risk with a potential impact above $100,000 must be reported to the steering committee. The team identifies a risk with a 5% probability and a $150,000 impact.
>
> Its expected monetary value is only $7,500.

What should the project manager do?

A Accept the risk without reporting it, because its EMV is small.

B Leave it out of the register, because it is unlikely.

C Add $7,500 to contingency and take no further action.

D Record it and report it to the steering committee, because its impact exceeds the threshold.

◆ PAUSE AND DECIDE

What should the project manager do?

A Accept quietly: small EMV.

The threshold is on impact, not EMV. The plan requires reporting.

B Leave it out.

Unlikely risks still belong in the register.

C Add contingency only.

May be part of the response, but skips the required escalation.

D Record and report per the threshold.

Follows the risk management plan, whatever the probability.

THE SIGNAL

A threshold defined on impact is crossed by impact alone. Follow the plan and escalate.

> A team must choose between building a component or buying one. Building costs $200,000, with a 60% chance of $500,000 benefit and a 40% chance of $300,000.
>
> Buying costs $300,000, with a 90% chance of $450,000 benefit and a 10% chance of $250,000.

Based on expected monetary value, which option should be chosen?

A. Build, with an EMV of $220,000.

B. Buy, with an EMV of $130,000.

C. Buy, because it has the higher chance of success.

D. Build, because it has the lower upfront cost.

◆ PAUSE AND DECIDE

Based on expected monetary value, which option should be chosen?

A Build, EMV $220,000.
-200 + (0.6 x 500) + (0.4 x 300) = 220. The higher EMV.

B Buy, EMV $130,000.
Correct figure for buy, but lower than build.

C Buy: higher success chance.
Probability alone ignores both cost and the size of benefits.

D Build: lower upfront cost.
Right choice, wrong reason. Cost alone ignores benefits.

THE SIGNAL

Compute each branch's EMV including its cost, and pick the highest. The question said 'based on EMV'.

> A risk workshop has identified sixty risks for a large infrastructure project. The steering committee meets in a week.
>
> The team has a Monte Carlo tool but limited time to prepare data.

What should the project manager do next?

A Run a quantitative analysis on all sixty risks.

B Plan detailed responses for all sixty risks straight away.

C Score all sixty qualitatively to prioritise them, then analyse the top few quantitatively.

D Focus only on the risks with the highest impact, whatever their probability.

◆ PAUSE AND DECIDE

What should the project manager do next?

A Quantify all sixty.
Far too much effort; most risks do not justify it.

B Respond to all sixty.
Skips analysis; effort goes equally to trivial and serious risks.

C Qualitative for all; quantitative for the top few.
The standard sequence, and it fits the week available.

D Highest impact only.
Ignores probability, the other half of the score.

THE SIGNAL

Qualitative analysis first, for everything. Then quantify only what matters most.

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> A project manager has written the risk register alone over a weekend, using a template from a previous project.
>
> The team and stakeholders have not yet been involved. The register is due to go to the sponsor.

What should the project manager do before submitting it?

A Submit it, since it follows the organisation's template.

B Hold a risk identification workshop with the team and key stakeholders, and update the register.

C Copy in the previous project's risks as well, to be thorough.

D Wait until issues arise, then add them to the register.

◆ PAUSE AND DECIDE

What should the project manager do before submitting it?

A Submit it.

One person's view misses the risks others can see.

B Workshop with team and stakeholders.

Identification is a group activity. It finds far more, and builds ownership.

C Copy old risks.

Useful input, but no substitute for this project's people.

D Wait for issues.

Turns risk management into issue management.

THE SIGNAL

Risks are identified with the team and stakeholders. A register written alone is half a register.

> A Monte Carlo simulation of a project's cost shows a 50% probability of finishing at or below $500,000, an 80% probability at or below $540,000, and a 100% probability at or below $600,000.
>
> The sponsor wants a budget with 80% confidence.

What budget should the project manager recommend?

A $500,000

B $600,000

C $480,000

D $540,000

◆ PAUSE AND DECIDE

What budget should the project manager recommend?

A $500,000

Only 50% confidence: a coin toss.

B $600,000

Near-certainty. More reserve than the sponsor asked for.

C $480,000

Below the median; well under 50% confidence.

D $540,000

The P80 figure: 80% chance of finishing at or below it.

THE SIGNAL

Read the simulation at the confidence level asked for. P80 here is $540,000.

Find risks widely. Write them clearly. **Score them all**, and quantify the few that matter.

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MODULE 37 · DOMAIN 3 - BUSINESS ENVIRONMENT

Risk II: respond, monitor, communicate

Response strategies for threats and opportunities, contingency and fallback plans, residual and secondary risks, monitoring, and reporting risk status.

DOMAIN III - Business Environment EXAM WEIGHT 26% MODULE 37 of 49

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A risk response is a decision taken before you need it.

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Strategies for threats

STRATEGY	WHAT IT DOES	EXAMPLE
Escalate	Pass it to the level that owns it	A regulatory change affecting the whole programme
Avoid	Remove the threat entirely	Drop a risky feature; change the approach
Transfer	Shift the impact to a third party	Insurance, a fixed-price contract, a warranty
Mitigate	Reduce probability or impact	Extra testing, a prototype, a backup supplier
Accept	Acknowledge it; act only if it occurs	Active: set aside contingency. Passive: do nothing

ON THE EXAM

Transfer moves the impact, not the risk itself. Someone else pays, usually for a price.

Strategies for opportunities

STRATEGY	WHAT IT DOES	EXAMPLE
Escalate	Pass it to the level that owns it	A benefit the wider organisation should capture
Exploit	Make sure it happens	Best people on it to secure an early bonus
Share	Partner with someone better placed	A joint venture; shared-savings clause
Enhance	Increase probability or impact	Add resources to raise the chance of finishing early
Accept	Take it if it comes	Welcome a lower price if the market falls

ON THE EXAM

Exploit mirrors avoid, share mirrors transfer, enhance mirrors mitigate.

Choosing a response

- **Proportionate**: the response should cost less than the risk it addresses.
- **Owned**: one named risk owner, with the authority to act.
- **Agreed** with stakeholders and funded, if it needs money.
- **Timely** and **realistic**: achievable in the time the risk allows.
- A $50,000 response to a risk with an EMV of $5,000 is poor value, unless a threshold demands it.

> ON THE EXAM
>
> The best response is the cheapest one that brings the risk within appetite, with an owner who can act.

Contingency plans, fallback plans and triggers

- A **contingency plan**: what to do if the risk occurs. Prepared in advance.
- A **fallback plan**: what to do if the contingency plan does not work.
- A **trigger**: the warning sign that a risk is about to occur, or has occurred.
- Trigger: supplier misses two delivery dates in a row. Contingency: switch to the backup supplier. Fallback: build in-house.

> ON THE EXAM
>
> When the trigger fires, the owner implements the contingency plan. No new analysis, no waiting.

Residual risk or secondary risk?

RESIDUAL RISK

What is left over

- Remains after a response
- Expected and accepted
- Often covered by contingency
- Monitored, not re-solved

SECONDARY RISK

A new risk the response creates

- Caused by the response itself
- Night shifts to recover time cause fatigue
- Must be identified and analysed
- Gets its own response

THE TIEBREAKER

Every response can leave residual risk and create secondary risk. Check for both.

Executing the risk management plan

- The outline asks you to **execute the risk management plan**, naming two examples.
- **Security responses**: access controls, patching, an incident response plan, rehearsed.
- **Sustainability risks**: supply chain disruption, climate impacts on sites, new environmental rules.
- Owners carry out agreed responses on time, and report progress and results.

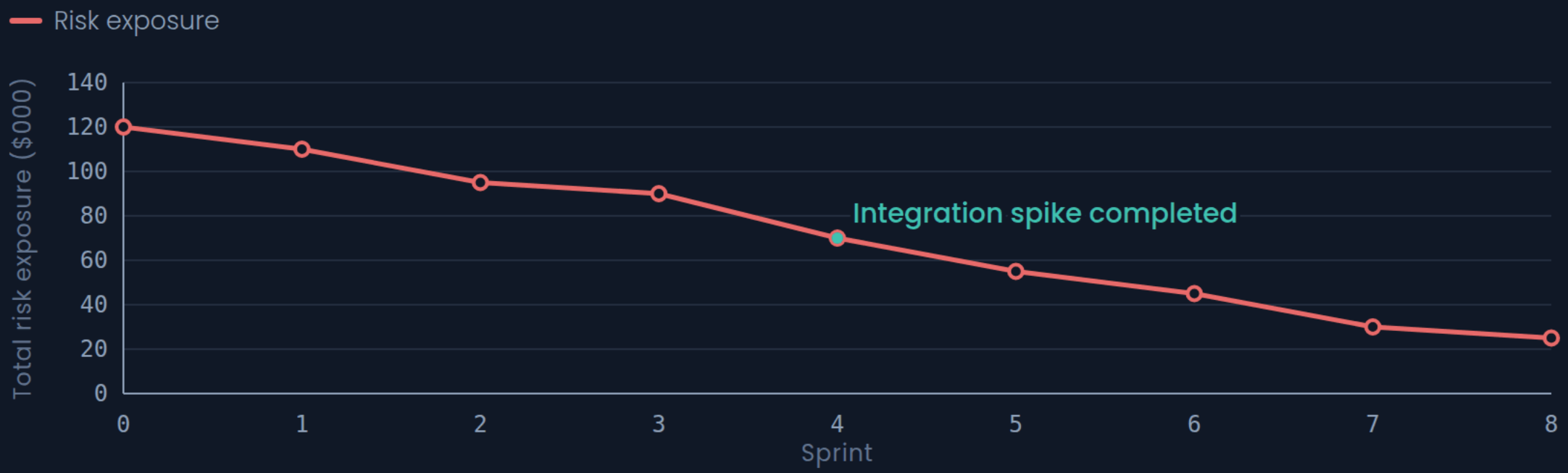
> ON THE EXAM
>
> Planned responses only help if they are carried out. Track them like any other work.

Monitoring and controlling risks

- Regular **risk reviews**: reassess existing risks, close expired ones, identify new ones.
- Watch **triggers** and check that responses are working.
- **Reserve analysis**: is enough contingency left for the risks that remain?
- **Risk audits** check that the risk process itself is being followed.
- Track overall risk exposure over time, and act on the trend.

> ON THE EXAM
>
> Risk management is continuous. A register last updated at kickoff is a history document.

A risk burndown



ON THE EXAM

A risk burndown shows total exposure falling as risky work is done early. Flat or rising means act.

Communicating risk impact status

- The outline asks you to **communicate risk impact status**.
- Share the **top risks**, their trend, and the responses in progress.
- Tailor the view: a heat map or top five for executives; the full register for the team.
- Keep the register **accessible** to those who need it, not on one person's laptop.
- Stakeholders surprised by a risk they were never told about lose trust fast.

> ON THE EXAM
>
> Risk status is communicated routinely, honestly and in a form each audience can use.

When to escalate a risk

- The risk lies **outside the project's scope**, or affects other projects.
- The response needs **authority** the project manager does not have.
- The risk crosses a **threshold** in the risk management plan.
- Escalate to the owner, such as the programme manager, and record it. They decide the response.

> ON THE EXAM
>
> Escalation is a real strategy when the risk belongs higher. It is not a way to avoid managing your own risks.

Project Lantern: responses in action

- **Transfer**: kiosk damage in transit covered by the supplier's insurance.
- **Mitigate**: a full migration rehearsal before each clinic go-live.
- **Enhance**: extra effort on the grant application, raising the chance of digital health funding.
- **Escalate**: a regional data-sharing rule change affecting all Northgate projects, passed to the portfolio board.
- Monthly risk review; top five risks and the risk burndown in every steering report.

> ON THE EXAM
>
> Threat and opportunity strategies used together, monitored monthly, reported routinely.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Buying insurance, or a fixed-price contract	Transfer
Make certain an opportunity happens	Exploit
The response itself creates a new risk	Secondary risk: identify, analyse, respond
The trigger has fired	The owner implements the contingency plan
The risk affects the whole programme	Escalate to the owner; record it
Stakeholders unaware of rising risk	Communicate risk status routinely, tailored

ON THE EXAM

Decide responses in advance, act on triggers, check for secondary risk, and keep people informed.

> A construction project is worried that expensive specialist equipment could be damaged on site. The project manager arranges an insurance policy that would pay for repairs or replacement.
>
> The risk register must record the response strategy.

Which risk response strategy is this?

A Avoid

B Mitigate

C Transfer

D Accept

◆ PAUSE AND DECIDE

Which risk response strategy is this?

A **Avoid**

Avoiding would remove the threat, such as not using the equipment.

B **Mitigate**

Mitigating would reduce the chance of damage, such as secure storage.

C **Transfer**

Insurance shifts the financial impact to a third party.

D **Accept**

Accepting would mean paying for damage ourselves if it happens.

THE SIGNAL

Insurance is the classic transfer. The damage can still happen; someone else pays.

> A contract offers a $200,000 bonus if a hospital wing is completed a month early. The project manager assigns the organisation's most experienced site team to the project to make sure the early date is met.
>
> The risk register must record the strategy.

Which opportunity response strategy is this?

A Exploit

B Enhance

C Share

D Accept

◆ PAUSE AND DECIDE

Which opportunity response strategy is this?

A Exploit

Acting to make sure the opportunity happens.

B Enhance

Raises the probability, without aiming to guarantee it.

C Share

Would involve a partner and splitting the benefit.

D Accept

Would take the bonus if it came, without extra effort.

THE SIGNAL

'Make sure it happens': exploit. 'Make it more likely': enhance.

> To recover a two-week delay, the project manager's approved response is to run night shifts on a construction site for a month.
>
> The site safety officer points out that tired crews working at night increase the risk of accidents.

What should the project manager do?

A Continue with the night shifts, since the response was approved.

B Cancel the night shifts and accept the delay.

C Record the accident risk as residual risk and accept it.

D Identify the accident risk as a secondary risk, analyse it, and plan a response.

◆ PAUSE AND DECIDE

What should the project manager do?

A Continue as approved.
Ignores a new safety risk the response itself created.

B Cancel and accept delay.
Premature. The new risk may be manageable.

C Call it residual; accept.
Wrong term, and safety risks are not simply accepted.

D Treat as secondary risk; analyse; respond.
A risk created by a response gets its own analysis and response.

THE SIGNAL

A risk caused by the response is a secondary risk. Analyse it and give it its own response.

> The risk register says: if the main supplier misses two consecutive delivery dates, the risk owner switches orders to the pre-approved backup supplier.
>
> The main supplier has just missed its second delivery date in a row.

What should happen next?

A Wait to see whether a third delivery is also missed.

B The risk owner implements the contingency plan and switches to the backup supplier.

C Carry out a new risk analysis before acting.

D Escalate the delays to the sponsor.

◆ PAUSE AND DECIDE

What should happen next?

A Wait for a third miss.

Ignores the agreed trigger; the delay grows.

B Owner implements the contingency plan.

The trigger has fired. The decision was already made.

C New analysis first.

The analysis was done when the plan was made.

D Escalate.

The response is pre-approved and within the owner's authority.

THE SIGNAL

Trigger fired, contingency plan agreed: the owner acts. That is what planning ahead buys.

> While planning, a project manager learns that a proposed national regulation could require changes to every one of the organisation's customer-facing systems, across six projects.
>
> The project manager's own project could adapt, but the other five would also be affected.

What should the project manager do?

A Mitigate the risk within this project, and leave the others to manage theirs.

B Accept the risk, since the regulation is only proposed.

C Escalate the risk to the programme or portfolio owner, and record the escalation.

D Take no action until the regulation is confirmed.

◆ PAUSE AND DECIDE

What should the project manager do?

A Mitigate locally only.

Protects one project; leaves a shared risk unmanaged across five others.

B Accept.

A proposed regulation affecting six projects is significant uncertainty.

C Escalate to the owner; record it.

The risk belongs to a higher level that can respond across all six projects.

D Wait for confirmation.

Loses the chance to prepare while the rule is still proposed.

THE SIGNAL

A risk spanning many projects belongs to the programme or portfolio. Escalate, and record it.

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> A project's overall risk exposure has risen for three months. The risk register exists only as a spreadsheet on the project manager's laptop.
>
> At a steering meeting, the sponsor is surprised to hear about two high risks for the first time.

What should the project manager do?

A Email the full risk register spreadsheet to all stakeholders.

B Report risks only when they become issues.

C Brief the sponsor verbally whenever a risk becomes high.

D Make the register accessible, and report top risks, trends and responses routinely, tailored to each audience.

◆ PAUSE AND DECIDE

What should the project manager do?

A Email the whole register.
Overload; the important trend is buried in detail.

B Report only issues.
Too late by definition. Risk reporting exists to warn early.

C Verbal briefings when high.
Ad hoc, unrecorded, and only for one stakeholder.

D Accessible register; routine tailored reporting.
Communicates risk impact status, as the outline asks.

THE SIGNAL

A surprised sponsor means risk status was not communicated. Make it routine, visible and tailored.

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Decide in advance. **Act on the trigger.** Check for secondary risk. Keep everyone informed.

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MODULE 38 · DOMAIN 3 - BUSINESS ENVIRONMENT

Continuous improvement and OPAs

Improvement at team, project and organisation level: retrospectives that change things, Lean waste, value stream mapping, and updating process assets.

DOMAIN III - Business Environment EXAM WEIGHT 26% MODULE 38 of 49

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Improvement that is not built into **how work is done** does not last.

Improvement at three levels

LEVEL	HOW IMPROVEMENT HAPPENS	WHERE IT IS KEPT
Team	Retrospectives; small experiments each iteration	Working agreements, definition of done
Project	Lessons applied during the project, not saved for the end	Plans, checklists, the lessons register
Organisation	Lessons turned into standard practice across projects	Process assets: templates, methods, benchmarks

ON THE EXAM

The outline's third enabler reaches the organisation level: update organisational process assets.

Retrospectives that change things

- Choose **one or two** improvements, not ten.
- Give each an **owner**, and put it in the backlog or plan so it gets done.
- Start the next retrospective by **checking** whether last time's actions worked.
- Vary the format so people keep thinking, rather than repeating the same answers.
- The same complaints at every retrospective mean the actions are not being done.

> ON THE EXAM
>
> A retrospective is judged by what changes afterwards, not by the quality of the discussion.

Retrospective formats

FORMAT	PROMPTS
Start, stop, continue	What should we start, stop, and keep doing?
Liked, learned, lacked, longed for	Four angles on the last iteration
Sailboat	Wind that pushed us, anchors that held us back, rocks ahead
Timeline	Map events and feelings across the period, then look for patterns
Five whys on one problem	Go deep on a single recurring issue

ON THE EXAM

The format matters less than the follow-through. Pick one that fits the team's mood and the period.

Kaizen and plan, do, check, act

- **Kaizen**: continuous, small improvements, by everyone, every day.
- Each change is an experiment: **plan** it, **do** it small, **check** the result, **act** to keep or drop it.
- Measure before and after, so you know the change helped.
- Once a change works, **standardise** it. Then improve from the new standard.

> ON THE EXAM
>
> Small, measured, repeated. Improve, standardise, improve again.

Lean: eight kinds of waste

WASTE	IN PROJECT WORK
Defects	Rework, bugs, errors found late
Overproduction	Features nobody uses; reports nobody reads
Waiting	Queues for approvals, environments, answers
Unused talent	People's skills and ideas ignored
Transport and handoffs	Work passed between many people or teams
Inventory	Half-finished work: too much in progress
Motion	Task switching; hunting for information
Extra processing	Steps, signatures or documents that add no value

ON THE EXAM

In knowledge work, waiting is usually the biggest waste. Look for it first.

Value stream mapping

- Map every step from **request** to **value delivered**, with the time each takes.
- Separate **work time** from **waiting time**.
- A change request takes 30 days end to end. The actual work is 4 days. The rest is waiting for approvals.
- Improve the waits first: they are usually where most of the time goes.

> ON THE EXAM
>
> When something takes too long, map the flow before speeding up the work. The delay is usually in the waiting.

Using lessons learned

- The outline's first enabler: **utilise lessons learned**, not just record them.
- At the start: search the repository and plan with what it says (module 16).
- During the project: apply this project's own lessons to the next phase or iteration.
- A lesson written down but not acted on is only a lesson **identified**. It is learned when behaviour changes.

> ON THE EXAM
>
> Lessons identified become lessons learned only when the next piece of work is done differently.

Updating process assets

LESSON FOUND	PROCESS ASSET UPDATED
Estimates for integration always run 40% low	Estimating benchmarks and templates
Go-lives kept missing a training step	The go-live checklist
The same supplier risk recurs on every project	The organisation's risk checklist
A new review step caught many defects	The standard quality procedure
Contracts lacked a data-return clause	The standard contract template

ON THE EXAM

The outline asks you to update organisational process assets. That is how one project's lesson helps every project.

Project Lantern: improving as it went

- After the pilot clinic, a retrospective produced two actions; both were in the next clinic's plan.
- A value stream map showed kiosk orders waiting eight days for sign-off; approval was delegated, cutting it to one.
- The go-live checklist was updated after each clinic, then adopted by Northgate's PMO for all projects.
- By clinic six, go-live issues had fallen from fourteen to two.

ON THE EXAM

Small actions, owned and followed up; waiting found and removed; a project checklist became an organisational asset.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Same problems at every retrospective	Fewer actions, with owners, checked next time
A lesson recorded but a new phase is starting	Apply it to the plan now
An organisational template is proven wrong	Update the process asset with the PMO
Work takes weeks but effort is days	Map the value stream; remove the waiting

ON THE EXAM

Use lessons, build them into the process, and make them part of the organisation's assets.

> At every retrospective for the last four sprints, the team has raised the same two problems and agreed eight or more actions. Few of the actions have been completed.
>
> Team members say retrospectives feel pointless.

What should the project leader do?

A. Help the team choose one or two actions, each with an owner, add them to the backlog, and review them at the next retrospective.

B. Make retrospectives longer, so more actions can be discussed.

C. Stop holding retrospectives until the team has more time.

D. Decide the improvement actions personally, to save the team's time.

◆ PAUSE AND DECIDE

What should the project leader do?

A Fewer actions, owned, in the backlog, reviewed.

Makes improvement actually happen, and restores the retrospective's value.

B Longer retrospectives.

More talk, more actions, the same lack of follow-through.

C Stop retrospectives.

Removes the team's main improvement mechanism.

D The leader decides.

Takes improvement away from the team that must own it.

THE SIGNAL

Repeated problems and unfinished actions: fewer, owned, visible actions, checked next time.

> The lessons learned register from phase one of a programme records that clashing bookings for the test environment caused two weeks of delay.
>
> Phase two, which needs the same environment, starts next week.

What should the project manager do?

A Add another note about test environment clashes to the register.

B Apply the lesson now: put a booking process for the test environment into the phase two plan.

C Wait to see whether clashes happen again in phase two.

D Email the lesson to the phase two team for awareness.

◆ PAUSE AND DECIDE

What should the project manager do?

A Note it again.
Records the lesson a second time without acting on it.

B Apply it: booking process in the plan.
Turns a lesson identified into a lesson learned, before it recurs.

C Wait and see.
Invites the same two-week delay.

D Email it.
Awareness alone rarely changes behaviour.

THE SIGNAL

A known lesson and the same situation coming: change the plan before it starts.

> A project manager has found that the PMO's standard estimating template underestimates data integration work by about forty percent. Three recent projects confirm the pattern.
>
> Other project managers across the organisation use the same template.

What should the project manager do?

A. Share the evidence with the PMO and work with them to update the template and benchmarks.

B. Adjust this project's own estimates, and leave the template as it is.

C. Email other project managers to warn them informally.

D. Keep using the template, since it is the organisation's standard.

◆ PAUSE AND DECIDE

What should the project manager do?

A Update the template with the PMO.
Updates an organisational process asset, so every project benefits.

B Fix only this project.
Helps one project; every other one keeps underestimating.

C Informal warning emails.
Unreliable, and the template stays wrong for new users.

D Keep using it.
Knowingly uses a flawed standard.

THE SIGNAL

A proven flaw in a shared template: fix the process asset itself, with its owner.

> A team's minor changes take about thirty days from request to release. A value stream map shows about four days of actual work, and twenty-six days waiting for three separate approvals.
>
> The sponsor asks how to speed up delivery.

What should the project manager recommend?

A. Add more developers to the team.

B. Work with the approvers to reduce the waiting, for example by delegating approval of minor changes.

C. Ask the developers to code faster.

D. Add a fourth approval to improve quality.

◆ PAUSE AND DECIDE

What should the project manager recommend?

A More developers.

Speeds up the four days of work, not the twenty-six of waiting.

B Reduce the approval waiting.

Targets the biggest waste the value stream map revealed.

C Code faster.

Same problem as A, with added pressure.

D Another approval.

Adds more of the waste that is causing the delay.

THE SIGNAL

The map shows where the time goes. Here, 26 of 30 days are waiting. Fix the waiting.

Use the lesson. Build it into the process.
Update the asset.

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MODULE 39 · DOMAIN 3 - BUSINESS ENVIRONMENT

Organisational change and culture

Assessing culture and readiness, change models, resistance and adoption, change fatigue, and what to do when the organisation itself changes around the project.

DOMAIN III - Business Environment EXAM WEIGHT 26% MODULE 39 of 49

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Projects deliver outputs. People adopting them deliver the benefits.

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Assessing organisational culture

- How are **decisions** made: quickly by one person, or slowly by consensus?
- What is the attitude to **risk** and to failure?
- How **hierarchical** is it? Can juniors challenge seniors?
- How much **trust** is there, and how did past changes go?
- Observe, ask, and read the history. Then tailor the plan, governance and engagement to fit.

> ON THE EXAM
>
> The outline's first enabler: assess organisational culture. A plan that ignores culture meets it later, as resistance.

Culture and what it means for the project

IF THE CULTURE IS	THEN THE PROJECT SHOULD
Consensus-driven and slow to decide	Allow time for decisions; involve people early; pre-discuss proposals
Risk-averse	Pilot first; show evidence; phase the change
Strongly hierarchical	Secure visible senior sponsorship; create safe routes for concerns
Low in trust after past failures	Start small, deliver quick wins, communicate honestly
Fast-moving and experimental	Iterate; release early; keep governance light

ON THE EXAM

Tailor to the culture you have, not the one you would like. Changing culture is rarely the project's job.

Assessing readiness and impact

- **Change impact assessment**: who is affected, what changes for them, and by how much.
- **Readiness assessment**: do they understand, want, and have the skills and time for the change?
- High-impact groups with low readiness need the most support, and the earliest.
- Assuming everyone is equally affected wastes effort on some groups and neglects others.

> ON THE EXAM
>
> Map impact and readiness by group. Put support where impact is high and readiness is low.

Change models

MODEL	IN ONE LINE
Lewin	Unfreeze, change, refreeze
Kotter's eight steps	Urgency, coalition, vision, remove obstacles, early wins, anchor it
ADKAR	Awareness, Desire, Knowledge, Ability, Reinforcement
Bridges' transition model	Endings, a neutral zone, new beginnings
Satir change model	Status quo, chaos, integration, new status quo

ON THE EXAM

The exam rarely asks you to name a model. It asks what to do next, and the models answer that.

ADKAR: change one person at a time

01
Awareness
Why is this change needed?

›

02
Desire
Do I want to take part?

›

03
Knowledge
Do I know how to change?

›

04
Ability
Can I actually do it now?

›

05
Reinforcement
What makes it stick?

ON THE EXAM

Find the first missing step and fix that. Training (knowledge) will not help someone who lacks desire.

Change or transition?

CHANGE

What happens outside

- A new system, structure or process
- Happens on a date
- Can be planned and delivered
- The project's output

TRANSITION

What people go through inside

- Letting go of the old way
- Takes time, and varies by person
- Includes a confusing 'neutral zone'
- Where adoption really happens

THE TIEBREAKER

A dip in morale and productivity after go-live is normal. Support people through it; do not reverse the change.

Understanding resistance

- Resistance is **normal**, and usually protects something real: status, workload, competence, a genuine risk.
- Listen first, privately. Ask what worries people about the change.
- Involve resisters in shaping the details. People support what they help create.
- Show what is in it for them, honestly, including what they will lose.
- Overriding resistance with authority gets compliance at best, and quiet sabotage at worst.

> ON THE EXAM
>
> Treat resistance as information, as module 12 said. Find the interest behind it.

Supporting adoption

- **Change champions** in each affected group: trusted peers who help and advocate.
- Training at the right time, close to go-live, with practice on real tasks.
- Communicate repeatedly, through leaders people trust, not just project emails.
- **Quick wins** early, and **reinforcement** afterwards: recognition, removing the old way.
- **Measure adoption**, such as usage, and act where it lags.

> ON THE EXAM
>
> Adoption is planned and measured like any deliverable. It is where the benefits come from.

Change fatigue

- People absorbing several changes at once stop engaging with any of them.
- Check what else is changing for the same people, at the same time.
- Coordinate through the PMO or portfolio: sequence changes, or combine training.
- Adjust timing if needed, and agree it with governance.

> ON THE EXAM
>
> A great change landed on exhausted people fails. Look across the portfolio, not just your project.

When the organisation changes around the project

- The outline's second enabler: **evaluate the impact of organisational change** on the project.
- A new sponsor. A restructure. A merger. New leadership, or a new strategy.
- Re-check the business case and objectives: do they still hold?
- Re-map stakeholders; engage new decision-makers early; update risks and resources.
- Carrying on unchanged after a major reorganisation assumes nothing that matters has moved.

> ON THE EXAM
>
> Organisational change is an environmental shift for the project. Evaluate its impact, then act.

Project Lantern: bringing people with it

- Culture assessment: consensus-driven, cautious. So a pilot clinic first, and every decision pre-discussed.
- Impact assessment: reception staff most affected; they got champions, early demos and hands-on training.
- A productivity dip after the pilot go-live was expected; floor walkers supported staff for two weeks.
- When Northgate appointed a new chief operating officer, the PM met her in week one to re-confirm objectives.

> ON THE EXAM
>
> Tailored to culture, support where impact was highest, the dip expected, and a leadership change handled early.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
The plan clashes with how the organisation works	Assess culture; tailor plan and governance
People know why, but do not want the change	ADKAR desire: understand concerns, involve them
A new sponsor or a reorganisation	Re-validate objectives; re-engage decision-makers
Several changes hitting the same people	Coordinate timing across the portfolio
Productivity dips after go-live	Normal transition: support, do not reverse
Delivered, but few people use it	Plan and measure adoption with the business owner

ON THE EXAM

Assess culture, find where people are stuck, support the transition, and watch the organisation around you.

A project manager joins an organisation where decisions are made by consensus across several committees, and people are cautious about risk. Her plan assumes approvals within two days.

After a month, every approval has taken two to three weeks.

What should the project manager do?

A Keep pushing the committees to meet the two-day approval target.

B Escalate the slow approvals to the sponsor as a performance problem.

C Assess the culture, and tailor the plan and engagement: allow for consensus time and discuss proposals early.

D Launch an initiative to make the organisation's culture faster.

◆ PAUSE AND DECIDE

What should the project manager do?

A Push for two days.

Fights the culture; approvals stay slow and relationships suffer.

B Escalate as a performance issue.

Treats the organisation's normal way of working as a fault.

C Assess culture; tailor the plan.

The outline's enabler: fit the plan to the culture you actually have.

D Change the culture.

Far beyond a project's remit, and no help to this plan.

THE SIGNAL

A plan that clashes with the culture: tailor the plan, not the organisation.

> Warehouse staff have attended briefings and understand why a new stock system is being introduced. However, many say openly that they do not want it, and prefer the old system.
>
> Training starts next week.

Using the ADKAR model, what should the project manager focus on?

A Add more training sessions, to build knowledge faster.

B Make use of the new system mandatory from go-live.

C Send more communications explaining why the change is needed.

D Build desire: understand their concerns, involve them, and use trusted champions before training.

◆ PAUSE AND DECIDE

Using the ADKAR model, what should the project manager focus on?

A More training.

Knowledge is step three. It does not fix missing desire.

B Mandate it.

Compliance without desire; workarounds and resistance follow.

C More awareness messages.

Awareness is already there. Repeating it fixes nothing.

D Build desire first.

The first missing ADKAR step is desire. Fix that before training.

THE SIGNAL

Awareness present, desire missing: work on desire. Training cannot create it.

> Halfway through a project, the organisation restructures. The project's sponsor moves to a new role, and a new sponsor from a different division takes over.
>
> The new sponsor was not involved in approving the business case.

What should the project manager do first?

A. Meet the new sponsor to review the business case, objectives and how she wants to be engaged.

B. Continue as planned, since the business case is already approved.

C. Pause the project until the restructure is complete.

D. Escalate the change of sponsor to the chief executive.

◆ PAUSE AND DECIDE

What should the project manager do first?

A Meet the new sponsor; review case and engagement.

Evaluates the impact of the change, and builds the key relationship early.

B Carry on as planned.

Assumes a sponsor from another division shares the old priorities.

C Pause the project.

Wastes time before anyone knows whether anything needs to change.

D Escalate to the chief executive.

A sponsor change is normal governance, not a crisis.

THE SIGNAL

A new sponsor: meet them first, and re-validate the business case and objectives together.

> A new customer system is due to go live in March to a customer service team. The project manager learns that the same team is also receiving a new telephony system and a new rota system in March.
>
> The team manager says her staff are already overwhelmed.

What should the project manager do?

A Go live in March as planned, since the other changes are not this project's concern.

B Work with the PMO and the other projects to assess readiness and coordinate the timing of the changes.

C Send more communications to prepare the team.

D Cancel this project's go-live.

◆ PAUSE AND DECIDE

What should the project manager do?

A Go live regardless.
Ignores change fatigue; adoption, and benefits, will suffer.

B Coordinate timing across projects.
Addresses change saturation with the people who can sequence it.

C More communications.
Adds to the overload rather than reducing it.

D Cancel the go-live.
Overreaction. Re-timing may be all that is needed.

THE SIGNAL

Several changes hitting the same team: coordinate across the portfolio. Adoption depends on capacity.

> Two weeks after a new finance system went live, staff are slower than before, frustrated, and say they miss the old system. No serious defects have been found.
>
> A department head asks whether the old system should be switched back on.

What should the project manager do?

A Switch back to the old system until staff are more comfortable.

B Repeat the full training programme for all staff.

C Explain that a temporary dip is normal, and increase hands-on support, quick fixes and feedback, without reversing the change.

D Take no action, since the project has delivered the system.

◆ PAUSE AND DECIDE

What should the project manager do?

A. Switch back.
Reverses a working change at the hardest moment; the next attempt is harder.

B. Repeat all training.
Classroom training again is not what people in the neutral zone need.

C. Normal dip: support, fixes, feedback.
Supports people through the transition, as Bridges and Satir describe.

D. No action.
The benefits depend on adoption, which is now at risk.

THE SIGNAL

A dip after go-live with no serious defects is transition, not failure. Support through it.

> A new expenses app went live three months ago on time and on budget. Only thirty percent of staff use it; the rest still submit paper forms, which finance continues to accept.
>
> The expected savings have not appeared.

What should the project manager do?

A Close the project, since the app was delivered on time and budget.

B Add new features to make the app more attractive.

C Report that the staff are refusing to use the app.

D Work with the business owner on adoption: champions, support, retiring the paper route, and measuring usage.

◆ PAUSE AND DECIDE

What should the project manager do?

A Close it.

Output delivered, benefit missing. The job is not finished.

B More features.

The barrier is adoption, not functionality.

C Blame the staff.

Ignores that the old way is still open and nobody is reinforcing the new one.

D Drive adoption with the business owner.

Reinforcement, support and measurement turn the output into benefits.

THE SIGNAL

Delivered but unused: the benefits depend on adoption. Reinforce the new way and retire the old one.

Outputs are delivered. Benefits are adopted. Plan for the people.

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MODULE 40 · DOMAIN 3 - BUSINESS ENVIRONMENT

The external business environment

Scanning for regulatory, technological, geopolitical and market change, assessing its impact on scope and the backlog, and reviewing it continually.

DOMAIN III - Business Environment EXAM WEIGHT 26% MODULE 40 of 49

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Projects live in a world that keeps moving.
Look outward, regularly.

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PESTLE: a prompt for scanning

FACTOR	EXAMPLES
Political	Elections, government priorities, trade policy, sanctions
Economic	Inflation, interest rates, exchange rates, recession
Social	Customer behaviour, demographics, workforce expectations
Technological	New tools such as AI, obsolescence, vendor end-of-life
Legal	New laws and regulations, court rulings, standards
Environmental	Climate impacts, environmental rules, resource scarcity

ON THE EXAM

PESTLE is a checklist, so nothing obvious is missed. The outline's four examples all fit inside it.

Surveying the external environment

- The outline's first enabler: **survey changes to the external business environment**.
- Sources: regulators and their consultations, industry bodies, news, suppliers, customers.
- Internal specialists: legal, compliance, procurement, finance, strategy teams.
- Assign an **owner** for scanning, and a regular rhythm, so it actually happens.

> ON THE EXAM
>
> Scanning that belongs to nobody does not happen. Give it an owner and a rhythm.

The outline's four examples

CHANGE	POSSIBLE IMPACT ON A PROJECT
Regulation	New requirements; features to add or change; compliance deadlines
Technology	A better tool to adopt; a platform going out of support
Geopolitical	Sanctions, tariffs or conflict disrupting suppliers and costs
Market	Competitors, customer demand or prices changing the value of scope

ON THE EXAM

Each can change scope, cost, schedule, risk, or the business case itself.

Responding to an external change

01
Detect
Through regular scanning.

02
Assess
What does it change for scope, cost, time, value?

03
Prioritise
How big, how likely, how soon?

04
Act
Backlog, change request, or risk response.

05
Communicate
Tell the decision-makers and those affected.

ON THE EXAM

Detect early, assess the impact, prioritise, act through the right channel, and tell people.

Assessing and prioritising impact

- The outline asks you to **assess and prioritise impact on project scope or backlog**.
- Has it happened, or might it? Happened is an issue; might is a risk.
- How large is the effect, and how soon must we act?
- Does it change the **value** of planned work, or the business case itself?
- Reacting to every headline is as harmful as ignoring the environment.

> ON THE EXAM
>
> Prioritise external changes by impact and timing. Act on the few that matter; watch the rest.

Responding in predictive or adaptive work

PREDICTIVE

Through change control

- Assess against the baselines
- Raise a change request
- The right authority decides
- Update plan and baselines

ADAPTIVE

Through the backlog

- Assess with the product owner
- Add or reorder backlog items
- Adjust the roadmap
- Deliver in the next iterations

THE TIEBREAKER

The outline says 'scope or backlog'. Use whichever mechanism the approach uses.

Geopolitical and supply chain change

- Sanctions, export controls, tariffs and conflict can cut off suppliers overnight.
- Know where critical components and services come from, including subcontractors.
- Identify alternative suppliers before you need them; consider buffer stock for critical items.
- Treat exposure as a risk, with procurement, legal and finance involved.

> ON THE EXAM
>
> Geopolitical risk is supply chain risk. Map dependencies, and plan alternatives early.

Technology change

- **Opportunity**: a new tool, such as AI assistance, may speed delivery or improve the product.
- **Threat**: a platform or component reaching end of support during or after the project.
- Evaluate new technology against value, risk and readiness; pilot before adopting.
- Chasing every new trend mid-project is as risky as ignoring obsolescence.

> ON THE EXAM
>
> Technology change can be an opportunity or a threat. Evaluate, pilot, then decide.

Market and regulatory change

- **Market**: a competitor launch, changing demand, inflation or interest rates can change the value of scope.
- **Regulation**: new or proposed laws bring requirements and deadlines, often with transition periods.
- Market changes may mean re-prioritising, or revisiting the business case with the sponsor.
- Regulatory changes flow into compliance requirements, as module 33 described.

> ON THE EXAM
>
> Market change challenges value; regulatory change adds obligations. Both go to the people who decide.

Reviewing continually

- The outline's third enabler: **continually review the external business environment**.
- Add an external scan to regular risk reviews, and to every phase gate or release review.
- Keep a short watch list of changes that are proposed but not yet certain.
- Feed findings into the risk register, the backlog, and stakeholder communication.

> ON THE EXAM
>
> External review is a routine, not a reaction. Build it into reviews you already hold.

Project Lantern: looking outward

- **Regulatory**: a proposed regional data-sharing rule, escalated to the portfolio board (module 37).
- **Geopolitical**: a chip shortage threatened kiosk supply; a second supplier was qualified early.
- **Technology**: a new text-message service cut reminder costs; piloted at one clinic, then adopted.
- **Routine**: an external scan is the first item of every monthly risk review.

> ON THE EXAM
>
> Regulatory, geopolitical and technology changes spotted early and acted on through the right channel.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
A new regulation is announced	Assess impact; add or reorder scope or backlog
Sanctions or tariffs threaten a supplier	Assess exposure; line up alternatives; update risks
A market shift changes the value of scope	Re-prioritise; revisit the business case with the sponsor
A platform goes out of support	Assess impact and options; decide through change control
An external change was missed until it hit	Set up regular scanning with an owner
Several external changes at once	Prioritise by impact and timing

ON THE EXAM

Scan regularly, assess impact on scope or backlog, prioritise, and act through the right channel.

A government announces an accessibility regulation for online services, taking effect in nine months. An agile team is building a public-facing booking service that will launch in six months.

The current backlog does not include accessibility work.

What should the project leader do?

A Ignore the regulation until it takes effect.

B Assess the impact with the compliance team, and work with the product owner to add and prioritise accessibility work in the backlog.

C Stop development until the regulation's full details are published.

D Escalate the regulation to the chief executive.

◆ PAUSE AND DECIDE

What should the project leader do?

A. Ignore it for now.
The service launches before the rule takes effect, and must still comply three months later.

B. Assess; add to the backlog with the product owner.
Assesses impact on the backlog and acts through the adaptive mechanism.

C. Stop development.
Wastes months; the direction of the rule is already clear.

D. Escalate to the chief executive.
A product-level response the team and product owner can handle.

THE SIGNAL

External regulation on agile work: assess the impact and put it into the backlog, now.

> New export restrictions have been announced on a country where the project's only supplier of a critical control unit manufactures it. Deliveries are due in four months.
>
> The supplier says it is 'looking into' the impact.

What should the project manager do?

A Wait for the supplier to confirm whether deliveries are affected.

B Order a large stock of the units immediately, without further analysis.

C Cancel the contract and redesign the product without the unit.

D Assess the exposure, identify alternative suppliers with procurement, and update the risk register and response plan.

◆ PAUSE AND DECIDE

What should the project manager do?

A Wait for the supplier.

Loses weeks of the four months available to prepare.

B Stockpile without analysis.

May be impossible under the restrictions, and commits money blindly.

C Cancel and redesign.

A drastic step before the impact is even known.

D Assess, find alternatives, update risks.

A proportionate response to a geopolitical supply risk, with time to act.

THE SIGNAL

A geopolitical threat to a single supplier: assess exposure and line up alternatives now.

> A bank is building a new car-loan app. Since the business case was approved, interest rates have risen sharply, and demand for car loans has fallen by a third.
>
> The team is building the loan features in the original order.

What should the project manager do?

A Assess the impact on the business case and backlog priorities, and take the analysis to the sponsor and product owner.

B Continue as planned, since the business case was approved.

C Lower the loan interest rate in the app to restore demand.

D Stop the project immediately.

◆ PAUSE AND DECIDE

What should the project manager do?

A Assess impact; take to sponsor and product owner.

Assesses a market change's effect on value, and puts it with the decision-makers.

B Continue as planned.

Ignores a change that cuts expected demand by a third.

C Lower the rate.

A commercial decision far outside the project manager's authority.

D Stop immediately.

Premature. The sponsor may re-scope rather than stop.

THE SIGNAL

A market shift that changes the value of scope: analyse it and bring it to the people who own value.

> A project is building a system on a software platform. The vendor announces that the platform will reach end of support in eighteen months. The project finishes in twelve.
>
> After that, the system would run on an unsupported platform.

What should the project manager do?

A Ignore it, since support ends after the project has closed.

B Switch to a different platform immediately, without analysis.

C Assess the impact and the options, such as migrating now or planning a later upgrade, and take them through change control.

D Ask the vendor for an extension, and take no other action.

◆ PAUSE AND DECIDE

What should the project manager do?

A Ignore it.

Hands operations a system six months from unsupported. Value and security suffer.

B Switch now, unanalysed.

A major change with cost and schedule impact, made blindly.

C Assess options; decide through change control.

Treats a technology change properly: analysis, options, the right decision-maker.

D Only ask for an extension.

Worth asking, but not a plan on its own.

THE SIGNAL

A technology change that hits after closure still matters. Assess it now and decide properly.

> A project's costs rose by eight percent after new import tariffs took effect. The tariffs had been publicly debated for four months, but nobody on the project noticed.
>
> The sponsor asks how this can be avoided in future.

What should the project manager recommend?

A. Hold the procurement team responsible for missing the tariffs.

B. Set up regular scanning of the external environment, with an owner, as part of the risk reviews.

C. Add a larger contingency reserve to future projects.

D. Accept that external changes cannot be predicted.

◆ PAUSE AND DECIDE

What should the project manager recommend?

A Blame procurement.
Nobody owned scanning. Blame fixes nothing.

B Regular scanning with an owner, in risk reviews.
Continual review of the external environment, as the outline asks.

C Bigger contingency.
Pays for surprises instead of spotting them.

D Accept unpredictability.
These tariffs were public for four months. They were predictable.

THE SIGNAL

A public change missed until it hit: nobody was scanning. Give scanning an owner and a rhythm.

> In one week, the project manager learns of three external changes: a regulation under consultation for two years' time, a competitor feature launched yesterday, and an exchange-rate swing affecting five percent of project costs.
>
> Time to respond is limited.

How should the project manager handle them?

A. Give all three equal attention and effort.

B. Focus only on the regulation, since compliance is mandatory.

C. Respond to the competitor launch first, since it is the newest.

D. Assess each for impact and timing on the project, prioritise, and respond to each proportionately.

◆ PAUSE AND DECIDE

How should the project manager handle them?

A Equal effort for all.
Wastes limited time on the least urgent changes.

B Regulation only.
It is two years away and still changing; the others are here now.

C Newest first.
Recency is not impact.

D Assess, prioritise, respond proportionately.
The outline's enabler: assess and prioritise impact.

THE SIGNAL

Several external changes: assess and prioritise by impact and timing. Watch the distant ones; act on the pressing ones.

Scan regularly. Assess the impact. Act through the right channel.

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MODULE 41 · FOUNDATIONS - ALL DOMAINS

AI and sustainability in project work

Using AI to automate, assist and augment, responsibly; and building sustainability into every stage of a project.

DOMAIN Cross-domain EXAM WEIGHT All domains MODULE 41 of 49

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Use AI as a capable assistant you check. Treat sustainability as a requirement, not a wish.

Automate, assist, augment

USE	WHAT AI DOES	EXAMPLE
Automate	Performs a routine task with little human effort	Transcribing meetings; filing actions to the tracker
Assist	Produces a first version that a person reviews	Drafting a status report or risk list for the PM to check
Augment	Extends what a person can see or analyse	Finding risk patterns across hundreds of past projects

ON THE EXAM

PMI describes project professionals using AI to automate, assist and augment their work. None removes accountability.

Where AI helps in project work

- **Drafting and summarising**: meeting notes, reports, first drafts of plans and user stories.
- **Analysing data**: spotting trends in schedule, cost, defects and flow metrics.
- **Risk and forecasting**: suggesting risks, testing scenarios, supporting estimates.
- **Finding knowledge**: searching lessons learned and documents in plain language.
- Most value comes from freeing time for the human work: leading, deciding, engaging.

> ON THE EXAM
>
> AI speeds up information work. The time it frees should go to people, judgement and relationships.

The risks of using AI

RISK	WHAT IT LOOKS LIKE
Errors and invented facts	Confident, plausible, wrong: a date, a figure, a source
Confidentiality and privacy	Sensitive data pasted into an unapproved public tool
Bias	Recommendations that disadvantage some groups
Intellectual property	Unclear ownership of generated content; copied material
Over-reliance	Skills fade; nobody checks; judgement is outsourced
Security	New tools and integrations as routes for attack

ON THE EXAM

Every one of these is a risk to identify and manage, like any other project risk.

Using AI responsibly

- **Human accountability**: the person who uses the output owns it.
- **Verify** facts, numbers and sources against the original data before anything goes out.
- **Protect data**: use approved tools, and never put confidential or personal data into unapproved ones.
- **Be transparent** about AI use where it matters, and follow organisational policy.
- Set simple team guidelines: what AI may be used for, with which tools, and how it is checked.

> ON THE EXAM
>
> Accountable, verified, protected, transparent. The Code's values applied to a new tool.

What AI can do, and what stays human

AI CAN

Speed up information work

- Draft, summarise, translate
- Analyse large amounts of data
- Suggest options and risks
- Answer questions from documents

THE PROJECT LEADER STILL

Leads, decides and is accountable

- Builds trust and motivates people
- Resolves conflict and negotiates
- Makes ethical judgements
- Owns decisions and their consequences

THE TIEBREAKER

On the exam, AI is never the answer to a people problem, and never the owner of a decision.

Introducing AI into a team

- Start with a **pilot** on a real, low-risk task, with volunteers.
- Agree **guidelines**: approved tools, allowed uses, and how output is checked.
- Address **concerns** openly, including worries about jobs and skills.
- **Measure** the value: time saved, quality, and whether people find it useful.
- It is an organisational change: module 39's approach applies.

> ON THE EXAM
>
> Adopt AI like any change: pilot, guide, listen, measure. Neither mandate it nor ban it outright.

Projects that build AI

- **Data first**: the quality and availability of data decides what is possible.
- **Uncertain by nature**: results emerge through experiments, so adaptive approaches fit.
- **New risks**: model accuracy, bias, drift over time, explainability.
- **Regulation and ethics**: AI-specific laws and standards are emerging in many regions.
- PMI offers a pattern-based approach for planning AI projects; the principles here still apply.

> ON THE EXAM
>
> An AI project is a high-uncertainty project with data, ethics and regulatory risks. Plan it as one.

Sustainability: the triple bottom line

THREE DIMENSIONS

People, planet, prosperity

- People: workers, communities, users
- Planet: emissions, materials, waste, energy
- Prosperity: long-term economic value
- All three, not one at the others' expense

IN THE 2026 EXAM

Named in several tasks

- Planning: sustainability as critical information
- Quality: cost of quality and sustainability
- Compliance: sustainability requirements
- Risk: sustainability risks

THE TIEBREAKER

PMBOK 8's principle: integrate sustainability within all project areas. The outline names it in four tasks.

Sustainability across the life cycle

STAGE	SUSTAINABILITY QUESTIONS
Business case	What are the environmental and social effects, good and bad?
Requirements	What measurable sustainability targets apply?
Design	Which materials, energy use, durability and repairability?
Procurement	Do selection criteria and contracts include sustainability?
Delivery	How is waste, travel and energy on the project reduced?
Handover and end of life	What does it cost to run, and what happens when it is retired?

ON THE EXAM

Sustainability decided early is cheap. Sustainability added at the end is expensive, or impossible.

Measuring sustainability, and making trade-offs

- Measure it like any objective: a **baseline**, a **target**, and an **owner**.
- Common measures: carbon emissions, energy, waste, water, social value.
- Compare options on **life-cycle cost**, not only purchase price.
- Where sustainability costs more now, make the trade-off **visible** and let the sponsor decide.
- Dropping sustainability quietly when budgets tighten breaks organisational commitments.

> ON THE EXAM
>
> Sustainability trade-offs are decisions for the sponsor, with the life-cycle numbers on the table.

Honest sustainability claims

- **Greenwashing**: claiming more environmental benefit than the evidence supports.
- Every claim should be specific, measured and evidenced.
- Involve legal and compliance: many regions now regulate environmental claims.
- Honesty, from the Code of Ethics, applies to sustainability claims as much as to status reports.

> ON THE EXAM
>
> An overstated sustainability claim is a false claim. Evidence first, then words.

Project Lantern: AI and sustainability

- **Assist**: AI drafts meeting summaries and the weekly report; the PM checks every figure before sending.
- **Augment**: AI analysis of past booking data found which clinics most needed reminders.
- **Data rule**: patient data never enters general AI tools; only the approved, secured one.
- **Sustainability**: energy-efficient kiosks, a take-back scheme, and fewer patient journeys from better booking.
- Missed-appointment reduction was also reported as a social and environmental benefit, with evidence.

> ON THE EXAM
>
> AI used to assist and augment, with checks and data rules; sustainability measured and reported honestly.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
AI produced a draft plan, report or register	Treat it as a draft; verify; the PM is accountable
Sensitive data and a public AI tool	Approved tools only; follow data policy
Team divided about adopting AI	Pilot, guidelines, address concerns, measure value
Sustainable option costs more	Life-cycle analysis; sponsor decides the trade-off
A sustainability claim goes beyond the evidence	Raise it; claims must be accurate and evidenced
Asked which use of AI: automate, assist, augment	Routine task, reviewed draft, or new insight

ON THE EXAM

Check AI's work, protect data, and treat sustainability as a measured, honest requirement.

> A project manager used an AI tool to draft the project's initial risk register. The draft includes several plausible but irrelevant risks, and misses a supplier risk the team discussed last week.
>
> The sponsor wants the register by tomorrow.

What should the project manager do?

A. Treat the AI output as a draft: review it with the team, remove irrelevant risks, add known ones, and then submit it.

B. Submit the AI draft as it is, since the deadline is tomorrow.

C. Ban AI tools from project work in future.

D. Ask the AI tool to regenerate the register until it looks right.

◆ PAUSE AND DECIDE

What should the project manager do?

A Draft; review with team; fix; submit.

AI assists; the team validates; the PM stays accountable.

B Submit as it is.

Sends known errors and omissions to the sponsor.

C Ban AI.

Overreacts. The draft saved time; the missing step was review.

D Regenerate until it looks right.

Looking right is not being right. Only the team knows its real risks.

THE SIGNAL

AI drafts; people verify. Risk identification still needs the team.

> An analyst wants to paste six months of customer complaint records, including names and contact details, into a free public AI chatbot to find themes.
>
> The organisation has a data protection policy and an approved, secured AI tool.

What should the project manager do?

A Allow it, since finding themes will help the project.

B Ban all AI analysis of customer data.

C Use the approved tool, remove personal details where possible, and follow the data protection policy.

D Ask customers for permission before any analysis.

◆ PAUSE AND DECIDE

What should the project manager do?

A Allow the public tool.

Puts personal data into an unapproved tool; likely breaches policy and law.

B Ban AI analysis.

Loses real value. A compliant route exists.

C Approved tool, anonymised data, per policy.

Gets the value while protecting personal data.

D Ask every customer first.

Unnecessary if the approved route and policy already cover it.

THE SIGNAL

Sensitive data plus AI: approved tools, minimum personal data, and the policy. Not a free public tool.

> A PMO uses AI to analyse records from three hundred past projects. It finds a pattern no one had noticed: projects with more than two supplier handoffs overrun far more often.
>
> The PMO now uses this insight in risk planning.

Which kind of AI use is this?

A Automate

B Augment

C Assist

D Replace

◆ PAUSE AND DECIDE

Which kind of AI use is this?

A **Automate**

Automation performs a routine task. This found something new.

B **Augment**

Extends what people can see, revealing a pattern they could not.

C **Assist**

Assist drafts something a person would otherwise produce.

D **Replace**

Not one of PMI's categories; people still decide what to do with it.

THE SIGNAL

Routine task done: automate. Draft for review: assist. New insight humans could not reach: augment.

> A project manager wants the team to use an AI tool for meeting notes and action tracking. Some team members are enthusiastic; others worry it will be used to monitor them, or replace their roles.
>
> The organisation has approved the tool.

What should the project manager do?

A Pilot it with volunteers, agree guidelines on use, discuss concerns openly, and measure the value.

B Make the tool mandatory for all meetings from next week.

C Drop the idea, since some team members are worried.

D Let each person decide whether and how to use it, with no guidelines.

◆ PAUSE AND DECIDE

What should the project manager do?

A Pilot, guidelines, open discussion, measure.

Treats AI adoption as a change: involves people and tests the value.

B Mandate it.

Confirms the fears and breeds resistance.

C Drop it.

Gives up real value without addressing the concerns.

D No guidelines.

Inconsistent use, and the monitoring worry is never answered.

THE SIGNAL

Adopting AI is an organisational change. Pilot, guide, listen, measure.

> A building project can use low-carbon concrete that costs six percent more than standard concrete. The organisation has a public net-zero commitment.
>
> The budget is tight, and the sponsor has not yet been asked.

What should the project manager do?

A. Choose standard concrete to protect the budget.

B. Choose low-carbon concrete and absorb the extra cost quietly.

C. Drop the net-zero target for this project.

D. Present the options with their life-cycle cost and carbon impact, and let the sponsor decide.

◆ PAUSE AND DECIDE

What should the project manager do?

A Standard concrete.
Quietly overrides a public organisational commitment.

B Low-carbon, absorbed quietly.
Hides a cost trade-off the sponsor should see.

C Drop the target.
Not the project manager's commitment to drop.

D Present options; sponsor decides.
Makes the trade-off visible, with the evidence, to the right person.

THE SIGNAL

A sustainability trade-off with cost: show the life-cycle numbers and let the sponsor decide.

> The marketing team wants to describe a new product as 'carbon neutral'. The project's data shows only its manufacturing emissions are offset; transport and packaging are not.
>
> Launch materials are due to be printed next week.

What should the project manager do?

A Approve the claim, since marketing owns the launch materials.

B Stay silent, since claims are not the project's responsibility.

C Raise the gap with marketing, and involve legal or compliance so the claim is accurate and evidenced.

D Adjust the project's emissions data so the claim is supported.

◆ PAUSE AND DECIDE

What should the project manager do?

A. Approve it.
Endorses a claim the project's own data does not support.

B. Stay silent.
Knowing a claim is false and saying nothing fails the honesty value.

C. Raise it; involve legal or compliance.
Honest, collaborative, and protects the organisation from a regulated false claim.

D. Adjust the data.
Falsifying data. A serious ethics breach.

THE SIGNAL

A sustainability claim beyond the evidence is greenwashing. Raise it, with the evidence, before it goes out.

AI drafts; people decide. Sustainability is measured, not promised.

MODULE 42 · FOUNDATIONS - ALL DOMAINS

Confusion pairs

Around fifty pairs of terms the exam uses to tempt you, each told apart in one line, grouped by theme, with eight scenarios on the most dangerous.

DOMAIN Cross-domain EXAM WEIGHT All domains MODULE 42 of 49

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Most wrong answers are right answers to a different question.

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How to use this module

- Read each pair's one-line difference, then cover it and test yourself.
- In a question, find the **detail** that decides which half applies: identified or not, before or after, who owns it.
- When two options feel equally right, check whether they are the two halves of a pair.
- Each pair points back to the module that taught it, if you need the fuller story.

> ON THE EXAM
>
> The deciding detail is usually in the scenario's second sentence. Read for it deliberately.

Who owns what

ROLE	OWNS
Sponsor	The business case, funding, and decisions beyond the PM's authority
Product owner	The product backlog order and what the product should do
Project manager	Delivery of the project within delegated authority, and integration
Scrum master	The Scrum process, coaching, and removing impediments
Functional manager	People and resources in their department, and their priorities
PMO	Standards, support, reporting, and sometimes control of projects

ON THE EXAM

Many answers turn on sending the decision to its owner. Backlog order: product owner. Business case: sponsor.

People and leadership

PAIR	THE DIFFERENCE IN ONE LINE
Task or relationship conflict	About the work, and healthy; or about the people, and harmful (11)
Collaborate or compromise	Meet both sides' real needs; or each gives something up, when time is short (11)
Smoothing or withdrawing	Play down differences to calm things; or step back and postpone (11)
Consensus or unanimity	Everyone can support it; or everyone actively agrees (10)
Explicit or tacit knowledge	Can be written down; or lives in people, moved by working together (16)
Servant or directive leadership	Remove obstacles and grow the team; or tell people what to do, for crises (06)

ON THE EXAM

The number after each line is the module that taught it.

Stakeholders and communication

PAIR	THE DIFFERENCE IN ONE LINE
Push or pull communication	Sent to specific people; or placed for people to fetch (14)
Interactive or push	Live exchange that confirms understanding; or delivery only (14)
Acknowledgement or understanding	'I got it'; or 'I understood it'. Neither means agreement (14)
Requirements or expectations	Stated and agreed; or often unstated assumptions (13)
Engagement plan or communications plan	Changing involvement and support; or moving information (13)
Internal or external customer	Inside the organisation, such as operations; or outside it (13)

ON THE EXAM

When understanding must be confirmed, interactive beats push. When a customer is unhappy despite requirements met, look for expectations.

Scope and value

PAIR	THE DIFFERENCE IN ONE LINE
Product or project scope	What is delivered; or all the work to deliver it (18)
Validate or control scope	Customer accepts deliverables; or managing changes to the baseline (18)
Definition of ready or done	Fit to start an iteration; or fit to count as finished (19)
Acceptance criteria or definition of done	Specific to one story; or applies to every item (19)
MVP or MMF	Smallest thing to learn from; or smallest thing of value on its own (20)
Output or outcome	What the project produces; or the change it causes (01, 20)

ON THE EXAM

Validate scope needs the customer; control scope needs the baseline. Acceptance criteria are per story; the DoD is for all.

Quality

PAIR	THE DIFFERENCE IN ONE LINE
Quality or grade	Meets requirements; or a level of features (28)
Manage or control quality	Checks the process; or checks the product (28)
Inspection or audit	Examines a deliverable; or examines whether processes are followed (28)
Control or specification limits	What the process normally does; or what the customer requires (28)
Conformance or non-conformance costs	Spent to get it right; or lost because it went wrong (28)
Internal or external failure	Found before release; or found by the customer, and costliest (28)

ON THE EXAM

Is the team following its process? Audit. Is this deliverable right? Inspection.

Schedule and estimating

PAIR	THE DIFFERENCE IN ONE LINE
Effort or duration	Work required; or calendar time it takes (21)
Crashing or fast tracking	Add resources, more cost; or overlap activities, more risk (22)
Levelling or smoothing	Can move the end date; or stays within float (22)
Total or free float	Slip without delaying the project; or without delaying the next activity (22)
Mandatory or discretionary dependency	Required by the work; or chosen, so can be overlapped (22)
Analogous or parametric	From one similar project; or from a unit rate times quantity (21)

ON THE EXAM

Money but no risk appetite: crash. Risk appetite but no money: fast track.

Cost and finance

PAIR	THE DIFFERENCE IN ONE LINE
Contingency or management reserve	Identified risks, in the baseline; or unforeseen, outside it (26)
Cost baseline or budget	Estimates plus contingency; or plus management reserve too (26)
Cost or schedule variance	EV minus AC; or EV minus PV. Negative is bad (27)
Typical or atypical EAC	Performance continues: BAC / CPI; or a one-off: AC + (BAC - EV) (27)
Sunk or opportunity cost	Already spent, ignore it; or value of the best option not chosen (20, 26)
Fixed price or cost-reimbursable	Seller carries cost risk; or buyer does (24)

ON THE EXAM

Identified risk occurs: contingency. Nobody foresaw it: management reserve, through change control.

Risk and issues

PAIR	THE DIFFERENCE IN ONE LINE
Risk or issue	Might happen; or is happening (35)
Planned response or workaround	Prepared for an identified risk; or improvised for a surprise (35)
Contingency or fallback plan	What to do if the risk occurs; or if the contingency fails (37)
Residual or secondary risk	Left after the response; or created by the response (37)
Transfer or mitigate	Someone else bears the impact; or reduce probability or impact (37)
Exploit or enhance	Make sure the opportunity happens; or make it more likely (37)
Appetite or threshold	General attitude to risk; or a number that triggers action (36)

ON THE EXAM

Risk occurred and a plan exists: use it. Contingency failed: fallback. Response made a new risk: secondary.

Governance, change and closure

PAIR	THE DIFFERENCE IN ONE LINE
Process asset or environmental factor	Our own, usable and updatable; or a condition to work within (31)
Portfolio or programme	Chooses the right work; or coordinates related projects (31)
Change request or backlog item	Changes a baseline; or product change inside the boundary (34)
Change log or configuration management	Requests and decisions; or which versions are current (34)
Lessons register or repository	This project's lessons; or the organisation's (16)
Phase or project closure	A checkpoint and decision; or the end (30)

ON THE EXAM

Agile product change inside budget and date: backlog. Baseline, budget or contract: change request.

Agile, charts and change

PAIR	THE DIFFERENCE IN ONE LINE
Sprint review or retrospective	Inspect the product with stakeholders; or improve how the team works (04)
Product or sprint backlog	Everything the product might need; or this sprint's plan (04)
Burndown or burnup	Work remaining, hides scope change; or work done and total scope (29)
Velocity or capacity	Points actually completed; or time available this iteration (04, 21)
Scrum or Kanban	Fixed sprints and roles; or continuous flow with WIP limits (04, 05)
Change or transition	What happens outside, on a date; or what people go through inside (39)

ON THE EXAM

Stakeholders inspect the product at the review. Only the team improves its process at the retrospective.

> On an agile project, a senior marketing manager asks the project manager to move her feature to the top of the product backlog for the next sprint.
>
> The team has a product owner.

What should the project manager do?

A Reorder the backlog, since the request comes from a senior stakeholder.

B Direct her to the product owner, who owns the backlog order.

C Ask the sponsor to decide the backlog order.

D Ask the development team to vote on the order.

◆ PAUSE AND DECIDE

What should the project manager do?

A PM reorders.

Takes the product owner's decision. Seniority does not transfer ownership.

B Send her to the product owner.

The product owner owns the backlog order.

C Sponsor decides.

The sponsor owns the business case, not day-to-day backlog order.

D Team votes.

Developers own how, not what order the product is built in.

THE PAIR

Pair: product owner or project manager. Backlog order belongs to the product owner.

> A story has been tested, reviewed and integrated, and it meets the team's definition of done. At the sprint review, the product owner rejects it, because it does not show the discount rules the story's acceptance criteria require.
>
> A developer argues it should count as done.

Which statement is correct?

A Meeting the definition of done is enough for the story to be accepted.

B Meeting the acceptance criteria is enough, even without the definition of done.

C The definition of done and acceptance criteria are the same thing.

D The story must meet both: the definition of done for all items, and its own acceptance criteria.

◆ PAUSE AND DECIDE

Which statement is correct?

A DoD is enough.

The DoD is a general quality bar; it cannot know this story's rules.

B Criteria are enough.

Without the DoD, quality standards are skipped.

C They are the same.

One applies to every item; the other to one story.

D Both are needed.

DoD for quality on every item; acceptance criteria for this story's behaviour.

THE PAIR

Pair: definition of done or acceptance criteria. Every story needs both.

> A control chart shows a machining process with all points randomly spread within its control limits. However, about one part in five falls outside the customer's specified tolerance.
>
> The line manager says the process is 'in control, so there is no problem'.

Which statement is correct?

A. The process is stable, but not capable of meeting the specification; it needs improving.

B. The process is out of control and must be investigated for a special cause.

C. The control limits should be moved to match the customer's specification.

D. There is no problem, because the process is in control.

◆ PAUSE AND DECIDE

Which statement is correct?

A Stable but not capable; improve it.

In control is not the same as meeting the customer's specification.

B Out of control.

No point is outside the control limits and there is no run. It is stable.

C Move the control limits.

Control limits come from the process data, not from the customer.

D No problem.

One part in five fails the customer's requirement.

THE PAIR

Pair: control limits or specification limits. Stable is not the same as good enough.

> An activity is delayed by three days. The schedule shows it has five days of total float and one day of free float.
>
> The project manager is explaining the effect to the team.

What is the effect of the delay?

A The project finish date is delayed by three days.

B There is no effect on any other activity or on the finish date.

C The finish date is not affected, but the next activity's early start is delayed by two days.

D The project finish date is delayed by two days.

◆ PAUSE AND DECIDE

What is the effect of the delay?

A Finish delayed three days.
Ignores the five days of total float.

B No effect on anything.
Ignores free float: only one day before the successor is affected.

C Finish safe; successor two days later.
3 is within total float 5; 3 minus free float 1 = 2 days for the successor.

D Finish delayed two days.
Applies the free-float arithmetic to the project end date.

THE PAIR

Pair: total float or free float. Total protects the end date; free protects the next activity.

> A portfolio board can fund only one of two projects. Project Alpha has an NPV of $120,000. Project Beta has an NPV of $80,000.
>
> The board chooses Alpha.

What is the opportunity cost of this decision?

A $0, since the better project was chosen.

B $120,000, the NPV of Alpha.

C $200,000, the combined NPV.

D $80,000, the NPV of Beta.

◆ PAUSE AND DECIDE

What is the opportunity cost of this decision?

A $0.
Choosing well does not remove the value given up.

B $120,000.
That is the value gained, not the value forgone.

C $200,000.
Both values together; not a cost of anything.

D $80,000.
The value of the best option not chosen.

THE PAIR

Pair: sunk cost or opportunity cost. Opportunity cost is the value of the option you gave up.

> When the main supplier failed, the risk owner switched to the backup supplier, as the contingency plan required. Two weeks later, the backup supplier also fails to deliver.
>
> The risk register includes a further plan for this case: build the component in-house.

What should happen next?

A Implement the fallback plan, and build the component in-house.

B Try the contingency plan again with the backup supplier.

C Carry out a new risk analysis before acting.

D Escalate the failure to the sponsor.

◆ PAUSE AND DECIDE

What should happen next?

A Implement the fallback plan.
The contingency has failed; the fallback exists for exactly this.

B Retry the contingency.
It has already failed.

C New analysis.
The fallback was analysed and agreed in advance.

D Escalate.
A planned response exists within the owner's authority.

THE PAIR

Pair: contingency or fallback plan. When the contingency fails, the fallback applies.

> At recent sprint reviews, the team has spent most of the time discussing its internal disagreements about code reviews and pairing. Invited stakeholders say they leave without seeing much of the product.
>
> The scrum master wants to fix this.

What should the scrum master do?

A Stop inviting stakeholders to sprint reviews.

B Lengthen the sprint review to cover both topics.

C Move process discussions to the retrospective, and use the review to inspect the increment with stakeholders.

D Replace the retrospective with a longer review.

◆ PAUSE AND DECIDE

What should the scrum master do?

A Drop the stakeholders.

The review exists to get stakeholder feedback on the product.

B Lengthen the review.

Keeps mixing two events with different purposes and audiences.

C Process to the retrospective; product to the review.

Each event does its own job, with its own audience.

D Replace the retrospective.

Removes the team's safe space to improve its own way of working.

THE PAIR

Pair: sprint review or retrospective. Product with stakeholders; process within the team.

> A new project manager is listing the inputs available for planning a data centre move.
>
> Which of these is an organisational process asset?

Which item is an organisational process asset?

A. The shortage of qualified network engineers in the local labour market.

B. The organisation's lessons learned repository from past data centre moves.

C. A government regulation on data residency.

D. The organisation's risk-averse culture.

◆ PAUSE AND DECIDE

Which item is an organisational process asset?

A Labour market shortage.
An external condition: an environmental factor.

B Lessons learned repository.
The organisation's own knowledge, usable and updatable.

C Government regulation.
An external condition the project must work within.

D Risk-averse culture.
Internal, but a condition, not a usable asset. An environmental factor.

THE PAIR

Pair: process asset or environmental factor. Can you use and update it? Asset. Must you work within it? Factor.

When two options both look right, find the pair, then find the detail that decides it.

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MODULE 43 · FOUNDATIONS - ALL DOMAINS

Numbers and graphs clinic

Every formula the exam uses on four slides, the traps built into the options, and a method for reading charts quickly, with exhibits to practise on.

DOMAIN Cross-domain EXAM WEIGHT All domains MODULE 43 of 49

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Most exam maths is one formula and one trap.

Estimating formulas

MEASURE	FORMULA	MODULE
Triangular average	(O + M + P) / 3	21
PERT, or beta, average	(O + 4M + P) / 6	21
PERT standard deviation	(P - O) / 6	21
Confidence ranges	+/-1 SD about 68%; +/-2 SD about 95%; +/-3 SD about 99.7%	21
Velocity forecast	Points remaining / average velocity = iterations	21
Little's Law	Cycle time = work in progress / throughput	05

ON THE EXAM

Read whether the question says PERT, beta or triangular. If it only gives three points, PERT is the usual default.

Earned value formulas

MEASURE	FORMULA	BAD NEWS WHEN
Cost variance	CV = EV - AC	Negative
Schedule variance	SV = EV - PV	Negative
Cost performance index	CPI = EV / AC	Below 1
Schedule performance index	SPI = EV / PV	Below 1
To-complete index	TCPI = (BAC - EV) / (BAC - AC)	Above 1, and above CPI

ON THE EXAM

Every formula starts with EV. Variance subtracts; index divides.

Forecasting formulas

MEASURE	FORMULA	USE WHEN
EAC, typical	BAC / CPI	Current performance will continue
EAC, atypical	AC + (BAC - EV)	The variance was a one-off
EAC, both	AC + (BAC - EV) / (CPI x SPI)	Schedule pressure also affects cost
Estimate to complete	ETC = EAC - AC	How much more will be spent
Variance at completion	VAC = BAC - EAC	Negative: forecast overrun

ON THE EXAM

The story in the question picks the EAC formula: 'continue', 'one-off', or 'both cost and schedule'.

Everything else

MEASURE	FORMULA OR RULE	MODULE
Communication channels	n x (n - 1) / 2	14
Expected monetary value	Probability x impact; threats negative	36
Decision tree	EMV of each branch, including its cost; choose highest	36
Float	Late start - early start; critical path float is zero	22
Incentive fee	Seller's share of underrun added to fee; of overrun, removed	24
Cost baseline and budget	Estimates + contingency; + management reserve	26

ON THE EXAM

Include the project manager in n. Keep EMV signs. Choose the decision branch with the highest EMV.

Selection rules

MEASURE	CHOOSE
Net present value	Highest. Ignore project length: timing is already in it
Internal rate of return	Highest
Return on investment	Highest
Benefit-cost ratio	Highest; above 1 means benefits exceed costs
Payback period	Shortest
Opportunity cost	The value of the best option not chosen

ON THE EXAM

Compare like with like. If every option gives NPV, decide on NPV and ignore distracting extra measures.

The traps in the options

- **Total or increase**: channels after a team grows; the new total, or how many more?
- **Units**: sprints or weeks; thousands or dollars; percent or points.
- **The wrong formula**: triangular for PERT; typical EAC for a one-off.
- **Signs**: an opportunity treated as a threat; a negative variance read as good.
- **Order and share**: buyer and seller shares swapped; TCPI upside down.
- **Intermediate numbers**: an option that is a step on the way, not the answer.

> ON THE EXAM
>
> After you calculate, check your answer is not one of these traps. Several options usually are.

Calculating under time pressure

- **Write the formula** first, on the whiteboard, before any numbers.
- **Plug in** the numbers, labelled, so you do not swap PV and AC.
- **Estimate** roughly before calculating: is the answer bigger or smaller than the budget?
- **Sanity-check** the result: a CPI below 1 must mean over budget.
- Use the on-screen calculator; spend about a minute; flag and move on if stuck.

> ON THE EXAM
>
> Formula, numbers, estimate, check. A minute each, then move on.

Reading any chart in five steps

01
Title and axes
What is measured, in what units, over what?

02
Legend
Which line or bar is which?

03
Key point
Find 'today', the end, or the threshold.

04
Compare
Which line is above which, and is the gap growing?

05
The question
Read what it asks, then take only that from the chart.

ON THE EXAM

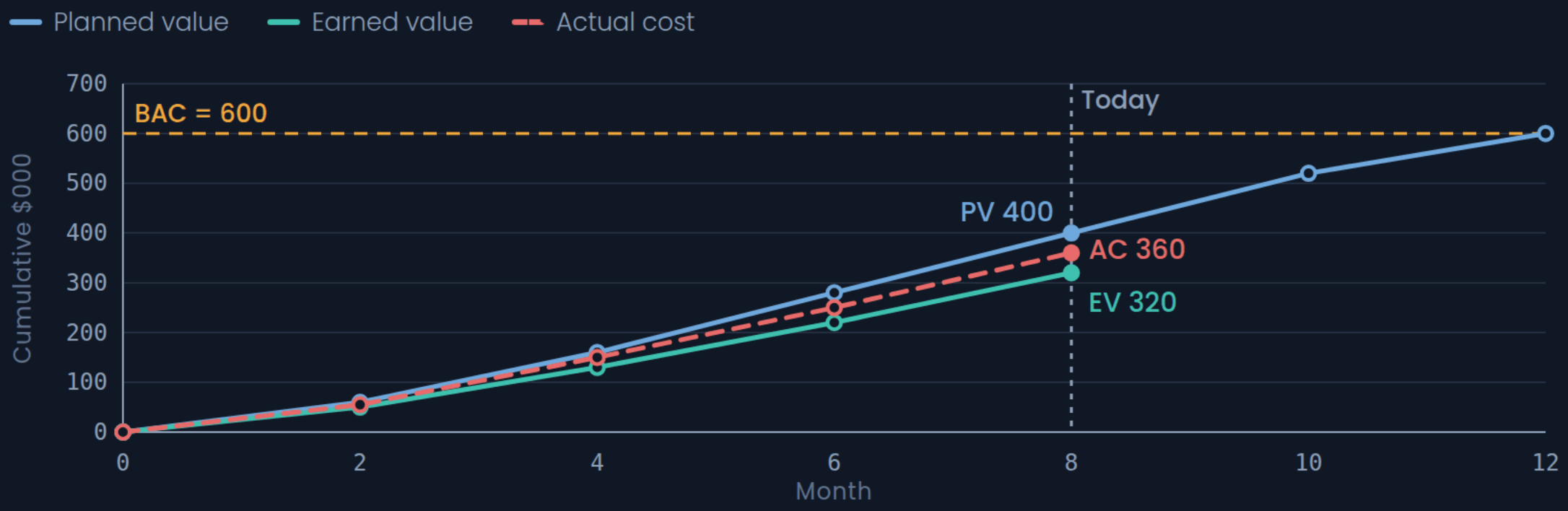
Axes and legend first. Most chart mistakes are reading the wrong line or the wrong unit.

Chart traps

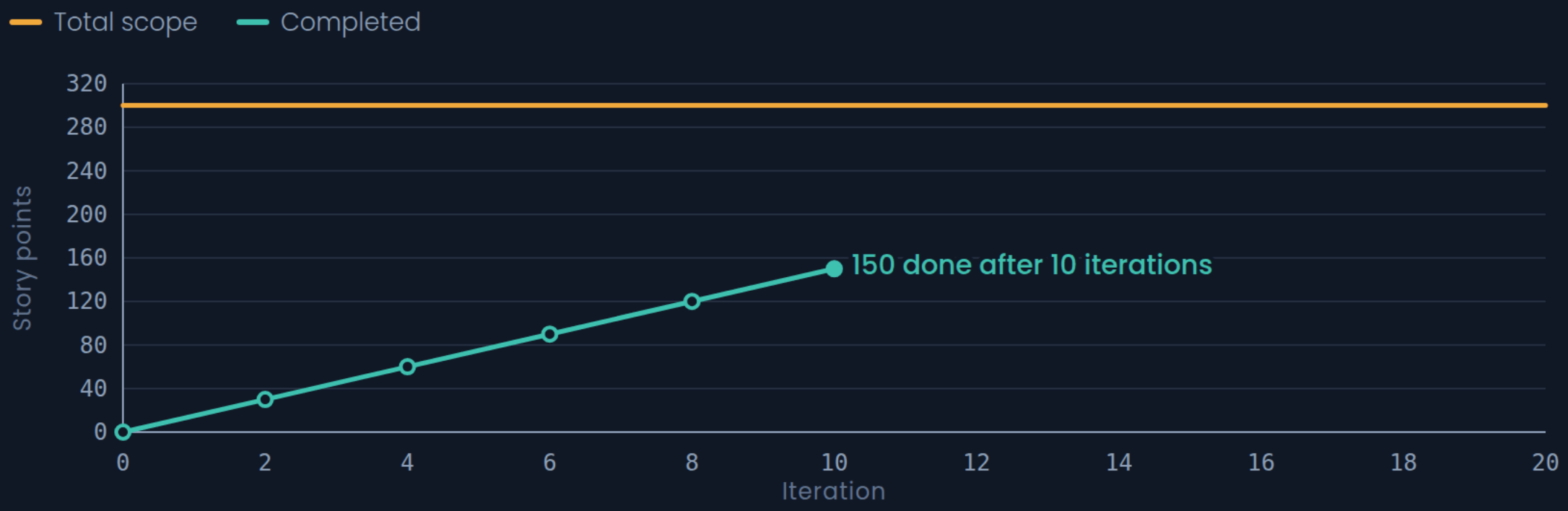
- An axis that does not start at zero makes small differences look large.
- **Cumulative** or **per-period**: an S-curve is cumulative; a velocity bar chart is not.
- Units in the axis title: '$000' means thousands.
- A label near a line is not always that line's value; read the axis.
- When in doubt, read the value at the exact point the question names.

> ON THE EXAM
>
> Check the axis start, the units, and whether the chart is cumulative, before you read a single value.

Earned value at month 8



Release burnup



> Refer to exhibit A. At month 8, planned value is $400,000, earned value is $320,000, and actual cost is $360,000.
>
> The sponsor asks for the project's status.

Based on the exhibit, what is the project's status?

A Ahead of schedule and under budget.

B Behind schedule but under budget.

C Behind schedule and over budget.

D Ahead of schedule but over budget.

◆ PAUSE AND DECIDE

Based on the exhibit, what is the project's status?

A Ahead and under.
EV is below both lines: both indices are below 1.

B Behind but under.
Right on schedule; AC is above EV, so over budget.

C Behind and over.
SPI = 320 / 400 = 0.8; CPI = 320 / 360 = 0.89.

D Ahead but over.
Right on cost; EV is below PV, so behind.

THE SIGNAL

On the chart: the green line below both others means behind and over. No arithmetic needed.

> Refer to exhibit A again. The budget at completion is $600,000.
>
> The team expects both the cost and the schedule performance so far to keep affecting the remaining work.

What is the estimate at completion, to the nearest thousand?

A $754,000

B $675,000

C $640,000

D $600,000

◆ PAUSE AND DECIDE

What is the estimate at completion, to the nearest thousand?

A $754,000

360 + (600 - 320) / (0.889 x 0.8) = 360 + 394 = 754.

B $675,000

BAC / CPI: the formula for cost performance only.

C $640,000

AC + (BAC - EV): the formula for a one-off variance.

D $600,000

The original budget; ignores all variance.

THE SIGNAL

'Both cost and schedule' picks the CPI x SPI formula. The other two EAC formulas are the distractors.

> Refer to exhibit B. The team has completed 150 of 300 story points in ten iterations, at a steady rate. Scope is not expected to change.
>
> The product owner asks for a forecast.

At the current rate, how many more iterations are needed to finish the release?

A 5

B 15

C 20

D 10

◆ PAUSE AND DECIDE

At the current rate, how many more iterations are needed to finish the release?

A 5

Would need 30 points per iteration, double the rate shown.

B 15

The velocity, not the number of iterations.

C 20

The total iterations for the whole release, not the remaining.

D 10

Velocity 150 / 10 = 15; remaining 150 / 15 = 10.

THE SIGNAL

'How many more' means remaining. Total and velocity both appear as distractors.

> A project team of six people, including the project manager, grows to ten people.
>
> The project manager is updating the communications plan.

How many communication channels does the team now have in total?

A 30

B 45

C 4

D 90

◆ PAUSE AND DECIDE

How many communication channels does the team now have in total?

A 30

The increase: 45 - 15. The question asks for the total.

B 45

10 x 9 / 2 = 45.

C 4

The number of new people.

D 90

Forgot to divide by 2.

THE SIGNAL

This time the question asks for the total. Module 14's scenario asked for the increase. Read the word.

> An activity has a PERT estimate of 20 days. Its optimistic estimate is 14 days and its pessimistic estimate is 32 days.
>
> The sponsor wants a range with about 95 percent confidence.

What range should the project manager give?

A 14 to 26 days

B 17 to 23 days

C 11 to 29 days

D 14 to 32 days

◆ PAUSE AND DECIDE

What range should the project manager give?

A 14 to 26 days

SD = (32 - 14) / 6 = 3; 20 +/- 2 x 3 = 14 to 26.

B 17 to 23 days

Plus or minus one SD: only about 68 percent.

C 11 to 29 days

Plus or minus three SD: about 99.7 percent.

D 14 to 32 days

The optimistic to pessimistic span, not a confidence range.

THE SIGNAL

95 percent means two standard deviations. The other ranges are one SD, three SD, and the raw span.

> A Kanban support team has about 30 items in progress at any time. It completes about 10 items a week.
>
> A stakeholder asks how long a typical item takes from start to finish.

Using Little's Law, what is the average cycle time?

A 300 weeks

B 0.33 weeks

C 10 weeks

D 3 weeks

◆ PAUSE AND DECIDE

Using Little's Law, what is the average cycle time?

A 300 weeks

Multiplied instead of divided.

B 0.33 weeks

Divided the wrong way: throughput over WIP.

C 10 weeks

That is the throughput per week, not a time.

D 3 weeks

Cycle time = WIP / throughput = 30 / 10 = 3.

THE SIGNAL

Cycle time = work in progress / throughput. To shorten it, reduce WIP.

Formula, numbers, estimate, check. **Axes and legend first.**

MODULE 44 · FOUNDATIONS - ALL DOMAINS

How PMP questions are written

The exam's format, its eight question types, the anatomy of a question, qualifiers and distractors, pacing, the breaks and scoring.

DOMAIN Cross-domain EXAM WEIGHT All domains MODULE 44 of 49

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Every question is a small project situation with **one decision** at its centre.

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The exam at a glance

FEATURE	FROM 9 JULY 2026
Questions	180: 170 scored, and 10 unscored pretest items you cannot tell apart
Time	240 minutes: about 80 seconds per question
Structure	A case-study section first, then independent questions
Breaks	Two optional 10-minute breaks
Domains	People 33%, Process 41%, Business Environment 26%
Approaches	About 40% predictive; 60% adaptive, agile and hybrid

ON THE EXAM

Sixty percent of questions sit in adaptive or hybrid settings. Read which approach each scenario uses.

The breaks

- The first break is offered after the **case-study section**.
- The second is offered about **midway** through the independent questions.
- Once you review your answers and take a break, you **cannot return** to the previous section.
- Review flagged questions in a section **before** starting its break.
- Take the breaks. Ten minutes of rest usually repays itself in accuracy.

> ON THE EXAM
>
> Finish reviewing each section before its break. There is no going back afterwards.

Eight question types

TYPE	WHAT YOU DO
Multiple choice	Choose one answer from four
Multiple response	Choose a stated number of answers, such as 'choose two'
Matching and enhanced matching	Pair items, such as terms with definitions
Point-and-click	Click the right place on an image or chart
Pull-down list	Choose the right option within a sentence or table
Case or scenario (new)	Answer several questions about one extended case
Graphic-based (new)	Interpret a chart, diagram or table to answer

ON THE EXAM

Two types are new in 2026: case or scenario, and graphic-based. The rest are established formats.

Case-study sets

- First pass: note the **approach**, the **phase**, and the **people**. They change what 'right' means.
- Read each question's call, then go back to the case for the **one detail** that decides it.
- Carry forward what you have learned. Each question adds to the picture.
- Re-reading the whole case for every question burns time you will need later.

> ON THE EXAM
>
> Map the case once: approach, phase, people. Then hunt for details question by question.

The anatomy of a question

01
Context
The project, the approach, the people.

02
Complication
What has happened or changed.

03
The detail
Often the second sentence: the fact that decides it.

04
The call
What exactly is asked: first, next, best, should have.

05
Options
One best answer, and three plausible distractors.

ON THE EXAM

Find the deciding detail and the call before reading the options.

Qualifiers change the question

THE CALL SAYS	WHAT IT MEANS
What should the PM do first?	The earliest right step, usually understanding or assessing
What should the PM do next?	Something has already happened; take the following step
What is the best action?	Several are reasonable; choose the most complete and appropriate
What should the PM have done?	Prevention: what would have stopped this happening
What is the most likely cause?	Diagnosis, not action

ON THE EXAM

'First' and 'next' often have different answers for the same scenario. Read the qualifier twice.

The distractors you will meet

DISTRACTOR PATTERN	WHY IT IS WRONG
Acts beyond authority	Decides what belongs to the sponsor, product owner or board
Escalates too early	Skips assessing and collaborating (module 02's ladder)
Ignores or waits	Leaves a real problem to grow
Blames or punishes	Treats a system problem as a person's fault
Only documents	Records the problem but does nothing about it
Right action, wrong time	Good in principle, but not the first or next step

ON THE EXAM

Most wrong answers fit one of these six patterns. Name the pattern, and the option falls away.

A method for every question

- **1.** Note the approach and your role.
- **2.** Find the real problem and the deciding detail.
- **3.** Read the call and its qualifier.
- **4.** Eliminate options that fit a distractor pattern.
- **5.** Between the last two, choose the one earlier on the ladder: understand, assess, collaborate, act, escalate.

> ON THE EXAM
>
> Approach, problem, call, eliminate, ladder. Practise it until it takes seconds.

Pacing

- About **80 seconds** per question on average.
- Case-study questions and graphics take longer; short situational questions take less.
- Check the clock every 40 to 50 questions, not after each one.
- Stuck after a minute and a half? Choose your best answer, **flag** it, and move on.
- Answer every question. An unanswered question is simply marked wrong.

> ON THE EXAM
>
> Never leave a question blank. Flag and move on; come back before the section's break.

Scoring

- PMI does not publish a numeric passing score or a pass rate.
- The ten pretest items do not count, and cannot be identified.
- Results show your performance in each domain, against a target level.
- Up to three attempts are allowed within your eligibility period.

> ON THE EXAM
>
> There is no target percentage to chase. Aim to be solid in all three domains.

Exam-day essentials

- Online or at a test centre: check the current rules in PMI's exam handbook.
- Valid identification that exactly matches your registration name.
- Use the on-screen calculator and the whiteboard for calculations and quick notes.
- Sleep, eat, and arrive early. Four hours of concentration is a physical task.

> ON THE EXAM
>
> Logistics change. Check PMI's current handbook in the week before your exam.

> A team member tells the project manager that a key vendor may miss its delivery date by about two weeks. The delivery is on the critical path.
>
> The project manager has not yet spoken to the vendor.

What should the project manager do FIRST?

A Update the schedule to show a two-week delay.

B Verify the situation with the vendor and assess the impact on the project.

C Inform the sponsor that the project will be two weeks late.

D Invoke the contract's late-delivery penalty clause.

◆ PAUSE AND DECIDE

What should the project manager do FIRST?

A Update the schedule.
Right action, wrong time: the delay is not yet confirmed.

B Verify and assess.
First means understand: confirm the facts and their impact.

C Tell the sponsor it will be late.
Reports unverified information as fact.

D Invoke the penalty clause.
Acts on a rumour, and damages the relationship.

THE QUALIFIER

'First', with unverified information: understand and assess before acting.

> The project manager has now confirmed with the vendor that delivery will be two weeks late, delaying the critical path. The risk register identified this risk, with a planned response: switch to a pre-qualified second vendor.
>
> The response is within the project manager's authority.

What should the project manager do NEXT?

A Escalate the delay to the sponsor for a decision.

B Crash the remaining critical path activities.

C Implement the planned response and switch to the second vendor.

D Wait to see whether the vendor recovers some of the time.

◆ PAUSE AND DECIDE

What should the project manager do NEXT?

A Escalate.

Escalates too early: a planned response exists within authority.

B Crash the schedule.

An unplanned, costly response when a planned one exists.

C Implement the planned response.

The facts are confirmed; the prepared response is the next step.

D Wait.

Ignores a confirmed delay on the critical path.

THE QUALIFIER

Same scenario, new qualifier, new answer. 'Next', with facts confirmed and a plan ready: act on the plan.

> At go-live, warehouse staff reject a new stock system because it does not match how they actually pick orders. The requirements were written by managers, and staff were not consulted.
>
> The system meets its documented requirements.

What should the project manager have done?

A Involved warehouse staff early, eliciting requirements from them and validating the workflow with prototypes.

B Written a more detailed contract with the software supplier.

C Carried out more system testing before go-live.

D Provided more training to warehouse staff before go-live.

◆ PAUSE AND DECIDE

What should the project manager have done?

A Involve users early; prototype.

Prevents the root cause: requirements that missed how work is really done.

B Tighter contract.

The supplier built what was specified. The specification was the problem.

C More testing.

Tests against the same wrong requirements, and passes.

D More training.

Teaches people to use a system that does not fit their work.

THE QUALIFIER

'Should have' asks for prevention at the root cause. Here, the missing stakeholders.

> An agile team's velocity has fallen for three sprints. In the retrospective, developers say several stakeholders contact them directly throughout each sprint with urgent requests.
>
> The team has a product owner.

What is the BEST action for the project leader?

A Ask the team to work longer hours to restore velocity.

B Report the falling velocity to the sponsor.

C Lengthen the sprints so there is more room for interruptions.

D Work with the stakeholders and product owner so requests go through the backlog, protecting the team's focus.

◆ PAUSE AND DECIDE

What is the BEST action for the project leader?

A Longer hours.
Treats the symptom, at the cost of sustainable pace.

B Report to the sponsor.
Only documents; the leader can remove this impediment.

C Longer sprints.
Accommodates the interruptions instead of removing them.

D Route requests through the product owner and backlog.
Removes the impediment at its cause, as a servant leader should.

THE QUALIFIER

'Best' among plausible options: the one that fixes the cause, by the right person, in the right way.

Find the decision. **Read the qualifier.** Name the distractor. Climb the ladder.

MODULE 45 · FOUNDATIONS - ALL DOMAINS

Case study: Project Lantern end to end

One extended case and ten linked questions across all three domains, including an exhibit, in the format the exam's case-study section uses.

DOMAIN Cross-domain EXAM WEIGHT All domains MODULE 45 of 49

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Map the case once. Then hunt for the one detail each question turns on.

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The case: Northgate Health and Project Lantern

ITEM	DETAIL
Organisation	Northgate Health: six community clinics; consensus-driven and cautious
Project	Replace patient booking with an app, kiosks and a new booking system
Approach	Hybrid: predictive for network, kiosks and data migration; Scrum for the app
Duration and budget	Fourteen months; a controlling PMO; monthly steering committee
Main benefit	Missed appointments down by a quarter within a year of the last go-live
Rollout plan	Pilot clinic live in month 6; one clinic a month after that

ON THE EXAM

Note the approach: hybrid. Predictive answers fit the kiosks and migration; agile answers fit the app.

The people

ROLE	WHO
Sponsor	Dana, the new chief operating officer, appointed in month 5
Project manager	Sam, who has authority up to 5 percent on cost and 2 weeks on schedule
Product owner	Priya, from the patient access team
Scrum master	Luis, for the app team of seven developers
Benefit owner	The head of patient access
Other stakeholders	Clinic managers, reception staff, a union representative, the IT operations lead

ON THE EXAM

Note who owns what: Priya owns the backlog order; Dana owns the business case; Sam's tolerances are stated.

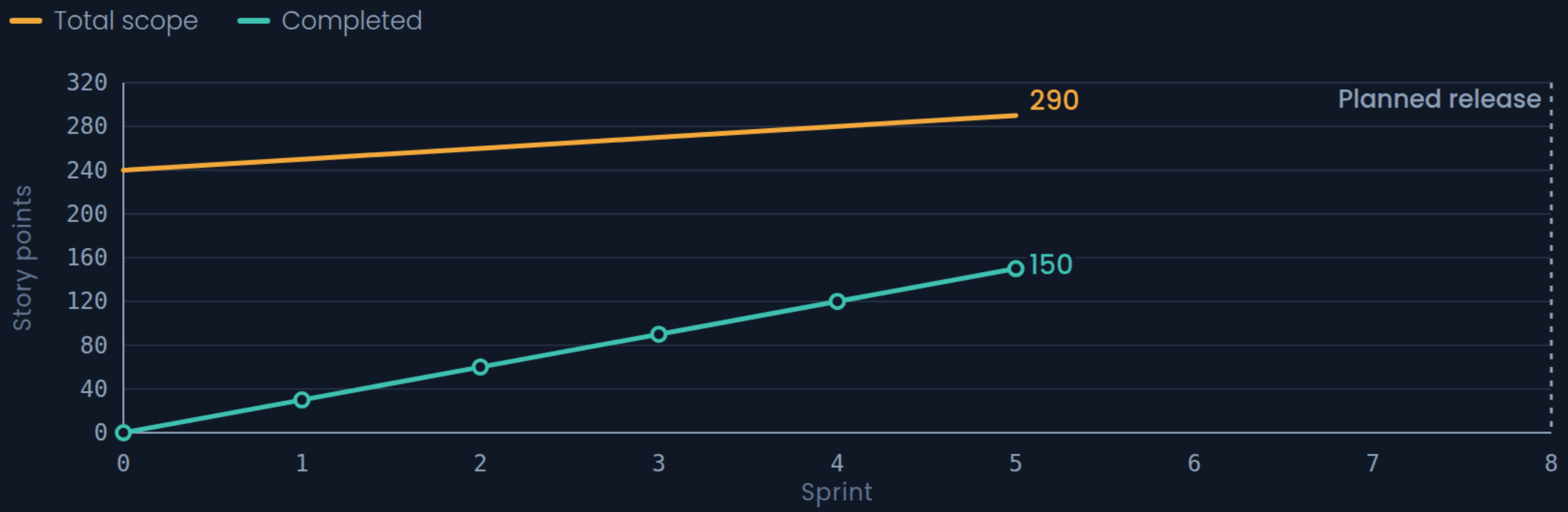
Where things stand at month 8

AREA	STATUS
Rollout	Pilot clinic live since month 6; clinic 2 went live this month
Predictive work	CPI 0.94 and SPI 0.97; the governance threshold for escalation is 0.92
App release	Planned to finish in 3 more sprints; see exhibit A
Clinic 2	Reception staff slower than before, frustrated, and using paper notes alongside the app
Risks	The kiosk supplier is on a firm fixed price contract; a planned response exists for late kiosks
Data	Clinic 3's migration rehearsal is next week; the error threshold is 2 percent

ON THE EXAM

Numbers here matter later: the 0.92 threshold, Sam's tolerances, and the 2 percent error threshold.

Lantern app: release burnup



> In her first steering meeting, Dana, the new sponsor, asks Sam to take the remaining four clinics live together in month 10, instead of one a month, to 'finish faster'.
>
> Clinic 2 is still settling in, and the culture is cautious.

What should Sam do FIRST?

A Agree, and re-plan for a combined go-live in month 10.

B Decline, because the approved plan is one clinic a month.

C Escalate Dana's request to the portfolio board.

D Assess the impact on readiness, risk and support with the team and clinics, and present options to Dana.

◆ PAUSE AND DECIDE

What should Sam do FIRST?

A Agree and re-plan.

Acts before understanding the risk to readiness and support.

B Decline.

The sponsor owns such decisions; the plan can change through proper analysis.

C Escalate above the sponsor.

Goes over the sponsor's head on her own decision.

D Assess impact; present options.

Understand and assess first; the sponsor then decides with evidence.

DRAWS ON

Modules 02, 13 and 39: a new sponsor's request is assessed and answered with options, not accepted or refused blind.

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> Refer to exhibit A. The app team has completed 150 story points in five sprints. Total scope has grown to 290 points.
>
> The release is planned for the end of sprint 8.

Based on the exhibit, which statement is most accurate?

A The team's velocity is falling, so the release is at risk.

B Velocity is steady at 30, but scope growth means about 5 more sprints are needed, not 3.

C The release will finish on time at the current velocity.

D The team has completed about two thirds of the total scope.

◆ PAUSE AND DECIDE

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Based on the exhibit, which statement is most accurate?

A Velocity is falling.
The completed line rises by a steady 30 each sprint.

B Steady velocity; scope growth; about 5 sprints.
140 remaining / 30 = 4.7 sprints, against 3 planned.

C On time.
Three sprints at 30 completes 90 of the 140 remaining.

D Two thirds done.
150 of 290 is about half, not two thirds. It reads the original 240 as the total.

DRAWS ON

Module 29: a burnup separates steady progress from scope growth. Module 43: remaining work over velocity.

> Following on from question 2, the release date is tied to clinic 3's go-live, which the clinics have already planned staff rotas around.
>
> Priya, the product owner, says every item in the backlog is important.

What should Sam do NEXT?

A Ask the team to increase its velocity to 47 points a sprint.

B Freeze all scope changes until the release.

C Help Priya reprioritise, for example with MoSCoW, so the release fits the date, and agree the approach with Dana.

D Move the release date without discussion, since scope has grown.

◆ PAUSE AND DECIDE

What should Sam do NEXT?

A Demand higher velocity.

A 57 percent jump on demand is unrealistic, and inflates points.

B Freeze scope.

Stops learning, and does not remove the 140 points already there.

C Reprioritise with the PO; agree with the sponsor.

Fixed date, flexing scope: the PO decides what fits; the sponsor agrees.

D Move the date unilaterally.

The date is tied to clinic rotas, and is not Sam's alone to move.

DRAWS ON

Modules 19 and 20: with time fixed, scope flexes. MoSCoW breaks 'everything is important'.

> Three weeks after clinic 2 went live, its manager reports that reception staff are slower, frustrated, and keeping paper notes alongside the app. There are no serious defects.
>
> The clinic manager asks whether the old booking system could be switched back on.

What should Sam do?

A. Explain that a temporary dip is normal, and increase on-site support, quick fixes and feedback, without reversing the change.

B. Switch the old system back on until staff are comfortable.

C. Repeat the full classroom training for all reception staff.

D. Ban paper notes so staff must use the app.

◆ PAUSE AND DECIDE

What should Sam do?

A Normal dip: support, fixes, feedback.

Supports people through the transition, and learns from their notes.

B Switch back.

Reverses a working change at its hardest point.

C Repeat classroom training.

People in the neutral zone need support at the desk, not a classroom.

D Ban paper notes.

Forces compliance, and hides what the notes reveal about the app's gaps.

DRAWS ON

Module 39: the dip after go-live is transition. Support through it; the paper notes are feedback.

> The union representative, recorded in month 2 as 'unaware', writes to Sam. Staff have heard that self-check-in kiosks will change reception shift patterns, and they are worried about their hours.
>
> Nobody from the project has met the union representative yet.

What should Sam do?

A Reply that shift patterns are a matter for human resources, not the project.

B Send the representative the project's communications plan.

C Escalate the letter to Dana, since it involves the union.

D Meet the representative to understand the concerns, involve staff in planning, and update the engagement plan.

◆ PAUSE AND DECIDE

What should Sam do?

A Not the project's matter.

The project's change is causing the concern. Deflecting builds resistance.

B Send the communications plan.

A document does not answer a worry about people's hours.

C Escalate to the sponsor.

Too early; nobody has even talked to the representative.

D Meet, understand, involve, update.

Engages a stakeholder who moved from unaware towards resistant, early and directly.

DRAWS ON

Modules 12 and 13: resistance is information. Meet, understand, involve, and update the engagement plan.

> The monthly report for the predictive workstreams shows a CPI of 0.94 and an SPI of 0.97. Last month they were 0.95 and 0.98.
>
> Dana asks whether this needs a steering committee decision.

What should Sam advise?

A. Both indices are above the 0.92 threshold, so no escalation is required; report the slight downward trend and keep monitoring.

B. Both indices are below 1, so the steering committee must decide on corrective action.

C. The project is on track, so the indices need not be reported.

D. Rebaseline the cost and schedule so both indices return to 1.

◆ PAUSE AND DECIDE

What should Sam advise?

A Within threshold; report the trend; monitor.

Governance set 0.92. Both are above it, but the downward trend deserves attention.

B Below 1, so escalate.

Escalation is triggered by the agreed threshold, not by 1.

C Do not report.

Honest reporting includes a worsening trend, even inside tolerance.

D Rebaseline.

Hides variance instead of managing it.

DRAWS ON

Modules 27 and 31: escalate against the agreed threshold, and always report the trend honestly.

> Luis reports that two senior developers have argued about the app's architecture in the last three daily meetings. The arguments have become personal, and other developers now stay silent.
>
> Both developers are highly skilled, and the team respects them.

What should happen?

A Sam decides the architecture question to end the argument.

B Remove one of the developers from the team.

C Luis meets them privately to separate the technical question from the personal friction, then helps the team decide the architecture together.

D Ban architecture discussions from daily meetings, and take no other action.

◆ PAUSE AND DECIDE

What should happen?

A Sam decides.
Forcing a technical decision that belongs to the developers.

B Remove a developer.
Disproportionate, and loses real skill and knowledge.

C Private conversation; team decides together.
Addresses relationship conflict privately, keeps healthy task debate.

D Ban the topic only.
Hides the conflict; the architecture still needs deciding.

DRAWS ON

Modules 10 and 11: turn relationship conflict back into task conflict, then decide as a team.

> Clinic 3's migration rehearsal finds errors in 7 percent of patient records, against a threshold of 2 percent. A Pareto analysis shows two error types cause 80 percent of the errors.
>
> Go-live is in three weeks, and Dana is keen to keep the date.

What should Sam do?

A Go live as planned, and fix the records afterwards.

B Fix the root causes of the two main error types, re-run the rehearsal, and go live only when errors are under the threshold.

C Raise the error threshold to 7 percent.

D Fix all error types equally, one record at a time.

◆ PAUSE AND DECIDE

What should Sam do?

A Go live; fix later.

Patient records wrong in live use: a quality and safety failure.

B Fix the vital few causes; re-rehearse.

Pareto points to two causes; fixing them gives the best chance of the date.

C Raise the threshold.

Changes the measure, not the quality.

D Fix everything equally.

Spreads three weeks thinly, and treats symptoms record by record.

DRAWS ON

Module 28: Pareto for the vital few; root causes, not records. Quality before the date on patient data.

> A regional regulator publishes a proposal requiring patient booking data to be stored within the region from next year. The new booking system's hosting is currently outside the region.
>
> The proposal is out for consultation for three months.

What should Sam do?

A Ignore it until the regulation is confirmed.

B Move the hosting into the region immediately.

C Assess the impact with compliance and IT, record it as a risk with options, and keep watching the consultation.

D Stop the remaining clinic go-lives until the regulation is final.

◆ PAUSE AND DECIDE

What should Sam do?

A Ignore it.
Loses months of preparation time on a likely obligation.

B Move hosting now.
A major, costly change before the rule is even final.

C Assess, record as a risk, prepare options, watch.
Proportionate response to a proposed external change.

D Stop the go-lives.
Halts all benefits for a rule that is still out for consultation.

DRAWS ON

Modules 33, 36 and 40: a proposed rule is a risk. Assess it, prepare options, and keep scanning.

> At month 14, all six clinics are live and accepted, contracts are closed, and lessons are captured. Missed appointments are down 12 percent so far, against a 25 percent target due within a year.
>
> Dana asks Sam to close the project.

What should Sam do?

A Keep the project open until the 25 percent target is reached.

B Close the project formally, with the final report showing benefit progress, and confirm the head of patient access will keep measuring it.

C Close the project and stop tracking the benefit, since the outputs are delivered.

D Report the project as a failure, since the target has not been met.

◆ PAUSE AND DECIDE

What should Sam do?

A Stay open for the benefit.

Benefits often arrive after closure; the project's work is done.

B Close formally; hand benefit tracking to its owner.

Outputs accepted; benefit measured by the named owner after closure.

C Stop tracking the benefit.

Loses the evidence of whether the project was worth it.

D Declare failure.

The target is due within a year of the last go-live, not at month 14.

DRAWS ON

Modules 20 and 30: close when the work is accepted; the benefit owner measures what arrives afterwards.

One case, ten decisions, every domain. The same habits answered them all.

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PRACTICE EXAM 1 OF 4 · FOUNDATIONS - ALL DOMAINS

Practice exam 1

Twenty questions across all three domains, in exam proportions, with a walkthrough after each one and a scoring guide at the end.

QUESTIONS 20 PEOPLE 7 PROCESS 8 BUS ENV 5

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How to take this exam

- Each question holds on screen for a few seconds after it is read. **Choose within that time**, or pause if you need to.
- **Write down your letter** before the answer is shown. No changing it afterwards.
- No notes and no rewinding. Treat it as the real thing.
- Use the method: approach, problem, call, eliminate, ladder.
- At the end, score yourself by domain, not just in total.

> ON THE EXAM
>
> Commit before you see the answer. A walkthrough you watch without committing teaches far less.

> An agile team has a new developer who joined from an organisation with a very hierarchical culture. In three retrospectives, he has not said a word, although his colleagues say he has good ideas.
>
> The scrum master wants his contribution.

What should the scrum master do?

A Talk with him privately, and use retrospective formats that start with silent, written ideas before discussion.

B Ask him directly for his views in front of the team at the next retrospective.

C Leave him alone; he will speak when he is ready.

D Ask his manager to encourage him to participate more.

◆ PAUSE AND DECIDE

What should the scrum master do?

A Private talk; silent-first formats.
Builds safety and gives him a way in that fits his background.

B Put him on the spot.
Likely to deepen the discomfort a hierarchical background creates.

C Leave him alone.
The team loses his ideas, and nothing changes.

D Ask his manager.
Turns a team matter into a management one; adds pressure, not safety.

THE SIGNAL

People domain. Psychological safety and cultural awareness: create safe ways in, don't force.

> On a predictive construction project, the monthly report shows a cost performance index of 1.15 and a schedule performance index of 0.85.
>
> The sponsor is pleased that the project is under budget.

What is the most accurate interpretation?

A The project is performing well on both cost and schedule.

B The project is over budget and behind schedule.

C The work done has cost less than planned, but the project is behind schedule, so the saving may partly reflect work not yet started.

D The project is ahead of schedule, which explains the low cost.

◆ PAUSE AND DECIDE

What is the most accurate interpretation?

A Good on both.
SPI 0.85 is well behind schedule.

B Over and behind.
CPI above 1 means under budget for the work done.

C Efficient on cost, behind on time.
Reads both indices correctly and questions the saving.

D Ahead of schedule.
SPI below 1 means behind, not ahead.

THE SIGNAL

Process domain. CPI above 1 is good; SPI below 1 is bad. Read both, and ask what links them.

> Two days before a public website release, testers find that key booking pages cannot be used with a screen reader. The organisation is legally required to meet an accessibility standard.
>
> The product owner wants to release on time and fix it later.

What should the project manager do?

A Release on time, as the product owner prefers, and log the defect.

B Remove the accessibility tests from the definition of done for this release.

C Delay the entire release indefinitely until every page is perfect.

D Explain the legal obligation to the product owner, and agree with compliance a way to release only compliant pages or fix them first.

◆ PAUSE AND DECIDE

What should the project manager do?

A Release and log it.
Knowingly releases a legal non-compliance.

B Drop the tests.
Hides a compliance failure by removing the check.

C Delay indefinitely.
Overreacts; a compliant partial release or quick fix may be possible.

D Explain; agree a compliant path.
Compliance is not negotiable, but the route to it can be.

THE SIGNAL

Business Environment domain. A legal requirement is not traded for a date; the approach to meeting it is.

> Midway through a sprint, a senior stakeholder asks a developer to add a new report. Doing it would put the sprint goal at risk.
>
> The developer asks the scrum master what to do.

What should happen?

A The developer adds the report, since the stakeholder is senior.

B The request goes to the product owner to add to the product backlog, protecting the current sprint goal.

C The sprint is cancelled so the report can be built.

D The scrum master rejects the request outright.

◆ PAUSE AND DECIDE

What should happen?

A Just add it.

Endangers the sprint goal and bypasses the product owner.

B To the product owner and backlog.

Protects the sprint and lets the owner order the request.

C Cancel the sprint.

Only the product owner can, and only if the goal becomes obsolete.

D Reject it.

The request may be valuable. It deserves ordering, not refusal.

THE SIGNAL

Process domain. Mid-sprint requests go to the product owner and the backlog. The sprint goal is protected.

> On a predictive project, two specialists each believe the other is accountable for validating migrated data. Validation has not been done, and the RACI chart lists both as 'responsible' but nobody as 'accountable'.
>
> Testing starts next week.

What should the project manager do?

A Clarify the roles with both specialists, name one person accountable, and update the RACI chart.

B Do the data validation personally to avoid delay.

C Ask both specialists' managers to decide between them.

D Make both specialists jointly accountable for the task.

◆ PAUSE AND DECIDE

What should the project manager do?

A Clarify; one accountable; update RACI.

Fixes the root cause: an unclear role, with one name accountable.

B Do it personally.

Solves this instance; the role gap remains.

C Ask their managers.

Escalates a gap the PM can clarify directly.

D Joint accountability.

A RACI needs exactly one accountable person. Joint means neither.

THE SIGNAL

People domain. One task, exactly one accountable person. A gap in the RACI is fixed with the team.

A usually reliable team member on a hybrid project has been working late every night for three weeks. Her work now has errors, and she seems exhausted and short-tempered in meetings.

She has not raised any problem.

What should the project manager do?

A Point out the errors in the next team meeting so they are corrected.

B Report her performance to her functional manager.

C Talk with her privately and supportively, understand what is driving the workload, and rebalance it.

D Wait until the next performance review to discuss it.

◆ PAUSE AND DECIDE

What should the project manager do?

A Raise errors in the meeting.
Public criticism of an exhausted person worsens things.

B Report to her manager.
Treats a wellbeing and workload problem as a conduct problem.

C Private, supportive talk; rebalance.
Emotional intelligence: find the cause and remove it.

D Wait for the review.
Leaves a burnout risk to grow for weeks.

THE SIGNAL

People domain. A reliable person's sudden change is a signal. Talk privately, find the cause, fix the load.

> A supplier on a firm fixed price contract writes that its home currency has fallen sharply, and asks for a 15 percent price increase. The contract has no economic price adjustment clause.
>
> The supplier hints that it may struggle to deliver without the increase.

What should the project manager do?

A. Agree the increase to keep the supplier happy.

B. Review the contract terms with procurement, assess the risk of supplier failure, and negotiate from there.

C. Terminate the contract immediately for threatening non-delivery.

D. Ignore the letter, since the price is fixed.

◆ PAUSE AND DECIDE

What should the project manager do?

A Agree the increase.

Gives away the protection the fixed price was bought for, without analysis.

B Review terms; assess risk; negotiate.

Holds the contract position while facing the real risk of supplier failure.

C Terminate.

Drastic, and may leave the project with no supplier at all.

D Ignore it.

The price is fixed, but the delivery risk is real and needs managing.

THE SIGNAL

Business Environment domain. External shock plus a contract: know the terms, weigh the risk, then negotiate.

> A junior developer on an agile team consistently underestimates his stories, and often needs to carry work into the next sprint. He is keen to improve.
>
> The rest of the team estimates well.

What is the best action for the scrum master?

A Stop him from estimating, and let senior developers estimate for him.

B Add 50 percent to every estimate he gives.

C Coach him, for example by pairing him with a senior developer and reviewing his estimates against actuals together.

D Raise his performance with the functional manager.

◆ PAUSE AND DECIDE

What is the best action for the scrum master?

A Stop him estimating.
Removes the learning, and his voice from planning poker.

B Pad his estimates.
Hides the problem instead of building the skill.

C Coach, pair, compare with actuals.
Builds the skill of a willing learner using his own history.

D Go to his manager.
Treats a development need as a performance problem.

THE SIGNAL

People domain. A willing learner with a skills gap needs coaching, not workarounds or reporting.

> On a predictive manufacturing project, the customer's acceptance inspections have rejected three batches for defects that the team's own quality control had passed.
>
> The team's inspectors say they followed the procedure.

What should the project manager do first?

A Review the quality control process, for example with an audit, and find the root cause of the missed defects.

B Add a second round of the same inspection.

C Replace the inspectors.

D Ask the customer to relax its acceptance criteria.

◆ PAUSE AND DECIDE

What should the project manager do first?

A Audit the process; find the root cause.

If the procedure was followed, the procedure itself may be the problem.

B Inspect twice.

Repeating a flawed inspection repeats the miss.

C Replace inspectors.

Blames people who followed the procedure.

D Relax the criteria.

Lowers quality to hide the defects.

THE SIGNAL

Process domain. Procedure followed but defects missed: the process is the suspect. Audit it.

> On a predictive project, a regional director with approval authority over design decisions regularly replies to approval emails days after the deadline, delaying work.
>
> Other stakeholders respond on time.

What should the project manager do?

A Send more frequent reminder emails.

B Treat missed deadlines as approval.

C Escalate the delays to the sponsor.

D Meet the director to understand how she prefers to engage, and agree an approach, such as short decision calls or a delegate.

◆ PAUSE AND DECIDE

What should the project manager do?

A More reminders.

More of the channel that is already failing.

B Silence as approval.

Assumes authority the PM does not have; risky and unfair.

C Escalate.

Premature. Nobody has asked her what would work.

D Meet; agree how she engages.

Tailors communication to the stakeholder's needs and constraints.

THE SIGNAL

People domain. When a channel keeps failing for one stakeholder, ask what works for them.

> A product owner wants to hold back an app's first release until all planned features are complete, in about five months. A smaller set of features, already done and tested, would solve customers' most common problem now.
>
> Customers have been asking for exactly that.

What should the project leader do?

A Support the product owner's decision without comment.

B Discuss with the product owner releasing the completed, valuable features now, to deliver value and gather feedback sooner.

C Release the features without the product owner's agreement.

D Escalate the release decision to the sponsor.

◆ PAUSE AND DECIDE

What should the project leader do?

A Stay silent.

Leaves five months of available value on the table without discussion.

B Discuss an early release with the PO.

Raises the value case with the person who owns the decision.

C Release anyway.

Overrides the product owner's authority.

D Escalate.

Premature; the product owner has not heard the case yet.

THE SIGNAL

Process domain. Finished, valuable work waiting months: make the case for releasing it, to its owner.

> During a hybrid project, a developer notices that an open-source library the product depends on has announced it will stop being maintained in six months.
>
> It is not in the risk register.

What should happen next?

A Replace the library immediately.

B Ignore it, since the library still works.

C Record it as a risk, analyse its impact, assign an owner, and plan a response.

D Escalate it to the sponsor as an issue.

◆ PAUSE AND DECIDE

What should happen next?

A Replace it now.

A major change before analysing impact or options.

B Ignore it.

Unmaintained code becomes a security and support risk.

C Record, analyse, own, respond.

A new risk goes into the register and through the risk process.

D Escalate as an issue.

It has not happened yet, and it is within the project's authority.

THE SIGNAL

Business Environment domain. A newly spotted future problem is a risk. Register, analyse, own, respond.

> On a predictive project, a supplier submits an invoice for a milestone payment. The contract says milestone payments are due on formal acceptance, and the deliverable has not yet passed acceptance testing.
>
> The supplier says it has done the work.

What should the project manager do?

A Check the invoice against the contract's acceptance terms, and pay only when the milestone is accepted, explaining this to the supplier.

B Pay the invoice to keep a good relationship.

C Pay half now and half on acceptance.

D Reject the invoice and stop communicating until testing is complete.

◆ PAUSE AND DECIDE

What should the project manager do?

A Pay on acceptance, per the contract; explain.
Follows the agreed terms, and keeps the relationship clear.

B Pay anyway.
Gives up the leverage the payment terms were designed to provide.

C Pay half.
Invents terms the contract does not contain.

D Reject and go silent.
Right on payment, wrong on the relationship.

THE SIGNAL

Process domain. Payment follows the contract's acceptance terms. Explain it; do not improvise.

> Several agile team members complain privately that one colleague often misses the daily meeting and does not finish work he commits to. The team's morale is suffering.
>
> The colleague is well liked and technically strong.

What should the scrum master do?

A Report the colleague to his functional manager.

B Ignore it, since he is technically strong.

C Raise his behaviour by name at the next daily meeting.

D Talk with him privately to understand what is happening, and help the team revisit its working agreement.

◆ PAUSE AND DECIDE

What should the scrum master do?

A Report him.
Escalates before anyone has asked him why.

B Ignore it.
Morale is already suffering; it will get worse.

C Name him in the meeting.
Public blame damages trust for everyone.

D Private conversation; revisit the working agreement.
Finds the cause, and lets the team own its norms.

THE SIGNAL

People domain. Private first to understand; then the team's own agreement sets the standard.

> A predictive project will use a technology the organisation has never used before. The sponsor wants a single, firm cost figure for the business case next week.
>
> There is no historical data, and the team is unsure how long the integration will take.

What should the project manager provide?

A. A single figure, as requested, based on the team's best guess.

B. A range, with its assumptions, and a plan to narrow it quickly, for example through a prototype.

C. A very high figure to be safe.

D. No estimate until the design is complete.

◆ PAUSE AND DECIDE

What should the project manager provide?

A One firm figure.

False precision; the sponsor would treat a guess as a commitment.

B A range, assumptions, and a plan to narrow it.

Honest about uncertainty, and useful for the business case.

C A padded figure.

Hidden padding distorts the business case.

D Nothing yet.

The sponsor needs something to decide with now.

THE SIGNAL

Process domain. High uncertainty: an honest range with assumptions, plus a plan to reduce it.

> In an organisation with four agile teams, the same kind of security defect has appeared in releases from three of them over two months. Each team fixed its own instance separately.
>
> The teams rarely talk to each other.

What should the project leader recommend?

A Ask each team to add more security testing on its own.

B Appoint one person to check every release for this defect.

C Share the lesson across teams, for example through a community of practice, and update the shared definition of done and standards.

D Record the defects in each team's lessons learned register.

◆ PAUSE AND DECIDE

What should the project leader recommend?

A Each team tests more alone.
Four separate fixes for one shared cause.

B One checker for all.
Inspection bottleneck; does not prevent the defect.

C Share across teams; update shared standards.
Turns one lesson into an organisational asset, preventing it everywhere.

D Record it in each register.
Records the lesson; does not spread or apply it.

THE SIGNAL

Business Environment domain. A lesson repeated across teams belongs in shared standards and assets.

A hybrid project team works across London, Chicago and Singapore. The weekly planning meeting is at a time convenient for London and Chicago, which is 10 p.m. in Singapore.

The Singapore members have begun missing it, and feel left out of decisions.

What should the project manager do?

A Rotate the meeting time so the inconvenience is shared, and record decisions for asynchronous review.

B Require the Singapore members to attend, since the time was agreed.

C Hold the meeting without Singapore and email them the outcome.

D Move all planning work to the Singapore team.

◆ PAUSE AND DECIDE

What should the project manager do?

A Rotate times; record for async review.
Shares the burden fairly and keeps everyone in decisions.

B Require attendance.
Keeps the unfair burden on one site.

C Meet without them.
Formalises the exclusion they already feel.

D Move planning to Singapore.
Shifts the problem to the other two sites.

THE SIGNAL

People domain. Distributed teams share the inconvenience, and design for asynchronous inclusion.

> A Kanban team's weekly throughput has ranged from 4 to 12 items over the last ten weeks. A stakeholder asks when 40 remaining items will be finished.
>
> The stakeholder wants a single date.

What should the project leader provide?

A. A single date based on the best week, 12 items.

B. A single date based on the worst week, 4 items.

C. No forecast, since Kanban does not use dates.

D. A forecast range based on the team's actual throughput history, with a likely date and a confidence level.

◆ PAUSE AND DECIDE

What should the project leader provide?

A Best week.
Optimistic; almost certainly missed.

B Worst week.
Pessimistic; understates what the team normally does.

C No forecast.
Kanban teams forecast from throughput all the time.

D Range from history, with confidence.
Honest about variation, and still useful for planning.

THE SIGNAL

Process domain. Variable throughput: forecast a range from history, with a confidence level.

> The organisation announces a merger with a competitor. The competitor already has a customer app that does much of what a hybrid project is building.
>
> The project is six months from completion.

What should the project manager do?

A Continue as planned until told otherwise.

B Evaluate the impact with the sponsor, and revalidate the business case and priorities in light of the merger.

C Stop the project immediately, since the competitor's app exists.

D Speed up delivery so the project finishes before the merger completes.

◆ PAUSE AND DECIDE

What should the project manager do?

A Continue until told.

Keeps building something that may duplicate the merged company's app.

B Evaluate impact; revalidate with the sponsor.

An organisational change affects value; the owner decides with evidence.

C Stop now.

Not the PM's decision, and the merged strategy may still need parts of it.

D Rush to finish.

Racing to complete possibly duplicate work adds cost and risk.

THE SIGNAL

Business Environment domain. An organisational change that affects value goes to the sponsor, with analysis.

> On a predictive project, the change control board approved a change adding a new feature, with extra cost and two weeks. The team updated the schedule baseline and started work.
>
> The cost baseline and the requirements traceability matrix were not updated.

What should the project manager do?

A Leave the cost baseline as it is, since the schedule was updated.

B Submit a new change request for the cost.

C Report the project as over budget at the next review.

D Update the cost baseline and all other affected documents to reflect the approved change.

◆ PAUSE AND DECIDE

What should the project manager do?

A Leave the cost baseline.
Performance will be measured against a baseline that is wrong.

B A new change request.
The cost was already approved as part of the same change.

C Report as over budget.
A reporting error caused by the missing update, not a real overrun.

D Update all affected baselines and documents.
An approved change is finished only when every document reflects it.

THE SIGNAL

Process domain. Integration: one approved change updates every affected baseline and document.

Scoring guide

DOMAIN	QUESTIONS	YOUR SCORE
People	1, 5, 6, 8, 10, 14, 17	out of 7
Process	2, 4, 9, 11, 13, 15, 18, 20	out of 8
Business Environment	3, 7, 12, 16, 19	out of 5
Total	20 questions	out of 20

ON THE EXAM

Which domain your misses cluster in matters more than the total. PMI publishes no pass mark; aim for consistency.

What to do with your result

- **Around 16 or more**: strong. Keep practising with the next exam.
- **Around 13 to 15**: close. Revisit the modules named in the walkthroughs you missed.
- **Below 13**: rewatch the modules for your weakest domain before the next exam.
- For every miss, name the distractor pattern you fell for. Patterns repeat.
- These bands are a study guide, not PMI's standard, which is not published.

> ON THE EXAM
>
> Name the pattern behind each miss. That is what carries into the next exam.

Score by domain. Name each miss's pattern. Then take exam 2.

PRACTICE EXAM 2 OF 4 · FOUNDATIONS - ALL DOMAINS

Practice exam 2

Twenty new questions in exam proportions, with a walkthrough after each and a scoring guide at the end.

QUESTIONS 20 PEOPLE 6 PROCESS 9 BUS ENV 5

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A construction schedule includes a seven-day lag for concrete to cure before walls can be built. This activity is on the critical path.

The site engineer says a proven additive would cut curing to three days, at a small extra cost within the project manager's authority.

What should the project manager do?

A Keep the seven-day lag, since it is in the approved schedule.

B Use the additive immediately without further checks.

C Confirm the additive meets quality and specification requirements, then approve it and update the schedule.

D Escalate the proposal to the sponsor.

◆ PAUSE AND DECIDE

What should the project manager do?

A Keep the lag.
Ignores an opportunity to shorten the critical path.

B Use it unchecked.
Skips the quality check a structural change needs.

C Check quality; approve; update schedule.
Exploits the opportunity safely, within authority.

D Escalate.
The cost is within the PM's authority.

THE SIGNAL

Process domain. An opportunity on the critical path: check quality, then take it, within authority.

> A new agile team keeps asking its scrum master to decide which story each developer should pick up next, and how to split tasks within the sprint.
>
> The scrum master has been answering to keep things moving.

What should the scrum master do?

A Coach the team to make these decisions itself, for example by answering with questions rather than instructions.

B Continue deciding, since the team is new.

C Ask the product owner to assign the tasks instead.

D Write a detailed task allocation for each sprint.

◆ PAUSE AND DECIDE

What should the scrum master do?

A Coach self-organisation.
Builds the team's ability to manage its own work.

B Keep deciding.
Creates dependence, and a bottleneck in one person.

C The PO assigns tasks.
The PO owns what, not who does which task.

D Detailed task allocation.
Command and control; the opposite of self-management.

THE SIGNAL

People domain. Agile teams self-manage their work. The leader coaches, rather than directs.

> A supplier on a fixed price contract asks for a price increase because the published steel index has risen 8 percent. Unlike practice exam 1's supplier, this contract includes an economic price adjustment clause tied to that index.
>
> The supplier has provided the index figures.

What should the project manager do?

A Refuse, since the price is fixed.

B Negotiate a new price from scratch.

C Escalate the request to the sponsor.

D Verify the index figures, and apply the adjustment exactly as the clause defines.

◆ PAUSE AND DECIDE

What should the project manager do?

A Refuse.
Ignores a clause both parties agreed.

B Renegotiate.
The clause already defines the answer.

C Escalate.
Applying an agreed clause is routine contract administration.

D Verify, then apply the clause.
The contract anticipated this; follow it exactly.

THE SIGNAL

Business Environment domain. Paired with exam 1, question 7: this time the contract has a clause, so apply it.

> An agile team's increments meet its definition of done, but the operations team repeatedly cannot deploy them without days of extra work on configuration and monitoring.
>
> Releases are being delayed.

What should the project leader do?

A Ask operations to work faster on deployments.

B Work with the team and operations to add deployment readiness to the definition of done.

C Add a separate deployment phase at the end of every release.

D Accept the delays as normal for agile work.

◆ PAUSE AND DECIDE

What should the project leader do?

A Operations work faster.

Blames the receivers for work that should have been done earlier.

B Deployment readiness in the DoD.

Makes 'done' mean usable, with operations as a customer.

C A deployment phase.

Batches the problem at the end instead of preventing it.

D Accept delays.

Agile aims for releasable increments, not delayed ones.

THE SIGNAL

Process domain. If 'done' is not deployable, the definition of done is incomplete. Operations is a customer.

> In steering meetings, a powerful department head repeatedly argues against a hybrid project. Privately, a colleague mentions he believes the project will cut his department's budget.
>
> The project is approved and funded.

What should the project manager do?

A Ask the sponsor to instruct him to support the project.

B Exclude him from future steering discussions.

C Meet him privately to understand his concerns, and look for ways the project can address his interests.

D Ignore the opposition, since the project is approved.

◆ PAUSE AND DECIDE

What should the project manager do?

A Sponsor instructs him.
Compliance, not support; the concern stays hidden.

B Exclude him.
Turns open opposition into something worse.

C Private meeting; address interests.
Finds the real interest behind the position.

D Ignore him.
A powerful opponent can still derail delivery.

THE SIGNAL

People domain. Behind a position is an interest. Find it privately before trying to change the position.

A risk owner responsible for eight risks on a predictive project has left the organisation. Two of those risks have triggers expected within the next month.

Nobody has taken over the risks.

What should the project manager do first?

A Assign new owners to the eight risks, starting with those near their triggers, and review their responses.

B Take personal ownership of all eight risks permanently.

C Remove the risks from the register until new owners are found.

D Report the departure to the steering committee.

◆ PAUSE AND DECIDE

What should the project manager do first?

A Reassign owners, urgent ones first.

A risk without an owner has no one to act on its trigger.

B PM owns all eight.

Owners should be the people best placed to act, not one person for everything.

C Remove them.

The risks still exist; removing them hides them.

D Report to steering.

Reporting does not assign anyone to act.

THE SIGNAL

Process domain. Every risk needs an owner who can act on its trigger. Fix ownership, urgent risks first.

> An agile team's velocity has averaged 30 points over the last six sprints. In sprint planning, the product owner pushes the team to commit to 50 points, to catch up on a release.
>
> The developers are uneasy but reluctant to argue.

What should the scrum master do?

A Support the 50-point commitment to meet the release.

B Tell the product owner the release date must move.

C Let the developers decide privately, without the product owner.

D Help the team plan to its measured capacity, and help the product owner adjust scope or expectations for the release.

◆ PAUSE AND DECIDE

What should the scrum master do?

A Commit to 50.
Two-thirds above measured velocity. almost certain failure, or cut corners.

B Move the date.
One option, but the scrum master does not decide it.

C Plan without the PO.
Sprint planning includes the product owner.

D Plan to capacity; help the PO adjust.
Honest planning, and a real conversation about scope or date.

THE SIGNAL

Process domain. Plan to measured velocity. The gap becomes a scope or date decision for the product owner.

> An internal audit finds that a hybrid project has been deleting test records after each release. A regulation requires such records to be kept for seven years.
>
> Six months of records are already gone.

What should the project manager do?

A Continue deleting records, since the audit is complete.

B Stop the deletion, put retention in place as the regulation requires, and report the gap through the compliance process.

C Recreate the deleted records from memory.

D Keep the finding within the project team.

◆ PAUSE AND DECIDE

What should the project manager do?

A Carry on deleting.
Continues a known breach.

B Stop, retain, report.
Contain, fix, and report a compliance gap honestly.

C Recreate from memory.
Fabricated records are worse than missing ones.

D Keep it quiet.
Concealment turns a gap into misconduct.

THE SIGNAL

Business Environment domain. A compliance gap: contain it, fix it, report it. Never fabricate or hide.

A predictive project team worked extra hours for several weeks and delivered a major milestone two weeks early. Delighted, the sponsor offers the project manager a personal bonus.

The team has not been recognised.

What should the project manager do?

A Thank the sponsor, and ask that the team's contribution be recognised, suggesting team-based recognition.

B Accept the bonus, since the project manager is accountable for the milestone.

C Decline the bonus and say nothing more.

D Accept the bonus and share it privately with the team.

◆ PAUSE AND DECIDE

What should the project manager do?

A Credit the team; suggest team recognition.
Recognises the people who did the work, openly and fairly.

B Accept it alone.
Takes individual credit for team effort; damages trust.

C Decline silently.
The team still goes unrecognised.

D Share it privately.
Well-meant, but informal and not the organisation's recognition.

THE SIGNAL

People domain. Recognition belongs to those who did the work. Make it visible and team-based.

> The architecture lead and product owner agreed in a corridor conversation to drop a planned integration. Nobody recorded or shared the decision.
>
> Two weeks later, part of the team is still building the integration.

What should the project manager do to prevent this recurring?

A Ask the architecture lead not to make decisions informally.

B Hold a daily meeting for all decisions.

C Ask the team to check with the architecture lead before starting any work.

D Agree that decisions are recorded in a decision log and shared through the agreed channels.

◆ PAUSE AND DECIDE

What should the project manager do to prevent this recurring?

A Ban informal decisions.
Decisions happen anywhere; recording them is the fix.

B Daily decision meeting.
Heavy, and still does not record or share decisions.

C Check with one person first.
Creates a bottleneck instead of transparency.

D Decision log, shared routinely.
Makes every decision visible to everyone affected.

THE SIGNAL

Process domain. Decisions nobody recorded get acted on twice. Log them and share them.

> A supplier on a predictive project reports a shortage of the specified pump model, and proposes a substitute it says is equivalent. The substitute's datasheet differs slightly on pressure rating.
>
> The installation date is five weeks away.

What should the project manager do?

A. Accept the substitute, since the supplier says it is equivalent.

B. Have the substitute evaluated against the specification, and if acceptable, agree it through the contract's change process.

C. Reject the substitute and wait for the original model.

D. Let the installation team decide on site.

◆ PAUSE AND DECIDE

What should the project manager do?

A Take the supplier's word.
The datasheet already shows a difference that matters.

B Evaluate; agree via contract change.
Checks quality, and changes the contract formally if accepted.

C Reject and wait.
Rejects a possibly acceptable option before evaluating it.

D Decide on site.
An informal, constructive change, made too late.

THE SIGNAL

Process domain. A substitute is a change to the contract and the specification. Evaluate, then change formally.

> Updated climate forecasts show a wetter rainy season than usual during the months a predictive project plans its excavation and foundations.
>
> The current risk register assumes normal rainfall.

What should the project manager do?

A Wait to see whether the forecast is accurate.

B Move all groundworks to the following year.

C Update the risk register, analyse the impact, and plan responses such as re-sequencing or extra drainage.

D Report the forecast to the sponsor and take no further action.

◆ PAUSE AND DECIDE

What should the project manager do?

A Wait.
Loses time to prepare for a foreseeable risk.

B Move everything a year.
A drastic response before analysis.

C Update the register; analyse; plan responses.
A changed external condition becomes a managed risk.

D Report only.
Informs without managing the risk.

THE SIGNAL

Business Environment domain. New external information changes a risk. Update, analyse, respond.

> An agile team says growing technical debt is slowing every sprint and causing defects. The product owner keeps ordering new features above all clean-up work.
>
> The disagreement has become tense.

What should the project leader do?

A Facilitate a discussion that makes the cost of the technical debt visible, so the product owner can decide with full information.

B Tell the team to fix the debt without telling the product owner.

C Support the product owner and ask the team to stop raising it.

D Escalate the disagreement to the sponsor.

◆ PAUSE AND DECIDE

What should the project leader do?

A Make the cost visible; PO decides.
Turns a tense disagreement into an informed decision.

B Fix it secretly.
Hides work from the product owner, and breaks transparency.

C Silence the team.
Suppresses useful task conflict and real information.

D Escalate.
A facilitated conversation has not been tried.

THE SIGNAL

People domain. Collaborate: make the trade-off visible, and let the owner decide with evidence.

> To meet sprint commitments, an agile team has started skipping some automated tests, even though its definition of done requires them. Escaped defects are rising.
>
> The team says it had no choice.

What should the scrum master do?

A Remove the automated tests from the definition of done.

B Ask a separate testing team to catch up on the tests later.

C Extend the sprint length to fit the tests.

D Reinforce the definition of done, and help the team commit to less work so quality is not traded away.

◆ PAUSE AND DECIDE

What should the scrum master do?

A Drop the tests.
Lowers the quality bar to match the shortcut.

B Test later.
Batches quality at the end; defects escape meanwhile.

C Longer sprints.
Treats a commitment problem with a timebox change.

D Hold the DoD; commit to less.
Quality stays fixed; scope per sprint adjusts.

THE SIGNAL

Process domain. The definition of done is not negotiable under pressure. Reduce commitment instead.

> A hybrid project's forecast now shows costs 11 percent over budget. The governance plan allows the sponsor to approve up to 10 percent; anything beyond goes to the steering committee.
>
> The sponsor asks the project manager to bring it to her alone.

What should the project manager do?

A. Bring it to the sponsor alone, as she asked.

B. Take it to the steering committee with analysis and options, as the governance plan requires, involving the sponsor.

C. Reduce scope quietly to bring the forecast under 10 percent.

D. Wait a month to see whether the forecast improves.

◆ PAUSE AND DECIDE

What should the project manager do?

A Sponsor only.

Beyond her tolerance; the agreed path is the steering committee.

B Steering committee, with the sponsor.

Follows the agreed governance, with analysis and options.

C Cut scope quietly.

Unauthorised change to hide a breach.

D Wait.

Delays a required escalation.

THE SIGNAL

Business Environment domain. Beyond the sponsor's tolerance: the governance path applies, even if asked otherwise.

> All subsidiary plans for a predictive project are drafted. The sponsor, keen to start, tells the project manager to begin execution tomorrow.
>
> The integrated plan and its baselines have not yet been approved.

What should the project manager do?

A Begin execution tomorrow, and seek approval later.

B Begin only the easy work packages while approval is pending.

C Explain that the plan and baselines need approval first, and arrange a quick approval with the sponsor.

D Refuse to start until the steering committee's next scheduled meeting.

◆ PAUSE AND DECIDE

What should the project manager do?

A Start now; approve later.
No approved baseline to measure or control against.

B Start easy work.
Still executing without an approved baseline.

C Explain; arrange quick approval.
Keeps the discipline and respects the sponsor's urgency.

D Wait for the next meeting.
Rigid; approval can be arranged quickly.

THE SIGNAL

Process domain. Approved baselines come before execution. Make approval fast, not optional.

> Two of an agile team's six developers will be on leave for most of the next sprint. The team's recent velocity has been 36 points.
>
> Sprint planning is tomorrow.

How should the team plan the sprint?

A Plan 36 points as usual, and work longer hours to cover.

B Cancel the sprint until everyone is back.

C Borrow two developers from another team for the sprint.

D Plan to the reduced capacity, for example about two thirds of normal, and tell the product owner.

◆ PAUSE AND DECIDE

How should the team plan the sprint?

A Same plan; longer hours.
Unsustainable, and likely to fail.

B Cancel the sprint.
Leave is normal; plan around it.

C Borrow two developers.
Disrupts two teams; new people slow a sprint at first.

D Plan to reduced capacity; inform the PO.
Honest capacity planning, and transparent expectations.

THE SIGNAL

People domain. Plan by capacity, which changes with availability. Keep pace sustainable.

> A project manager is preparing their company's bid for a public contract. A former colleague, now at a competing bidder, emails them the competitor's confidential pricing, saying 'this might help you'.
>
> The bid is due in three days.

What should the project manager do?

A. Not use the information, and report the email through the organisation's ethics or legal process.

B. Use the pricing, since it was sent voluntarily.

C. Delete the email and say nothing.

D. Use it only to check that the bid is not too high.

◆ PAUSE AND DECIDE

What should the project manager do?

A Don't use; report.

Protects fairness and the company; reporting is the responsible step.

B Use it.

Unfair, and likely unlawful in public procurement.

C Delete silently.

Right not to use it; wrong to hide a possible breach.

D Use it a little.

Any use is use.

THE SIGNAL

Business Environment domain. Confidential competitor data: do not use it, and report it. Fairness and responsibility.

> On a hybrid project, the app team's next release depends on a hardware milestone from the predictive workstream. The hardware milestone is now forecast to slip by two weeks.
>
> The app team has not yet been told.

What should the project manager do?

A Let each team manage its own schedule.

B Tell both teams at once, and work with the product owner to reorder the backlog around the dependency, informing stakeholders of any release impact.

C Crash the hardware work to protect the release.

D Move the release date without discussing options.

◆ PAUSE AND DECIDE

What should the project manager do?

A Leave it to each team.
Integration is the PM's job; the dependency needs coordinating.

B Inform; reorder around it; tell stakeholders.
Integrates the two workstreams at the dependency.

C Crash the hardware.
Commits cost before exploring cheaper options.

D Move the date unilaterally.
Skips options and the people who decide.

THE SIGNAL

Process domain. Hybrid integration: a slipping dependency is coordinated across both workstreams.

> A developer on an agile team tells the project leader she wants to grow into a leadership role. She is respected by the team and skilled at facilitation.
>
> There is no formal leadership vacancy.

What is the best action for the project leader?

A Tell her to wait until a leadership role becomes available.

B Promote her to team lead immediately.

C Create development opportunities, such as facilitating some retrospectives, with mentoring and feedback.

D Suggest she moves to another team with a vacancy.

◆ PAUSE AND DECIDE

What is the best action for the project leader?

A Wait for a vacancy.
Misses the chance to develop her now.

B Promote now.
No vacancy, and not the project leader's decision alone.

C Stretch opportunities, mentoring, feedback.
Develops her skills in real work, within the team.

D Move teams.
Loses a respected member without trying to grow her here.

THE SIGNAL

People domain. Develop people through real opportunities, mentoring and feedback, not just titles.

Scoring guide

DOMAIN	QUESTIONS	YOUR SCORE
People	2, 5, 9, 13, 17, 20	out of 6
Process	1, 4, 6, 7, 10, 11, 14, 16, 19	out of 9
Business Environment	3, 8, 12, 15, 18	out of 5
Total	20 questions	out of 20

ON THE EXAM

Compare with exam 1, domain by domain. A domain weak in both exams is your priority.

What to do with your result

- **Around 16 or more**: strong. Move on to exam 3.
- **Around 13 to 15**: revisit the modules named in the walkthroughs you missed.
- **Below 13**: rewatch your weakest domain's modules before exam 3.
- Did you get question 3 right? It tested whether you noticed the clause that changed exam 1's answer.
- These bands are a study guide, not PMI's standard.

> ON THE EXAM
>
> Look for the same distractor pattern across both exams. That is the habit to fix.

Compare domains across exams. **Fix the repeating pattern.** Then take exam 3.

PRACTICE EXAM 3 OF 4 · FOUNDATIONS - ALL DOMAINS

Practice exam 3

Twenty new questions in exam proportions, with a walkthrough after each and a scoring guide at the end.

QUESTIONS 20 PEOPLE 7 PROCESS 8 BUS ENV 5

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> After two departments merged, a hybrid project team includes people from both. Six weeks in, the two groups sit apart, hold separate meetings, and rarely share information.
>
> Handoffs between them are causing rework.

What should the project manager do?

A Split the project into two separate teams, one per former department.

B Bring both groups together around a shared goal and team charter, and create cross-group working such as mixed pairs.

C Ask both former department heads to tell their staff to cooperate.

D Wait for the groups to integrate naturally over time.

◆ PAUSE AND DECIDE

What should the project manager do?

A Split the team.
Formalises the divide, and multiplies handoffs.

B Shared goal, charter, mixed pairs.
Builds one team through shared purpose and real collaboration.

C Department heads instruct.
Instructions rarely create trust between groups.

D Wait.
Six weeks of rework already; it will not fix itself.

THE SIGNAL

People domain. Two groups become one team through a shared goal, a charter, and working together.

> Midway through a sprint, the government bans the payment method the sprint goal was built around, with immediate effect. The product owner says the sprint goal is now obsolete.
>
> In practice exam 1, a mid-sprint request went to the backlog. This time, the goal itself has lost its value.

What should happen?

A The team finishes the sprint as planned, then reconsiders.

B The team adds replacement work alongside the current sprint backlog.

C The product owner cancels the sprint, and the team holds a new sprint planning.

D The scrum master escalates the ban to the sponsor for a decision.

◆ PAUSE AND DECIDE

What should happen?

A Finish anyway.

Completes work for a goal that is now worthless.

B Add work alongside.

Overloads a sprint whose goal no longer makes sense.

C PO cancels; new planning.

The Scrum Guide lets the product owner cancel a sprint whose goal is obsolete.

D Escalate.

The product owner has the authority, and the facts are clear.

THE SIGNAL

Paired with exam 1, question 4. A request goes to the backlog; an obsolete goal means the PO cancels the sprint.

On a predictive building project, the contingency reserve has been fully used on identified risks. A flash flood, which nobody had identified, then damages stored materials worth $60,000.

The project manager has no reserve left in the cost baseline.

What should the project manager do?

A Request use of the management reserve through the change control process.

B Take the money from another project's budget.

C Reduce quality elsewhere to absorb the cost.

D Record the cost as a variance and take no other action.

◆ PAUSE AND DECIDE

What should the project manager do?

A Management reserve via change control.
Unforeseen event, outside the baseline: exactly what it is for.

B Borrow from another project.
Not the PM's money to move.

C Cut quality.
Hides a cost by damaging the product.

D Just record it.
Leaves the project unfunded for the recovery.

THE SIGNAL

Process domain. Unidentified risk plus no contingency: management reserve, through change control.

> Two days before the financial year end, the sponsor asks the project manager to certify a predictive project's major deliverable as complete, so the payment can be recorded this year.
>
> The deliverable still has two weeks of work remaining and has not passed acceptance.

What should the project manager do?

A Certify it, since it will be complete in two weeks.

B Certify it, but note the remaining work in a private file.

C Ask a team member to sign the certificate instead.

D Decline to certify it, explain the actual status honestly, and discuss legitimate options with the sponsor.

◆ PAUSE AND DECIDE

What should the project manager do?

A Certify: nearly done.
A false certification, whatever happens in two weeks.

B Certify with a private note.
Still false; the note only proves you knew.

C Someone else signs.
Passes the dishonesty to a colleague.

D Decline; be honest; discuss options.
Honesty, delivered with care and a search for legitimate routes.

THE SIGNAL

Business Environment domain. A false certification is dishonest, whoever asks. Decline, and help find a legitimate route.

> At a sprint review, several stakeholders criticised an agile team's increment harshly and personally. The team's morale has dropped sharply.
>
> Some of the feedback contained useful points.

What should the scrum master do?

A Stop inviting those stakeholders to sprint reviews.

B Acknowledge the team's feelings, draw out the useful feedback with them, and agree constructive feedback norms with stakeholders.

C Tell the team to ignore the criticism.

D Complain to the stakeholders' managers.

◆ PAUSE AND DECIDE

What should the scrum master do?

A Exclude the stakeholders.
Loses the feedback the review exists for.

B Acknowledge; extract value; set norms.
Supports the team, keeps the feedback, and fixes how it is given.

C Ignore it.
Throws away the useful points with the harsh ones.

D Complain to managers.
Escalates before trying to improve the review itself.

THE SIGNAL

People domain. Protect the team's morale and the feedback loop. Set norms for how feedback is given.

> On a hybrid project, finance wants a new report to show transactions by cost centre, while operations wants the same report organised by site. Both insist their layout is essential.
>
> Only one layout fits the planned release.

What should the project manager do?

A Choose the finance layout, since finance funds the project.

B Build both layouts, adding the extra work to the release.

C Bring both groups together to trace each need to the business objectives, and let the product owner decide the order.

D Ask the sponsor to choose between the two departments.

◆ PAUSE AND DECIDE

What should the project manager do?

A Finance wins.
Decides by budget power, not value.

B Build both now.
Scope creep; the release cannot absorb it.

C Joint session; trace to objectives; PO orders.
Resolves conflicting requirements by value, with the right decision-maker.

D Sponsor picks.
Escalates before the groups have worked it through.

THE SIGNAL

Process domain. Conflicting requirements: trace both to objectives, together, and let the owner decide.

> On a predictive project in a matrix organisation, a functional manager tells the project manager she is moving a key engineer to another project next week. The engineer is halfway through a critical path activity.
>
> The project manager has no authority over the engineer's assignment.

What should the project manager do first?

A Meet the functional manager to explain the impact, and negotiate timing or alternatives.

B Escalate immediately to the sponsor to block the move.

C Accept the move and re-plan the schedule.

D Ask the engineer to refuse the new assignment.

◆ PAUSE AND DECIDE

What should the project manager do first?

A Explain impact; negotiate.

In a matrix, the functional manager owns assignment. Negotiate with her first.

B Escalate at once.

Skips the conversation with the person who decides.

C Accept and re-plan.

Gives up without testing timing or alternatives.

D Engineer refuses.

Puts the engineer in an impossible position.

THE SIGNAL

People domain. In a matrix, negotiate resources with the functional manager before escalating.

> An agile team's board shows 18 items in progress for a team of six, and only two items finished this week. Developers say they switch between tasks all day.
>
> Stakeholders complain nothing seems to get finished.

What should the project leader do?

A Add more developers to the team.

B Ask developers to work longer hours to finish more items.

C Ask the product owner to stop adding items.

D Work with the team to introduce work-in-progress limits and finish items before starting new ones.

◆ PAUSE AND DECIDE

What should the project leader do?

A More developers.
More people starting more work adds to the problem.

B Longer hours.
Tired people switching more does not finish more.

C Stop new items.
The backlog is not the problem; too much started work is.

D WIP limits; finish before starting.
Cuts task switching; flow and throughput improve.

THE SIGNAL

Process domain. Too much in progress, too little done: limit WIP. Stop starting, start finishing.

A cloud provider announces a 30 percent price increase from next quarter. A hybrid project's business case assumed the old price for five years of running costs.

The increase would cut the project's forecast net present value substantially.

What should the project manager do?

A Ignore it, since running costs occur after the project closes.

B Switch to a different cloud provider immediately.

C Assess the impact on the business case and benefits, and present options to the sponsor.

D Reduce the project's scope to compensate.

◆ PAUSE AND DECIDE

What should the project manager do?

A Ignore it.
The business case includes running costs; its value has changed.

B Switch now.
A major decision before analysis.

C Assess the business case; options to sponsor.
An external change to value goes to the case's owner, with evidence.

D Cut scope.
Changes the product without analysis or authority.

THE SIGNAL

Business Environment domain. A market change that alters value: reassess the business case with its owner.

During a stakeholder meeting on a hybrid project, the product owner and the project manager openly disagreed about whether the release date could hold. Stakeholders left confused about the real position.

The two respect each other.

What should the project manager do?

A Send stakeholders an email setting out the project manager's own view.

B Meet the product owner privately to understand each other's reasoning, agree one position, and then update stakeholders together.

C Ask the sponsor to decide who is right.

D Let the product owner's view stand, to avoid further conflict.

◆ PAUSE AND DECIDE

What should the project manager do?

A Email own view.
Deepens the public split.

B Private talk; one position; joint update.
Collaborates to resolve, then restores a single clear message.

C Sponsor decides.
Escalates before two respectful colleagues have talked.

D Give way.
Smoothing: the real disagreement about the date stays unresolved.

THE SIGNAL

People domain. Disagree in private, align, then speak with one voice.

A customer on a predictive project insists on 'zero defects' in delivered software. Achieving that would require testing that roughly doubles the remaining budget.

The contract's acceptance criteria are vague on quality levels.

What should the project manager do?

A Promise zero defects and add the testing.

B Ignore the demand, since the contract is vague.

C Reduce testing to protect the budget.

D Discuss quality levels and their cost with the customer, and agree measurable acceptance criteria.

◆ PAUSE AND DECIDE

What should the project manager do?

A Promise zero defects.
An unrealistic promise at double the cost, without agreement.

B Ignore it.
Leaves a dispute waiting at acceptance.

C Cut testing.
Moves the opposite way from the customer's need.

D Agree measurable quality and its cost.
Turns a vague demand into agreed acceptance criteria.

THE SIGNAL

Process domain. Vague quality demands become measurable acceptance criteria, agreed with their cost.

> One of the only two suppliers of a critical sensor for a predictive project has just been bought by the other. The project now effectively has a single source.
>
> Current orders are unaffected for now.

What should the project manager do?

A Record the new single-source risk, assess its impact, and work with procurement on alternatives or protective terms.

B Take no action, since current orders are unaffected.

C Cancel all orders with the merged supplier.

D Redesign the product to remove the sensor.

◆ PAUSE AND DECIDE

What should the project manager do?

A Record, assess, work on alternatives.
A market change creates a risk; manage it early.

B No action.
Single-source exposure is a real, growing risk.

C Cancel orders.
Drastic, and leaves no supply at all.

D Redesign.
A major change before the risk is even analysed.

THE SIGNAL

Business Environment domain. A market shift that creates a single source: register it, analyse it, act early.

> A regulator announces an inspection of a predictive project's construction site next week, focused on safety and environmental permits.
>
> The site manager suggests tidying away a few incomplete safety records before the visit.

What should the project manager do?

A Agree to tidy away the incomplete records.

B Ask the regulator to postpone the inspection.

C Limit what the inspectors are shown.

D Prepare honestly with the compliance officer: complete what can properly be completed, and present records as they are.

◆ PAUSE AND DECIDE

What should the project manager do?

A Hide incomplete records.
Concealment from a regulator is far worse than the gap.

B Postpone.
Delays rather than prepares; may look evasive.

C Limit access.
Obstructs the inspection.

D Prepare honestly with compliance.
Fixes what can be fixed properly, and stays transparent.

THE SIGNAL

Business Environment domain. With regulators, prepare properly and stay transparent. Never conceal.

> A new developer joins an agile team mid-project. The sponsor wants her productive as soon as possible.
>
> The team has a charter, a definition of done, and a complex codebase.

What should the scrum master do?

A Give her the product documentation to read for her first two weeks.

B Assign her a large story on her own, so she learns quickly.

C Prepare her access in advance, share the charter and definition of done, and pair her with an experienced developer on real work.

D Have her observe the team for a sprint before contributing.

◆ PAUSE AND DECIDE

What should the scrum master do?

A Two weeks of reading.
Slow; tacit knowledge is not in the documents.

B A big solo story.
Sink or swim, in a complex codebase.

C Access ready; charter; pairing.
Fast, supported start, with the team's norms from day one.

D Observe for a sprint.
Delays contribution with little real learning.

THE SIGNAL

People domain. Onboard with access ready, the team's norms, and pairing on real work.

> At a sprint review, stakeholders say the increment meets its acceptance criteria, but would not actually solve their problem in daily use. They explain why.
>
> The team is frustrated, because it built exactly what was specified.

What is the best response?

A The product owner uses the feedback to update the backlog and acceptance criteria. This is the review working as intended.

B Insist the stakeholders accept the increment, since it met the criteria.

C Raise a formal change request before changing anything.

D Treat it as a team failure and review how the work was done.

◆ PAUSE AND DECIDE

What is the best response?

A Update the backlog; the review worked.

Finding this early, cheaply, is the purpose of the review.

B Insist on acceptance.

Wins the argument, loses the value.

C Formal change request.

Backlog changes need no change request inside the boundary.

D Team failure.

The team did what was asked; the learning is about the need.

THE SIGNAL

Process domain. Feedback that the product misses the need is the review succeeding. Adapt the backlog.

> An agile team has started using an AI coding assistant for production code. The legal team says the tool's licence terms on ownership of generated code have not been reviewed.
>
> The organisation's policy requires legal review of new software licences.

What should the project leader do?

A Continue, since the tool saves time and the risk seems small.

B Pause its use for production code, and arrange the required legal review of the licence.

C Ban all AI tools permanently.

D Let each developer decide whether to use it.

◆ PAUSE AND DECIDE

What should the project leader do?

A Carry on.

Breaches policy, with unclear ownership of the product's code.

B Pause for production; get the review.

Follows policy and protects intellectual property, without banning the tool.

C Ban AI permanently.

Overreacts to an unreviewed licence.

D Each decides.

Leaves a policy requirement to individual choice.

THE SIGNAL

Business Environment domain. New AI tools still follow licence and IP policy. Pause, review, then decide.

> A project manager takes over a predictive project. The previous project manager reported green status for three months, but the new project manager finds the schedule is actually six weeks behind.
>
> The next steering meeting is in two days.

What should the new project manager do?

A. Keep reporting green while trying to recover the six weeks.

B. Report amber for now, and reveal the full delay later.

C. Blame the previous project manager in the steering meeting.

D. Report the true status promptly to the sponsor, with the evidence and a recovery plan.

◆ PAUSE AND DECIDE

What should the new project manager do?

A Keep reporting green.
Continues a false report, now knowingly.

B Amber, full truth later.
Still misleading; drip-feeding bad news erodes trust.

C Blame the predecessor.
Blame does not help anyone decide what to do now.

D True status, evidence, recovery plan.
Honest, prompt, and constructive: rebuilds trust.

THE SIGNAL

People domain. Trust starts with honest status. Bring the truth early, with evidence and a plan.

> A predictive project is evaluating bids for a data platform. The lowest-priced bid fails a mandatory security criterion stated in the request for proposal. The next bid meets every criterion but costs 8 percent more.
>
> The sponsor asks whether the cheaper bid can be accepted anyway.

What should the project manager advise?

A. Accept the cheaper bid and ask the seller to fix security later.

B. Re-tender the whole contract.

C. Exclude the non-compliant bid, as the stated criteria require, and select the best compliant bid.

D. Drop the security criterion for all bidders.

◆ PAUSE AND DECIDE

What should the project manager advise?

A Accept and fix later.
Breaks the stated criteria, and unfair to compliant bidders.

B Re-tender.
Unnecessary; a compliant bid exists.

C Exclude it; choose best compliant bid.
Applies the published criteria fairly and consistently.

D Drop the criterion.
Changes the rules after bids are in; unfair and risky.

THE SIGNAL

Process domain. Mandatory criteria are applied as published. Fairness to every bidder, and the security need.

> During sprint planning, the team realises one story, 'customers can manage their whole account online', is too large to finish in a sprint.
>
> The product owner wants some value from it this sprint.

What should the team do?

A Split it into design, build and test tasks across three sprints.

B Split it into smaller stories, each delivering a usable slice of value, such as 'customers can update their address'.

C Extend the sprint to fit the whole story.

D Start the whole story and carry the remainder into the next sprint.

◆ PAUSE AND DECIDE

What should the team do?

A Split by phase.
Horizontal slices deliver no usable value until the end.

B Split into vertical, valuable slices.
Each small story is usable and testable on its own.

C Extend the sprint.
Breaks the timebox.

D Start it all; carry over.
Nothing finished, nothing of value this sprint.

THE SIGNAL

Process domain. Split stories vertically, so each slice delivers usable value.

> An agile team has debated for two weeks which test automation tool to adopt. Two options remain, both acceptable, and the debate has become circular.
>
> Work that depends on the choice is waiting.

What should the scrum master do?

A Help the team timebox the decision, agree criteria and a decision method, with a fallback if no consensus is reached.

B Choose a tool for the team to end the debate.

C Let the debate continue until everyone agrees.

D Ask the product owner to choose the tool.

◆ PAUSE AND DECIDE

What should the scrum master do?

A Timebox, criteria, method, fallback.

Facilitates a decision the team owns, without endless debate.

B Scrum master chooses.

Takes a technical decision away from the developers.

C Wait for unanimity.

Two weeks lost already; blocked work keeps waiting.

D Product owner chooses.

Tooling is how, not what: the developers' call.

THE SIGNAL

People domain. Circular debate between acceptable options: timebox, agree a method, keep a fallback.

Scoring guide

DOMAIN	QUESTIONS	YOUR SCORE
People	1, 5, 7, 10, 14, 17, 20	out of 7
Process	2, 3, 6, 8, 11, 15, 18, 19	out of 8
Business Environment	4, 9, 12, 13, 16	out of 5
Total	20 questions	out of 20

ON THE EXAM

Add this to your exam 1 and 2 results. Three data points show a trend; exam 4 reads it.

What to do with your result

- **Around 16 or more**: strong. Take exam 4 under full exam conditions.
- **Around 13 to 15**: revisit the modules named in your missed walkthroughs.
- **Below 13**: focus on your weakest domain across all three exams.
- Check question 2: did you spot that an obsolete goal changes exam 1's answer?
- These bands are a study guide, not PMI's standard.

> ON THE EXAM
>
> Write down your three domain scores for each exam. Exam 4 will ask you to compare them.

Three exams, three data points. **Record your domain scores**, then take exam 4.

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PRACTICE EXAM 4 OF 4 · FOUNDATIONS - ALL DOMAINS

Practice exam 4

The final twenty questions, then a guide to reading your results across all four practice exams, and a close to the course.

QUESTIONS 20 PEOPLE 6 PROCESS 8 BUS ENV 6

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> At the end of the design phase of a predictive project, updated market research shows the expected benefits have fallen by half. At that level, the business case no longer meets the organisation's investment threshold.
>
> The phase gate review is next week.

What should the project manager do?

A Leave the market research out of the gate pack, since design is complete.

B Stop the project immediately.

C Present the updated business case at the gate with options, including stopping or re-scoping, for governance to decide.

D Proceed to build, and review the benefits after launch.

◆ PAUSE AND DECIDE

What should the project manager do?

A Omit the research.
Hides decisive information from the people deciding.

B Stop now.
Not the PM's decision; that is what the gate is for.

C Updated case and options at the gate.
Gives governance the evidence and choices it needs.

D Build, review later.
Spends the build budget on a case that no longer holds.

THE SIGNAL

Business Environment domain. A gate is a decision point. Bring the updated case, honestly, with options.

> A senior developer's code review comments are technically correct but blunt and dismissive. Two junior developers have become reluctant to submit their code for review.
>
> The senior developer seems unaware of the effect.

What should the project leader do?

A. Give the senior developer private, specific feedback on the effect of the comments, and agree constructive review norms with the team.

B. Remove the senior developer from code reviews.

C. Tell the junior developers to be less sensitive.

D. Stop requiring code reviews for junior developers.

◆ PAUSE AND DECIDE

What should the project leader do?

A Private feedback; review norms.

Addresses the behaviour and its impact, and sets shared standards.

B Remove him from reviews.

Loses his expertise, and he never learns why.

C Juniors toughen up.

Dismisses a real effect on quality and safety.

D Drop reviews for juniors.

Removes a quality control from the people who need it most.

THE SIGNAL

People domain. Specific, private feedback on impact; then team norms. Keep the expertise, fix the delivery.

> A predictive project manager learns that a key supplier closes for two weeks every August. The schedule shows the supplier delivering critical components during those weeks.
>
> It is now April.

What should the project manager do?

A Ask the supplier to stay open in August for this project.

B Wait until July to see whether the supplier's plans change.

C Find a new supplier immediately.

D Update the resource calendar, re-sequence or move the deliveries, and communicate the revised plan.

◆ PAUSE AND DECIDE

What should the project manager do?

A Ask them to stay open.
Unlikely, and the calendar is a known constraint.

B Wait until July.
Loses months of room to re-plan.

C New supplier.
Drastic for a predictable, plannable closure.

D Update the calendar; re-sequence; communicate.
Plans around a known constraint while there is time.

THE SIGNAL

Process domain. A known availability gap belongs in the resource calendar, and the schedule is planned around it.

> Sprint planning sessions on an agile team now take six hours, because most stories arrive unclear and the team spends the time working out what they mean.
>
> The team is frustrated.

What should the scrum master do?

A Shorten sprint planning to two hours, regardless of clarity.

B Help the product owner and team hold regular backlog refinement, so stories are ready before planning.

C Ask the product owner to write every story in full detail alone.

D Skip sprint planning and let developers pick stories as they go.

◆ PAUSE AND DECIDE

What should the scrum master do?

A Cut the time only.
Unclear stories still enter the sprint.

B Regular backlog refinement.
Stories arrive ready, so planning is quick.

C PO writes everything alone.
Loses the shared understanding refinement builds.

D Skip planning.
No sprint goal, no plan.

THE SIGNAL

Process domain. Long planning means unready stories. Refine the backlog together, before planning.

A hybrid project's product owner asks the team to add a feature that records customers' precise locations. No privacy impact assessment has been done for it.

The organisation's data policy requires one for new personal data.

What should the project manager do?

A Arrange the privacy impact assessment with the data protection officer before the feature is built.

B Build the feature, and assess privacy before release.

C Refuse the feature, since location data is sensitive.

D Leave the decision to the development team.

◆ PAUSE AND DECIDE

What should the project manager do?

A Assessment before building.
Privacy by design: the policy's requirement, applied early.

B Build, then assess.
Risks building something that must be redesigned.

C Refuse it.
The feature may be fine with the right safeguards.

D Team decides.
A policy requirement is not a team preference.

THE SIGNAL

Business Environment domain. New personal data: assess privacy first. Compliance is designed in, not added.

> A hybrid project needs advice from a security architect in another division. She has no obligation to help, and her own work is busy.
>
> The project manager has no authority over her.

What is the best approach?

A Ask the sponsor to order her to help.

B Proceed without security advice.

C Meet her, explain the project and what is in it for her division, ask for a small, specific amount of help, and involve her manager if needed.

D Copy her manager on every request to create pressure.

◆ PAUSE AND DECIDE

What is the best approach?

A Sponsor orders her.

Forces compliance, and strains a relationship you will need again.

B Go without advice.

Takes on security risk for no reason.

C Relationship, mutual benefit, specific ask.

Influence without authority, done well.

D Pressure via her manager.

Damages trust before any real conversation.

THE SIGNAL

People domain. Influence without authority: build the relationship, show mutual benefit, make a small, clear ask.

A predictive project team is split between building a document management system in-house and buying a software-as-a-service product. The debate is based on opinions.

The sponsor wants a recommendation.

What should the project manager do?

A Choose the in-house option, since the team prefers it.

B Carry out a make-or-buy analysis covering total cost, capability, risk, time and strategic fit, and recommend from it.

C Choose the cheapest upfront option.

D Ask the software vendor for its recommendation.

◆ PAUSE AND DECIDE

What should the project manager do?

A Team preference.
Opinion, not analysis.

B Make-or-buy analysis.
Replaces opinion with evidence across the factors that matter.

C Cheapest upfront.
Ignores running costs, risk and fit.

D Ask the vendor.
A seller is not a neutral adviser.

THE SIGNAL

Process domain. Build or buy is decided by make-or-buy analysis, not preference or upfront price.

> A fully remote agile team member has become quiet. She keeps her camera off, contributes little in meetings, and recently said she feels 'out of the loop'.
>
> Her work quality is still good.

What should the scrum master do?

A Require all team members to keep cameras on.

B Leave her alone, since her work is good.

C Raise her low participation in the retrospective.

D Check in with her one to one, and increase connection, such as virtual pairing and informal team time.

◆ PAUSE AND DECIDE

What should the scrum master do?

A Cameras mandatory.
A rule, not a remedy for isolation.

B Leave her.
She has said she feels out of the loop; that will grow.

C Raise it in the retrospective.
Puts a personal concern on public display.

D One-to-one; build connection.
Understands her, and reduces isolation in practical ways.

THE SIGNAL

People domain. Remote isolation is solved with connection: one-to-ones, pairing and informal time.

> A national transport strike is announced for next month. A predictive project depends on deliveries of heavy equipment by road during that period.
>
> The strike could last one to three weeks.

What should the project manager do?

A Assume the strike will be called off.

B Postpone the whole project until after the strike.

C Assess the impact, update the risk register, and plan responses such as earlier delivery or alternative transport.

D Ask the supplier to absorb any delay.

◆ PAUSE AND DECIDE

What should the project manager do?

A Assume it is called off.
Hope is not a response.

B Postpone everything.
Disproportionate for a one-to-three-week risk.

C Assess; update risks; plan responses.
A proportionate response to a known external threat.

D Supplier absorbs it.
Shifts blame, but the equipment still does not arrive.

THE SIGNAL

Business Environment domain. A known external disruption: assess, register, and plan responses early.

> Stakeholders ask an agile team for a twelve-month roadmap showing exactly which features will ship on which dates. The team works in two-week sprints and expects priorities to change.
>
> The stakeholders need to plan marketing and training.

What should the project leader provide?

A. A roadmap with near-term detail and later themes, showing date ranges, updated at each review.

B. A fixed, feature-by-date plan for all twelve months.

C. No roadmap, since agile teams do not plan that far ahead.

D. Only the next sprint's plan.

◆ PAUSE AND DECIDE

What should the project leader provide?

A Detailed near, themed far, with ranges.

Useful for planning, honest about uncertainty, kept current.

B Fixed twelve-month plan.

False precision that will be wrong within weeks.

C No roadmap.

Agile teams do roadmap; stakeholders have real planning needs.

D Next sprint only.

Too short for marketing and training plans.

THE SIGNAL

Process domain. Agile roadmaps: detail near, themes far, ranges not promises, updated often.

> A predictive data migration project has a team spending weeks correcting old records by hand. The work is repetitive, and morale and accuracy are falling.
>
> The deadline cannot move.

What should the project manager do?

A Offer overtime pay to keep people going.

B Explain why the work matters, rotate tasks, automate repetitive checks where possible, and recognise progress.

C Replace the team with contractors.

D Monitor each person's error rate and report it weekly.

◆ PAUSE AND DECIDE

What should the project manager do?

A Overtime pay.
More hours of tedium; accuracy falls further.

B Purpose, variety, automation, recognition.
Tackles both motivation and the repetitive work itself.

C Replace the team.
Loses knowledge, and the work stays just as tedious.

D Individual error reports.
Surveillance lowers morale further.

THE SIGNAL

People domain. Repetitive work: give purpose, variety and recognition, and automate what you can.

> In a hybrid project, the team discovers problems but reports them only when they can no longer be hidden. Team members say the organisation punishes people who bring bad news.
>
> The project manager is new to the organisation.

What should the project manager do?

A Report the team's behaviour to the sponsor.

B Introduce penalties for hiding problems.

C Accept this as the organisation's culture.

D Make it safe to raise problems within the team, report honestly upwards, and work with the sponsor on how bad news is received.

◆ PAUSE AND DECIDE

What should the project manager do?

A Report the team.
Confirms their fear of punishment.

B Penalties for hiding.
More fear, which drives problems further underground.

C Accept it.
Late bad news will keep hurting the project.

D Safety within; honesty upwards; work with sponsor.
Changes what the PM can, and influences what they cannot.

THE SIGNAL

Business Environment domain. Assess the culture; build safety where you can, and influence the rest honestly.

> A supplier offers a predictive project a 10 percent discount on materials if the order is placed two months early. The site has secure storage, and cash flow allows early payment.
>
> The discount would save more than the storage costs.

What should the project manager do?

A. Decline, since the order is not due for two months.

B. Place the order immediately without checking anything else.

C. Confirm the materials' specification is stable and any storage risks are covered, then take the opportunity with the necessary approval.

D. Ask the supplier for a 20 percent discount instead.

◆ PAUSE AND DECIDE

What should the project manager do?

A Decline.
Ignores a real saving for no reason.

B Order unchecked.
Risks buying materials whose specification may still change.

C Check specification and storage; take it.
Exploits an opportunity after a quick check of its risks.

D Push for 20 percent.
May lose a good offer on a gamble.

THE SIGNAL

Process domain. Opportunities are managed like risks: check them quickly, then act to capture the value.

> A hybrid project must choose a new collaboration tool that the team will use every day. Team members ask to have a say in the choice.
>
> The budget and security requirements are fixed.

What should the project manager do?

A Involve the team in setting evaluation criteria and testing shortlisted tools, within the fixed budget and security requirements.

B Choose the tool personally to save time.

C Let the team choose any tool it likes.

D Ask the IT department to choose without consulting the team.

◆ PAUSE AND DECIDE

What should the project manager do?

A Involve the team within constraints.
Empowers the users, while keeping fixed requirements fixed.

B PM chooses.
Quick, but the daily users have no say.

C Any tool.
Ignores the fixed budget and security requirements.

D IT chooses alone.
Excludes the people who will use it all day.

THE SIGNAL

People domain. Empower within boundaries: involve users in the choice, inside fixed constraints.

> A supplier on a predictive project claims an extra $80,000 for work it says was outside the contract's scope. The project manager believes the work was included.
>
> The contract has a dispute resolution clause requiring negotiation, then mediation.

What should the project manager do?

A. Pay the claim to avoid a dispute.

B. Review the contract and records with procurement, negotiate, and follow the dispute resolution clause if needed.

C. Start legal action immediately.

D. Withhold all payments to the supplier until it drops the claim.

◆ PAUSE AND DECIDE

What should the project manager do?

A Pay it.
Pays for work that may already be included.

B Review, negotiate, follow the clause.
Evidence first, then the contract's own route.

C Sue now.
Skips the negotiation and mediation the contract requires.

D Withhold everything.
May breach the contract on undisputed payments.

THE SIGNAL

Business Environment domain. Claims are settled from the evidence, by the contract's dispute route, in order.

> An agile team integrates everyone's code only at the end of each sprint. Each time, two or three days are lost resolving conflicts and broken builds.
>
> The sprint review is often rushed as a result.

What should the team change?

A Lengthen sprints to allow more integration time.

B Appoint one person to handle all integration.

C Integrate once a week instead of once a sprint.

D Adopt continuous integration, merging and testing small changes many times a day.

◆ PAUSE AND DECIDE

What should the team change?

A Longer sprints.
Bigger batches, bigger integration pain.

B One integrator.
A bottleneck, and still big-batch merging.

C Weekly integration.
Better, but still batching the problem.

D Continuous integration.
Small, frequent merges make conflicts small and early.

THE SIGNAL

Process domain. Integration pain grows with batch size. Integrate continuously.

> A predictive project has a detailed, approved scope and a complete work breakdown structure. The sponsor needs the most accurate possible cost estimate to secure funding.
>
> The team has time to prepare it.

Which estimating approach should the project manager use?

A Analogous estimating from a similar past project.

B An order-of-magnitude estimate.

C Bottom-up estimating from the work packages, rolled up.

D Ask the sponsor what figure the funding committee expects.

◆ PAUSE AND DECIDE

Which estimating approach should the project manager use?

A Analogous.
Quick, but less accurate; suits early stages.

B Order of magnitude.
The least accurate kind, for very early decisions.

C Bottom-up from work packages.
The most accurate when a full WBS exists and time allows.

D Ask what they expect.
Anchoring, not estimating.

THE SIGNAL

Process domain. Full WBS, time available, accuracy needed: bottom-up estimating.

> A portfolio board cuts a hybrid project's remaining funding by 20 percent to fund a higher-priority initiative. The end date is unchanged.
>
> The sponsor asks the project manager to respond.

What should the project manager do?

A. Work with the sponsor and product owner to prioritise the highest-value scope within the new budget, and update the baselines through change control.

B. Keep the full scope and hope to deliver it for less.

C. Appeal to the portfolio board to reverse its decision.

D. Cut 20 percent from every workstream equally.

◆ PAUSE AND DECIDE

What should the project manager do?

A Prioritise by value; rebaseline properly.
Keeps the most value within the new constraint, formally.

B Same scope, less money.
Hope, not a plan; quality or schedule will suffer.

C Appeal.
The board has made a portfolio decision for strategic reasons.

D Equal cuts.
Treats high- and low-value work the same.

THE SIGNAL

Business Environment domain. A portfolio decision changes the constraint. Re-prioritise by value; rebaseline formally.

> On a hybrid project, an offshore team always agrees to proposed deadlines in meetings, but often misses them. A team lead explains that in their culture, saying 'no' to a manager is considered disrespectful.
>
> The project manager wants reliable commitments.

What should the project manager do?

A Insist that team members say 'no' directly when a deadline is unrealistic.

B Create safer ways to raise concerns, such as asking open questions, private check-ins and team-owned estimates.

C Add a buffer to every deadline the offshore team agrees.

D Escalate the missed deadlines to the offshore team's management.

◆ PAUSE AND DECIDE

What should the project manager do?

A Insist on direct 'no'.
Asks people to act against their culture; unlikely to work.

B Safer ways to raise concerns.
Adapts to the culture so real information surfaces.

C Hidden buffers.
Plans around the problem instead of fixing communication.

D Escalate.
Treats a cultural difference as poor performance.

THE SIGNAL

People domain. Cultural awareness: change how you ask, so the truth can be told comfortably.

> Two days after a hybrid project's release, customers report a defect that stops some payments. The fix is small, and the organisation has an emergency change process.
>
> The team wants to fix it quickly.

What should the project manager do?

A Wait for the next planned release to include the fix.

B Let a developer fix it directly in production, without recording it.

C Roll back the entire release.

D Fix it through the emergency change process, then find the root cause and prevent recurrence.

◆ PAUSE AND DECIDE

What should the project manager do?

A Wait.
Customers cannot pay in the meantime.

B Unrecorded fix.
Uncontrolled change in production; no record, no review.

C Roll back everything.
Disproportionate for a small, fixable defect.

D Emergency change; then root cause.
Fast and controlled, then prevention.

THE SIGNAL

Process domain. Urgent production fixes use the emergency change route, then root cause analysis.

Scoring guide

DOMAIN	QUESTIONS	YOUR SCORE
People	2, 6, 8, 11, 14, 19	out of 6
Process	3, 4, 7, 10, 13, 16, 17, 20	out of 8
Business Environment	1, 5, 9, 12, 15, 18	out of 6
Total	20 questions	out of 20

ON THE EXAM

Across all four exams: People 26 questions, Process 33, Business Environment 21. Add up each domain.

Reading your trend across four exams

IF YOUR RESULTS SHOW	THEN
Scores rising from exam 1 to 4	The method is working. Keep practising; book your exam
One domain low in three or four exams	Rewatch that domain: 06-16, 17-30, or 31-40
The same distractor pattern, repeatedly	Rewatch modules 02 and 44; name the pattern every time
Misses on the paired questions	Slow down on the deciding detail; rewatch module 42
Flat, middling scores	Re-read missed walkthroughs; retake that exam in a week

ON THE EXAM

Where your misses cluster, and the pattern they share, matters more than the total.

Before your exam

- Aim for consistent results across **all three domains**, not one high score. PMI publishes no pass mark.
- Practise with a **full-length timed exam** from a 2026-aligned source at least once.
- Re-read **PMI's exam handbook** in the final week, and check your ID and appointment details.
- On the day: find the decision, read the qualifier, name the distractor, climb the ladder.

> ON THE EXAM
>
> Consistency across domains, a full-length rehearsal, and the handbook in the final week.

Understand, assess, collaborate, act, escalate. **Lead the people. Deliver the value.**